

Structure and Reliability in Interactive Decision Making

by

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This thesis studies how structural assumptions determine what can be learned reliably in interactive learning. Here a learner acts, receives feedback, updates its behavior, and thereby changes the data it sees in the future. This feedback loop drives settings such as bandits, reinforcement learning, and the post-training of foundation models. In high stakes settings, it is important for algorithms targeting these settings to be backed by guarantees. The central claim of this thesis is that challenges in such settings become tractable only when the right structural object is identified.

The first part of the thesis develops fundamental theory for multi-agent interactive learning. In the second chapter, we study decentralized multi-agent Lipschitz bandits in continuous action spaces with hard collisions. We introduce a communication-free, multi-phase protocol that achieves the single-agent rate through maxima-directed identification. In the third chapter, we study multi-agent reinforcement learning under networked dependence. We show that locality in such scenarios can be a policy-induced property in addition to depending on the environment.

The second part turns to modern questions in interactive learning applied to foundation-model post-training. In the fourth chapter, we analyze post-training of autoregressive models through the lens of generator access. We prove that when interaction is limited to root-start rollouts, rich observations such as token log probabilities remain bottlenecked by on-policy reachability, whereas even weak prefix reset breaks this barrier. In the fifth chapter, we study data attribution in interactive learning, where perturbing a training occurrence can update the learner and change future data. We formalize targets for algorithms in these settings to estimate, prove that purely offline information is insufficient to identify these in general, and show where identification becomes possible. Together, these results provide a unified theoretical account of structure, locality, access, and reliability in modern interactive learning.

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- 1.1 An overview of the thesis. The work bridges the foundational mathematical principles of multi-agent interactive learning (Part I) to the modern complexities of aligning large language models (Part II), united by a focus on structural assumptions and reliability in interactive learning. 3

Chapter 1

Introduction

In machine learning, the traditional supervised learning paradigm relies on a fixed and static dataset. A learning algorithm learns passively by identifying patterns in data that was collected and curated long before the training process began. In **Interactive Learning**, the learner acts as an autonomous agent in an environment and generates its own data through trial and error. The agent takes an action, observes the response from the environment, and updates its internal behavior. This creates a continuous feedback loop where the agent's current decisions dictate the data it will see tomorrow, and that new data will shape its decisions the day after.

The simplest mathematical framework to study this interaction is the **multi-armed bandit problem**. Imagine a gambler facing a row of slot machines where each machine, or arm, yields a reward sampled from an unknown probability distribution. The gambler must decide which arm to pull at each time step. This introduces the core challenge of interactive learning known as the **exploration-exploitation dilemma**. The agent must exploit the actions that have yielded high rewards in the past, but it must also explore unknown actions to discover if they offer even greater payoffs. We measure the success of an algorithm in such settings using a metric called **Regret**. It is defined as the difference between the total reward the agent actually accumulated and the total theoretical reward it could have accumulated had it known the single best action from the very beginning.

The bandit setting models the exploration-exploitation dilemma by isolating the choice of action from its effect on the state of the agent itself, i.e., actions are assumed to be independent at

each time step. For instance, in the slot machine example, the setup assumes that if the gambler chose action 1 at time step t and got a low reward, their internal state remains unaffected and thus they will choose the next action independently. This need not hold in practical settings.

Reinforcement Learning extends the concept of interactive learning to sequential decision making where an action taken today impacts the situations the agent will face tomorrow by adding the concept of a state. This is formalized using the language of **Markov Decision Processes**. In this framework, the environment exists in a specific state. The agent observes this state and chooses an action. The environment then transitions to a new state and provides a reward based on the action taken. The ultimate goal of the agent is to learn a **policy**, which is a mathematical mapping from states to actions, that maximizes long-term cumulative reward.

As these environments become more complex, the number of possible states and actions grows exponentially. For instance, a robot deployed to operate on a factory floor might discretize the floor into a grid and treat each position as a distinct state, resulting in a massive state space. Similarly, the robot deciding exactly how to move its actuators faces a massive action space. These large spaces also appear in multi-agent scenarios. While each individual agent might have a manageable number of states and actions, their combined joint state and action space is overwhelmingly large. In a more modern context, reinforcement learning has been applied to language models. Here, predicting the next word is treated as taking an action. Because a modern language model can generate a practically infinite number of sentences, the resulting environment is incredibly vast. This exponential explosion in complexity causes learning algorithms to become drastically slower and is known as the **curse of dimensionality**.

In practice, algorithm designers often rely on empirical heuristics to navigate these massive spaces and be able to apply traditional algorithms. Deep neural networks and clever engineering techniques frequently yield impressive results and drive modern applications. However, if we want to deploy these systems in high-stakes environments, empirical success is not enough. We need systems that are reliable and trustworthy. We must understand exactly what mathematical assumptions guarantee safe and optimal behavior.

This thesis approaches this challenge by focusing on structure and reliability. By identifying the underlying mathematical structure of an environment, one hopes to pick the correct assumptions and make theoretically intractable problems tractable. By designing algorithms that respect this structure, we can provide rigorous theoretical guarantees regarding their reliability. The thesis is divided into two main parts. The first part focuses on foundational aspects of bandits and reinforcement learning theory. The second part turns to more modern applications of interactive learning in the context of training large language models. The roadmap has one central theme, interactive learning, which branches into a foundations part covering multi-agent bandits and MARL and a modern part covering post-training interfaces and data attribution. Figure 1.1 summarizes this organization.

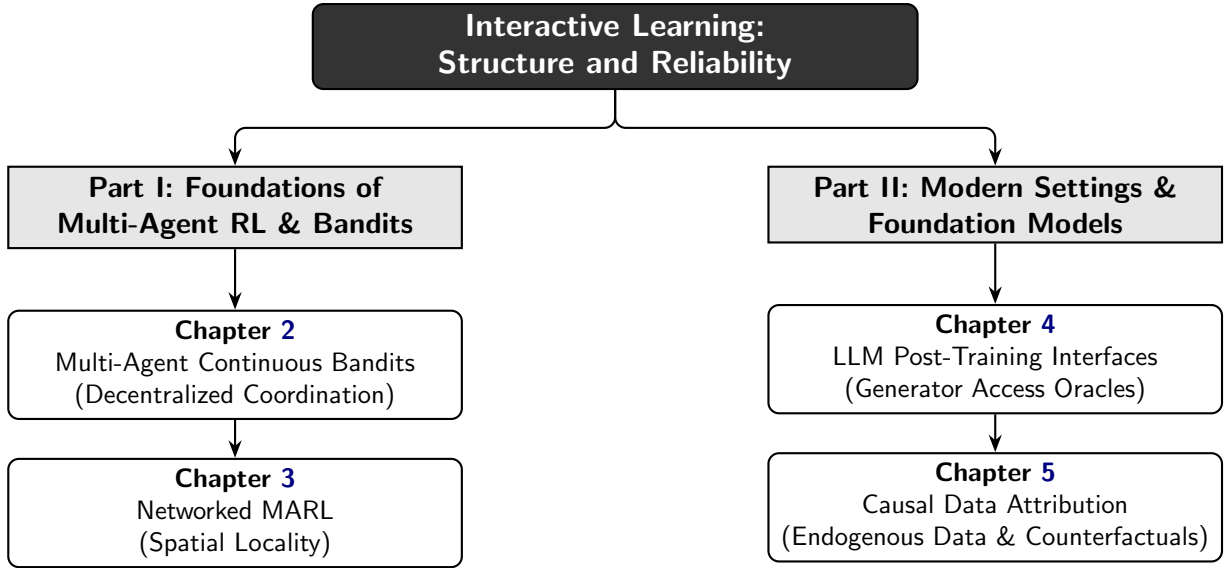


Figure 1.1: An overview of the thesis. The work bridges the foundational mathematical principles of multi-agent interactive learning (Part I) to the modern complexities of aligning large language models (Part II), united by a focus on structural assumptions and reliability in interactive learning.

1.1 Foundations of Multi-Agent Bandits and Reinforcement Learning

While classical frameworks provide a strong foundation for a single learner, deploying these algorithms in the real world often involves multiple autonomous agents interacting within the exact

same environment. In a multi-agent setting, the fundamental assumption of a purely responsive environment breaks down. As each agent explores and updates its own behavior, it alters the environment for everyone else. This creates a highly dynamic and non-stationary system. The agents must not only learn to maximize their individual rewards but also navigate the complex dynamics of coordination, interference, and exponentially larger joint spaces. This first part of the thesis explores these foundational challenges and focuses on how we can design reliable algorithms that scale from localized continuous interactions to complex networked systems.

Chapter 2 investigates the challenge of decentralization in continuous action spaces. We study a scenario where multiple autonomous agents operate in a shared physical environment and face the multi-armed bandit problem cooperatively. When multiple agents choose the exact same action or spatial region at the same time, they interfere with each other. This interference is called a collision and is traditionally modeled by a zero reward for everyone involved, thereby ruining the learning opportunity for agents. While multi-agent bandits in discrete action spaces possess a rich literature, we initiate the study of multi-agent scenarios in the continuous setting. This continuous approach is arguably more practical because it avoids the need to arbitrarily discretize the physical world.

In a basic discrete setting with a few distinct choices, avoiding collisions is relatively straightforward. When the action space is a continuous domain like physical space or radio frequencies, defining and avoiding collisions becomes highly complex. A naive approach might divide the space into a simple grid, but this creates biases if the true optimal actions lie near the artificial boundaries of the grid cells. To solve this, we introduce a protocol that breaks the problem into distinct phases, separating the task of coordinating the agents from the task of optimizing their individual rewards. By deploying local exploratory probes inside candidate regions, the agents can identify the true optimal actions without needing to communicate with one another. We combine this with a decentralized seating protocol to prove that agents can achieve safe coordination in an expected time that scales linearly with the number of players. We establish a gap-free end-to-end regret bound that matches the optimal single-player learning rate. This proves that we can relegate the

costs of decentralized coordination to a constant factor that is completely independent of the time horizon.

Chapter 3 transitions from the mobile agents and continuous action spaces of our single-step bandit setting to full reinforcement learning within multi-agent networks. Unlike the previous scenario where agents could move freely around a shared physical space, here the agents occupy fixed relative locations within a spatial network. Here the challenge is not interference but rather that the joint state and action spaces are exponential in the number of agents. In contrast, Chapter 2 only dealt with a large action space, since bandit problems have no state.

This massive joint state space renders traditional reinforcement learning algorithms far too slow for practical use. To make learning mathematically tractable, researchers often assume that the environment satisfies some continuity assumption. One common assumption in the networked multi-agent reinforcement learning (MARL) setting is that of spatial locality. This means the influence of distant agents decays rapidly as the distance between them increases on the network graph. While this exponential decay property is heavily used to justify truncating algorithms to local neighborhoods, it is traditionally treated as a worst-case assumption about the environment itself.

We argue that this traditional view is too conservative. Locality is not merely a property of the environment’s physics. It is an emergent property that depends heavily on the behavioral logic of the policies the agents execute. We introduce a new theoretical decomposition that separates the environment’s coupling from the policy’s sensitivity. We prove that a smooth and exploratory policy can induce strict locality even if the underlying environment is highly interconnected. This provides a formal and deterministic justification for localized algorithms. It also exposes a direct mathematical tradeoff between how optimal a policy is and how locally it can be computed.

1.2 Modern Settings and the Post-Training of Foundation Models

The second part of the thesis bridges these foundational principles to the modern era of foundation models. Large language models operate by predicting the next word or token in a

sequence based on the previous tokens. We can naturally view this sequence generation process as taking an action in a reinforcement learning environment. The state is the current context (prompt and generated tokens so far), the action is the next generated token, and the transition moves the system to a new context with the appended token. Because these models are initially trained to simply mimic vast amounts of internet text, they must undergo a post-training phase using reinforcement learning to align their behavior with human preferences and safety guidelines.

Chapter 4 examines the structural interfaces used during the autoregressive post-training of large language models. While the initial training phase (called pretraining) teaches a model to mimic internet text, the post-training phase is where we actually align the model to follow instructions and act safely. To do this efficiently, learning algorithms need to interact with the base model. However, the overall efficiency of these algorithms depends heavily on the type of information and access they are granted. Researchers often debate the optimal design for this interface. A common question is whether the learning agent needs access to the full internal probability scores of the model or if simply receiving the generated text samples is enough.

We argue that this debate misses a more important source of difficulty. What matters most is how much control the algorithm has over the generation process. We ask a critical question regarding whether the learner can force the model to generate text starting from a specific chosen prefix or whether it is strictly restricted to observing the natural trajectory of the model. This distinction between active control and passive observation is vital for understanding how to design algorithms that can learn from feedback without requiring an impractical amount of data.

We prove a universal mathematical barrier that clearly illustrates this dynamic. Our findings show that all purely observation-based interfaces suffer from the exact same exponential sample complexity as long as they lack prefix control. This means that without the ability to guide the starting point of the text generation, learning becomes overwhelmingly slow. Conversely, we demonstrate that granting the learner even a weak ability to reset the model to a chosen prefix completely shatters this barrier and restores efficiency. This proves that the fundamental difficulty of aligning language models is not a universal constant of the environment. Instead, it is strictly

relative to the specific oracle interface provided to the learner. In the remainder of the chapter, we define a taxonomy of oracle models and characterize their relative power.

Chapter 5 tackles the critical problem of causal data attribution in adaptive systems. When a deployed foundation model exhibits a specific behavior or failure, developers need a way to trace that outcome back to the exact moments in its training history that caused it. This process of linking final model behavior to individual training examples is known as data attribution. In traditional supervised learning, this is relatively straightforward. Researchers can mathematically simulate removing a single data point and calculate how the final model would change. This works perfectly because the training dataset is entirely fixed and independent of the learning algorithm.

In the realm of interactive learning and post-training, this standard approach completely breaks down due to a circular dependence between the model and its data. In these dynamic settings, the data is endogenous. This means the system generates its own training data as it learns. If we try to ask what would happen if we altered or removed a past interaction, we cannot simply replay the rest of the fixed training log. Altering that past interaction changes the learned policy of the agent at that exact moment. That slightly altered policy will then go on to make different decisions and collect a completely different set of future data. The data shapes the policy, and the policy actively shapes the future data.

We formalize this divergence between simply replaying a fixed future log and properly intervening to let the counterfactual future naturally evolve. We establish the exact mathematical conditions under which we can actually identify these counterfactual targets using only logged data. Furthermore, we prove impossibility theorems showing that a single observed training run is inherently insufficient to perform accurate data attribution without relying on additional structural assumptions.

By formally defining the boundaries of structure and reliability across continuous bandits, networked multi-agent systems, and generative language models, this thesis provides a rigorous theoretical foundation for the next generation of interactive learning systems.

Chapter 2

Multi-Agent Lipschitz Bandits

This chapter is based on the following paper:

Sourav Chakraborty, Amit Kiran Rege, Claire Monteleoni, and Lijun Chen. Multi-agent lipschitz bandits, 2026.

Abstract: We study the decentralized multi-player stochastic bandit problem over a continuous, Lipschitz-structured action space where hard collisions yield zero reward. Our objective is to design a communication-free policy that maximizes collective reward, with coordination costs that are independent of the time horizon T . We propose a modular protocol that first solves the multi-agent coordination problem—identifying and seating players on distinct, high-value regions via a novel maxima-directed search—and then decouples the problem into N independent single-player Lipschitz bandits. We establish a near-optimal regret bound of $\tilde{O}(T^{(d+1)/(d+2)})$ plus a T -independent coordination cost, matching the single-player rate. Our framework is, to our knowledge, the first to provide such guarantees and extends to general distance-threshold collision models.

The corresponding supplementary material can be found in [Appendix A](#).

2.1 Introduction

Many sequential decision-making problems involve multiple autonomous agents operating in a shared environment without a central controller. Consider a team of cognitive radios searching for unoccupied, high-quality frequency bands, or a fleet of drones coordinating to survey distinct,

high-value areas. In these scenarios, agents must learn the value of different actions from stochastic feedback, a classic exploration-exploitation dilemma. However, three fundamental challenges arise: the action space is often continuous, agents may interfere with each other through hard collisions, and they must act without direct communication.

This chapter addresses the confluence of these three challenges within the framework of multi-player stochastic bandits. We consider a cooperative setting where N players share a continuous action domain. The environment is partitioned into a finite set of regions. If two or more players choose actions in the same region at the same time, a "hard collision" occurs, and all colliding players receive zero reward and no information. This models contention for a rivalrous resource. The reward structure of the continuous domain is governed by an unknown but smooth function, which we model as being Lipschitz continuous. The goal for the collective is to maximize the total reward over a time horizon T .

Most prior work on multi-player bandits has focused on settings with discrete, finite action sets. While foundational, these models do not capture applications where actions are inherently continuous, such as setting a price, tuning a physical parameter, or choosing a location. The introduction of a continuous action space, combined with decentralization and collisions, presents a formidable challenge. A naive discretization of the action space would be computationally intractable and statistically inefficient. The Lipschitz structure is key, as it allows for generalization: the reward at one point provides information about rewards at nearby points. However, this structure also introduces a new subtlety: the value at the center of a region can be a poor proxy for the maximum value achievable within it, especially if the reward function has sharp peaks near region boundaries. An effective strategy must therefore be sensitive to the maxima of regions, not just their centers.

Our work provides a principled, end-to-end solution for this problem. We propose a fully decentralized, multi-phase algorithm that strategically decouples the problem of coordination from the problem of continuous optimization. The core idea is to invest a fixed, time-independent budget to solve the multi-agent coordination problem first: identifying the best set of N distinct

regions and assigning one player to each. Once this seating is achieved, the problem factorizes into N independent single-player Lipschitz bandit problems, which can be solved with near-optimal efficiency for the remaining duration.

Our contributions establish the first comprehensive and near-optimal solution for this problem. First, we introduce a modular, communication-free protocol that decouples coordination from learning. Its core innovation is a maxima-directed identification phase that performs a “local peek” inside candidate regions to bracket true cell suprema, provably avoiding the biases of simpler center-based rankings. Second, we provide the first analysis of the practical, decentralized Musical Chairs protocol where players are unaware of which cells are free, proving it seats all N players in an efficient $O(N)$ expected time. Third, we establish an end-to-end regret guarantee of $\tilde{O}(T^{(d+1)/(d+2)})$. Crucially, this optimal rate is achieved in a fully **gap-free** manner, removing the common assumption of a reward margin between optimal and suboptimal actions under a public scheduling assumption. Our bound cleanly decomposes regret into a horizon-independent coordination cost and a learning term that matches the optimal single-player rate. To prove the optimality of our approach, we derive a matching **lower bound**, confirming that our algorithm’s asymptotic performance cannot be improved. Finally, we demonstrate the framework’s versatility by extending it to handle more general distance-threshold collision models. To our knowledge, this is the first work to provide such guarantees for multi-player bandits in continuous domains.

2.2 Related Work

Multi-armed bandits have been studied extensively for decades [38, 2]. The literature on multi-agent bandits is vast; we highlight the main works introducing collisions, decentralization, and cooperative settings. In the discrete-arm case, Rosenski et al. [66] introduced the Musical Chairs protocol for decentralized coordination, later improved by Wang et al. [76] with optimal regret guarantees. Variants with delayed feedback, fairness, or corruption have also been explored [8, 19].

Our setting departs by considering a continuum of actions. Single-player Lipschitz bandits

have been studied via discretization, zooming, and hierarchical methods [47, 9, 36]. These works provide minimax rates but do not address multi-agent collisions.

Multi-agent bandits in continuous domains remain underexplored. Existing works often assume stronger structure (e.g., convexity, linearity) [4], or allow explicit communication [39].

In contrast, to our knowledge, our work is the first fully decentralized, communication-free solution for multi-player Lipschitz bandits with hard collisions. It achieves near-optimal regret with only a constant coordination overhead.

2.3 Preliminaries

We build upon three established areas in sequential decision-making: the stochastic multi-armed bandit problem, its extension to continuous arms via the Lipschitz assumption, and the multi-player variant with collisions. We briefly review each to establish notation and context.

2.3.1 Stochastic Multi-Armed Bandits

The canonical multi-armed bandit (MAB) problem involves a single player sequentially choosing from a set of K discrete actions, or “arms”. At each time step t , the player selects an arm $i_t \in \{1, \dots, K\}$ and receives a stochastic reward drawn from an unknown distribution with mean μ_i . The player’s goal is to maximize the cumulative expected reward over a horizon T . Performance is measured by the cumulative regret, defined as the expected difference between the reward from always playing the single best arm and the reward accumulated by the player’s policy:

$$R(T) = T \cdot \mu^* - \sum_{t=1}^T \mathbb{E}[\mu_{i_t}],$$

where $\mu^* = \max_{i \in \{1, \dots, K\}} \mu_i$. Minimizing regret requires balancing the exploration of arms to learn their mean rewards with the exploitation of the arm that currently seems best.

2.3.2 Lipschitz Bandits in Continuous Domains

When the set of actions is a continuous domain, such as $\mathcal{X} = [0, 1]^d$, the problem becomes intractable without further assumptions, as there are infinitely many arms to explore. The Lipschitz bandit model introduces structural smoothness. The unknown mean-reward function $\mu : \mathcal{X} \rightarrow [0, 1]$ is assumed to be L -Lipschitz with respect to a norm, typically the Euclidean norm $\|\cdot\|_2$:

$$|\mu(x) - \mu(y)| \leq L\|x - y\|_2 \quad \text{for all } x, y \in \mathcal{X}.$$

This condition ensures that the mean rewards of nearby points are similar, allowing an algorithm to generalize from a finite number of samples to the entire space. This structure makes the problem tractable, and algorithms for this setting can achieve near-optimal regret that scales as $\tilde{O}(T^{(d+1)/(d+2)})$, where the exponent depends on the dimension d of the action space.

2.3.3 Cooperative Multi-Player Bandits and Collisions

In the multi-player MAB setting, N players simultaneously choose from a common set of arms. We focus on the cooperative goal, where the objective is to maximize the sum of rewards across all players. A central challenge is handling collisions. In the "hard collision" model, if two or more players select the same arm (or region) in the same round, all colliding players receive zero reward. This creates an incentive for players to coordinate on distinct, high-value arms.

A simple and effective communication-free protocol for this coordination task is known as Musical Chairs. Once a set of N high-quality arms has been identified, the players must assign themselves to these arms without conflict. In the Musical Chairs protocol, each unassigned player repeatedly samples an arm uniformly from the target set. If a player lands on an arm that no one else chose in that round, they "seat" there and play that arm for the remainder of the game. This process continues until all players are seated. A crucial aspect of our analysis considers the practical implementation where players do not know which of the N target arms are already occupied, making their sampling choices over the full set of N target arms.

2.4 Problem Setup and Benchmarks

We consider a decentralized, cooperative stochastic bandit problem with N players acting over a shared, continuous action domain. Our goal is to design a communication-free policy that allows players to achieve near-optimal collective reward, where the costs of coordination are provably independent of the time horizon T .

2.4.1 Actions, Rewards, and Lipschitz Structure

The action space for each of the N players is the compact set $\mathcal{X} = [0, 1]^d$, endowed with the Euclidean norm $\|\cdot\|_2$. At each round $t = 1, 2, \dots, T$, every player $j \in [N]$ chooses an action $X_t^{(j)} \in \mathcal{X}$. The rewards are governed by an unknown mean-reward function $\mu : \mathcal{X} \rightarrow [0, 1]$ that is assumed to be L -Lipschitz continuous:

$$|\mu(x) - \mu(y)| \leq L\|x - y\|_2 \quad \text{for all } x, y \in \mathcal{X}.$$

When a player's action does not result in a collision, they observe a stochastic reward drawn from a distribution with mean $\mu(X_t^{(j)})$ and independent 1-sub-Gaussian noise. The Lipschitz property provides the essential smoothness structure that makes learning over a continuous space feasible.

2.4.2 A Tractable Collision Model for Continuous Spaces

Defining a meaningful collision model is a primary conceptual hurdle in continuous domains. If a collision were defined as two players selecting the exact same point, such an event would occur with zero probability, rendering the notion trivial. A practical model must instead capture the idea that players interfere when they operate in "proximate" regions of the action space.

As one of the first approaches for this new setting, we introduce a collision geometry based on a fixed, discretized partition of the space. We assume the action space $\mathcal{X}$ is partitioned into a set $\mathcal{P} = \{C_1, \dots, C_K\}$ of $K = \lceil 1/h \rceil^d$ disjoint hypercubic cells, each of side-length h . We assume there are enough cells to accommodate all players, $K \geq N$. A **hard collision** occurs for a set of

players if, at the same round, they all choose actions within the **same** cell $C \in \mathcal{P}$. When a collision occurs in a cell, every player involved receives a null observation, denoted $\perp$, and a reward of zero.

This partition-based model is a natural and analyzable abstraction for many real-world systems where operational zones are discrete by design. For example, in cognitive radio networks, the spectrum is divided into discrete channels; in logistics, a city is divided into service zones for delivery drones; and in cloud computing, resources may be allocated from discrete server clusters. In these cases, interference is determined by co-location within a predefined region, not just by continuous proximity.

While this partition model provides a clean and practical foundation, our algorithmic framework is sufficiently general to accommodate other geometries. In later sections, we will show how our approach extends to a distance-threshold model, where a collision occurs if any two players' actions are within a certain Euclidean distance ρ of each other. That model is more suited to applications like mobile robotics or sensor networks where interference is governed by physical proximity. By first solving the partition-based problem, we establish the core algorithmic principles in a clear and simple setting.

2.4.3 Performance Benchmark and Objective

The collision model imposes a fundamental constraint: at most one player can earn a non-zero reward from any given cell C in a single round. It is therefore inappropriate to compare the system's performance to the ideal single-player benchmark of $N \cdot \sup_{x \in \mathcal{X}} \mu(x)$, as this might require all N players to occupy the same infinitesimally small region. A principled comparator must respect this feasibility constraint.

We therefore define the benchmark based on the best possible static assignment of players to N distinct cells. For each cell $C \in \mathcal{P}$, let its optimal value be its cell-wise supremum, $\mu^*(C) := \sup_{x \in C} \mu(x)$. Let $\mu_{(1)}^* \geq \mu_{(2)}^* \geq \dots \geq \mu_{(K)}^*$ be these values sorted in nonincreasing order. The optimal collision-feasible reward in a single round is the sum of the top N cell maxima: $\text{OPT}_{\text{cont}}(\mathcal{P}, N) := \sum_{m=1}^N \mu_{(m)}^*$.

The cumulative regret of a decentralized policy π over a horizon T is the difference between this optimal benchmark and the expected total reward collected by all players:

$$R_{\text{cont}}(T; \pi, \mathcal{P}, N) := T \cdot \text{OPT}_{\text{cont}}(\mathcal{P}, N) - \mathbb{E}_{\pi} \left[\sum_{t=1}^T \sum_{j=1}^N r_t^{(j)} \right], \quad (2.1)$$

where $r_t^{(j)}$ is the realized reward for player j at round t . Note that the benchmark depends on the partition $\mathcal{P}$. This is an intentional feature of the model, not an artifact of the analysis; the partition defines the physical collision geometry of the environment.

A subtlety in this continuous setting is the difference between a cell's center value, $\mu(x_C)$, and its true maximum, $\mu^*(C)$. While the Lipschitz property guarantees they are close - $|\mu^*(C) - \mu(x_C)| \leq Lh\sqrt{d}/2$ - their relative ordering across cells can be completely different. For instance, a cell with a modest center value might contain a sharp peak near its boundary, making it more valuable than a cell with a higher center value that is relatively flat. Any successful algorithm must therefore be designed to identify cells based on their maxima, not just their centers.

We use $[m]$ for the set $\{1, \dots, m\}$. We prove high-probability correctness of identification phases; the final regret bound is in expectation with an explicit failure term. We define a total failure probability budget $\delta_{\text{sys}} \in (0, 1/4)$ and allocating portions of it, such as δ_{I} and δ_{II} , to different phases of the algorithm. This ensures the entire system behaves as expected with probability at least $1 - \delta_{\text{sys}}$.

Since our work is the first in extending multi-agent bandits to the Lipschitz, setting we have chosen the most fundamental of settings to study (decentralized, zero-reward collisions etc). We leave extensions to more advanced settings to future work.

2.5 Our Approach: A Multi-Phase Decentralized Protocol

Before going into technical details, we present a high-level overview of our strategy. The core idea is to decouple the multi-agent coordination problem from the single-agent continuous optimization problem. We achieve this with a four-phase protocol where the first three phases are dedicated to coordination and incur a total cost that is **independent** of the time horizon T .

- **Phase I: Coarse Identification.** For a fixed duration T_0 , all players explore the space by sampling cell centers uniformly at random. This process is intentionally chaotic and communication-free. While many samples will result in collisions, we show that with high probability, every player obtains a sufficient number of successful, non-colliding observations from every cell to construct coarse but statistically valid confidence bounds on each cell’s maximum value. The purpose of this phase is to safely prune the vast majority of suboptimal cells.
- **Phase II: Using Maxima for Refinement.** Using the candidate cells identified in Phase I, players perform a localized “peek” inside each one. They sample from a fine grid of points within each candidate cell to build high-resolution confidence bounds that tightly bracket the true cell maximum $\mu^*(C)$. This critical step corrects for the center-versus-maximum bias and allows us to identify the top- N cells, even without a gap between the best and the rest.
- **Phase II $_{\frac{1}{2}}$: Decentralized Seating.** Having agreed upon a common set of N target cells, players must assign themselves to these cells without conflict. They use the Musical Chairs protocol, where unseated players repeatedly sample from the target set until they land on a free cell. We analyze the practical version of this protocol and show that all players are seated in expected $O(N)$ time.
- **Phase III: Within-Cell Optimization.** Once each player is uniquely assigned to a high-quality cell, the multi-agent problem factorizes. For the remainder of the horizon, each player independently runs a single-player Lipschitz bandit algorithm confined to their assigned cell, efficiently optimizing their local reward.

This modular structure allows us to isolate and solve the challenges of coordination and learning sequentially, leading to a cleaner analysis.

2.6 Phase I: Coarse Identification

The first phase aims to solve a difficult task with a simple tool: uniform random exploration. The goal is for every player to obtain a coarse but reliable confidence bracket on each cell's maximum value, $\mu^*(C)$. The exploration is “collision-censored”—players make no attempt to avoid each other and instead rely on randomness to provide a sufficient number of non-collision events to learn from.

2.6.1 Phase I Protocol

This phase runs for a fixed budget of T_0 rounds. In each round $t \in [T_0]$, every player j independently draws a cell $C_t^{(j)}$ uniformly at random from the partition $\mathcal{P}$ and probes its center, $x_{C_t^{(j)}}$. If player j is the unique occupant of cell $C_t^{(j)}$, she observes a reward $Y_t^{(j)}$; otherwise, she observes the null symbol $\perp$.

For each player j and cell C , we track the number of successful observations, or the **success count**

$$o_{j,C}(T_0) := \sum_{t=1}^{T_0} U_j(C, t), \quad U_j(C, t) := \mathbf{1}\{j \text{ uniquely occupies cell } C \text{ at round } t\}.$$

The probability of such a success in any given round for a specific player-cell pair is constant. A player chooses cell C with probability $1/K$, and the other $N-1$ players all avoid it with probability $(1 - 1/K)^{N-1}$. Thus, the per-round success probability is $p_K := \frac{1}{K} (1 - \frac{1}{K})^{N-1}$.

From these successful observations, each player computes an empirical mean for each cell's center, $\hat{\mu}_j^{(0)}(C)$. This estimate, combined with a concentration radius $r_j^{(0)}(C)$ and the geometric bracket from the Lipschitz property, yields initial lower and upper confidence bounds on the cell's true maximum value, $\mu^*(C)$. Formally, the bounds are:

$$\begin{aligned} \text{LCB}_j^{(0)}(C) &:= \hat{\mu}_j^{(0)}(C) - r_j^{(0)}(C) \\ \text{UCB}_j^{(0)}(C) &:= \hat{\mu}_j^{(0)}(C) + r_j^{(0)}(C) + \frac{Lh\sqrt{d}}{2}, \end{aligned}$$

where the radius $r_j^{(0)}(C)$ is defined to ensure high-probability concentration with $\beta_0 :=$

$$\log \frac{4NK(T_0+1)}{\delta_I}.$$

$$r_j^{(0)}(C) := \sqrt{\frac{\beta_0}{2 \max\{1, o_{j,C}(T_0)\}}} \quad (2.2)$$

2.6.2 Phase I Guarantees

Despite the collision-prone nature of the exploration, standard concentration inequalities show that this simple protocol is highly effective. The first result establishes that the number of successes, $o_{j,C}(T_0)$, is sharply concentrated around its mean for all players and cells simultaneously. The second shows that the empirical means are accurate estimates of the true center values.

Lemma 1 (Success Counts Under Collisions). *For any $\eta \in (0, 1)$, with probability at least $1 - \delta_I/2$, the success count for every player $j \in [N]$ and every cell $C \in \mathcal{P}$ is bounded by $(1 \pm \eta)T_0 p_K$ provided $T_0 \geq \frac{3}{\eta^2 p_K} \log \left(\frac{4NK}{\delta_I} \right) ..$*

Lemma 2 (Anytime Concentration for Center Means). *With probability at least $1 - \delta_I/2$, for every player $j \in [N]$ and every cell $C \in \mathcal{P}$, the empirical mean is close to the true mean: $|\hat{\mu}_j^{(0)}(C) - \mu(x_C)| \leq r_j^{(0)}(C)$. This holds for any realized value of the success count $o_{j,C}(T_0)$.*

The proofs of these lemmas, which involve standard applications of Chernoff and Hoeffding bounds with a union bound over all players and cells, are deferred to the appendix. By combining these statistical guarantees with the geometric bound derived from the Lipschitz property, we arrive at the main result of Phase I: with high probability, every player constructs a valid confidence interval for every cell's true maximum value.

Proposition 1 (Phase-I Maxima Brackets). *With probability at least $1 - \delta_I$, for every player $j \in [N]$ and every cell $C \in \mathcal{P}$, the computed bounds are valid: $\text{LCB}_j^{(0)}(C) \leq \mu^*(C) \leq \text{UCB}_j^{(0)}(C)$.*

2.7 Phase II: Zooming in on cells

Phase I provides each player with valid but wide confidence brackets on each cell's potential. This initial map is crucial for pruning the search space, but it is not sharp enough for final decision-making, primarily due to the center-versus-maximum bias.

The purpose of Phase II is to resolve this ambiguity. Here, players “zoom in” on the most promising regions identified in Phase I, conducting a localized exploration - which we call a “local peek” - to refine their estimates and construct confidence bounds on the true cell maxima, $\mu^*(C)$, down to a pre-specified target accuracy, $\varepsilon > 0$.

The phase begins with each player independently forming a smaller “active set” of candidate cells, $\mathcal{S}_{\text{act}_j}^{(0)}$. This is a safe elimination step: using their Phase I brackets, players discard any cell whose most optimistic outcome (its upper bound) cannot compete with the most pessimistic outcome (the lower bound) of the top- N candidate cells. Players may form different active sets due to the randomness in their Phase I observations; our analysis handles this by considering the maximum active set size, $M_{\text{act}} := \max_j |\mathcal{S}_{\text{act}_j}^{(0)}|$.

Within each of their active cells, players then conduct the local peek. For a fixed duration T_1 , they sample from a fine η -net of probe points laid out inside each cell. An η -net is a grid of points so dense that any point in the cell is within a distance η of some grid point. This strategy allows us to approximate the supremum of the Lipschitz function over the continuous cell by taking the maximum over a finite set of points. A probe at one of these points is successful only if no other player samples any point within the same cell in that round. While still subject to collisions, this exploration is highly focused on the regions that matter most.

2.7.1 Collision-Tolerant Probe Sampling

A key technical challenge is to ensure that, despite collisions and decentralization, every player gathers enough information at **every** probe point. Our analysis shows that a carefully chosen phase duration T_1 is sufficient to guarantee this with high probability.

Lemma 3 (Phase-II Probe Coverage). *Let*

$$q_{M_{\text{act}}, \eta} \gtrsim \frac{1}{M_{\text{act}}} \left(\frac{\eta}{h} \right)^d \left(1 - \frac{1}{N} \right)^{N-1}$$

be the per-round success probability for a (player, cell, probe) triple, and let N_{probe} be the total

number of probe points across all players' active sets. Choose integers

$$b \geq 4 \log(2N_{\text{probe}}/\delta_{II}) \quad \text{and} \quad T_1 \geq 2b/q_{M_{\text{act}}, \eta}.$$

Then, with probability at least $1 - \delta_{II}/2$, every triple attains at least b non-collision samples.

With a guaranteed budget of at least b successful samples per probe point, standard concentration inequalities ensure that the empirical mean at each point is a highly accurate estimate of its true mean. This accuracy at the probe points translates directly into a tight bound on the cell's maximum value.

Proposition 2 (Refined Maxima Brackets). *At the end of Phase II, by choosing parameters η and b appropriately for a target accuracy ε , each player j constructs a new, refined bracket $[\text{LCB}_j^{(1)}(C), \text{UCB}_j^{(1)}(C)]$ for each of her active cells. With high probability, this bracket contains the true maximum $\mu^*(C)$ and has a width of at most ε .*

2.7.2 Selecting the Top- N Cells

With these tight and reliable brackets on the true cell maxima, players are equipped to make their final selection. We adopt a deterministic rule to select N cells: each player selects the N cells corresponding to their largest lower confidence bounds, $\text{LCB}_j^{(1)}(C)$, breaking any ties with a fixed, public ordering of the cells (e.g. say lexicographic). This guarantees every player outputs a set, $S_\varepsilon^{(j)}$, of size exactly N .

Theorem 1 (Gap-Free ε -Optimality). *With high probability, the set $S_\varepsilon^{(j)}$ chosen by any player j is ε -optimal: every cell $C \in S_\varepsilon^{(j)}$ satisfies $\mu^*(C) \geq \mu_{(N)}^* - \varepsilon$.*

This powerful guarantee holds for any reward function, irrespective of the gaps between cell values. It ensures our procedure is robust even when the decision is difficult. Remarkably, this simple, decentralized rule leads to a powerful emergent behavior: under a mild separation condition on the reward function, it forces all players to agree on the exact same set of top- N cells, achieving consensus without any communication.

Definition 1 (ε -Uniqueness at the Top- N). *A mean function μ is ε -unique at the top- N if there is a unique set $S^\dagger \subset \mathcal{P}$ of size N such that $\min_{C \in S^\dagger} \mu^*(C) \geq \max_{C \notin S^\dagger} \mu^*(C) + 2\varepsilon$.*

Lemma 4 (Consensus under ε -Uniqueness). *If the instance is ε -unique and the bracket width is at most ε , then on our high-probability event, all players select the exact same set of cells: $S_\varepsilon^{(j)} = S^\dagger$ for all $j \in [N]$.*

This decentralized agreement is a cornerstone of our protocol’s success since it allows the final seating phase to happen.

Example 1 (Center-vs-maximum pathology in 1D). *Let $d = 1$, $h = \frac{1}{2}$, so $\mathcal{P} = \{C_1 = [0, \frac{1}{2}], C_2 = [\frac{1}{2}, 1]\}$. Define a Lipschitz mean $\mu(x) = x + \alpha\phi(x)$ with a narrow bump ϕ of height 1 supported in $[\frac{1}{2} - \delta, \frac{1}{2}]$, with $\delta \ll h$ and $\alpha > 0$ small so that L is finite. Then $\mu(x_{C_1}) < \mu(x_{C_2})$ (center ranking favors C_2) but $\mu^*(C_1) > \mu^*(C_2)$ (the bump near the boundary makes C_1 optimal). Phase I center estimates therefore mis-rank the cells, while Phase II’s local peek in C_1 identifies the bump and restores the correct top- N set.*

2.8 Phase II $\frac{1}{2}$: Musical Chairs

With a set of N high-quality target cells identified, the players must perform the final coordination step: assigning themselves to these cells so that each is occupied by exactly one player. This is achieved the Musical Chairs algorithm [66]. A key strength of our analysis is that we model the practical, challenging version of this protocol where players have no side-channel telling them which cells are already occupied; they must discover free cells through trial and error.

2.8.1 The Seating Protocol

The seating phase proceeds in rounds. Initially, all N players are “unseated.” In each round, every unseated player samples a cell uniformly at random from the entire N -cell target set. A collision occurs at a cell if it is chosen by more than one unseated player, or if an unseated player chooses a cell that is already occupied by a seated player. If, however, a single unseated player

chooses a currently unoccupied cell, that player becomes “seated” at that cell. They cease to participate in the sampling process and will occupy that cell for the remainder of the horizon. The process terminates when all N players are seated.

2.8.2 Expected Seating Time and Regret

Let U_t be the number of unseated players at the start of round t . We define the expected number of players that become seated per round as “drift”. When u players remain unseated, this drift is given by $\Delta(u) := \mathbb{E}[U_t - U_{t+1} \mid U_t = u] = \frac{u^2}{N}(1 - \frac{1}{N})^{u-1}$. The quadratic dependence on u means that progress is rapid when many players are searching for spots. This positive drift allows us to bound the total expected time for the dance to conclude.

Theorem 2 (Expected Seating Time). *The expected time, T_{MC} , for all N players to become seated is linear in the number of players: $\mathbb{E}[T_{MC}] = O(N)$.*

This result demonstrates that this simple, decentralized procedure is highly efficient and scales gracefully, far better than a naive $O(N \log N)$ coupon-collector analysis might suggest. The total regret incurred during this phase is therefore a fixed cost, dependent on N but crucially, independent of the total time horizon T . This confirms that the entire coordination and seating process can be completed for a small, one-time price (not dependent on the time horizon).

Corollary 1 (Seating-Phase Regret). *The expected cumulative regret from Musical Chairs is bounded by a horizon-independent constant: $\mathbb{E}[R_{MC}] = O(N^2)$.*

2.9 Phase III: Optimization within Cells

With the completion of Phase II, the multi-agent coordination problem is solved. Each player is now the sole occupant of a distinct cell. For the remainder of the horizon, $T' = T - T_0 - T_1 - T_{MC}$, the problem decouples entirely into N independent, single-player bandit problems. Collisions are no longer a concern, and each player’s objective is simply to cultivate the maximum possible reward from within their own cell.

Each player must now solve a standard Lipschitz bandit problem over their assigned domain C_j . A provably near-optimal and standard approach for this task is an epoch-based, discretize-and-explore algorithm like Zooming [36]. In each epoch, they create a grid of points within their cell, with the grid resolution becoming finer over time.

The performance of such single-player strategies is well-established in the bandit literature [10, 69]. The regret incurred by each player is known to follow the classical minimax rate for a d -dimensional Lipschitz problem.

Proposition 3 (In-Cell Regret). *The expected regret for any player j during the Phase III optimization within their cell C_j over a duration of T' rounds is bounded by:*

$$\mathbb{E}[R_{\text{in}}^{(C_j)}(T')] \leq c_d (Lh)^{\frac{d}{d+2}} (T')^{\frac{d+1}{d+2}} + c'_d,$$

where c_d and c'_d are constants that depend only on the dimension d .

The total regret from Phase III is the sum of these individual regrets across all N players. This term represents the primary, horizon-dependent component of our overall regret bound.

2.10 End-to-End Regret Guarantees

With the components of our multi-phase protocol established, we combine them to get end-to-end performance guarantees. Our main result is that the significant upfront cost of decentralized coordination is carefully managed to ensure that the long-term performance is dictated by the optimal rate of single-agent learning. We build to this conclusion by first presenting a clean result for well-separated problem instances, and then stating our main guarantee that holds for any instance.

2.10.1 Global Regret Under a Reward Gap

Our first result quantifies the performance of the protocol in an ideal setting where the top- N cells are clearly better than the rest. This scenario is formalized by the ε -uniqueness condition

(Definition 1), which assumes a sufficiently large “reward gap” between the N -th best cell and the $(N + 1)$ -th best cell. When this gap exists, Phase II is guaranteed to lead to consensus, where all players identify the exact same set of top- N cells. This allows for a seamless transition into seating and optimization.

Theorem 3 (Global Regret with Consensus). *Assume the ε -uniqueness condition holds (i.e., a sufficient reward gap exists). For any horizon T , the expected total regret is bounded by:*

$$\mathbb{E}[R_{cont}(T)] \leq \underbrace{N(T_0 + T_1) + c_{MC}N^2}_{\text{Coordination Cost}} + \underbrace{c_d N(Lh)^{\frac{d}{d+2}} T^{\frac{d+1}{d+2}}}_{\text{Learning Cost}} + \delta_{sys}T,$$

where c_{MC} and c_d are universal constants.

This theorem formalizes our core architectural claim: the total regret decomposes into a horizon-independent “Coordination Cost” and a horizon-dependent “Learning Cost.” In well-behaved instances, the initial price of coordination becomes negligible over a long horizon.

2.10.2 The Main Result: A Gap-Free Guarantee

The true strength of a decentralized protocol lies in its ability to perform well without favorable structural assumptions. A natural and challenging scenario arises when the reward gap is small or even zero, making the top- N cells statistically indistinguishable from the next best. In this “gap-free” setting, consensus is no longer guaranteed; different players might identify slightly different (but still high-quality) sets of target cells.

Our main result shows that our algorithm is robust to this challenge under a public coverage assumption. This is achieved by running the protocol in epochs of doubling length ($T_k = 2^k$), with the precision ε_k recalibrated for each epoch. This standard “doubling trick” [11] allows the algorithm to control regret from potential sub-optimality without knowing the reward gaps or the horizon T in advance.

Corollary 2 (Epochic, Gap-Free Global Regret). *By running the multi-phase protocol in epochs, with precision $\varepsilon_k \propto 2^{-k/(d+2)}$ for epoch k , the algorithm achieves, for any L -Lipschitz reward*

function and any horizon T , an expected total regret of

$$\mathbb{E}[R_{\text{cont}}(T)] \leq \tilde{O}\left(N(Lh)^{\frac{d}{d+2}}T^{\frac{d+1}{d+2}}\right).$$

This result demonstrates that by dynamically adapting the precision of the identification phases, our algorithm robustly achieves the optimal regret rate without requiring any separation between the values of good and bad cells. The costs of repeated coordination in each epoch are controlled and are ultimately subsumed by the dominant learning cost.

2.11 Distance-Threshold Collisions

Our analysis has centered on a partition-based collision model, a useful abstraction for systems with predefined operational zones. This section demonstrates the modularity of our framework by showing how it naturally extends to a more physically-motivated model where collisions are governed by proximity.

In the distance-threshold model, a collision occurs if any two players' actions $X_t^{(j)}$ and $X_t^{(j')}$ are within a distance $\rho > 0$ of each other. To handle this, we reduce the problem to our existing framework. We first discretize the space $\mathcal{X}$ into a set of “safe centers” $\{z_1, \dots, z_M\}$ that form an r -packing with $r > \rho$. We then associate each center z_i with a “safe ball” B_i of radius $\sigma < (r - \rho)/2$. By construction, any two points chosen from two different safe balls are guaranteed to be separated by a distance greater than ρ , making inter-ball collisions impossible.

Our entire multi-phase protocol (Phase I - III) can then be applied directly, treating the family of safe balls $\{B_i\}$ as if they were the cells of the partition. Players identify, seat themselves upon, and perform optimization within these balls. The performance guarantees translate directly, with regret measured against the optimal assignment to these safe balls, $\text{OPT}_{\text{pack}}(r, \sigma, N)$.

Theorem 4 (Regret in the Distance-Threshold Model). *When applied to a set of safe balls derived from an r -packing, our protocol's expected total regret is bounded by an expression identical in form to that in Theorem 3, with the partition parameters (K, h) replaced by the packing parameters (M, σ) .*

2.12 A Lower Bound

Having established an upper bound on our algorithm’s regret, we now confirm its asymptotic optimality. The following lower bound shows that no decentralized algorithm can achieve a fundamentally better dependence on the horizon T . This proves that the challenges of decentralization and collisions do not degrade the problem’s core statistical difficulty.

Theorem 5 (Lower Bound). *For the decentralized multi-player bandit problem in a partition-based collision model, for any decentralized algorithm, there exists an L -Lipschitz mean-reward function such that the expected regret is bounded below by:*

$$\mathbb{E}[R_{\text{cont}}(T)] \geq c \cdot N(Lh)^{\frac{d}{d+2}} T^{\frac{d+1}{d+2}},$$

where c is a constant depending only on dimension d .

The proof relies on a standard reduction from the well-established lower bound for finite-armed bandits [40, 10]. We construct a challenging reward function that embeds N independent, hard, finite-armed bandit problems into N disjoint cells.

2.13 Conclusion

We have introduced and provided the first end-to-end solution for the cooperative, decentralized multi-player stochastic bandit problem in continuous domains with hard collisions. Our central contribution is a modular, multi-phase protocol that requires no communication between players. The key insight is to decouple the problem into a horizon-independent coordination stage and a horizon-dependent learning stage. A crucial innovation is our maxima-directed identification phase, which performs a localized search to avoid systemic biases inherent in simpler center-based discretizations. Our analysis culminates in a near-optimal, gap-free regret bound, under a public scheduling assumption, of $\tilde{O}(T^{(d+1)/(d+2)})$, which matches the single-player minimax rate for Lipschitz bandits.

Chapter 3

A Unified Framework for Locality in Scalable MARL

This chapter is based on the following paper:

Sourav Chakraborty, Amit Kiran Rege, Claire Monteleoni, and Lijun Chen. A unified framework for locality in scalable marl, 2026.

Abstract: Scalable Multi-Agent Reinforcement Learning (MARL) is fundamentally challenged by the curse of dimensionality. A common solution is to exploit locality, which hinges on an Exponential Decay Property (EDP) of the value function. However, existing conditions that guarantee the EDP are often conservative, as they are based on worst-case, environment-only bounds (e.g., supremums over actions) and fail to capture the regularizing effect of the policy itself. In this work, we establish that locality can also be a **policy-dependent** phenomenon. Our central contribution is a novel decomposition of the policy-induced interdependence matrix, H^π , which decouples the environment’s sensitivity to state (E^s) and action (E^a) from the policy’s sensitivity to state ($\Pi(\pi)$). This decomposition reveals that locality can be induced by a smooth policy (small $\Pi(\pi)$) even when the environment is strongly action-coupled, exposing a fundamental locality-optimality tradeoff. We use this framework to derive a general spectral condition $\rho(E^s + E^a\Pi(\pi)) < 1$ for exponential decay, which is strictly tighter than prior norm-based conditions. Finally, we leverage this theory to analyze a provably-sound localized block-coordinate policy improvement framework with guarantees tied directly to this spectral radius.

The corresponding supplementary material can be found in [Appendix B](#).

3.1 Introduction

Multi-agent reinforcement learning (MARL) is a powerful paradigm for controlling complex, decentralized systems. However, its application is fundamentally challenged by the curse of dimensionality: the joint state and action spaces ($\mathcal{S} = \prod_i \mathcal{S}_i$, $\mathcal{A} = \prod_i \mathcal{A}_i$) grow exponentially with the number of agents, n . This problem places MARL in the category of “succinctly described” but intractable MDPs [5, 58], rendering standard, centralized learning algorithms computationally infeasible.

A highly effective approach to overcome this barrier is to exploit the local interaction structure inherent in many networked systems. This line of work, often termed Networked MARL [60, 62, 41], assumes agents are situated on a graph and that their local transitions and rewards depend only on a local neighborhood. These foundational papers show that such systems often exhibit an Exponential Decay Property (EDP), where the influence of distant agents on an agent’s value function decays exponentially with graph distance. This property is the theoretical bedrock that justifies truncating computations to a small κ -hop neighborhood, enabling scalable algorithms whose complexity depends on the neighborhood size, not the full network.

A critical question, however, is when this EDP holds. In the discounted setting, one gets EDP for “free” due to a $\gamma < 1$ multiplicative factor. In the more challenging average-reward ($\gamma = 1$) setting, prior work has established the EDP by proving a Dobrushin-style weak coupling condition [60]. This condition, however, is policy-independent - it relies on a worst-case (action-supremum) bound on the environment’s influence. This bound is inherently conservative. It can fail to certify locality in a system where the environment is strongly action-coupled, even if the agents’ policies are “smooth” (e.g., highly exploratory) and effectively break this strong coupling.

This chapter’s central thesis is that locality is best viewed depending on both the environment and the policy. Our main contribution is a new theoretical framework that formally decouples the sources of influence. We introduce a novel decomposition of the one-step interdependence matrix,

C^π , that separates the environment’s “physics” from the policy’s “logic”:

$$C^\pi \preceq E^s + E^a \Pi(\pi)$$

Here, E^s and E^a measure the environment’s sensitivity to changes in state and action, respectively, while $\Pi(\pi)$ measures the policy’s sensitivity to changes in state (Proposition 4). This decomposition reveals that a smooth policy (a small $\Pi(\pi)$) can induce locality even in an environment with strong action-based coupling (a large E^a).

This framework allows us to derive a new, more general condition for locality: the system exhibits exponential decay if the spectral radius $\rho(H^\pi) < 1$ (Theorem 6). This is strictly tighter than prior norm-based, policy-independent tests. We connect this theory to practice by showing that for standard entropy-regularized policies [18], the temperature τ serves as a direct, practical knob to control $\Pi(\pi)$ and thus manage the system’s locality (Lemma 20). This exposes a fundamental locality-optimality tradeoff.

Finally, we leverage this theory to analyze a deterministic, oracle-based policy improvement framework (Section 3.5). By decoupling the analysis of spatial approximation from the complex analysis of stochastic sampling, we provide a clean, provably-sound justification for localized algorithms. We prove that the error of our localized framework is bounded deterministically by the tail of a Neumann series governed by $\rho(H^\pi)$, completing the link from our fundamental theory to a provable algorithmic guarantee. We describe additional related work in Appendix B.1.

3.2 Setup and preliminaries

We consider n agents with local state spaces $\mathcal{S}_i$ and local action spaces $\mathcal{A}_i$, both finite and nonempty. The joint state and action spaces are $\mathcal{S} = \prod_{i=1}^n \mathcal{S}_i$ and $\mathcal{A} = \prod_{i=1}^n \mathcal{A}_i$. At each discrete time t , the system is in state $S_t = (S_t^{(1)}, \dots, S_t^{(n)}) \in \mathcal{S}$. We denote a state excluding an agent i by s_{-i} meaning $s = (s_i, s_{-i})$. Under a policy π , a joint action $A_t \in \mathcal{A}$ is sampled from $\pi(\cdot | S_t)$.

The next state S_{t+1} is drawn from a transition kernel $P(\cdot | S_t, A_t)$ that factorizes $P(s' | s, a) = \prod_{j=1}^n P_j(s'_j | s, a)$, where $P_j(\cdot | s, a)$ is a probability distribution on $\mathcal{S}_j$. This is a general formulation

that, notably, allows the next state of agent j , s'_j , to depend on the entire global state s and the entire joint action a . This setup is more general than prior work on theoretical networked MARL (e.g., [61, 42]), which often assume a priori structural locality. For instance, prior models restrict the kernel to $P_j(s'_j \mid s_{N_j}, a_j)$, thereby assuming that j 's next state is affected only by states of agents in its neighborhood on a network.

We adopt this more general model precisely because our goal is to derive locality from first principles, not to assume it. We do not impose a graph on the environment's dynamics. Instead, our analysis will be phrased in terms of operator bounds to show that locality - and the exponential decay property it implies - is an emergent property of the system. This allows us to recover prior networked models as a special case (i.e., where the interaction matrices are sparse), while also revealing new, policy-dependent sources of locality.

A policy is of product form. For each agent k , there is a stochastic map $\pi_k(\cdot \mid s)$ on $\mathcal{A}_k$ and the global policy $\pi(a \mid s) = \prod_{k=1}^n \pi_k(a_k \mid s)$. If the policy depends on a subset of state coordinates (instead of global state s), we specify the subset using observation scopes $O_k \subseteq [n]$ such that $\pi(a \mid s) = \prod_{k=1}^n \pi_k(a_k \mid s_{O_k})$. When $O_k = [n]$, we note that $s = s_{O_k}$. In some examples we will consider localized observation scopes, but the general theory does not require this.

The policy-induced state transition kernel is

$$P^\pi(s' \mid s) = \sum_{a \in \mathcal{A}} \left(\prod_{j=1}^n P_j(s'_j \mid s, a) \right) \left(\prod_{k=1}^n \pi_k(a_k \mid s_{O_k}) \right).$$

We write $\mathbb{E}_\pi[\cdot \mid S_t = s]$ for expectation with respect to one step of this kernel. We work in the more challenging average reward setting and prove guarantees for the discounted setting as a corollary in the Appendix.

We measure sensitivity using coordinatewise oscillations. For a bounded function $f : \mathcal{S} \rightarrow \mathbb{R}$, the i -oscillation is

$$\delta_i(f) = \sup_{x, y \in \mathcal{S}, x_{-i} = y_{-i}} |f(x) - f(y)|.$$

The vector $\delta(f) = (\delta_1(f), \dots, \delta_n(f))$ will be used to track how changes in individual coordinates propagate through linear operators and use the shorthand $\|\delta(f)\|_\infty = \max_i \delta_i(f)$. This seminorm -

often used in Dobrushin-type arguments for interacting particle systems and Glauber dynamics [15, 49] - has a simple interpretation: it is the Lipschitz constant of f with respect to Hamming changes at a single coordinate. (See B.5.3 for details)

We consider three families of sensitivities. The first measures how much changing one coordinate of the current state can change the next-state marginal of a given agent when the action is held fixed. For $j, i \in [n]$, we denote the total variation distance between two probability measures using $TV(\cdot, \cdot)$ and define

$$E_{j \leftarrow i}^s = \sup_{\substack{s, s' \in \mathcal{S}: s_{-i} = s'_{-i} \\ a \in \mathcal{A}}} TV(P_j(\cdot \mid s, a), P_j(\cdot \mid s', a))$$

The second measures how much changing one agent's action can change the next-state marginal of another agent when the state is held fixed. For $j, k \in [n]$ define

$$E_{j \leftarrow k}^a = \sup_{\substack{s \in \mathcal{S} \\ a, a' \in \mathcal{A}: a_{-k} = a'_{-k}}} TV(P_j(\cdot \mid s, a), P_j(\cdot \mid s, a'))$$

The third measures how sensitive each agent's local action distribution is to a change in one coordinate of the state. For $k, i \in [n]$ define

$$\Pi_{k \leftarrow i}(\pi) = \sup_{s, s' \in \mathcal{S}: s_{-i} = s'_{-i}} TV(\pi_k(\cdot \mid s_{O_k}), \pi_k(\cdot \mid s'_{O_k}))$$

The $n \times n$ matrices E^s and E^a are properties of the environment. The $n \times n$ matrix $\Pi(\pi)$ depends on the policy. We will sometimes suppress the policy in the notation when there is no ambiguity. Note that if agent k does not observe i then the corresponding entry of Π is zero.

The policy-induced one-step marginal influence of i on the next-state marginal of agent j is

$$C_{j \leftarrow i}^\pi = \sup_{s, s' \in \mathcal{S}: s_{-i} = s'_{-i}} TV(P_j^\pi(\cdot \mid s), P_j^\pi(\cdot \mid s')), \quad P_j^\pi(\cdot \mid s) = \sum_{a \in \mathcal{A}} P_j(\cdot \mid s, a) \prod_{k=1}^n \pi_k(a_k \mid s_{O_k}).$$

The matrix $C^\pi = [C_{j \leftarrow i}^\pi]$ is the policy-induced interdependence matrix. It governs how one step of the kernel maps oscillations of functions.

This policy-dependent definition is a critical departure from prior approaches. A common alternative for measuring influence is to define a single, policy-independent matrix that captures

the worst-case coupling of the environment. This is typically done by taking a supremum over all joint actions, rather than an expectation (or “mixture”) over the policy $\pi(\cdot \mid s)$. While this action-supremum approach has the advantage of being a fixed property of the environment, it is fundamentally conservative. It conflates the “physics” of the environment with the “logic” of the policy.

Consider an environment where the next state P_j is extremely sensitive to a neighbor’s action a_k . An action-supremum bound would conclude the system is tightly coupled (i.e., not local), even if the agent’s policy π_k is “dumb” and never chooses that action, or is completely insensitive to the state s_i . By defining C^π as an expectation over the policy, our definition correctly measures the actual one-step influence of the closed-loop system (environment + policy). This precision is what allows us to decouple these effects, which is the subject of the next section.

3.3 Policy-induced influence and locality

In this section, we introduce the policy-induced influence operator and show it leads to exponential decay. Proofs are deferred to the appendix.

The first result is a decomposition of the policy-induced interdependence. This inequality formalizes the informal reasoning from the introduction. A perturbation at i propagates to the next state of coordinate j along two one-step routes. The direct route is through the environment: even if the action is fixed, the conditional update $P_j(\cdot \mid s, a)$ may change when s_i changes. The indirect route goes through the policy and back into the environment: changing s_i changes the distribution of each local action a_k , and changing a_k may alter the next-state distribution of S'_j . The matrices $E^s, E^a, \Pi(\pi)$ quantify these two routes. The policy-induced interdependence matrix C^π aggregates their net effect. The next proposition makes this decomposition precise.

Proposition 4 (Decomposition of policy-induced influence). *For any policy π of product form and any synchronous dynamics $P(\cdot \mid s, a)$ on a finite state space, the interdependence matrix C^π of the policy-induced kernel satisfies the entrywise bound $C^\pi \preceq E^s + E^a \Pi$. In particular, for each pair*

of indices i, j , $C_{j \leftarrow i}^\pi \leq E_{j \leftarrow i}^s + \sum_{k=1}^n E_{j \leftarrow k}^a \Pi_{k \leftarrow i}(\pi)$.

Informally, the proof separates the two routes of influence using the triangle inequality. A detailed construction using a coupling argument is provided in Appendix B.2.1.

To build intuition, we discuss a quantitative example before moving ahead. Let $n = 2$ agents have binary states and actions, $\mathcal{S}_i = \mathcal{A}_i = \{0, 1\}$. Suppose agent 1 is unaffected by anything (P_1 is constant). Agent 2’s next state is **deterministically** set by agent 1’s action, regardless of state: $P_2(s'_2 = 1 \mid s, a) = \mathbf{1}\{a_1 = 1\}$. This environment has maximal action-based coupling. The sensitivities are:

- $E^s = 0$ (state s has no direct effect on the next state).
- $E_{2 \leftarrow 1}^a = 1$ (flipping a_1 from 0 to 1 flips the next state s'_2 from 0 to 1, a TV distance of 1).

All other E^a entries are 0.

An “action-supremum” analysis (from prior work) would stop here. It sees $E_{2 \leftarrow 1}^a = 1$ and concludes the system is globally coupled, as the worst-case influence is 1.

Now, consider a policy π where agent 1 is “sleepy” and barely reacts to its state, s_1 . Let $\pi_1(\cdot \mid s_1)$ change by a small amount α (in TV distance) when s_1 flips. Assume π_1 doesn’t observe s_2 , so $\Pi_{1 \leftarrow 2} = 0$.

$$\Pi_{1 \leftarrow 1}(\pi) = \sup_{s_{-1}=s'_{-1}} \text{TV}(\pi_1(\cdot \mid s), \pi_1(\cdot \mid s')) = \alpha.$$

We can now compute a bound on the true policy-induced influence $C_{2 \leftarrow 1}^\pi$ (measuring how a change in s_1 affects s'_2):

$$C_{2 \leftarrow 1}^\pi \leq E_{2 \leftarrow 1}^s + \sum_{k=1}^2 E_{2 \leftarrow k}^a \Pi_{k \leftarrow 1}(\pi) = 0 + (E_{2 \leftarrow 1}^a \Pi_{1 \leftarrow 1} + E_{2 \leftarrow 2}^a \Pi_{2 \leftarrow 1}) = (1 \cdot \alpha) + (0 \cdot 0) = \alpha.$$

Thus, if the policy is insensitive to the state (small α), our bound certifies that the system is only weakly coupled, even when the environment’s “action channel” E^a is not. Our bound also captures the opposite failure mode, where smoothing the policy is ineffective (See Appendix B.5.1).

The second result states that a bound on the interdependence matrix controls how one step of the synchronous kernel acts on coordinatewise oscillations. This is the synchronous analogue of classical Dobrushin–type bounds.

Lemma 5 (Oscillation bound via H^π). *Let $T^\pi f(s) = \mathbb{E}[f(S_{t+1}) \mid S_t = s]$ under the synchronous update with product-form policy $\pi(a \mid s) = \prod_k \pi_k(a_k \mid s_{O_k})$ and factorized kernel $P(s' \mid s, a) = \prod_j P_j(s'_j \mid s, a)$.*

For $i \in [n]$ and bounded $f : \mathcal{S} \rightarrow \mathbb{R}$, $\delta_i(T^\pi f) \leq \sum_{j=1}^n H_{j \leftarrow i}^\pi \delta_j(f)$ where $H^\pi := E^s + E^a \Pi(\pi)$. Equivalently, in vector form, we write $\delta(T^\pi f) \leq (H^\pi)^\top \delta(f)$.

The next theorem turns one–step bounds into global decay. It gives a spectral condition on the matrix $H^\pi = E^s + E^a \Pi(\pi)$ that, if satisfied uniformly over a policy class, ensures contraction and average–reward solvability. This solvability is centered on the average–reward Poisson equation, which is the Bellman equation for this setting. This equation helps us find a relative value function (or differential value function), h^π , which represents the long-term, transient advantage of starting in a particular state relative to the long-term average reward $\bar{r}^\pi$. The equation is given by:

$$h^\pi(s) = (r^\pi(s) - \bar{r}^\pi) + \mathbb{E}[h^\pi(S_{t+1}) \mid S_t = s],$$

or, written in operator form, $h^\pi - T^\pi h^\pi = r^\pi - \bar{r}^\pi$.

Our Poisson decay result in the following theorem shows that if the system is well-behaved (i.e., $\rho(H^\pi) < 1$), then the solution h^π to this equation inherits the locality of the reward. Specifically, the sensitivity of the value function at one coordinate, $\delta_i(h^\pi)$, is bounded by the (decayed) sensitivities of the reward function, $\delta(r^\pi)$.

Theorem 6 (Policy–uniform synchronous contraction and Poisson decay). *Suppose there is a policy class Π_{pol} such that for some $\lambda \in [0, 1)$, $\rho(E^s + E^a \Pi(\pi)) \leq \lambda$ for all $\pi \in \Pi_{\text{pol}}$.*

Then for all $\pi \in \Pi_{\text{pol}}$ and all bounded f , $\delta((T^\pi)^t f) \leq ((E^s + E^a \Pi(\pi))^\top)^t \delta(f)$.

In particular, there is $C \geq 1$ independent of t and π such that $\|\delta((T^\pi)^t f)\|_\infty \leq C \lambda^t \|\delta(f)\|_\infty$

for all $t \geq 0$. The average-reward Poisson equation

$$h^\pi - T^\pi h^\pi = r^\pi - \bar{r}^\pi \quad \text{with } r^\pi(s) = \sum_a r(s, a) \prod_k \pi_k(a_k | s),$$

has a solution unique up to an additive constant, and

$$\delta(h^\pi) \leq (I - (E^s + E^a \Pi(\pi))^\top)^{-1} \delta(r^\pi).$$

This theorem is geometry-free in the sense that it provides a global contraction condition, $\rho(H^\pi) < 1$, without assuming any underlying network structure. The connection to spatial locality becomes clear when we assume H^π is sparse. If we assume the system has an interaction graph G such that E^s , E^a , and $\Pi(\pi)$ are all sparse with respect to G , then H^π is also sparse. In this case, the matrix power $((H^\pi)^\top)^t$ has non-zero entries (i, j) only if there is a path of length at most t from j to i in G .

This directly implies the exponential decay property:

- The solution to the Poisson equation, $\delta(h^\pi)$, is bounded by the Neumann series

$$\sum_t \left((H^\pi)^\top \right)^t \delta(r^\pi).$$

- The influence of an agent j on i is a sum over paths, which decays at the spectral rate λ^t with path length t .
- Therefore, the κ -local truncation error - the error from ignoring agents more than κ hops away - is bounded by the tail of this series, which decays exponentially in κ .

This framework recovers and generalizes the locality results in prior work [61, 42]. Those analyses start with a graph G and propose a sufficient condition for locality based on a policy-independent, worst-case (action-supremum) matrix C , such as requiring $\|C\|_\infty < 1$ (details in Appendix B.2.4).

Our result reveals this to be a conservative special case. Their norm-based condition ($\|C\|_\infty < 1$) is simply one way to certify that our more fundamental spectral radius condition ($\rho(H^\pi) < 1$) holds. Our analysis is sharper, as it correctly identifies that locality can be policy-induced (via a small $\Pi(\pi)$) even when the environment's worst-case coupling (a large E^a) is strong.

We briefly discuss alternatives to our choice of definition. One could take a supremum over actions in the definition of policy-induced influence. This means that the matrix is now policy independent but ignores the actual policy-state dependence and is too conservative whenever action coupling is strong but the policy changes slowly with the state (as discussed in Section 3.2). One could also attempt to analyze the average-reward Poisson equation starting from discounted problems and letting the discount tend to one. This route hides the role of the environment and the policy in the contraction and yields weaker constants in problems where discounting is not a modeling choice. The direct contraction analysis is both sharper and closer to the algorithms we ultimately care about.

Our framework’s core finding that locality is governed by the spectral radius of H^π is robust. As we demonstrate in Appendix B.2.6, the same analysis extends directly to the asynchronous update model (e.g., Glauber dynamics), where only one agent updates at a time step.

Extension: Discounted setting We have focused on the average-reward setting ($\gamma = 1$), as this is the most challenging case where no “automatic” contraction is given. Our entire framework, however, applies directly to the standard γ -discounted setting. In that case, the one-step Bellman operator T_γ^π contracts oscillations by $\gamma(H^\pi)^\top$. The governing spectral radius is therefore $\lambda = \gamma \cdot \rho(H^\pi)$. This is a strictly tighter bound than the standard $\lambda = \gamma$ used in most discounted analyses. Our framework thus shows that the true decay rate is a product of the “trivial” temporal decay (γ) and the “structural” decay ($\rho(H^\pi)$) we have characterized. This confirms that our analysis is not just a special case for average-reward, but a more general theory of locality that sharpens the bounds for the discounted setting as well.

Let’s use an example: Assume $\gamma = 0.99$ (a very patient agent, so the default bound is weak). Assume our system is very weakly coupled, so $\rho(H^\pi) = 0.5$. Prior work would say to get an error of 0.01, we need $0.99^\kappa \leq 0.01$, so $\kappa \approx 458$. They would require a 458-hop neighborhood. Our work would say, “The true decay rate is $\lambda = 0.99 \times 0.5 = 0.495$. To get an error of 0.01, we need $0.495^\kappa \leq 0.01$, so $\kappa \approx 7$. Thus, we say “We can certify the algorithm with a 7-hop neighborhood, while the standard analysis would (incorrectly) demand 458 hops.”

3.4 The Algorithmic Link: Controlling Interdependence

Our analysis so far has revealed a fundamental, geometry-free condition for locality: the system is well-behaved if the spectral radius of the policy-induced influence matrix is less than one. This raises the question: how can an algorithm use this insight to keep the system well-behaved (local)?

The environment’s “physics” matrices, E^s (state-based influence) and E^a (action-based influence), are fixed. The only knob an algorithm can turn is the policy sensitivity matrix, $\Pi(\pi)$. If we can control $\Pi(\pi)$, we can **induce** locality even when E^a is large. All proofs are in Appendix B.3.

We now formally introduce the agent’s objective. We consider the standard entropy-regularized average reward objective. The goal is to find a policy π that maximizes:

$$J_\tau(\pi) = \mathbb{E}_{s \sim d^\pi, a \sim \pi(\cdot|s)} \left[r(s, a) - \tau \sum_{k=1}^n \log \pi_k(a_k | s_{O_k}) \right]$$

Here, d^π is the stationary distribution of the system under π , $r(s, a)$ is the global reward, and $\tau > 0$ is the temperature parameter, which weighs the importance of maximizing reward against maximizing policy entropy (i.e., exploration).

The well-known solution to this problem is the softmax policy [18, 25]. The policy for each agent k is parameterized by a logit function g_k , which in practice will be our estimate of the local Q-function or advantage function.

Definition 2 (Softmax Policy). *A policy π is a temperature- τ softmax policy if for each agent k , there is a logit function $g_k : \mathcal{S}_{O_k} \times \mathcal{A}_k \rightarrow \mathbb{R}$ such that: $\pi_k(a_k | s_{O_k}) \propto \exp(g_k(s_{O_k}, a_k)/\tau)$*

This policy class gives us a controllable knob: temperature τ mediates the policy’s randomness.

Intuitively, a “random” (high τ) policy should be less sensitive to small changes in its state input. If all actions are almost equally likely, why care if the state changes slightly? The following lemma makes this intuition precise. We first define the “logit Lipschitz constant,” $L_{k \leftarrow i}$, which measures the maximum sensitivity of agent k ’s logits to a change in agent i ’s state.

Definition 3 (Logit Lipschitz Constant). *For a logit function $g_k(s_{O_k}, \cdot)$, the one-coordinate logit Lipschitz constant with respect to state i is: $L_{k \leftarrow i} := \sup_{s_{-i}=s'_{-i}} \|g_k(s_{O_k}, \cdot) - g_k(s'_{O_k}, \cdot)\|_\infty$ where the supremum is over all states s, s' that differ only on coordinate i .*

This $L_{k \leftarrow i}$ term is an intrinsic property of the logit function g_k . The next lemma, proved in Appendix B.3, connects this to the policy sensitivity $\Pi(\pi)$ via the temperature τ .

Lemma 6 (Softmax Temperature Controls $\Pi(\pi)$). *For a temperature- τ softmax policy, the policy sensitivity matrix $\Pi(\pi)$ is bounded for all $k, i \in [n]$ by: $\Pi_{k \leftarrow i}(\pi) \leq \min \left\{ 1, \frac{L_{k \leftarrow i}}{2\tau} \right\}$*

This lemma provides the link between theory and algorithm design by giving an upper bound on $\rho(H^\pi)$. Let $L = [L_{k \leftarrow i}]_{k,i}$. Since $H^\pi = E^s + E^a \Pi(\pi)$ and all matrices are nonnegative,

$$H^\pi \preceq E^s + E^a \frac{L}{2\tau} \implies \rho(H^\pi) \leq \rho\left(E^s + E^a \frac{L}{2\tau}\right).$$

The spectral monotonicity follows from Perron–Frobenius. Hence, we have essentially shown the abstract problem of controlling $\Pi(\pi)$ to be equivalent to tuning τ for the entropy-regularized objective. This reveals a locality-optimality tradeoff:

- We can simply make τ large. As $\tau \rightarrow \infty$, we have $\Pi(\pi) \rightarrow 0$, which in turn means $H^\pi \rightarrow E^s$.
If we assume the E^s is stable (i.e., $\rho(E^s) < 1$), then a high-entropy policy guarantees locality.
This justifies using a small, computationally cheap truncation radius κ .
- Letting $\tau \rightarrow 0$ allows the policy to become deterministic and greedy. The policy becomes sensitive to its inputs, which can cause the entire system to become globally coupled ($\rho(H^\pi) \geq 1$), breaking the exponential decay property.

A practical, localized algorithm can start with a high τ (system will be local; can run cheap algorithms on $\kappa = 1$) and then anneal τ towards zero, potentially needing to increase κ as the policy becomes more optimized and the system’s coupling increases. See Appendix B.5.2 for a quantitative example.

3.5 A Provable Framework for Localized Policy Improvement

We now integrate the results developed so far into a complete framework for **localized policy improvement**. The starting point is conceptual rather than algorithmic. Most analyses of multi-agent reinforcement learning (MARL)—especially stochastic ones such as actor–critic—intertwine spatial truncation error (from limiting attention to local neighborhoods) with stochastic approximation error (from finite samples and noisy updates). This coupling makes it difficult to tell whether a poor bound arises from the algorithm or from the structure of the environment. We therefore take a step back and study an **oracle model** in which all expectations under π (i.e., r^π , H^π) are known exactly. This idealization isolates the purely structural aspect of locality. Within this setting, we can quantify precisely how the truncation radius κ , the decay rate λ , and the policy smoothness (via τ) interact.

For each policy π , we associate a directed **support graph** G^π on vertices $[n]$ by drawing an edge $i \rightarrow j$ whenever $H_{j \leftarrow i}^\pi > 0$. This graph captures which coordinates can directly influence each other in one step of the policy–environment dynamics. When H^π is sparse—e.g., due to limited observation scopes or weak couplings—this graph has small degree, and a κ -hop neighborhood covers only a small subset of agents. All our algorithms will exploit this structure to compute local approximations by message passing on G^π .

Phase 1: Localized evaluation via message passing

The first phase aims to approximate the (generally global) Poisson solution h^π , which encodes the long-term value landscape of the policy. The challenge is that h^π depends on all agents through the transition kernel. Our approach constructs a **localized surrogate** $\hat{h}_\kappa^\pi$ that agrees with h^π up to an exponentially small bias outside a κ -hop neighborhood, and provides an explicit certificate for its accuracy.

Step 1: computing a locality certificate. Recall from Theorem 6 that the oscillation vector $\delta(h^\pi)$ can be written as a Neumann series in powers of $(H^\pi)^\top$ applied to $b^\pi = \delta(r^\pi)$. Truncating

this series at depth κ yields the computable **certificate**

$$\widehat{\delta}^{(\kappa)} := \sum_{t=0}^{\kappa} ((H^\pi)^\top)^t b^\pi.$$

Intuitively, $\widehat{\delta}_i^{(\kappa)}$ measures how far the effect of a reward perturbation can propagate to coordinate i within κ steps. Since $(H^\pi)^\top$ has nonzeros only along edges of G^π , this truncated sum can be computed by κ rounds of local message passing on G^π .

Step 2: constructing a surrogate value. Using this certificate, we define a truncated Poisson approximation

$$\widehat{h}_\kappa^\pi = \sum_{t=0}^{\kappa} (T^\pi)^t (r^\pi - \bar{r}^\pi) - c_\kappa,$$

where c_κ is an arbitrary constant (the value function is defined up to shift). The surrogate satisfies $\delta(\widehat{h}_\kappa^\pi) \leq \widehat{\delta}^{(\kappa)}$ by construction.

Theorem 7 (Localized evaluation: certificate and bias). *Let π be a product policy with $H^\pi = (E^s + E^a \Pi(\pi))^\top$ and assume $\|(H^\pi)^t\|_{\infty \rightarrow \infty} \leq C\lambda^t$ for some $\lambda < 1$. Then:*

(1) **Certificate.** *The true sensitivity satisfies $\delta(h^\pi) \leq \sum_{t=0}^{\infty} (H^\pi)^t b^\pi$, and the truncated certificate $\widehat{\delta}^{(\kappa)}$ gives a uniform bound*

$$\|\delta(h^\pi) - \widehat{\delta}^{(\kappa)}\|_\infty \leq \frac{C}{1-\lambda} \lambda^{\kappa+1} \|b^\pi\|_\infty.$$

(2) **Locality.** *Each component $\widehat{\delta}_i^{(\kappa)}$ depends only on nodes within $\text{Ball}_\kappa(i)$ in G^π and can be computed by κ message-passing rounds.*

(3) **Bias.** *The corresponding truncated Poisson solution satisfies*

$$\|\widehat{h}_\kappa^\pi - h^\pi\|_\infty \leq \frac{C}{2(1-\lambda)} \lambda^{\kappa+1} \|\delta(r^\pi)\|_1.$$

Hence, increasing κ expands the radius of communication, while increasing λ (tighter coupling) slows exponential decay. Together they control the bias-cost tradeoff in local evaluation.

Given $\hat{h}_\kappa^\pi$, the **localized surrogate advantage**

$$\hat{A}_\kappa^\pi(s, a) = r(s, a) - \bar{r}^\pi + \sum_y P(y | s, a) \hat{h}_\kappa^\pi(y) - \hat{h}_\kappa^\pi(s)$$

acts as a locally computable substitute for A^π .

Phase 2: Localized policy improvement via block-coordinate KL updates

The second phase performs policy improvement. Updating all agents simultaneously can cause instability because each agent's update changes the stationary distribution d^π that the others depend on. To ensure monotonic progress, we adopt a **block-coordinate** approach, updating one agent at a time while keeping others fixed.

For a given agent k , define the expected local advantage logits

$$g_k^\pi(s, a_k) = \mathbb{E}_{a_{-k} \sim \prod_{j \neq k} \pi_j(\cdot | s_{O_j})} [\hat{A}_\kappa^\pi(s, (a_k, a_{-k}))].$$

We then perform a **KL-proximal** update around the current policy:

$$\pi_k^{\text{new}}(\cdot | s_{O_k}) := \arg \max_{\mu \in \Delta(\mathcal{A}_k)} \{ \langle \mu, g_k^\pi(s, \cdot) \rangle - \tau \text{KL}(\mu \| \pi_k(\cdot | s_{O_k})) \}. \quad (3.1)$$

This has the closed-form solution $\pi_k^{\text{new}}(\cdot | s_{O_k}) \propto \pi_k(\cdot | s_{O_k}) \exp(g_k^\pi(s, \cdot)/\tau)$, which preserves the product structure and provides an interpretable temperature-controlled step size.

Algorithm 1 Localized Policy Improvement (LPI) – oracle iteration

- 1: **Inputs:** baseline policy $\pi^{(0)}$, truncation radius κ , temperature τ
 - 2: **for** outer iteration $\ell = 0, 1, 2, \dots$ **do**
 - 3: **Phase 1 (evaluation):** compute $b^{(\ell)} = \delta(r^{\pi^{(\ell)}})$, $H^{(\ell)} = E^s + E^a \Pi(\pi^{(\ell)})$
 - 4: Message passing for κ rounds to get $\hat{\delta}^{(\kappa, \ell)} = \sum_{t=0}^{\kappa} ((H^{(\ell)})^\top)^t b^{(\ell)}$
 - 5: Construct $\hat{h}_\kappa^{(\ell)} = \sum_{t=0}^{\kappa} (T^{\pi^{(\ell)}})^t (r^{\pi^{(\ell)}} - \bar{r}^{\pi^{(\ell)}}) - c_\kappa$
 - 6: **Phase 2 (block updates):** set $\pi_{\text{temp}} \leftarrow \pi^{(\ell)}$
 - 7: **for** each agent $k = 1, \dots, n$ **do**
 - 8: compute $g_k^{(\ell)}(s, a_k) = \mathbb{E}_{a_{-k} \sim \prod_{j \neq k} \pi_j^{(\ell)}(\cdot | s_{O_j})} [\hat{A}_\kappa^{(\ell)}(s, (a_k, a_{-k}))]$
 - 9: update $\pi_{\text{temp}, k}(\cdot | s_{O_k}) \propto \pi_k^{(\ell)}(\cdot | s_{O_k}) \exp(g_k^{(\ell)}(s, \cdot)/\tau)$
 - 10: **end for**
 - 11: set $\pi^{(\ell+1)} \leftarrow \pi_{\text{temp}}$
 - 12: **end for**
-

Each block step is the exact maximizer of a local surrogate that upper bounds the global

KL-anchored objective

$$\bar{r}_{\tau,\pi}(\mu) = \mathbb{E}_{d^\mu,\mu}[r(S, A)] - \tau \mathbb{E}_{S \sim d^\mu} \text{KL}(\mu_k(\cdot \mid S_{O_k}) \parallel \pi_k(\cdot \mid S_{O_k})).$$

Thus a one-block update satisfies

$$\bar{r}_{\tau,\pi}(\mu) - \bar{r}_{\tau,\pi}(\pi) \geq \tau \mathbb{E}_{S \sim d^\mu} \text{KL}(\pi_k^{\text{new}}(\cdot \mid S_{O_k}) \parallel \pi_k(\cdot \mid S_{O_k})) - 2 \|\hat{h}_\kappa^\pi - h^\pi\|_\infty,$$

where the first term ensures monotonic improvement and the second captures the truncation bias from Phase 1. Under $\rho(H^\pi) \leq \lambda < 1$, this bias decays as $O(\lambda^{\kappa+1})$ by Theorem 7. Conceptually, the theorem guarantees that **local improvement is globally sound up to exponentially small error**: if the system is sufficiently local ($\rho(H^\pi) < 1$), then even approximate, κ -hop updates produce predictable progress toward the global regularized optimum.

Chapter 4

The Role of Generator Access in Autoregressive Post-Training

This chapter is based on the following paper:

Amit Kiran Rege. The role of generator access in autoregressive post-training, 2026.

Abstract: We study how generator access constrains autoregressive post-training. The central question is whether the learner is confined to fresh root-start rollouts or can return to previously built prefixes and query the next-token rule there. In the root-start regime, output sampling, generated-token log probabilities, top- k reports, and full next-token distributions along sampled trajectories all reduce to one canonical experiment, limited by the on-policy probability of reaching informative prefixes. Weak prefix control breaks this barrier, and once control is available, richer observations such as conditional sampling or logits can outperform top-1 access. Changing only the generator interface creates an exponential gap for KL-regularized outcome-reward post-training.

The corresponding supplementary material can be found in [Appendix C](#).

4.1 Introduction

Large language models have transformed modern AI, and recent reasoning-oriented systems push that progress further through post-training, reinforcement learning, and inference-time search [55, 24, 30]. At the same time, a recurring empirical theme is that standard post-training often sharpens trajectories that are already present in the base model, rather than uncovering genuinely new ones [77, 34, 72]. That raises a basic question which is easy to state but has not been cleanly

separated in existing theory. When post-training runs into a barrier, how much of that barrier comes from the base distribution itself, and how much comes from the way the learner is allowed to interrogate the generator?

A simple thought experiment already shows why this matters. Imagine a reasoning task whose successful solution requires a long chain of correct intermediate decisions. Along the one correct chain, the base model gives the right next token only a mild advantage over its competitors. After a single wrong turn, however, the distribution becomes uninformative. In that situation, ordinary sampling has to survive the entire chain on policy. Generated-token log probabilities, top- k reports, or even full next-token distributions on visited prefixes do not change that basic geometry: they still reveal information only along whatever trajectory the model happened to generate. By contrast, if the learner can return to a partially constructed prefix and inspect the next-token rule there, then it can test the t th step directly instead of re-earning the first $t - 1$ steps every time. This gap is about the interface to the generator and not about a more sophisticated optimizer.

This paper studies that interface through two notions. Prefix control asks which prefixes the learner may place the generator at before making a query. Prefix observation asks what the learner sees once that prefix has been fixed. The first distinction is about where the learner may look; the second is about what the learner sees there. Our main claim is that, for autoregressive post-training, these two distinctions do not sit on equal footing. Prefix control is the first boundary. Observation richness becomes the second boundary only after control has been granted. Table 4.1 summarizes the access taxonomy studied in this chapter.

The key takeaway is that the decisive split is between interfaces that are limited to root-start rollouts and interfaces that let the learner choose which prefix to inspect.

The paper develops this point in four steps. First, we characterize the no-reset regime, where each generator query begins with a fresh rollout from the root and may depend only on the sampled trajectory and the next-token distributions encountered along that trajectory. In this regime there is a single canonical experiment. Output-only sampling, generated-token log probabilities, top- k reports on sampled tokens, and even full next-token distributions on sampled prefixes are all just

Control regime	Limited observation	Rich observation
No prefix control	Output-only sampling	Generated-token log probabilities, top-log-probability lists, or full next-token distributions on visited prefixes
Prefix control	Top token at a chosen prefix	Chosen-prefix sampling, chosen-prefix logits, or teacher-forced sequence scores

Table 4.1: The taxonomy studied in this paper. The main point is that the horizontal distinction comes first. Once the learner is confined to root-start rollouts, richer trajectory-local observations do not overcome the need to reach informative prefixes on policy.

randomized post-processings of the same object. Second, once the learner is allowed even weak local reset, meaning that it may revisit previously queried prefixes and extend them one token at a time, the picture changes sharply: the universal no-reset barrier disappears on a hidden-path family. Third, after control has been supplied, observation richness becomes meaningful again. On a branching family, top-1 access can be much weaker than chosen-prefix sampling or logits. Fourth, this generator-side distinction can be lifted into the standard post-training objective. Holding the reward model, the KL regularizer, and the policy class fixed, changing only generator access can create an exponential gap in the number of generator queries required for outcome-reward post-training.

This perspective complements two lines of recent work. One line fixes the access model and shows that coverage of good responses under the base model governs the difficulty of sampling-style alignment and inference-time selection [17, 30, 14, 29]. Another line fixes generator access and studies the feedback channel, showing for example that process feedback can avoid outcome-reward barriers [54]. Our contribution is to isolate the remaining axis. We ask what changes when the reward and update class are held fixed and only the generator-side interface is allowed to vary.

The rest of the paper is organized as follows. Section 4.2 formalizes prompt-conditioned autoregressive generators, the prefix-tree view, and the access models studied here. Section 4.3 proves the collapse of the no-reset regime and the resulting reachability barrier. Later sections show how weak local reset breaks that barrier, why branching makes observation matter again once

control is available, and how the same boundary reappears in KL-regularized post-training.

4.2 Setup

4.2.1 Prompt-conditioned autoregressive generators

Let $\mathcal{X}$ be a prompt space with prompt distribution ρ . Fix a finite vocabulary $\Sigma = \{1, 2, \dots, K\}$ with $K \geq 2$ and a horizon $H \geq 1$. For $t \geq 0$, let Σ^t denote the set of strings of length t , write $\Sigma^{<H} := \bigcup_{t=0}^{H-1} \Sigma^t$ and $\Sigma^{\leq H} := \bigcup_{t=0}^H \Sigma^t$, and let $\emptyset$ denote the unique element of Σ^0 .

A prompt-conditioned autoregressive generator is a family of next-token distributions

$$M(\cdot \mid x, p) \in \Delta(\Sigma), \quad x \in \mathcal{X}, p \in \Sigma^{<H}.$$

Given a prompt x , a response $Y = (Y_1, \dots, Y_H) \in \Sigma^H$ is generated sequentially by drawing $Y_t \sim M(\cdot \mid x, Y_{<t})$ for $t = 1, \dots, H$. Equivalently,

$$\Pr_M(Y = y \mid x) = \prod_{t=1}^H M(y_t \mid x, y_{<t}) \quad \text{for every } y \in \Sigma^H.$$

The structural results hold at a fixed prompt; the prompt distribution enters only in the last section, where we study a prompt-averaged post-training objective. The single-hard-prompt case corresponds to ρ being a point mass.

4.2.2 Prefix trees and generator access

For a fixed prompt x , autoregressive generation can be viewed as a depth- H tree whose internal nodes are prefixes in $\Sigma^{<H}$. From a prefix p , each token $a \in \Sigma$ leads to the child pa , and the generator attaches to p the next-token distribution $M(\cdot \mid x, p)$. In this picture, a complete response is a root-to-leaf path.

This viewpoint makes the access distinctions concrete. A learner might only be allowed to start at the root, sample a rollout, and inspect whatever prefixes that rollout happened to visit. Or it might be allowed to return to a previously constructed prefix and ask for the next-token rule there. The first kind of interaction is the analogue of ordinary episodic access in reinforcement

learning; the second is the analogue of reset-like access [52, 65]. For autoregressive models, the natural state space is the prefix tree itself.

We separate generator access into two parts. Prefix control asks which prefixes the learner may interrogate. Prefix observation asks what the learner receives once the relevant prefix has been fixed. A standard API that returns output tokens together with generated-token log probabilities has richer observation than plain sampling, but no extra control: both are tied to a single root-start rollout. A chosen-prefix sampling oracle and a chosen-prefix logit oracle have the same control but different observations. Teacher-forced sequence scoring is score-like, but it lies on the control-rich side because the learner is allowed to choose the entire completion whose probability is evaluated.

The weakest regime studied in the paper is no-reset access. Informally, one query chooses a parameter q , draws a fresh root-start rollout $Y \sim M(\cdot \mid x)$, forms the visited-prefix distributions $\mu_t := M(\cdot \mid x, Y_{<t})$, and returns a reply that may depend on q , on the sampled trajectory Y , and on the visited-prefix distributions $(\mu_1, \dots, \mu_H)$, but not on any unvisited prefix. This covers output-only sampling, generated-token log probabilities, sampled-token top- k reports, and even full next-token distributions on sampled prefixes.

The stronger regime used later is weak local reset. A sequence of queried prefixes $p_1, \dots, p_q \in \Sigma^{<H}$ obeys the local-reset discipline if $p_1 = \emptyset$ and, for every round $r \geq 2$, the new query p_r is either a previously queried prefix or a one-token extension of a previously queried prefix. The learner is not allowed to teleport to an unrelated deep node, but it is allowed to build a prefix one token at a time and to revisit prefixes it has already constructed.

Once local reset is available, we consider three concrete chosen-prefix interfaces. Under **PrefixSample**, a query prefix p returns an independent draw $A \sim M(\cdot \mid x, p)$. Under **PrefixTop**, a query prefix p returns the unique most likely next token under $M(\cdot \mid x, p)$ when that maximizer is unique, and returns a distinguished symbol $\perp$ otherwise. Under **PrefixLogit $_{\xi}$** , a query prefix p returns a vector $\tilde{\ell}_p \in \mathbb{R}^{\Sigma}$ satisfying $\|\tilde{\ell}_p - \log M(\cdot \mid x, p)\|_{\infty} \leq \xi$. We write **PrefixLogit** for the exact case $\xi = 0$.

We also include teacher-forced sequence scoring. Under **SeqScore $_{\xi}$** , a query completion $y \in \Sigma^H$

returns a number $\tilde{s}(y)$ satisfying

$$|\tilde{s}(y) - \log \Pr_M(Y = y \mid x)| \leq \xi.$$

Operationally, this is just teacher forcing followed by reading off the sequence log-likelihood. It is not a literal chosen-prefix oracle, but it still lies on the control-rich side of the taxonomy because the learner is allowed to evaluate a counterfactual completion of its choice rather than waiting for that completion to arise on policy.

These interfaces are close to what current systems expose. Generated-token log probabilities and top-log-probability lists appear in widely used APIs, while open-weight causal language models expose next-token logits and sequence scores through ordinary forward passes [57, 56, 33, 31, 32]. The taxonomy organizes access patterns that already appear in practice.

In practical terms, weak local reset corresponds to a very modest form of counterfactual control. The learner does not get to jump to an arbitrary deep prefix. It can only keep a partial continuation, revisit it, and extend it one token at a time. This is the prefix-tree analogue of stepping back to a previously constructed partial solution and resuming from there, rather than resampling the entire trajectory from the beginning.

4.2.3 KL-regularized outcome-reward post-training

The last part of the paper returns from access models to a standard post-training objective. Let $r : \mathcal{X} \times \Sigma^H \rightarrow \mathbb{R}$ be an outcome reward, let $M(\cdot \mid x)$ be the base generator, and let $\pi(\cdot \mid x)$ be a prompt-conditioned policy. For $\beta > 0$, define

$$J_\beta(\pi) := \mathbb{E}_{x \sim \rho} [\mathbb{E}_{y \sim \pi(\cdot \mid x)} r(x, y) - \beta D_{\text{KL}}(\pi(\cdot \mid x) \parallel M(\cdot \mid x))].$$

The final theorem keeps this reward model, this KL-regularized objective, and the policy class fixed, and varies only the generator-side interface. The resulting separation shows that the access taxonomy is not merely a statement about synthetic search problems. It already changes the complexity of a standard post-training objective.

4.3 The no-reset regime

The no-reset regime is the root-start side of the taxonomy. Each query begins with a fresh rollout from the root, and whatever the learner sees is constrained to the prefixes visited by that rollout. At first sight, this leaves many different interfaces: the sampled completion, generated-token log probabilities, a top- k list, or even the full next-token distributions on sampled prefixes. Statistically, however, these are all equivalent.

Throughout this section we fix a prompt $x \in \mathcal{X}$. Prompt-distributed versions follow by averaging over $x \sim \rho$.

Definition 4. *A generator experiment at prompt x is called no-reset if one query is answered as follows. The learner chooses a query parameter q . The oracle then draws a fresh rollout $Y = (Y_1, \dots, Y_H) \sim M(\cdot \mid x)$, forms the visited-prefix distributions $\mu_t := M(\cdot \mid x, Y_{<t})$ for $t = 1, \dots, H$, and returns a reply drawn from a kernel that may depend on q , on Y , and on $(\mu_1, \dots, \mu_H)$, but not on any unvisited prefix.*

Definition 4 is deliberately broad. It allows arbitrary measurable summaries of the sampled trajectory and of the next-token distributions encountered along that trajectory, including arbitrary internal randomization. What it forbids is counterfactual inspection of prefixes that the rollout did not visit.

The canonical experiment in this regime simply returns everything that a no-reset interface is ever allowed to depend on.

Definition 5. *For a fixed prompt x , let PathFull_x denote the experiment that returns*

$$W_x := (Y, \mu_1, \dots, \mu_H),$$

where $Y \sim M(\cdot \mid x)$ and $\mu_t = M(\cdot \mid x, Y_{<t})$ for $t = 1, \dots, H$.

Theorem 8. *Fix a prompt x . An experiment is no-reset if and only if it is a randomized post-processing of PathFull_x . In the usual comparison-of-experiments sense, PathFull_x is therefore the maximal no-reset experiment.*

Once the learner is confined to one root-start rollout per query, there is no deeper hierarchy among trajectory-local interfaces. Output-only sampling, generated-token log probabilities, sampled-token top- k reports, and full next-token distributions on sampled prefixes all sit inside the same experiment class; richer replies simply retain more information from the same trajectory.

This means that a lower bound on PathFull_x applies simultaneously to every trajectory-local interface. The key quantity governing that bound is on-policy reachability.

Definition 6. For a prefix set $U \subseteq \Sigma^{<H}$, define the reachability event

$$E_U(x) := \{\exists t \in \{1, \dots, H\} : Y_{<t} \in U\}, \quad Y \sim M(\cdot | x),$$

and the corresponding prefix reachability

$$\text{Reach}_M(x, U) := \Pr(E_U(x)).$$

The interpretation is immediate: $\text{Reach}_M(x, U)$ is the probability that an on-policy rollout ever visits the informative region U .

Now suppose two generators $M(\cdot | x)$ and $M'(\cdot | x)$ agree outside a prefix set U , meaning that $M(\cdot | x, p) = M'(\cdot | x, p)$ for every $p \notin U$. If a rollout never enters U , then the two models generate exactly the same transitions along that rollout, and the canonical no-reset reply is identical in the two worlds. The only way to tell the models apart is to reach U on policy.

Theorem 9. Fix a prompt x and a prefix set $U \subseteq \Sigma^{<H}$. Let $M(\cdot | x)$ and $M'(\cdot | x)$ agree outside U . Let A be any randomized adaptive algorithm that makes at most q queries to any no-reset experiment at prompt x , and let $T_q^A(M, x)$ denote its full transcript under M . Then

$$\text{TV}\left(\mathcal{L}(T_q^A(M, x)), \mathcal{L}(T_q^A(M', x))\right) \leq q \text{Reach}_M(x, U).$$

The same reachability value appears for both models. Agreement outside U implies

$$\text{Reach}_M(x, U) = \text{Reach}_{M'}(x, U).$$

Theorem 9 is the core lower bound on the no-reset side. It says that the real bottleneck is not whether the API reveals text, scores, or full distributions along a sampled path. The bottleneck is that the learner cannot inspect prefixes that the base model fails to reach on its own.

A direct consequence is that the standard trajectory-local interfaces used in practice all obey the same barrier.

Corollary 3. *Under the assumptions of Theorem 9, the same bound holds for output-only sampling, for output together with generated-token log probabilities, for output together with sampled-token top-log-probability lists or top-k reports, and for replies that include the full next-token distributions on sampled prefixes.*

This corollary is worth pausing over. Generated-token log probabilities are often treated as substantially stronger than sampling, and in a narrow sense they are. They reveal more about the prefixes that were actually visited. The point of Theorem 9 is that this extra information does not change the need to reach the informative part of the prefix tree in the first place.

The next sections confirm that this is indeed a control barrier. On the hidden-path family, every no-reset interface faces the same exponential bottleneck, while weak local reset breaks it. Once that boundary is crossed, a second distinction emerges: on a branching family, richer observations such as conditional sampling or logits outperform top-1 access even at chosen prefixes.

4.4 Escaping the barrier with weak local reset

The previous section shows that every no-reset interface is governed by the same on-policy reachability quantity. The question is whether this bottleneck comes from the absence of prefix control specifically, or reflects a more general hardness. The hidden-path family answers this by isolating the long-horizon cost as directly as possible.

We use a hidden path. This family is meant to model a fragile chain of reasoning: there is one long sequence of mildly favored correct steps, and once the rollout leaves that sequence the model stops providing useful guidance. Ordinary sampling then has to survive the whole chain on

policy, while local reset lets the learner inspect one step at a time. In this section and the next, all witness families are conditioned on a single fixed prompt, which we suppress from the notation.

Fix a signal parameter $\lambda > 0$ and define

$$p_+ := \frac{e^\lambda}{e^\lambda + K - 1}, \quad p_- := \frac{1}{e^\lambda + K - 1}, \quad \Delta := p_+ - p_-.$$

Thus $p_+ > 1/K > p_-$ and $\Delta > 0$.

Definition 7. For each hidden string $z = (z_1, \dots, z_H) \in \Sigma^H$, the hidden-path generator M_z is defined by

$$M_z(a \mid p) = \begin{cases} p_+, & \text{if } p = z_{<t} \text{ for some } t \in \{1, \dots, H\} \text{ and } a = z_t, \\ p_-, & \text{if } p = z_{<t} \text{ for some } t \in \{1, \dots, H\} \text{ and } a \neq z_t, \\ 1/K, & \text{if } p \text{ is not a prefix of } z. \end{cases}$$

Along the hidden path, the correct next token is only mildly preferred. Away from the hidden path, the model is uniform and carries no information about z . That is exactly the feature we want: the family isolates the cost of repeatedly re-reaching a useful prefix from the root.

Theorem 10. Fix $K \geq 2$, $H \geq 1$, and $\lambda > 0$. Let $z, z' \in \Sigma^H$ satisfy $z_{1:H-1} = z'_{1:H-1}$ and $z_H \neq z'_H$. Let A be any randomized adaptive algorithm that makes at most q queries to any no-reset experiment at the fixed prompt. Then

$$\text{TV}\left(\mathcal{L}(T_q^A(M_z)), \mathcal{L}(T_q^A(M_{z'}))\right) \leq q p_+^{H-1}.$$

Consequently, any such algorithm that distinguishes M_z from $M_{z'}$ with worst-case success probability at least $2/3$ must satisfy

$$q \geq \frac{1}{3p_+^{H-1}}.$$

The reason is simple. The two models differ only at the depth- $H - 1$ prefix $z_{1:H-1}$. A root-start rollout reaches that prefix only if it follows the hidden path correctly for $H - 1$ consecutive steps, which happens with probability p_+^{H-1} . The reachability barrier from Section 4.3 then applies immediately. For fixed K and λ , this is exponential in the horizon.

Algorithm 2 Recovering a hidden path with chosen-prefix samples

```

1: Initialize  $\hat{z}_{<1} := \emptyset$ 
2: for  $t = 1$  to  $H$  do
3:   for  $j = 1$  to  $m$  do
4:     Query PrefixSample( $\hat{z}_{<t}$ ) and receive  $A_j^{(t)} \in \Sigma$ 
5:   end for
6:   for each  $a \in \Sigma$  do
7:     Set  $N_a^{(t)} := \sum_{j=1}^m \mathbf{1}\{A_j^{(t)} = a\}$ 
8:   end for
9:   Set  $\hat{z}_t \in \arg \max_{a \in \Sigma} N_a^{(t)}$ 
10: end for
11: Return  $\hat{z} = (\hat{z}_1, \dots, \hat{z}_H)$ 

```

The point of the family is that weak local reset breaks this exponential barrier. The learner does not need arbitrary teleportation to deep nodes. It is enough to build the prefix one token at a time and revisit prefixes it has already constructed. The next algorithm does exactly that with chosen-prefix next-token samples.

The algorithm starts at the root and keeps a running reconstruction of the hidden prefix. At stage t , it repeatedly queries the next-token distribution at the current reconstructed prefix $\hat{z}_{<t}$. If the reconstruction so far is correct, then the true next token z_t appears with probability p_+ while every rival appears with probability p_- . The empirical majority therefore identifies z_t once m is large enough. The next round moves to the one-token extension that was just recovered. No deep jump is ever made, so the procedure obeys the local-reset discipline exactly as defined in Section 4.2.

Theorem 11. Fix $K \geq 2$, $H \geq 1$, $\lambda > 0$, and $\delta \in (0, 1)$. Let

$$m := \left\lceil \frac{2}{\Delta^2} \log \left(\frac{H(K-1)}{\delta} \right) \right\rceil.$$

Then Algorithm 2 recovers the hidden path z with probability at least $1 - \delta$ using at most Hm chosen-prefix samples. The queried prefixes obey the local-reset discipline.

For fixed K and λ , the gap Δ is a positive constant, so the query complexity is

$$O\left(H \log \frac{H}{\delta}\right).$$

On the same family, no-reset interaction requires exponentially many queries on some pair of models, while weak local reset with conditional next-token samples already gives a polynomial reconstruction.

The hidden path is the right family for the first axis, but not for the second. Once control is available, even top-1 access is enough on this family, because each informative prefix has a unique best child. To see a genuine observation gap after control has been granted, we need branching.

4.5 Observation matters after control

The hidden path shows that prefix control is the first boundary. It says little about what richer observations add once control is available, since there is only one informative continuation at each useful prefix. To isolate the observation boundary, we need a family where the top token is unique at every prefix yet still reveals nothing about the branching structure.

We work with a family of prefix trees in which one continuation is a stable default suggestion, shared across all models, while the genuinely informative branch is encoded in the lower-ranked coordinates. This is still a branching family, but it is branching in a way that does not disappear under a unique top token.

Assume $K \geq 3$, and let token 1 play the role of a distinguished leader token.

Definition 8. *A leader trie of depth H is a prefix-closed set $T \subseteq \Sigma^{\leq H}$ such that $\emptyset \in T$, every leaf of T has depth exactly H , and every internal node $p \in I(T) := T \cap \Sigma^{<H}$ has exactly two children in T , namely the leader child $p1$ and one additional child $pb_T(p)$ for some $b_T(p) \in \{2, \dots, K\}$.*

Thus every internal node has a common leader child and one hidden child. The tree is still branching, but the identity of the leader is the same everywhere.

Now define four probabilities

$$\alpha := \frac{4}{K+4}, \quad \beta := \frac{2}{K+4}, \quad \gamma := \frac{1}{K+4}, \quad \gamma_0 := \frac{1}{K+3}.$$

These satisfy

$$\alpha > \beta > \gamma_0 > \gamma.$$

The associated generator is chosen so that the leader token is always the unique top token, whether or not the queried prefix is informative.

Definition 9. *Given a leader trie T , the generator M_T is defined as follows. If $p \in I(T)$, then*

$$M_T(1 \mid p) = \alpha, \quad M_T(b_T(p) \mid p) = \beta, \quad M_T(a \mid p) = \gamma \text{ for } a \notin \{1, b_T(p)\}.$$

If $p \notin I(T)$, then

$$M_T(1 \mid p) = \frac{4}{K+3}, \quad M_T(a \mid p) = \gamma_0 \text{ for } a \neq 1.$$

At an internal node, the leader token is uniquely best, but one nonleader token is also elevated. At an off-trie prefix, the leader remains uniquely best and all nonleaders sit at the same lower baseline. The informative part of the model is therefore not the top token. It is the identity of the elevated nonleader.

Theorem 12. *Assume $K \geq 3$. For every leader trie T and every queried prefix $p \in \Sigma^{<H}$, the oracle $\text{PrefixTop}(p)$ returns the token 1. Consequently, for any adaptive algorithm interacting only with PrefixTop , the full transcript law is the same for every leader trie of depth H . In particular, no such algorithm can distinguish two different leader tries with success probability greater than $1/2$.*

The theorem is stronger than the tie-based example because it does not rely on any ambiguity in the maximizer. The top token is unique everywhere. It is simply the wrong statistic.

The hidden branch, however, can be recovered from richer observations. The relevant margins are

$$\Gamma_{\text{lead}} := \frac{\beta - \gamma_0}{2} = \frac{K+2}{2(K+4)(K+3)}, \quad \gamma_{\text{lead}} := \frac{\log \beta - \log \gamma_0}{2} = \frac{1}{2} \log \left(\frac{2(K+3)}{K+4} \right).$$

The probability margin Γ_{lead} separates the hidden child from every nonhidden nonleader. The log-probability margin γ_{lead} is the corresponding gap in log space.

The first recovery result uses chosen-prefix logits and a breadth-first traversal of the trie.

The queue stores prefixes whose children still need to be inspected. At a queried prefix p , line 4 asks for the full next-token score vector. Line 5 looks only at nonleader coordinates and

Algorithm 3 Recovering a leader trie with chosen-prefix logits

```

1: Initialize  $\hat{T} := \{\emptyset\}$  and queue  $Q := (\emptyset)$ 
2: while  $Q$  is nonempty do
3:   Pop the first prefix  $p$  from  $Q$ 
4:   Query  $\text{PrefixLogit}_\xi(p)$  and receive  $\tilde{\ell}_p \in \mathbb{R}^\Sigma$ 
5:   Set  $\hat{B}(p) := \{a \in \{2, \dots, K\} : \tilde{\ell}_p(a) > \log \gamma_0 + \gamma_{\text{lead}}\}$ 
6:   if  $\hat{B}(p)$  is a singleton  $\{\hat{b}(p)\}$  then
7:     Add  $p1$  and  $p\hat{b}(p)$  to  $\hat{T}$ 
8:     if  $|p| + 1 < H$  then
9:       Append  $p1$  and  $p\hat{b}(p)$  to  $Q$ 
10:    end if
11:  end if
12: end while
13: Return  $\hat{T}$ 

```

checks which of them lie above the fixed threshold halfway between $\log \beta$ and $\log \gamma_0$. If exactly one nonleader coordinate clears that threshold, then p is an internal node and the corresponding token is the hidden child. Lines 6 through 8 then add both active children, the common leader child and the hidden child, and place them in the queue if they have not yet reached depth H . If no nonleader clears the threshold, then p is not an internal node and the algorithm stops expanding it.

Theorem 13. *Let T be a leader trie. If $\xi < \gamma_{\text{lead}}$, then Algorithm 3 recovers T exactly from $|I(T)|$ chosen-prefix logit queries. The queried prefixes obey the local-reset discipline.*

The same breadth-first idea works with chosen-prefix samples. At each queried prefix, instead of reading the logit vector once, the learner takes repeated conditional samples and identifies the hidden child from empirical frequencies.

Theorem 14. *Let T be a leader trie and write $s := |I(T)|$. Suppose the learner knows an upper bound $S \geq s$. Fix $\delta \in (0, 1)$, and set*

$$m := \left\lceil \frac{1}{2\Gamma_{\text{lead}}^2} \log \left(\frac{2(K-1)S}{\delta} \right) \right\rceil.$$

Then the local-reset algorithm obtained by replacing each query in Algorithm 3 with m calls to

`PrefixSample(p)` and replacing line 4 by the empirical test

$$\hat{B}(p) := \{a \in \{2, \dots, K\} : \hat{P}_p(a) > \gamma_0 + \Gamma_{\text{lead}}\}$$

recovers T with probability at least $1 - \delta$ using at most Sm chosen-prefix samples.

Once control is available, the observation model matters. The branching structure is invisible to top-1 access but recoverable from conditional samples or logits.

One score-based interface deserves separate mention. Teacher-forced sequence scoring lies on the control-rich side of the taxonomy for the same reason: it evaluates a completion chosen by the learner rather than one generated on policy.

Corollary 4. *In the hidden-path family, exact SeqScore recovers the hidden path using HK chosen-completion score queries.*

At stage t , one simply fixes an arbitrary suffix of length $H - t$, compares the scores of the K one-token extensions of the current prefix followed by that suffix, and keeps the best extension. This is just teacher forcing. It is not a literal chosen-prefix oracle, but it belongs on the same side of the control boundary.

The two witness families can now be summarized in one statement.

Corollary 5. *Fix $K \geq 3$ and $\lambda > 0$. On the hidden-path family, every no-reset generator interface requires exponentially many queries on some pair of models by Theorem 10, while chosen-prefix sampling recovers the hidden path with*

$$O\left(H \log \frac{H}{\delta}\right)$$

local-reset queries by Theorem 11. On the leader-trie family, chosen-prefix top-token access is useless by Theorem 12, while chosen-prefix logits and chosen-prefix sampling recover the trie efficiently under the local-reset discipline by Theorems 13 and 14. Prefix control is therefore the first boundary in autoregressive generator access, and observation richness becomes a second boundary only after control is available.

The next section shows that the same distinction changes the complexity of a standard post-training objective. The reward model and the KL regularizer stay fixed. Only the generator-side interface changes.

4.6 Implications for KL-regularized post-training

The same boundary that separates the identification regimes also appears in a standard post-training objective. The reward is outcome-only, the objective is KL-regularized, and the policy class is the full family of prompt-conditioned policies; only the generator access changes.

Fix a prompt distribution ρ on $\mathcal{X}$, and let $x^* \in \mathcal{X}$ be a designated hard prompt with mass $\eta := \rho(x^*) > 0$. Choose integers $D, L \geq 1$ and set $H := D + L + 1$. Let $v \in \Sigma^D$ be a known scaffold, and let $\tau_0, \tau_1 \in \Sigma$ be two distinct terminal tokens. The hidden state of the instance consists of a suffix $s \in \Sigma^L$ and a reward bit $b \in \{0, 1\}$. Write

$$z_{s,b} := v s \tau_b \in \Sigma^H$$

for the target completion.

On the hard prompt x^* , the base generator Q_s behaves like the hidden-path family from Section 4.4 along the string vs : for the first $D + L$ steps it mildly favors the next token that stays on that path, and once the full prefix vs has been reached the final next-token distribution is uniform. On every prompt $x \neq x^*$, the base generator is a fixed easy model, identical across the whole family. The outcome reward is concentrated on the single target completion:

$$r_{s,b}(x, y) := R \mathbf{1}\{x = x^*, y = z_{s,b}\}.$$

The scaffold v is known, the hidden suffix s determines where the informative subtree lies, and the reward bit b is hidden even after s is known, because the final token is uniform under the base generator.

We evaluate policies with the KL-regularized objective from Section 4.2,

$$J_\beta(\pi) = \mathbb{E}_{x \sim \rho} \left[\mathbb{E}_{y \sim \pi(\cdot | x)} r_{s,b}(x, y) - \beta D_{\text{KL}}(\pi(\cdot | x) \| Q_s(\cdot | x)) \right].$$

As before, a reward query chooses a prompt and a policy, samples a completion from that policy, and observes the resulting outcome reward.

The exact optimizer on each instance has the standard Gibbs form. On every easy prompt $x \neq x^*$, the reward is zero and the optimizer agrees with the base generator. At the hard prompt,

$$\pi_{s,b}^*(y \mid x^*) \propto Q_s(y \mid x^*) \exp(r_{s,b}(x^*, y)/\beta).$$

So once the hidden state (s, b) has been identified, the optimal policy can be written down exactly.

The upper bound is straightforward once chosen-prefix sampling is available.

Theorem 15. *Fix $\delta \in (0, 1)$, and let*

$$m := \left\lceil \frac{2}{\Delta^2} \log \left(\frac{L(K-1)}{\delta} \right) \right\rceil.$$

For the family above, there is an algorithm with chosen-prefix next-token sampling access to the base generator and one outcome-reward query that outputs the exact maximizer of J_β with probability at least $1 - \delta$. The algorithm uses at most $H + Lm$ generator queries and one reward query.

The algorithm first walks down the known scaffold one token at a time so that the queried prefixes obey the local-reset discipline. It then treats the hidden suffix as a length- L hidden path and recovers it with Algorithm 2. Once s is known, one reward query to the deterministic policy concentrated on $v s \tau_0$ identifies the reward bit b . At that point both the base generator and the rewarded completion are known, so the Gibbs formula gives the exact optimizer.

The lower bound shows that no-reset access leaves the problem exponentially hard on some instance, even though the reward model and objective have not changed.

Let $q_0 := Q_s(z_{s,b} \mid x^*) = p_+^{D+L}/K$ be the base-model mass of the rewarding completion, and choose

$$R := \beta \log \left(\frac{4}{q_0} \right).$$

Also write $N := 2K^L$ for the number of possible target completions $v s \tau_b$.

Theorem 16. *Assume the reward scale above. Consider any algorithm that may interleave at most q_g no-reset generator queries with at most $q_r < N$ outcome-reward queries and then outputs a prompt-conditioned policy $\hat{\pi}$. Then there exists a choice of hidden suffix $s \in \Sigma^L$ and reward bit $b \in \{0, 1\}$ such that*

$$\Pr\left(J_\beta(\hat{\pi}) \geq J_\beta(\pi_{s,b}^*) - \frac{\eta\beta}{4}\right) \leq q_g p_+^D + \frac{q_r}{N} + \frac{4}{N - q_r}.$$

Each term has a clear meaning. The factor $q_g p_+^D$ is the chance that one of the no-reset generator queries even reaches the scaffold and enters the informative part of the tree. If that never happens, then generator interaction reveals nothing about the hidden suffix. The term q_r/N is the chance that an outcome-reward query simply guesses the rewarded completion by brute force. If that never happens either, then the remaining reward queries can do no more than rule out candidates one by one, which leads to the last term $4/(N - q_r)$.

All of the hardness is concentrated on a single prompt x^* , so this is already a prompt-distributed post-training lower bound rather than a separate single-prompt phenomenon. The average-regret threshold scales with the prompt mass η .

The cleanest summary comes from keeping the number of reward queries constant.

Corollary 6. *Fix $K \geq 2$, $\lambda > 0$, $\beta > 0$, and any constant $q_r \geq 1$. There exists a constant $c = c(K, \lambda) > 1$ such that for every sufficiently large horizon H and every prompt distribution ρ with a designated prompt x^* of mass $\eta > 0$, there is a family of KL-regularized outcome-reward post-training instances of horizon H with the following property. With chosen-prefix next-token sampling access, one reward query and*

$$O\left(H \log \frac{H}{\delta}\right)$$

generator queries suffice to output the exact optimizer with probability at least $1 - \delta$. With only no-reset generator access, any algorithm that uses at most q_r reward queries and succeeds with probability at least $2/3$ in outputting a policy of average regret at most $\eta\beta/4$ must make

$$\Omega(c^H)$$

generator queries.

The reward, KL objective, and policy class are identical across the two access regimes. Changing only the generator interface turns a polynomial problem into an exponentially hard one.

4.7 Related work

Recent theory already clarifies two adjacent parts of the picture. Under fixed sampling-style access, coverage of good responses under the base model governs the complexity of exploration, inference-time alignment, and sharpening-style self-improvement [17, 30, 14, 29]. Our no-reset theorem fits naturally into that line of work. It identifies the generator-side reason those coverage quantities appear there: once interaction is confined to root-start rollouts, all trajectory-local APIs collapse to a single experiment whose power is controlled by on-policy reachability.

A second line of work studies stronger access models in RL and related sequential settings. Reset-style interaction changes what can be learned or planned efficiently in reinforcement learning [52, 65], while conditional queries can make otherwise hard sequential distributions learnable [48, 44]. Our prefix-tree formalism is the autoregressive analogue of that distinction. For generators, the first distinction is whether the learner can return to a chosen prefix at all.

The paper is also complementary to work on richer feedback and richer intermediate signals. Mousavi-Hosseini and Erdogdu show that process rewards can avoid outcome-reward barriers under fixed generator access [54]. Botta et al., Amani et al., and Tuyls et al. study verifier guidance, partial rationale exposure, and hidden-state-based exploration [6, 1, 73]. Our KL theorem keeps the reward model fixed and varies only generator access, while the structural results operate on the distributional generator side, without hidden states or external verifiers.

Recent empirical work on reasoning post-training points to the same tension between support, exploration, and intermediate control [77, 34, 72]. The present paper does not explain all of those phenomena, but it does show that root-start interaction and reset-like prefix control are genuinely different computational regimes.

4.8 Conclusion

The main lesson is that generator access is part of the computational problem. If interaction is confined to root-start rollouts, then output sampling, generated-token log probabilities, top- k reports, and full visited-prefix distributions all collapse to one experiment governed by on-policy reachability. Weak local reset crosses that boundary, and once across it, richer observations become genuinely useful. That distinction is already strong enough to create an exponential gap for KL-regularized outcome-reward post-training.

Chapter 5

Data Attribution in Adaptive Learning

This chapter is based on the following paper:

Amit Kiran Rege. Data attribution in adaptive learning, 2026.

Abstract: Machine learning models increasingly generate their own training data – online bandits, reinforcement learning, and post-training pipelines for language models are leading examples. In these adaptive settings, a single training observation both updates the learner and shifts the distribution of future data the learner will collect. Standard attribution methods, designed for static datasets, ignore this feedback. We formalize occurrence-level attribution for finite-horizon adaptive learning via a conditional interventional target, prove that replay-side information cannot recover it in general, and identify a structural class in which the target is identified from logged data.

The corresponding supplementary material can be found in [Appendix D](#).

5.1 Introduction

Modern machine learning systems are trained on large datasets, and the composition of those datasets matters. Questions around data privacy, model debugging, and data curation all reduce to a common underlying question: how much does a specific training example actually shape what a model learns? Data attribution makes this precise. Given a trained model, it asks how the model would have changed if a particular training point had been removed, reweighted, or modified. In

the standard supervised setting this is a well-studied problem. Influence functions, TracIn, Data Shapley, and related methods estimate how the final predictor changes when a sample is perturbed in a fixed training set, and a large literature has developed around the approximation strategies and theoretical foundations behind these estimates. The shared structure underlying all of them is that after one perturbs a training point, the future training data do not themselves change.

Adaptive learning complicates the picture in a way these methods are not designed to handle. In online bandits, reinforcement learning, and post-training procedures where later data depend on the current policy, training data is not collected from a fixed source. The learner interacts with an environment, and what it observes next depends on what it has already learned. A single training observation therefore does two things at once: it updates the learner, and through the updated learner it reshapes which data will be collected going forward. Perturbing that observation can change the final model along two distinct channels—through the learning dynamics on the data that follows, and through the future data distribution itself.

A two-step bandit illustrates the point. Suppose the learner observes a reward at round 1, updates its policy, and then acts again at round 2. If one downweights the first interaction, there are two different counterfactuals one might study. One can hold the realized second-round interaction fixed and ask how the final policy changes along that fixed continuation. Or one can hold fixed only what happened through round 1, allow the perturbed learner to act again, and compare the outcome. In a fixed-dataset problem these two constructions coincide. In adaptive learning they do not.

Attribution in adaptive learning should therefore attach to a realized occurrence rather than to an abstract sample identity. The same nominal content can appear at different times and under different learner states, and in an adaptive system those are causally distinct objects. Once one takes this view, two choices must be made before any approximation method enters the picture: whether to study replay on a fixed realized continuation or recollection under the perturbed learner, and how much of the realized history to condition on when defining the target.

The target studied here conditions on the realized prefix through the perturbed round and rec-

collects the future under the perturbed learner. This identifies which occurrence is being attributed and leaves the full post-occurrence future free to adapt. The perturbation is a one-parameter reweighting: negative values downweight the learning effect of the occurrence, and the endpoint suppresses its direct update contribution while leaving it part of the observed prefix.

The paper develops several consequences of this choice. The structural decomposition shows that the interventional target equals conditional expected replay plus a correction term driven by the perturbation-induced shift in the future data law. A stronger negative result shows this gap cannot be closed by enriching the replay information: even knowing the baseline future law and the full family of fixed-log replay responses for every continuation, replay-side data does not identify the interventional target in general. On the positive side, in a contextual-bandit model where the learner affects future data only through a known action law, the perturbed future law has an exact change-of-measure form and the target is identified under overlap; if unknown learner-state dependence enters context or reward generation, identification fails. The paper also gives a bandit example in which replay and intervention have opposite signs, and a depth- L truncation framework that gives local attribution windows a precise interpretation and quantitative error bounds.

The connection to the causal literature on sequential interventions is that conditioning on an evolving history while modifying later treatment mechanisms is exactly the structure needed here. The contribution is to bring that framework to occurrence-level attribution in adaptive learning and to map out the boundary between identifiable and non-identifiable cases.

The rest of the paper is organized as follows. Section 5.3 defines the adaptive learning model, the replay and interventional targets, the replay oracle, and the action-only class that later marks the positive side of the identification frontier. Section 5.2 positions the paper relative to static data attribution, trajectory-specific attribution on fixed training streams, recent work on sequential attribution in RL and post-training, and the broader causal literature on sequential interventions. Later sections develop the structural decomposition, the replay-oracle insufficiency theorem, the identification frontier, exact model-based computation, and the bandit examples.

5.2 Related work

The paper sits at the intersection of several literatures that ask different questions under the broad heading of data influence. The most direct ancestor is the fixed-dataset attribution literature. Influence functions, TracIn, and valuation-based methods such as Data Shapley ask how a model would have changed if part of an exogenous training set had been removed or reweighted [37, 59, 20]. Those works differ in approximation strategy and scale, but they all study learning from a data source that is fixed independently of the learner.

Recent work has sharpened the estimand question within that fixed-data setting. Bae et al. show that practical influence-function calculations often approximate a response functional different from literal leave-one-out retraining, while Schioppa et al. analyze why the usual leave-one-out reading can fail in nonlinear training [3, 67]. That line of work is close in spirit to the present paper. The common lesson is that before one asks whether an approximation is accurate, one has to say what object is being approximated.

Another nearby line keeps the data source exogenous but ties attribution to the realized training trajectory rather than to a permutation-invariant sample identity. Hara et al. introduced an SGD-based notion of occurrence-specific influence along a training run, Wang et al. formalized trajectory-specific leave-one-out influence, and In-Run Data Shapley studies run-specific attribution for a target model produced by a single training run [26, 75, 74]. These papers are especially relevant because they take realized occurrences seriously. The key difference is that perturbing one occurrence still does not change the law of future data. The training trajectory changes, but the data source remains exogenous. In the present paper, replay is the adaptive analogue of that fixed-log viewpoint, while conditional intervention is the target that appears once the learner can reshape the future data stream.

Recent work on sequential learning and RL moves closer to the setting of this paper but studies different targets. Gottesman et al. analyze influential transitions for off-policy evaluation, which is a sensitivity question about an evaluation estimator rather than about the learned system after

perturbing part of its own training history [22]. Hiraoka et al. study the influence of experiences in replay-based RL agents [27]. Hu et al. study online RL directly and emphasize that each experience affects both the policy update and future data collection, but their framework is intentionally local, tying attribution to a recent PPO buffer and a nearby checkpoint [28]. The present paper instead studies a global occurrence-level target that conditions on the realized prefix through the perturbed round and lets the whole post-occurrence future be recollected under the perturbed learner.

There is also a broader causal literature on sequential interventions. Longitudinal causal inference has long studied interventions that change later treatment mechanisms while conditioning on evolving histories, together with identification by g-computation, weighting, and doubly robust methods. Modern work on longitudinal modified treatment policies is a particularly relevant point of contact because it treats stochastic, history-dependent interventions without requiring point-mass treatment assignments [16]. The present paper uses that framework to formulate occurrence-level attribution in adaptive learning and then asks which replay-side objects fail to identify the resulting target and which structural restrictions make it identifiable from logged data.

Finally, the paper is close in spirit to the counterfactual-learning-systems perspective of Bottou et al., who stressed that replaying logged data is not the same as allowing the system and its future observations to evolve under an intervention [7]. The difference is one of focus. Bottou et al. study counterfactual reasoning for learning systems in general. The present paper specializes that viewpoint to training-data attribution, treats attribution as an occurrence-level question, and develops the structural decomposition, replay-oracle insufficiency result, and identification frontier that arise from that specialization.

5.2.1 A three-axis taxonomy of attribution targets

Papers on data attribution in adaptive learning often study different objects while using similar language, and the disagreement is usually not about estimator quality but about what counterfactual is being studied. An attribution question has three independent components: the unit of attribution (what is being perturbed—a dataset item, a realized occurrence, a replay-buffer experi-

ence, a rollout episode), the future semantics (what happens to later data after the perturbation—a frozen log, a truncated recollection, full recollection under the perturbed learner, an off-policy surrogate), and the target functional (what downstream quantity is being evaluated—the final learned system, a checkpoint, an evaluation estimator, the final training reward). None of these choices determines the others, and a method is not fully specified until all three are fixed. Table 5.1 organizes the nearby literature along these axes.

The table’s main point is that these methods differ along three independent axes; the present paper occupies the realized-occurrence, full-recollection, final-system corner.

The present paper fixes one particular corner of this space: attribution to a **realized occurrence**, full **recollection** of the post-occurrence future under the perturbed learner, and evaluation of the **final learned system**. The table is valuable because it lays out the three axes. That lets us ask whether a given method targets the same global object, a principled surrogate, or a different influence question altogether.

Work family	Unit of attribution	Future semantics	Target functional	Relation to the present paper
Trajectory-specific attribution [26, 75]	realized training occurrence	frozen / exogenous future	final trained model	replay-side special case of the exogenous limit
Replay-buffer influence [27]	stored replay-buffer experience	fixed replay buffer / replay-side	final agent performance	same broad problem family; literal subsumption needs an extension in which the learner state includes buffer contents and replay-sampling events naturally viewed as proxying checkpoint-local or short-horizon truncated targets
Local online RL attribution [28, 68]	recent rollout-buffer record or rollout episode	truncated local window	checkpoint action / checkpoint return / nearby post-training behavior	
Replay-based non-local on-policy attribution [45]	realized rollout record	replay over fixed future buffers	non-local replay-LOO response	non-local in time, but still replay-side rather than full recollection
OPE influence [22]	transition in an offline batch	frozen batch	evaluation estimator	different target class: estimator sensitivity rather than learned-system attribution
RLVR off-policy influence [79]	prompt / trajectory / training sample	off-policy surrogate	RLVR training objective	best viewed as an estimation surrogate near the action-only frontier, not as the same estimand
RFT sample influence [70]	dataset sample or repeated sample presentation	optimization-path retracing / proxy	final training reward	related global influence target at a different attribution unit
Present paper	realized occurrence	full recollection under the perturbed learner	final learned system	canonical global target for occurrence-level attribution in adaptive learning

Table 5.1: A three-axis taxonomy of attribution targets in and around adaptive learning. The present paper occupies one specific corner of this space. Nearby empirical methods often live in neighboring corners that should be understood as replay-side, truncated, or off-policy surrogates rather than as direct estimands of the global interventional target.

5.3 Problem Formulation

We introduce the setting in full generality and then specialize to a specific example. The finite-horizon assumption lets us present the main ideas without measure-theoretic machinery.

5.3.1 Finite-horizon adaptive learning

Fix a horizon $T \geq 1$. For each round $t \in \{1, \dots, T\}$, let $\mathcal{Z}_t$ be a finite interaction space. We write

$$\mathcal{H}_t := \mathcal{Z}_1 \times \dots \times \mathcal{Z}_t, \quad \mathcal{H}_0 := \{\emptyset\},$$

and denote a realized prefix by $z_{1:t} \in \mathcal{H}_t$.

The learner has internal state $\theta_t \in \Theta$, where Θ is an arbitrary state space. The environment is indexed by ν . For each round t , learner state $\theta \in \Theta$, and history prefix $z_{1:t-1} \in \mathcal{H}_{t-1}$, let

$$K_{\nu,t}^\theta(\cdot \mid z_{1:t-1})$$

be a probability distribution on $\mathcal{Z}_t$. This is the law of the next interaction under the current learner state and the current history.

The learner updates through maps

$$U_t : \Theta \times \mathcal{Z}_t \times [0, 1 + \rho] \rightarrow \Theta, \quad t = 1, \dots, T,$$

for some fixed $\rho > 0$. The third argument is a nonnegative **weight** that scales the direct learning effect of the t -th interaction. Under a standard run the weight is 1; setting it to 0 asks what would have happened if the learner observed the interaction but did not update from it. The range $[0, 1 + \rho]$ allows both downweighting and upweighting.

Given a weight vector $w = (w_1, \dots, w_T) \in [0, 1 + \rho]^T$, we write $\theta_t^{(w)}$ for the learner state at time t under weights w . The system starts at $\theta_1^{(w)}(\emptyset) := \theta_1$. Online, the data at step t is $Z_t \sim K_{\nu,t}^{\theta_t^{(w)}}(\cdot \mid Z_{1:t-1})$, and the state updates via $\theta_{t+1}^{(w)} := U_t(\theta_t^{(w)}, Z_t, w_t)$.

Given a fixed realized log $z_{1:T}$, the **replay operator** pushes a perturbed learner through that same sequence without allowing the data to change:

$$\theta_1^{(w)}(z_{1:T}) := \theta_1, \quad \theta_{t+1}^{(w)}(z_{1:T}) := U_t(\theta_t^{(w)}(z_{1:T}), z_t, w_t). \quad (5.1)$$

We fix an evaluation functional F_ν mapping the final learner state to a real number, for instance the expected reward of the terminal policy. For now F_ν is treated as a known abstract function; the question of identifying it from logged data is taken up in Section 5.6.

For each weight vector w , the recursion in $Z_t \sim K_{\nu,t}^{\theta_t^{(w)}}(\cdot \mid Z_{1:t-1})$ induces a law on full histories:

$$\mathbb{P}_\nu^{(w)}(z_{1:T}) = \prod_{t=1}^T K_{\nu,t}^{\theta_t^{(w)}(z_{1:t-1})}(z_t \mid z_{1:t-1}). \quad (5.2)$$

The baseline run corresponds to the all-ones vector $\mathbf{1} := (1, \dots, 1)$, and we write $\mathbb{P}_\nu := \mathbb{P}_\nu^{(\mathbf{1})}$.

We also abbreviate $\theta_s(z_{1:T}) := \theta_s^{(\mathbf{1})}(z_{1:T})$ for the baseline replayed state along a fixed realized history.

To perturb one realized occurrence at round t , we use the one-coordinate family

$$w_s^{(t,\epsilon)} := 1 + \epsilon \mathbf{1}\{s = t\}, \quad \epsilon \in [-1, \rho]. \quad (5.3)$$

Positive ϵ upweights the direct learning effect of the realized occurrence, negative ϵ downweights it, and $\epsilon = -1$ suppresses its direct update contribution.

5.3.2 Example: Instantiating the Model

To make the above formulation concrete, consider a standard two-armed Bernoulli bandit trained via online gradient ascent.

The learner's state is its current policy, so $\theta_t \in (0, 1)$ is the probability p_t of pulling Arm 1. Each interaction records both the action and the reward: $Z_t = (A_t, R_t) \in \{0, 1\} \times \{0, 1\}$. The environment ν is defined by the true reward probabilities μ_0 and μ_1 for each arm; the kernel $K_{\nu,t}^{\theta_t}$ first samples $A_t \sim \text{Bernoulli}(\theta_t)$ and then $R_t \sim \text{Bernoulli}(\mu_{A_t})$.

The learner updates via entropic mirror descent with an importance-weighted reward estimate. Given the current policy, the observed pair, and a learning weight w_t , the next policy is

$$\text{logit}(\theta_{t+1}) = \text{logit}(\theta_t) + \eta w_t R_t \left(\frac{\mathbf{1}\{A_t = 1\}}{\theta_t} - \frac{\mathbf{1}\{A_t = 0\}}{1 - \theta_t} \right), \quad (5.4)$$

with $\theta_{t+1} = \sigma(\text{logit}(\theta_{t+1}))$. In a standard training run $w = (1, 1, \dots, 1)$. Setting $w_1 = 0$ evaluates the counterfactual in which the learner observed the interaction at $t = 1$ but did not update its policy from it.

5.3.3 Replay and conditional intervention

When asking how the final model would have changed under a perturbation at round t , one must decide what happens to the data collected afterward. There are two natural answers.

The first is replay: hold the future data fixed. For a fully realized history $z_{1:T}$, the finite replay effect and the replay influence are

$$\begin{aligned} \Delta_{t,\epsilon}^{\text{rep}}(F_\nu; z_{1:T}) &:= F_\nu\left(\theta_{T+1}^{(w^{(t,\epsilon)})}(z_{1:T})\right) - F_\nu(\theta_{T+1}(z_{1:T})), \\ \mathcal{I}_t^{\text{rep}}(F_\nu; z_{1:T}) &:= \left. \frac{d}{d\epsilon} F_\nu\left(\theta_{T+1}^{(w^{(t,\epsilon)})}(z_{1:T})\right) \right|_{\epsilon=0}. \end{aligned}$$

Replay is the natural object when the training set is fixed—it asks how the terminal model changes as the perturbed learner is pushed through the same sequence of data.

Before defining the second target, we note a basic causal fact: perturbing the learning weight at round t does not change the past, because the learner only updates its state after observing Z_t . Therefore

$$\mathbb{P}_\nu^{(w^{(t,\epsilon)})}(z_{1:t}) = \mathbb{P}_\nu(z_{1:t}).$$

This prefix invariance means that conditioning on a realized prefix $z_{1:t}$ of positive baseline probability remains a valid event under the perturbation.

The second target, conditional intervention, holds reality fixed only through round t and then lets the perturbed learner recollect the future. Fix a realized prefix $z_{1:t} \in \mathcal{H}_t$ with $\mathbb{P}_\nu(z_{1:t}) > 0$. The perturbed conditional future law is

$$Q_{\nu,t}^\epsilon(z_{t+1:T} \mid z_{1:t}) := \mathbb{P}_\nu^{(w^{(t,\epsilon)})}(z_{t+1:T} \mid z_{1:t}), \quad \epsilon \in [-1, \rho].$$

The expected terminal target under this law is

$$\Psi_t^\epsilon(z_{1:t}) := \sum_{z_{t+1:T}} Q_{\nu,t}^\epsilon(z_{t+1:T} \mid z_{1:t}) F_\nu \left(\theta_{T+1}^{(w^{(t,\epsilon)})}(z_{1:T}) \right),$$

and the finite interventional effect and conditional interventional influence are

$$\Delta_{t,\epsilon}^{\text{int}}(F_\nu; z_{1:t}) := \Psi_t^\epsilon(z_{1:t}) - \Psi_t^0(z_{1:t}), \quad \mathcal{I}_t^{\text{int}}(F_\nu; z_{1:t}) := \left. \frac{d}{d\epsilon} \Psi_t^\epsilon(z_{1:t}) \right|_{\epsilon=0}.$$

This is the primary attribution target of the paper. It attributes the impact of a specific realized occurrence while allowing the post-occurrence data stream to adapt to the altered learner.

Averaging over all realized prefixes at round t gives a slot-level influence:

$$\bar{\mathcal{I}}_t^{\text{int}}(F_\nu) := \left. \frac{d}{d\epsilon} \sum_{z_{1:T} \in \mathcal{H}_T} \mathbb{P}_\nu^{(w^{(t,\epsilon)})}(z_{1:T}) F_\nu \left(\theta_{T+1}^{(w^{(t,\epsilon)})}(z_{1:T}) \right) \right|_{\epsilon=0}.$$

The choice to condition on $z_{1:t}$ specifically is deliberate. Conditioning on the full history $z_{1:T}$ freezes all future randomness, collapsing the interventional target into replay. Conditioning on a strict prefix $z_{1:k}$ with $k < t$ loses the identity of the perturbed occurrence, averaging over all possible histories through round t . Conditioning on $z_{1:t}$ isolates the exact occurrence being attributed while leaving the causal future free to respond.

5.3.4 When recollection collapses to replay

Fix a round t and a realized prefix $h = z_{1:t}$. Suppose that after round t , the interaction law no longer depends on the learner state. Then a perturbation at round t can still change the terminal learner along any fixed continuation, but it cannot change which continuation is collected. Recollection and replay therefore coincide at the level of the future data law. The only remaining choice is how much of the realized history one conditions on.

Proposition 5 (Exogenous reduction). *Fix $t \in \{1, \dots, T\}$ and a realized prefix $h = z_{1:t} \in \mathcal{H}_t$ with $\mathbb{P}_\nu(h) > 0$. Assume that for every future round $s \in \{t+1, \dots, T\}$, every history prefix $z_{1:s-1} \in \mathcal{H}_{s-1}$, and every pair of learner states $\theta, \vartheta \in \Theta$,*

$$K_{\nu,s}^\theta(\cdot \mid z_{1:s-1}) = K_{\nu,s}^\vartheta(\cdot \mid z_{1:s-1}).$$

Then for every $\epsilon \in [-1, \rho]$ and every continuation $c \in \mathcal{C}_t$,

$$Q_{\nu,t}^\epsilon(c \mid h) = Q_{\nu,t}^0(c \mid h) = \mathbb{P}_\nu(c \mid h).$$

Consequently,

$$\Psi_t^\epsilon(h) = \mathbb{E}_{\mathbb{P}_\nu} \left[F_\nu \left(\theta_{T+1}^{(w^{(t,\epsilon)})}(Z_{1:T}) \right) \mid Z_{1:t} = h \right].$$

Hence the finite conditional interventional effect reduces to conditional expected replay:

$$\Delta_{t,\epsilon}^{\text{int}}(F_\nu; h) = \mathbb{E}_{\mathbb{P}_\nu} \left[\Delta_{t,\epsilon}^{\text{rep}}(F_\nu; Z_{1:T}) \mid Z_{1:t} = h \right].$$

If the derivative at 0 exists, then

$$\mathcal{I}_t^{\text{int}}(F_\nu; h) = \mathbb{E}_{\mathbb{P}_\nu} [\mathcal{I}_t^{\text{rep}}(F_\nu; Z_{1:T}) \mid Z_{1:t} = h].$$

Proposition 5 is the formal bridge to occurrence-specific attribution on exogenous training streams. In that class, the future-law correction disappears identically. The adaptive target therefore reduces to conditional expected replay. If one conditions further on the full realized continuation $Z_{1:T} = z_{1:T}$, one recovers the fixed-log, trajectory-specific viewpoint studied in exogenous run-specific attribution.

5.4 Deconstructing the Adaptive Gap

5.4.1 The Structural Decomposition

The interventional target $\Psi_t^\epsilon(h)$ is a sum over all possible continuations of the probability of that continuation times the terminal value under the perturbed learner. Taking the derivative in ϵ by the product rule decomposes the interventional influence into two terms. Fix a realized prefix $h = z_{1:t}$ with $\mathbb{P}_\nu(h) > 0$, and let

$$\dot{Q}_{\nu,t}(c \mid h) := \left. \frac{d}{d\epsilon} Q_{\nu,t}^\epsilon(c \mid h) \right|_{\epsilon=0} \tag{5.5}$$

denote the first-order shift in the conditional future law at the perturbation.

Theorem 17 (Structural decomposition). *Fix $t \in \{1, \dots, T\}$ and a realized prefix $h = z_{1:t}$ with positive baseline probability. Assuming the maps $\epsilon \mapsto Q_{\nu,t}^\epsilon$ and $\epsilon \mapsto F_\nu$ are differentiable at 0,*

$$\mathcal{I}_t^{\text{int}}(F_\nu; h) = \mathbb{E}_{\mathbb{P}_\nu}[\mathcal{I}_t^{\text{rep}}(F_\nu; Z_{1:T}) \mid Z_{1:t} = h] + \sum_{c \in \mathcal{C}_t} \dot{Q}_{\nu,t}(c \mid h) F_\nu(\theta_{T+1}(h, c)). \quad (5.6)$$

The first term is conditional expected replay: it accounts for how the perturbed learner changes the terminal model when pushed through the realized future data. In a fixed-dataset setting the future data law does not depend on the learner, so $\dot{Q} = 0$ and this term is all there is. In an adaptive setting the future data law does depend on the learner, and the second term—the future-law correction—captures the change in which continuations the perturbed learner is likely to encounter. It is this term that replay ignores.

5.4.2 The Centered Form

Because $\dot{Q}_{\nu,t}(\cdot \mid h)$ sums to zero over all continuations (the total probability mass is always 1), one can subtract a baseline constant from each summand in the correction term without changing the total. This gives a more interpretable form.

Proposition 6 (Centered form of the future-law term). *Under the assumptions of Theorem 17,*

$$\sum_{c \in \mathcal{C}_t} \dot{Q}_{\nu,t}(c \mid h) = 0,$$

and consequently

$$\mathcal{I}_t^{\text{int}}(F_\nu; h) - \mathbb{E}_{\mathbb{P}_\nu}[\mathcal{I}_t^{\text{rep}}(F_\nu; Z_{1:T}) \mid Z_{1:t} = h] = \sum_{c \in \mathcal{C}_t} \dot{Q}_{\nu,t}(c \mid h) \left[F_\nu(\theta_{T+1}(h, c)) - F_\nu(\theta_{t+1}(h)) \right]. \quad (5.7)$$

Equation (5.7) shows that the gap behaves like a covariance between the shift in the future data law and the terminal value of the shifted continuations. The gap is zero when the perturbation does not change which continuations the learner encounters ($\dot{Q} = 0$), and also when all continuations lead to the same terminal value regardless of which ones become more or less likely. The gap is large when the perturbation meaningfully shifts the data distribution toward continuations with substantially different terminal outcomes.

5.5 The Insufficiency of Replay-Side Information

The structural decomposition shows that replay misses the future-law correction. A natural question is whether richer replay-side information could close the gap. To answer this, we consider the most informative replay-side object one could construct.

For any continuation $c \in \mathcal{C}_t := \mathcal{Z}_{t+1} \times \cdots \times \mathcal{Z}_T$, define the fixed-log replay response curve

$$\phi_{\nu,t,c}(\epsilon; z_{1:t}) := F_\nu\left(\theta_{T+1}^{(w^{(t,\epsilon)})}(z_{1:t}, c)\right),$$

which records how the terminal evaluation changes as the perturbation level ϵ varies, with the learner forced through the specific continuation c . The replay oracle at prefix $z_{1:t}$ is then

$$\mathcal{R}_{\nu,t}(z_{1:t}) := \left(Q_{\nu,t}^0(\cdot \mid z_{1:t}), \{\phi_{\nu,t,c}(\cdot; z_{1:t}) : c \in \mathcal{C}_t\}\right),$$

pairing the baseline probability of every continuation with the full family of fixed-log response curves for every continuation. This is the richest conceivable replay summary.

Theorem 18 (Replay-oracle insufficiency). *There exists a smooth horizon-2 adaptive learning class and a realized prefix z_1^* of positive probability such that for every pair $\alpha \neq \beta$, the environments share the same replay oracle:*

$$\mathcal{R}_{\nu_\alpha,1}(z_1^*) = \mathcal{R}_{\nu_\beta,1}(z_1^*),$$

but their interventional targets differ:

$$\Psi_{\nu_\alpha,1}^\epsilon(z_1^*) \neq \Psi_{\nu_\beta,1}^\epsilon(z_1^*) \quad \text{for every } \epsilon \text{ with } \alpha\epsilon \neq \beta\epsilon.$$

In particular, $\mathcal{I}_{1,\nu_\alpha}^{\text{int}}(F; z_1^) \neq \mathcal{I}_{1,\nu_\beta}^{\text{int}}(F; z_1^*)$, and the interventional target is not identified by any functional of the replay oracle over this class.*

The reason is that the replay oracle, however rich, contains no information about $\dot{Q}$ —the derivative of the future data law with respect to the perturbation. Two environments can agree on every fixed-log replay response curve while disagreeing on how the environment’s data-generation mechanism responds to a changed learner state. The oracle captures how the perturbed learner

behaves on every conceivable historical log; it cannot capture how the perturbation changes which log the learner would actually encounter.

5.6 Identification under Action-Only Learner State Dependence

The negative result rules out identification in arbitrary environments. To recover it, one must restrict how the learner and the environment interact. The natural restriction for contextual bandit and RL settings is that the learner affects the future only through its action policy, not through the context or reward distributions. We call this the action-only class.

Formally, assume each interaction decomposes as $Z_s = (X_s, A_s, R_s)$ and that the transition kernel factors as

$$K_{\nu,s}^\theta(x, a, r \mid z_{1:s-1}) = D_s(x \mid z_{1:s-1}) \pi_\theta(a \mid x) P_s(r \mid x, a, z_{1:s-1}), \quad (5.8)$$

where D_s and P_s do not depend on the learner state θ . The environment controls contexts and rewards; the learner controls only its action distribution. Under this structure, perturbing the learner at round t changes the future data law only through the policy, and the change can be tracked exactly. For a realized prefix $h = z_{1:t}$, define the pathwise policy ratio

$$\Lambda_t^\epsilon(c; h) := \prod_{s=t+1}^T \frac{\pi_{\theta_s^{(w(t,\epsilon))}}(h, c_{t+1:s-1})(a_s \mid x_s)}{\pi_{\theta_s}(h, c_{t+1:s-1})(a_s \mid x_s)}. \quad (5.9)$$

Because the context and reward terms cancel in the ratio of perturbed to baseline future laws, Λ_t^ϵ gives an exact change-of-measure representation.

Theorem 19 (Exact change of measure in the action-only class). *Assume the factorization (5.8).*

Fix $h = z_{1:t}$ with $\mathbb{P}_\nu(h) > 0$. Then for every $\epsilon \in [-1, \rho]$ and every continuation $c \in \mathcal{C}_t$ with $Q_{\nu,t}^0(c \mid h) > 0$, the perturbed future probability is simply the baseline probability scaled by the policy ratio:

$$Q_{\nu,t}^\epsilon(c \mid h) = \Lambda_t^\epsilon(c; h) Q_{\nu,t}^0(c \mid h). \quad (5.10)$$

Suppose in addition that there exists $\epsilon_0 > 0$ such that for every $\epsilon \in (-\epsilon_0, \epsilon_0)$ and every

continuation $c \in \mathcal{C}_t$,

$$Q_{\nu,t}^\epsilon(c \mid h) > 0 \implies Q_{\nu,t}^0(c \mid h) > 0. \quad (5.11)$$

Then for every $\epsilon \in (-\epsilon_0, \epsilon_0)$,

$$\Psi_t^\epsilon(h) = \mathbb{E}_{\mathbb{P}_\nu} \left[\Lambda_t^\epsilon(Z_{t+1:T}; h) F_\nu \left(\theta_{T+1}^{(w^{(t,\epsilon)})}(Z_{1:T}) \right) \mid Z_{1:t} = h \right]. \quad (5.12)$$

The representation (5.12) is an importance-sampling identity: the interventional target can be evaluated on baseline data by reweighting each realized continuation by the ratio of the perturbed policy's likelihood to the baseline policy's likelihood along that continuation.

Identification then follows if the terminal evaluation F_ν is a known functional of the baseline law.

Corollary 7 (Identification in the action-only class). *Assume the hypotheses of Theorem 19, assume that the learner update maps $U_1, \dots, U_T$ and the action law π_θ are known, and assume that there exists a measurable functional $\mathfrak{F}$ such that for every $\vartheta \in \mathcal{R}_{\nu,t}^{\epsilon_0}(h)$,*

$$F_\nu(\vartheta) = \mathfrak{F}(\mathbb{P}_\nu, \vartheta).$$

Then for every $\epsilon \in (-\epsilon_0, \epsilon_0)$, the finite target $\Psi_t^\epsilon(h)$ is identified from the baseline law. If the derivative at 0 exists, then $\mathcal{I}_t^{\text{int}}(F_\nu; h)$ is identified as well.

If the action-only assumption is relaxed and the environment uses the learner's state to generate future contexts or rewards in an unknown way, identification fails.

Proposition 7 (Unknown context-state dependence destroys identification). *Consider the class of finite-horizon environments with factorization*

$$K_{\nu,s}^\theta(x, a, r \mid z_{1:s-1}) = D_s^\theta(x \mid z_{1:s-1}) \pi_\theta(a \mid x) P_s(r \mid x, a, z_{1:s-1}),$$

where D_s^θ is otherwise unrestricted. Even when the learner update maps and the action law are known, the conditional interventional target is not identified from the baseline law over this class.

When unknown learner-state dependence enters context or reward generation, the baseline law does not carry enough information to predict how the future data stream will shift under a perturbation. The action-only class is the structural boundary between identifiable and non-identifiable cases when attribution is done from logged data alone.

5.7 Quantifying the Gap between Replay and Intervention

We examine how large the gap can be and when it collapses, using a two-armed Bernoulli bandit trained via online entropic mirror descent as a concrete model.

5.7.1 The Directional Failure of Replay

Theorem 20 (Strong separation). *There exist reward configurations in a horizon-2 bandit such that the conditional expected replay is strictly negative while the conditional interventional influence is strictly positive.*

The sign flip arises from the self-correcting nature of adaptive learning. Suppose the learner pulls a sub-optimal arm and receives a zero reward. Replay looks at this event on its fixed log and concludes that downweighting it would have improved the final model—a negative influence. But the zero reward is what drove the learner to shift toward the better arm for the next round. Intervention sees this: the perturbed model, having downweighted that corrective signal, actually explores less efficiently and ends up worse. The two targets therefore land on opposite sides of zero. Replay, which freezes the future, cannot see the self-correction.

5.7.2 The Anatomy of the Gap

The stagewise bounds derived in the appendix (and stated in the controlled-approximation section) show that the gap is driven by three compounding factors: how strongly the perturbation propagates forward through the learner’s parameter updates (model propagation), how much the environment’s data distribution shifts in response to a changed learner state (environment sensi-

tivity), and how much the terminal value varies across the resulting trajectories (value oscillation). The sign flip occurs when all three are large simultaneously.

5.7.3 The Stable Small-Step Regime

Controlling these three factors forces the gap to collapse.

Theorem 21 (Replay in a stable regime). *When the learning rate is sufficiently small and the policy is bounded away from zero and one, the gap between conditional interventional influence and conditional expected replay is $\mathcal{O}(\eta^2)$, where η is the learning rate.*

A small learning rate limits how far the perturbation can propagate and how much the environment can react. In this regime the sign-flip scenario disappears and replay is an accurate first-order approximation of the interventional target.

5.8 Controlled local approximations

The negative results above show that the global interventional target is not, in general, recoverable from replay-side information alone. At the same time, much of the recent empirical literature does not try to reconstruct the full adaptive future. Instead it works with recent buffers, checkpoint-local targets, replay-side windows, or off-policy surrogates. The right foundational response is therefore to place locality on the same counterfactual map and ask what exactly is being approximated.

We begin with the most basic controlled approximation: truncate the **recollection depth**. The idea is to let the perturbation change the future data law for only the next L rounds after the realized occurrence and then freeze the interaction law at its baseline form. Importantly, this truncates only the **future-law channel**. The perturbed learner still continues to update on the later sampled log all the way to time T . Thus the approximation is local in **recollection depth**, not local in learning dynamics.

Fix $t \in \{1, \dots, T\}$, a realized prefix $h = z_{1:t} \in \mathcal{H}_t$ with $\mathbb{P}_\nu(h) > 0$, and an integer $L \in$

$\{0, \dots, T - t\}$. For a continuation $c = z_{t+1:T} \in \mathcal{C}_t$, define the depth- L mixed future law by

$$Q_{\nu,t}^{\epsilon,\text{tr},L}(c \mid h) := \prod_{s=t+1}^{t+L} K_{\nu,s}^{\theta_s^{(w(t,\epsilon))}}(h, c_{t+1:s-1})(z_s \mid h, c_{t+1:s-1}) \prod_{s=t+L+1}^T K_{\nu,s}^{\theta_s^{(h, c_{t+1:s-1})}}(z_s \mid h, c_{t+1:s-1}). \quad (5.13)$$

The first product is empty when $L = 0$, and the second product is empty when $L = T - t$.

The corresponding depth- L finite target is

$$\Psi_t^{\epsilon,\text{tr},L}(h) := \sum_{c \in \mathcal{C}_t} Q_{\nu,t}^{\epsilon,\text{tr},L}(c \mid h) F_{\nu} \left(\theta_{T+1}^{(w(t,\epsilon))}(h, c) \right), \quad (5.14)$$

with finite effect

$$\Delta_{t,\epsilon}^{\text{tr},L}(F_{\nu}; h) := \Psi_t^{\epsilon,\text{tr},L}(h) - \Psi_t^0(h),$$

and infinitesimal form

$$\mathcal{I}_t^{\text{tr},L}(F_{\nu}; h) := \left. \frac{d}{d\epsilon} \Psi_t^{\epsilon,\text{tr},L}(h) \right|_{\epsilon=0}. \quad (5.15)$$

This family interpolates exactly between replay and full recollection. When $L = 0$, the future law is baseline all the way to time T , so $\Psi_t^{\epsilon,\text{tr},0}$ is the conditional expected replay target. When $L = T - t$, the mixed law is the fully perturbed future law, so $\Psi_t^{\epsilon,\text{tr},T-t} = \Psi_t^{\epsilon}$.

To state the next theorem compactly, let Γ_s and V_s denote the forward state-sensitivity and baseline continuation-value objects from the exact model-based recursion, and define the stagewise contribution of round s to the replay-intervention gap by

$$\Xi_s(g) := \sum_{z_s \in \mathcal{Z}_s} \left(\nabla_{\theta} K_{\nu,s}^{\theta}(z_s \mid g) \Big|_{\theta=\theta_s(g)}^{\top} \Gamma_s(g) \right) (V_{s+1}(g, z_s) - V_s(g)). \quad (5.16)$$

Theorem 22 (Depth- L recollection identity). *For every $L \in \{0, \dots, T - t\}$,*

$$\mathcal{I}_t^{\text{tr},L}(F_{\nu}; h) = \mathbb{E}_{\mathbb{P}_{\nu}}[\mathcal{I}_t^{\text{rep}}(F_{\nu}; Z_{1:T}) \mid Z_{1:t} = h] + \sum_{s=t+1}^{t+L} \mathbb{E}_{\mathbb{P}_{\nu}}[\Xi_s(Z_{1:s-1}) \mid Z_{1:t} = h], \quad (5.17)$$

$$\mathcal{I}_t^{\text{int}}(F_{\nu}; h) - \mathcal{I}_t^{\text{tr},L}(F_{\nu}; h) = \sum_{s=t+L+1}^T \mathbb{E}_{\mathbb{P}_{\nu}}[\Xi_s(Z_{1:s-1}) \mid Z_{1:t} = h]. \quad (5.18)$$

In particular, $L = 0$ recovers conditional expected replay, and $L = T - t$ recovers full conditional interventional influence.

The theorem gives a precise meaning to local attribution windows. A small value of L does not merely say “look nearby.” It says: allow the adaptive future to respond to the perturbation for L further rounds, then freeze the future data law and continue only the replayed learning dynamics. The bias of this approximation is exactly the omitted tail in (5.18).

To convert that identity into an approximation prescription, recall the total-variation sensitivity and value oscillation quantities

$$L_s^{\text{TV}}(g) := \sup_{\|u\|=1} \frac{1}{2} \sum_{z_s \in \mathcal{Z}_s} \left| \nabla_{\theta} K_{\nu,s}^{\theta}(z_s \mid g) \Big|_{\theta=\theta_s(g)}^{\top} u \right|, \quad \text{osc}_s(g) := \max_{z_s} V_{s+1}(g, z_s) - \min_{z_s} V_{s+1}(g, z_s).$$

Corollary 8 (Adaptive-horizon truncation). *For every $L \in \{0, \dots, T-t\}$,*

$$\left| \mathcal{I}_t^{\text{int}}(F_{\nu}; h) - \mathcal{I}_t^{\text{tr},L}(F_{\nu}; h) \right| \leq \sum_{s=t+L+1}^T \mathbb{E}_{\mathbb{P}_{\nu}} \left[L_s^{\text{TV}}(Z_{1:s-1}) \|\Gamma_s(Z_{1:s-1})\| \text{osc}_s(Z_{1:s-1}) \mid Z_{1:t} = h \right]. \quad (5.19)$$

Under the deterministic bounds

$$\|\partial_w U_t\| \leq \bar{B}_t, \quad \|\partial_{\theta} U_u\|_{\text{op}} \leq \bar{\rho}_u, \quad L_s^{\text{TV}} \leq \bar{L}_s, \quad \text{osc}_s \leq \bar{\Delta}_s,$$

this simplifies to

$$\left| \mathcal{I}_t^{\text{int}}(F_{\nu}; h) - \mathcal{I}_t^{\text{tr},L}(F_{\nu}; h) \right| \leq \bar{B}_t \sum_{s=t+L+1}^T \bar{L}_s \bar{\Delta}_s \prod_{u=t+1}^{s-1} \bar{\rho}_u. \quad (5.20)$$

Consequently, for any tolerance $\tau > 0$, any choice of L satisfying

$$\bar{B}_t \sum_{s=t+L+1}^T \bar{L}_s \bar{\Delta}_s \prod_{u=t+1}^{s-1} \bar{\rho}_u \leq \tau$$

guarantees a depth- L approximation error at most τ .

Corollary 8 gives a clean prescription for choosing how local an approximation may be. One first upper-bounds the omitted tail and then selects the smallest horizon L whose tail falls below a desired tolerance. This lets us view recent-buffer or checkpoint-local attribution as approximations to a depth- L recollection target, with error controlled by propagation, environment sensitivity, and downstream value oscillation.

This also clarifies how several nearby empirical directions fit into the present theory. Snapshot-style recent-buffer methods are naturally interpreted as scalable proxies for checkpoint-local variants

of $\mathcal{I}_t^{\text{tr},L}$, obtained by replacing F_ν with a checkpoint functional and taking small L [28]. Replay-buffer and replay-LOO methods instead stay on the frozen-future side after the switch, even when they become non-local in time [27, 45]. Off-policy influence methods for RLVR live on a different approximation axis: they replace online recollection with an off-policy surrogate [79]. Sample-level post-training influence methods such as RFT-Inf change the attribution unit and target functional rather than only the horizon length [70].

Once we say how a method treats the future, we can distinguish methods that target the same object, methods that use a controlled local truncation, and methods that solve a different surrogate problem.

5.9 Conclusion

Data attribution in adaptive learning requires distinguishing two counterfactuals that coincide in static learning but diverge once the learner shapes its own future data. The conditional interventional target defined captures this by conditioning on what has already occurred and allows the perturbed learner to recollect the future. The main results show that replay-side information cannot recover this target in general, even with full knowledge of the baseline future law and all fixed-log responses, while the action-only class marks the boundary where identification from logged data becomes possible. Whether one targets the full recollection or a depth- L truncation, the choice of future semantics determines what attribution question is actually being asked.

Chapter 6

Conclusion

This thesis studied a family of sequential problems that look quite different at first sight. Despite these differences, this thesis was organized around the common claim that, to make theoretical guarantees tractable in sequential settings, the main mathematical work often lies in identifying the right reduced object to study. That reduced object must be rich enough to preserve the real source of difficulty and simple enough to make the problem analyzable. It must also match the way the learner is allowed to interact with the system.

The need for such reductions is easiest to appreciate by thinking about what goes wrong without them. In the continuous multi-player bandit problem of Chapter 2, a brute-force view of the action space is useless, because the domain is continuous and because the players destroy one another’s observations through collisions. In the multi-agent reinforcement-learning problem of Chapter 3, a brute-force view of the global state space is impossible, because the state and action spaces grow exponentially with the number of agents. In the post-training setting of Chapter 4, a brute-force view of the policy is misleading, because the policy may be mathematically well defined while the algorithm can only sample from it or query it through a restricted oracle. In the attribution problem of Chapter 5, a brute-force static notion of influence misses the point, because the effect of a trajectory can unfold only through the future states, observations, and actions that it helps create. This thesis argued that these four problems are manifestations of the same phenomenon. Sequential interaction produces large global objects, but the learner never has direct access to those objects in full generality. A successful theory therefore has to begin by respecting

both the structure of the problem and the interface through which the problem is exposed.

One of the main lessons of the thesis is that tractability rarely comes from shrinking a problem arbitrarily. In Chapter 2, the right reduction is to a set of collision-feasible regions whose values are determined by their maxima, together with a coordination mechanism that assigns one player to each region. Reducing instead to a fixed finite set of representative points would erase the geometric property that makes the continuous problem learnable. In Chapter 3, the right reduction is to summarize dependence through a policy-dependent influence operator that measures how local perturbations propagate through the closed-loop dynamics. These chapters differ substantially in model and technique, but they support the same conclusion. The critical step is to identify an intermediate object that preserves the dependence in the problem while discarding the parts of the global description that cannot be exploited algorithmically.

Another lesson is that there is value in separating a hard sequential problem into components that live on different time scales. Chapter 2 provided the most concrete instance of this idea. In that setting, coordination and learning are entangled at the level of raw interaction, because collisions both reduce reward and censor information. The chapter showed that the right way to organize the problem is to pay a finite upfront price to identify good collision-free regions and to seat the players on distinct ones. Once that has been done, the dominant long-run difficulty is no longer multi-agent coordination. It is ordinary continuous exploration inside each assigned region. This separation is mathematically powerful because it turns a permanently coupled problem into one with a transient coordination phase and an asymptotically dominant learning phase. Chapter 3 contains a closely related idea in a different form. There the coupling is not due to collisions but due to the dynamical dependence of one agent on the others. The relevant question is whether that dependence accumulates or decays as one moves farther through the network or farther through time. When the closed-loop influence operator is contractive, distant effects can be truncated and the remaining approximation error is controlled by a decaying tail. Across the thesis, difficult forms of coupling become manageable once the analysis distinguishes the short-run mechanism that creates the dependence from the long-run term that actually determines performance.

A third lesson is that the interface should be treated as part of the formal problem statement, not as an afterthought. This is especially visible in Chapter 4, but it is present in all of the chapters. There, the central difficulty often comes from the fact that a large generative policy can only be queried through a restricted set of operations. Sampling access, richer score access, and token-level interaction do not merely lead to different algorithms. They lead to different computational regimes. A related issue appears in Chapter 5. An attribution theory that depends on counterfactual quantities which the data interface never reveals may be elegant, but it is not yet a usable theory of attribution in a sequential system. Taken together, these chapters support a methodological stance that in modern sequential learning problems, the notion of access to the environment, to the policy, or to the data is often every bit as important as the loss function or the reward objective. A theory that omits that notion of access, thus, risks proving loose upper and lower bounds.

The thesis also points to a different view of locality. In many areas of machine learning, locality is introduced as a computational convenience, often motivated by an intuitive belief that distant variables matter less. The chapters here suggest a stronger and more precise view. Locality is often not a primitive assumption but an emergent property of a properly chosen representation. Chapter 3 made this point directly by showing that locality in a many-agent control system is a property of the closed-loop combination of environment and policy, not just of the environment in isolation. Chapter 4 suggests an analogous idea in the setting of post-training and reasoning models. A sequence-level problem may look hopeless if it is viewed through whole outputs alone, while the same problem becomes much more structured once the token-level factorization of the model is taken seriously. This means that the search for tractable algorithms should often begin with the search for the correct notion of locality, and that notion may arise from geometry, from dynamics, from factorization, or from causal propagation through time.

These findings suggest several directions for future work. The first set of directions comes from the continuous multi-player bandit setting of Chapter 2. That chapter treated one of the most fundamental models in which continuous actions, decentralization, and collisions appear together.

Several extensions are immediately compelling. One is to move beyond a fixed collision partition and allow the collision geometry itself to be unknown or state dependent. Another is to study settings in which the number of players changes over time, the players are heterogeneous, or the reward landscape is nonstationary. A third is to go beyond the cooperative objective and allow strategic or partially aligned agents, where the identification and seating phases would interact with incentives rather than with pure coordination alone. A broader structural question connects Chapter 2 to the later chapters. The chapter achieved a clean decoupling between coordination and within-region optimization in a bandit setting. It would be interesting to know how far that decoupling can be pushed in full reinforcement-learning models with continuous actions, collisions, and state dynamics. Such models would combine the geometric issues of Chapter 2 with the dependence and localization issues of Chapter 3.

The second set of directions comes from the theory of policy-dependent locality developed in Chapter 3. That chapter isolated the role of the influence operator and separated the question of spatial truncation from the separate and difficult question of stochastic approximation under finite samples. The next step is to combine those pieces. A mature theory of scalable multi-agent reinforcement learning will need to connect the locality guarantees of the chapter to concrete sampled algorithms, to partial observability, to function approximation, and to regimes in which the system structure is not known in advance. Proving that a smooth enough policy induces decay of influence is different from estimating, adapting to, and exploiting that decay online in a noisy learning process. The chapter also suggests a conceptual direction. The work showed that locality is a property of the closed-loop system rather than of the environment alone. A natural continuation would be to develop algorithms that actively shape that locality. Entropy regularization already acts as one such knob. More generally, one can ask whether policies, observation scopes, or communication patterns can be chosen so as to preserve tractable locality without sacrificing too much reward.

The third set of directions comes from the post-training and reasoning-model perspective of Chapter 4. The chapter treated model access as a central mathematical object, and that viewpoint is likely to become even more important. Large generative policies are now used in settings where

the line between training-time optimization, search at inference time, and interaction with external tools is increasingly blurred. In such settings, a theory that treats all forms of model access as equivalent is unlikely to capture the real computational bottlenecks. One natural direction is therefore to develop a fuller hierarchy of access models and to understand which complexity measures govern each level of that hierarchy. Another is to understand more precisely when a token-level or step-level decomposition truly replaces a sequence-level coverage bottleneck and when it merely moves the difficulty elsewhere. A related question is how reasoning traces and verifiers fit into this picture. Once the output of a model is itself a structured sequential object, both the reward mechanism and the access model become more intricate. The broader lesson of Chapter 4 is that computational tractability in these problems cannot be read off from an objective alone. That lesson is likely to remain central as post-training pipelines become more interactive and more dependent on structured feedback.

The final set of directions arises from the attribution questions of Chapter 5. Attribution in sequential problems is still much less settled than attribution in static supervised learning, and several challenges are likely to remain important. One concerns counterfactual reasoning. An attribution score is most useful when it says something about what would have changed had a trajectory or piece of data been different, yet in sequential settings such counterfactuals alter future visitation and future feedback as well as immediate reward. Another concerns exploration. A data point may be valuable because it redirected exploration toward states that later turned out to be crucial, not just because it directly improved a policy at the state where it was collected. Capturing that effect rigorously requires attribution notions that are sensitive to the downstream consequences of data. Sequential attribution also has an important operational role. It can guide data curation, replay-buffer design, logging policies, safety interventions, and the diagnosis of failure modes in long-horizon systems. A strong theory would therefore do more than assign scores after the fact. It would connect attribution to actionable decisions about what to store, what to revisit, and what to trust.

Beyond the chapter-specific directions, the thesis suggests a more general research program

built around the joint study of structure and interfaces. Many current machine-learning systems are large enough that one cannot hope to reason about them directly in their raw form. At the same time, they are interactive enough that their statistical and computational properties depend heavily on the access channel through which learning happens. The chapters of the thesis suggest that these two issues should be studied together. Structure without interface can lead to elegant models that assume away the hard part of the problem. Interface without structure can lead to impossibility results that are correct but uninformative because they do not identify the regularity that makes the intended application tractable. A more mature theory of large sequential systems will often need to ask both questions at once. Which reduced object captures the right dependence, and through which operations can that object actually be learned, controlled, or interrogated?

There is a second broad direction that cuts across the thesis and concerns adaptivity. In each chapter, some useful regularity was assumed or revealed. The reward function was Lipschitz, the closed-loop influence decayed, the generative policy factorized in a useful way, or influence propagated through time in a trackable manner. A natural next question is how much of that structure can be discovered rather than postulated. Can a learner infer the right partition or effective collision geometry as it acts? Can it estimate which parts of a many-agent system are genuinely influential and which are negligible? Can it detect when a weak model interface is the real bottleneck and when richer sequential structure can be exploited instead? Can attribution itself reveal the structural bottlenecks that later chapters turn into algorithms? Questions of this kind would move the thesis-wide perspective from analysis of fixed models toward adaptive procedures that learn which abstraction is appropriate for the problem instance they face.

The technical results also point to a design question. The interface is often treated as an external constraint, something handed to the learner by the application. In practice, interfaces are frequently designed. Communication protocols in multi-agent systems are engineered. Logging and storage choices in sequential data systems are chosen by practitioners. Model APIs determine which queries are cheap and which are impossible. Reward and verifier pipelines in post-training are themselves artifacts of system design. The results of the thesis suggest that these choices deserve

a more central role in learning theory. Once it is recognized that the interface changes the problem, interface design becomes a legitimate algorithmic question rather than a purely systems-level concern. A communication budget, a logging scheme, or a model API is part of the mathematical environment in which learning takes place.

The broader view developed in the thesis is that large sequential systems should be analyzed through abstractions that are matched to the operations the learner is actually allowed to perform. The central difficulty is usually that the full mathematical object is too large, too coupled, or too opaque to be handled directly. What made progress possible in the chapters of the thesis was the choice of an intermediate representation that respected both the structure of the problem and the interface through which the problem was exposed. That perspective does not solve every sequential problem, but it gives a principled way to ask where the real difficulty lies and what kind of reduction has a chance of preserving it.

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Appendix A

Supplementary Material: Multi-Agent Lipschitz Bandits

In this chapter, we provide supplementary material corresponding to Chapter 2.

A.1 Appendix A: Proofs for Phase I

We collect here the proofs of Lemmas 1 and 2 and Proposition 1. Throughout, probabilities and expectations are taken with respect to all sources of randomness (players' exploration, collision process, and reward noise). We begin by restating the standing assumptions, notation, and two identities that are used repeatedly.

Standing assumptions and notation.

- The action domain is $\mathcal{X} = [0, 1]^d$, partitioned into $\mathcal{P} = \{C_1, \dots, C_K\}$, where each $C \in \mathcal{P}$ is a (closed) axis-aligned hypercube of side length h ; the cells have disjoint interiors and cover $\mathcal{X}$. For each $C \in \mathcal{P}$, we denote by x_C its geometric center and by

$$D_h := h\sqrt{d}$$

the Euclidean diameter of any such cell. (Boundary cells may be smaller; using D_h is conservative and simplifies the presentation.)

- The unknown mean-reward function $\mu : \mathcal{X} \rightarrow [0, 1]$ is L -Lipschitz w.r.t. $\|\cdot\|_2$:

$$|\mu(x) - \mu(y)| \leq L\|x - y\|_2 \quad \forall x, y \in \mathcal{X}.$$

- In Phase I, at each round $t \in [T_0]$ and for each player $j \in [N]$ independently, a cell $C_t^{(j)}$ is sampled uniformly from $\mathcal{P}$; the player probes the cell center $x_{C_t^{(j)}}$.
- A collision occurs in cell C at round t if at least two players sample C in that round. If player j is the unique occupant of C at round t , a noisy reward $Y_t^{(j)}$ with mean $\mu(x_C)$ is observed; otherwise a null symbol $\perp$ is observed and no reward is recorded.
- We analyze the standard bounded-reward model: whenever a reward is observed, $Y_t^{(j)} \in [0, 1]$ with $\mathbb{E}[Y_t^{(j)} \mid C_t^{(j)} = C, \text{ no collision}] = \mu(x_C)$. This implies the centered noise is σ -sub-Gaussian with $\sigma \leq \frac{1}{2}$.
- Players observe whether a collision occurred (via $\perp$). The collision/missingness process depends only on the independent exploration choices and is independent of the reward noise.

Two basic identities.

- (i) **Single-round success probability.** For a fixed player j and cell $C \in \mathcal{P}$, the probability that j samples C and no other player samples C in the same round is

$$p_K := \frac{1}{K} \left(1 - \frac{1}{K}\right)^{N-1}. \quad (\text{A.1})$$

Proof. $\Pr[C_t^{(j)} = C] = 1/K$ by uniform sampling; for each other player $j' \neq j$, $\Pr[C_t^{(j')} \neq C] = 1 - 1/K$ independently; multiply over $N - 1$ players.

- (ii) **Center vs. maximum (Lipschitz geometry).** For any cell $C \in \mathcal{P}$,

$$\mu(x_C) \leq \mu^*(C) \leq \mu(x_C) + \frac{L}{2}D_h = \mu(x_C) + \frac{Lh\sqrt{d}}{2}. \quad (\text{A.2})$$

Proof. For any $x \in C$, $\|x - x_C\|_2 \leq D_h/2$, hence $\mu(x) \leq \mu(x_C) + L\|x - x_C\|_2 \leq \mu(x_C) + \frac{L}{2}D_h$.

Taking $\sup_{x \in C}$ yields the RHS; the LHS is trivial since $\mu^*(C) = \sup_{x \in C} \mu(x) \geq \mu(x_C)$.

We write $U_j(C, t) := \mathbf{1}\{\text{player } j \text{ is the unique occupant of cell } C \text{ at round } t\}$. The success count over Phase I is

$$o_{j,C}(T_0) := \sum_{t=1}^{T_0} U_j(C, t).$$

When $o_{j,C}(T_0) \geq 1$, the empirical mean at the cell center is

$$\hat{\mu}_j^{(0)}(C) := \frac{1}{o_{j,C}(T_0)} \sum_{t=1}^{T_0} U_j(C, t) Y_t^{(j)};$$

For $o_{j,C}(T_0) = 0$, we define $\hat{\mu}_j^{(0)}(C) := 0$ (this choice is harmless because $\mu(x_C) \in [0, 1]$ and our confidence radius below will be ≥ 1 in that case). We recall the Phase I radii from the main text:

$$r_j^{(0)}(C) := \sqrt{\frac{\beta_0}{2 \max\{1, o_{j,C}(T_0)\}}}, \quad \beta_0 := \log \frac{4NK(T_0 + 1)}{\delta_I}, \quad (\text{A.3})$$

and

$$\text{LCB}_j^{(0)}(C) := \hat{\mu}_j^{(0)}(C) - r_j^{(0)}(C), \quad \text{UCB}_j^{(0)}(C) := \hat{\mu}_j^{(0)}(C) + r_j^{(0)}(C) + \frac{Lh\sqrt{d}}{2}. \quad (\text{A.4})$$

A.1.1 A.1 Success counts under collisions (Lemma 1)

Lemma 7 (Restatement of Lemma 1). *Fix $\eta \in (0, 1)$. With probability at least $1 - \delta_I/2$, simultaneously for all $j \in [N]$ and $C \in \mathcal{P}$,*

$$(1 - \eta) T_0 p_K \leq o_{j,C}(T_0) \leq (1 + \eta) T_0 p_K.$$

Proof. Fix $j \in [N]$ and $C \in \mathcal{P}$. By the sampling protocol, across rounds $t = 1, \dots, T_0$ the random variables $\{C_t^{(j)}\}_{t=1}^{T_0}$ are i.i.d., and for each fixed round the choices $\{C_t^{(j')}\}_{j'=1}^N$ are mutually independent. Therefore, for this fixed (j, C) , the indicators $\{U_j(C, t)\}_{t=1}^{T_0}$ are i.i.d. Bernoulli(p_K) with p_K given by (A.1). Consequently,

$$o_{j,C}(T_0) = \sum_{t=1}^{T_0} U_j(C, t) \sim \text{Bin}(T_0, p_K).$$

Let $\mu := \mathbb{E}[o_{j,C}(T_0)] = T_0 p_K$. The two-sided multiplicative Chernoff bound for binomial variables states that for any $\eta \in (0, 1)$,

$$\Pr(|o_{j,C}(T_0) - \mu| > \eta\mu) \leq 2 \exp\left(-\frac{\eta^2 \mu}{3}\right) = 2 \exp\left(-\frac{\eta^2 T_0 p_K}{3}\right).$$

Applying a union bound over all N players and all K cells yields

$$\Pr(\exists(j, C) \in [N] \times \mathcal{P} : |o_{j,C}(T_0) - T_0 p_K| > \eta T_0 p_K) \leq 2NK \exp\left(-\frac{\eta^2 T_0 p_K}{3}\right).$$

Hence, if T_0 is chosen to satisfy

$$2NK \exp\left(-\frac{\eta^2 T_0 p_K}{3}\right) \leq \frac{\delta_I}{2} \iff T_0 \geq \frac{3}{\eta^2 p_K} \log \frac{4NK}{\delta_I}, \quad (\text{A.5})$$

then the claimed event holds with probability at least $1 - \delta_I/2$. This is precisely the prerequisite on T_0 used later (see §A.1.4). $\square$

A.1.2 A.2 Anytime concentration for center means (Lemma 2)

Lemma 8 (Restatement of Lemma 2). *With probability at least $1 - \delta_I/2$, simultaneously for all $j \in [N]$ and $C \in \mathcal{P}$,*

$$|\hat{\mu}_j^{(0)}(C) - \mu(x_C)| \leq r_j^{(0)}(C), \quad r_j^{(0)}(C) = \sqrt{\frac{\beta_0}{2 \max\{1, o_{j,C}(T_0)\}}}, \quad \beta_0 = \log \frac{4NK(T_0 + 1)}{\delta_I}.$$

Proof. Fix (j, C) and let $n := o_{j,C}(T_0)$. Condition on the sigma-field generated by all exploration choices and collisions, and on the realized value of n .

Conditional on n , and because the reward noise is independent of the exploration/collision process (see assumptions), the n observed rewards associated with (j, C) are i.i.d., lie in $[0, 1]$, and have mean $\mu(x_C)$. Denote their average by $\hat{\mu}_j^{(0)}(C)$ (for $n = 0$ we keep the convention $\hat{\mu}_j^{(0)}(C) = 0$).

Case $n \geq 1$. Hoeffding's inequality for $[0, 1]$ -bounded i.i.d. variables yields, for any $\epsilon > 0$,

$$\Pr\left(|\hat{\mu}_j^{(0)}(C) - \mu(x_C)| > \epsilon \mid o_{j,C}(T_0) = n\right) \leq 2 \exp(-2n\epsilon^2).$$

Choosing $\epsilon = \sqrt{\beta_0/(2n)}$ gives

$$\Pr\left(|\hat{\mu}_j^{(0)}(C) - \mu(x_C)| > \sqrt{\beta_0/(2n)} \mid o_{j,C}(T_0) = n\right) \leq 2e^{-\beta_0}. \quad (\text{A.6})$$

Case $n = 0$. Then $\hat{\mu}_j^{(0)}(C) = 0$ by definition. Since $\mu(x_C) \in [0, 1]$, the event

$$\left\{ |\hat{\mu}_j^{(0)}(C) - \mu(x_C)| > \sqrt{\beta_0/2} \right\}$$

is **impossible** whenever $\sqrt{\beta_0/2} \geq 1$. With our choice $\beta_0 = \log \frac{4NK(T_0+1)}{\delta_I}$ and global budget $\delta_I \leq \delta_{\text{sys}} < 1/4$, we have

$$\beta_0 \geq \log \frac{4 \cdot 1 \cdot 1 \cdot 2}{1/4} = \log(32) > 2,$$

so indeed $\sqrt{\beta_0/2} > 1$ and the $n = 0$ failure probability equals 0.

Let $E_{j,C}$ be the (unconditional) failure event for this (j, C) , namely

$$E_{j,C} := \left\{ |\hat{\mu}_j^{(0)}(C) - \mu(x_C)| > \sqrt{\frac{\beta_0}{2 \max\{1, o_{j,C}(T_0)\}}} \right\}.$$

By the law of total probability and the discussion above,

$$\begin{aligned} \Pr(E_{j,C}) &= \sum_{n=0}^{T_0} \Pr(E_{j,C} \mid o_{j,C}(T_0) = n) \Pr(o_{j,C}(T_0) = n) \\ &\leq \sum_{n=1}^{T_0} 2e^{-\beta_0} \Pr(o_{j,C}(T_0) = n) \\ &\leq 2e^{-\beta_0}. \end{aligned}$$

Finally, apply a union bound over all players and cells:

$$\begin{aligned} \Pr(\exists(j, C) \in [N] \times \mathcal{P} : E_{j,C}) &\leq 2NK e^{-\beta_0} \\ &= \frac{\delta_I}{2(T_0 + 1)} \\ &\leq \frac{\delta_I}{2}. \end{aligned}$$

Hence, with probability at least $1 - \delta_I/2$, the desired deviation bound holds simultaneously for all (j, C) , i.e., the lemma. $\square$

A.1.3 A.3 Phase-I maxima brackets (Proposition 1)

Proposition 8 (Restatement of Proposition 1). *On an event of probability at least $1 - \delta_I$, simultaneously for all $j \in [N]$ and $C \in \mathcal{P}$,*

$$\text{LCB}_j^{(0)}(C) \leq \mu^*(C) \leq \text{UCB}_j^{(0)}(C),$$

with $\text{LCB}_j^{(0)}, \text{UCB}_j^{(0)}$ as in (A.4).

Proof. By Lemma 2, on an event $\mathcal{E}_{\text{means}}$ of probability at least $1 - \delta_I/2$, we have for all (j, C)

$$\mu(x_C) \in [\hat{\mu}_j^{(0)}(C) - r_j^{(0)}(C), \hat{\mu}_j^{(0)}(C) + r_j^{(0)}(C)].$$

Combining this with (A.2) gives, for every (j, C) ,

$$\hat{\mu}_j^{(0)}(C) - r_j^{(0)}(C) \leq \mu^*(C) \leq \hat{\mu}_j^{(0)}(C) + r_j^{(0)}(C) + \frac{L}{2}D_h,$$

which is exactly $\text{LCB}_j^{(0)}(C) \leq \mu^*(C) \leq \text{UCB}_j^{(0)}(C)$ by (A.4). Thus, the validity holds on $\mathcal{E}_{\text{means}}$.

For later phases we also record the success-count event $\mathcal{E}_{\text{counts}}$ of Lemma 1, which holds with probability at least $1 - \delta_I/2$ provided T_0 satisfies (A.5). While $\mathcal{E}_{\text{counts}}$ is not needed for the validity of the intervals, it controls their **width**. On the intersection

$$\mathcal{E}_I := \mathcal{E}_{\text{means}} \cap \mathcal{E}_{\text{counts}},$$

the bracket validity still holds, and by a union bound

$$\Pr(\mathcal{E}_I) \geq 1 - \frac{\delta_I}{2} - \frac{\delta_I}{2} = 1 - \delta_I.$$

This proves the proposition as stated. $\square$

A.1.4 A.4 Choice of T_0 and bracket precision

We record explicit (mild) lower bounds on the Phase I budget T_0 that guarantee both Lemma 1 and a desired bracket accuracy.

Prerequisite for Lemma 1. As shown in (A.5), for any fixed $\eta \in (0, 1)$ it suffices to take

$$T_0 \geq \frac{3}{\eta^2 p_K} \log \frac{4NK}{\delta_I}. \quad (\text{A.7})$$

Uniform accuracy target. Fix $\alpha \in (0, 1)$. On $\mathcal{E}_{\text{counts}}$ we have $o_{j,C}(T_0) \geq (1 - \eta)T_0 p_K$ for all (j, C) . Using (A.3),

$$r_j^{(0)}(C) \leq \sqrt{\frac{\beta_0}{2(1 - \eta)T_0 p_K}}, \quad \beta_0 = \log \frac{4NK(T_0 + 1)}{\delta_I}.$$

Therefore a sufficient condition for $r_j^{(0)}(C) \leq \alpha$ uniformly over (j, C) on $\mathcal{E}_{\text{counts}}$ is

$$T_0 \geq \frac{\beta_0}{2\alpha^2(1 - \eta)p_K}. \quad (\text{A.8})$$

A convenient consolidated choice. Since $\beta_0 \geq \log \frac{4NK}{\delta_I}$ (as $T_0 + 1 \geq 1$), the single implicit requirement

$$T_0 \geq \frac{\beta_0}{p_K} \cdot \max\left\{\frac{1}{\alpha^2}, 12\right\} \quad \text{with } \beta_0 = \log \frac{4NK(T_0 + 1)}{\delta_I} \quad (\text{A.9})$$

implies both (A.7) (taking $\eta = \frac{1}{2}$ so that $\frac{3}{\eta^2} = 12$) and (A.8) (since $\frac{1}{2(1-\eta)} = 1$ for $\eta = \frac{1}{2}$). The dependence on T_0 inside β_0 is benign and monotone; any T_0 large enough to satisfy (A.9) is acceptable. In our subsequent analysis we only need the validity of the Phase I brackets (which already holds under Lemma 2); the accuracy parameter α can thus be taken moderate.

A.2 Appendix B: Proof Details for Phase II

Throughout this appendix we condition on the Phase I clean event $\mathcal{E}_I$ from Appendix A: (i) all Phase I brackets are valid for all players and cells; and (ii) the Phase I count concentration holds when invoked. We keep the Phase II failure budget $\delta_{II} \in (0, 1/4)$ and allocate it across events.

Standing assumptions and notation (Phase II).

- As in Appendix A, observed rewards lie in $[0, 1]$. Hence the centered noise is σ -sub-Gaussian with $\sigma \leq \frac{1}{2}$, and Hoeffding-type concentration bounds apply.
- **Collision observability** Players observe a collision bit (null symbol $\perp$). The missingness process (collisions) depends only on the independent exploration choices and is independent of the reward noise. We will condition on counts of successful observations without affecting noise distributions.
- **Active sets (safe elimination).** For player j , let $\theta_j^{(0)}$ be the N -th order statistic of $\{\text{LCB}_j^{(0)}(C) : C \in \mathcal{P}\}$, and define

$$\mathcal{S}_{\text{act}_j}^{(0)} := \left\{ C \in \mathcal{P} : \text{UCB}_j^{(0)}(C) \geq \theta_j^{(0)} \right\}. \quad (\text{A.10})$$

We will show that on $\mathcal{E}_I$ the active set is safe and has size at least N for every player.

- **Probe nets.** For each cell C , fix an η -net $Z_C \subset C$ built as an axis-aligned grid of spacing $s := \eta/\sqrt{d}$ in each coordinate. Then for every $x \in C$ there exists $z \in Z_C$ with $\|x - z\|_\infty \leq s/2$ and hence $\|x - z\|_2 \leq (s/2)\sqrt{d} = \eta/2 < \eta$. Its cardinality satisfies

$$|Z_C| \leq \left(2 + \frac{h\sqrt{d}}{\eta}\right)^d \leq c_d \left(1 + \frac{h}{\eta}\right)^d \leq c'_d \left(1 + \left(\frac{h}{\eta}\right)^d\right) \leq C_d \left(\frac{h}{\eta}\right)^d, \quad (\text{A.11})$$

where $c_d := (2 + \sqrt{d})^d$ and $C_d := \max\{c_d, 2^d\}$ is an absolute constant depending only on d . For simplicity we use the conservative bound $|Z_C| \leq C_d(h/\eta)^d$ below. Define $P_{\max} := \max_{C \in \mathcal{P}} |Z_C|$, so $P_{\max} \leq C_d(h/\eta)^d$.

- **Phase II sampling protocol (fixed for T_1 rounds).** At each round $t = 1, \dots, T_1$, independently across rounds and players:

- (1) Player j samples a cell $C_t^{(j)}$ uniformly from $\mathcal{S}_{\text{act}_j}^{(0)}$.
- (2) Given $C_t^{(j)}$, player j samples a probe $Z_t^{(j)}$ uniformly from $Z_{C_t^{(j)}}$ and probes $Z_t^{(j)}$.
- (3) A collision in a cell C occurs if at least two players select any probes in the same C at round t . A probe is successful if no other player selects cell C in that round.

The active sets $\{\mathcal{S}_{\text{act}_j}^{(0)}\}$ and nets $\{Z_C\}$ are fixed throughout Phase II.

- **Counting notation.** For a triple (j, C, z) with $C \in \mathcal{S}_{\text{act}_j}^{(0)}$ and $z \in Z_C$, define the per-round success indicator

$$V_{j,C,z}(t) := \mathbf{1}\{C_t^{(j)} = C, Z_t^{(j)} = z, \text{ and no other player selects } C \text{ at round } t\},$$

and total successes $s_{j,C,z}(T_1) := \sum_{t=1}^{T_1} V_{j,C,z}(t)$. Let

$$N_{\text{probe}} := \sum_{j=1}^N \sum_{C \in \mathcal{S}_{\text{act}_j}^{(0)}} |Z_C| \quad \text{and} \quad M_{\text{act}} := \max_{j \in [N]} |\mathcal{S}_{\text{act}_j}^{(0)}|.$$

A.2.1 B.1. Safe active sets (size and completeness)

Lemma 9 (Active-set safety and size). *On $\mathcal{E}_I$, for every player j :*

$$(1) \quad |\mathcal{S}_{\text{act}_j}^{(0)}| \geq N;$$

(2) every true top- N cell belongs to $\mathcal{S}_{\text{act}_j}^{(0)}$, i.e., if T^* denotes the set of N cells with the largest $\mu^*(\cdot)$ values, then $T^* \subseteq \mathcal{S}_{\text{act}_j}^{(0)}$.

Proof. On $\mathcal{E}_I$, $\text{LCB}_j^{(0)}(C) \leq \mu^*(C) \leq \text{UCB}_j^{(0)}(C)$ for every C .

Let $\mu_{(1)}^* \geq \dots \geq \mu_{(N)}^*$ be the sorted cell maxima and T^* the set of corresponding cells with ranks 1 to N .

Since each $\text{LCB}_j^{(0)}(C) \leq \mu^*(C)$, the N -th LCB order statistic satisfies $\theta_j^{(0)} \leq \mu_{(N)}^*$.

For any $C \in T^*$, we have $\text{UCB}_j^{(0)}(C) \geq \mu^*(C) \geq \mu_{(N)}^* \geq \theta_j^{(0)}$, hence $C \in \mathcal{S}_{\text{act}_j}^{(0)}$ by (A.10).

Therefore $T^* \subseteq \mathcal{S}_{\text{act}_j}^{(0)}$ and $|\mathcal{S}_{\text{act}_j}^{(0)}| \geq |T^*| = N$. $\square$

A.2.2 B.2. Coverage: uniform lower bound on per-round probe success

Lemma 10 (Per-round success probability). *Under the Phase II protocol and Lemma 9, for every triple (j, C, z) and every round t ,*

$$\Pr[V_{j,C,z}(t) = 1] \geq q_{M_{\text{act}},\eta} \quad \text{with} \quad q_{M_{\text{act}},\eta} := \frac{1}{M_{\text{act}} P_{\max}} \left(1 - \frac{1}{N}\right)^{N-1},$$

where $P_{\max} := \max_C |Z_C| \leq C_d(h/\eta)^d$.

Proof. Fix (j, C, z) and t .

Player j picks C with probability $1/|\mathcal{S}_{\text{act}_j}^{(0)}| \geq 1/M_{\text{act}}$ and, given C , picks z with probability $1/|Z_C| \geq 1/P_{\max}$.

For each other player $k \neq j$, either $C \notin \mathcal{S}_{\text{act}_k}^{(0)}$ (so $\Pr(C_t^{(k)} = C) = 0$) or $C \in \mathcal{S}_{\text{act}_k}^{(0)}$ and then $\Pr(C_t^{(k)} = C) = 1/|\mathcal{S}_{\text{act}_k}^{(0)}| \leq 1/N$ by Lemma 9.

Independence across players implies

$$\Pr[\text{no other player selects } C] \geq \left(1 - \frac{1}{N}\right)^{N-1}.$$

Multiplying completes the proof. $\square$

Lemma 11 (Phase-II Probe Coverage). *Let $b \in \mathbb{N}$ and $T_1 \in \mathbb{N}$ satisfy*

$$b \geq 4 \log \frac{2 N_{\text{probe}}}{\delta_{II}} \quad \text{and} \quad T_1 \geq \frac{2b}{q_{M_{\text{act}},\eta}}.$$

Then, with probability at least $1 - \delta_{II}/2$, every triple achieves at least b successes:

$$\min_{(j,C,z)} s_{j,C,z}(T_1) \geq b.$$

Proof. For any fixed triple, $\{V_{j,C,z}(t)\}_{t=1}^{T_1}$ are i.i.d. Bernoulli with success probability $q_{j,C,z} := \Pr[V_{j,C,z}(t) = 1] \geq q_{M_{act},\eta}$ by Lemma 10.

Thus $\mathbb{E}[s_{j,C,z}(T_1)] = T_1 q_{j,C,z} \geq 2b$.

By the Chernoff lower-tail bound with $\lambda = \frac{1}{2}$,

$$\Pr[s_{j,C,z}(T_1) < b] \leq \exp\left(-\frac{\mathbb{E}[s_{j,C,z}(T_1)]}{8}\right) \leq e^{-b/4}.$$

A union bound over all N_{probe} triples gives $\Pr(\exists(j, C, z) : s_{j,C,z}(T_1) < b) \leq N_{\text{probe}} e^{-b/4} \leq \delta_{II}/2$ by the choice of b . $\square$

A.2.3 B.3. Uniform probe accuracy

Lemma 12 (Uniform probe accuracy). *Let*

$$\beta_1 := \log \frac{4 N_{\text{probe}}}{\delta_{II}} \quad \text{and} \quad r_1 := \sqrt{\frac{\beta_1}{2b}}.$$

On the event of Lemma 11 (so $s_{j,C,z}(T_1) \geq b$ for all triples), we have

$$\Pr\left[\exists(j, C, z) : |\hat{\mu}_j(C, z) - \mu(z)| > r_1\right] \leq \frac{\delta_{II}}{2},$$

where $\hat{\mu}_j(C, z)$ is the empirical mean over the $s_{j,C,z}(T_1)$ successful observations at (j, C, z) . Equivalently, with probability at least $1 - \delta_{II}/2$,

$$|\hat{\mu}_j(C, z) - \mu(z)| \leq r_1 \quad \text{simultaneously for all } (j, C, z).$$

Proof. Fix (j, C, z) . Condition on $s := s_{j,C,z}(T_1) \geq b$ and on the exploration/collision sigma-field.

By our assumption on the collisions, the s observed rewards at (j, C, z) are i.i.d. in $[0, 1]$ with mean $\mu(z)$. For any $n \geq b$, Hoeffding yields

$$\Pr\left(|\hat{\mu}_j(C, z) - \mu(z)| > \sqrt{\beta_1/(2n)} \mid s = n\right) \leq 2e^{-\beta_1}.$$

Hence, by the law of total probability restricted to $n \geq b$,

$$\Pr(|\hat{\mu}_j(C, z) - \mu(z)| > r_1 \mid s \geq b) \leq 2e^{-\beta_1}.$$

A union bound over the N_{probe} triples yields the claim with the chosen β_1 . $\square$

Remark 1 (Anytime variant (not used in parameterization)). *If one prefers a radius that adapts to the realized count, set $\beta_1^{\text{any}} := \log \frac{4N_{\text{probe}}(T_1+1)}{\delta_{II}}$ and $r^{\text{any}}(s) := \sqrt{\beta_1^{\text{any}}/(2\max\{1, s\})}$. Then, by the same Hoeffding+ law-of-total-probability argument and a union bound over $n \in \{1, \dots, T_1\}$,*

$$\Pr(\exists(j, C, z) : |\hat{\mu}_j(C, z) - \mu(z)| > r^{\text{any}}(s_{j,C,z}(T_1))) \leq \frac{\delta_{II}}{2}.$$

We do not use this form below; our fixed- b parameterization already yields the target width with fewer log factors.

A.2.4 B.4. Refined maxima brackets

For player j and active cell $C \in \mathcal{S}_{\text{act}_j}^{(0)}$, define

$$\text{LCB}_j^{(1)}(C) := \max_{z \in Z_C} \{\hat{\mu}_j(C, z) - r_1\}, \quad \text{UCB}_j^{(1)}(C) := \max_{z \in Z_C} \{\hat{\mu}_j(C, z) + r_1\} + L\eta. \quad (\text{A.12})$$

Proposition 9 (Refined Maxima Brackets). *On the intersection of the events in Lemmas 11 and 12 (probability at least $1 - \delta_{II}$), simultaneously for all players j and active cells $C \in \mathcal{S}_{\text{act}_j}^{(0)}$,*

$$\text{LCB}_j^{(1)}(C) \leq \mu^*(C) \leq \text{UCB}_j^{(1)}(C) \quad \text{and} \quad \text{UCB}_j^{(1)}(C) - \text{LCB}_j^{(1)}(C) \leq 2r_1 + L\eta.$$

Proof. Work on the event of Lemma 12.

For any $z \in Z_C$, $\mu(z) \in [\hat{\mu}_j(C, z) - r_1, \hat{\mu}_j(C, z) + r_1]$.

Taking maxima over $z \in Z_C$ gives

$$\max_{z \in Z_C} \mu(z) \in \left[\max_{z \in Z_C} (\hat{\mu}_j(C, z) - r_1), \max_{z \in Z_C} (\hat{\mu}_j(C, z) + r_1) \right].$$

By the η -net property and L -Lipschitzness, $\mu^*(C) \in [\max_{z \in Z_C} \mu(z), \max_{z \in Z_C} \mu(z) + L\eta]$.

Combining yields the validity and the width bound stated. $\square$

A.2.5 B.5. Selecting the top- N cells: gap-free ε -optimality and consensus

Recall that each player j selects the N cells with the largest $\text{LCB}_j^{(1)}(C)$ (breaking ties by a fixed public ordering), and denote the selected set by $S_\varepsilon^{(j)}$.

Theorem 23 (Gap-Free ε -Optimality). *Let $\varepsilon > 0$. If Phase II parameters are such that $2r_1 + L\eta \leq \varepsilon$, then on the intersection of the events in Lemmas 11 and 12 (probability at least $1 - \delta_{II}$), for every player j and every $C \in S_\varepsilon^{(j)}$,*

$$\mu^*(C) \geq \mu_{(N)}^* - \varepsilon.$$

Proof. On the event of Proposition 9, for any true top- N cell C^* we have $\text{LCB}_j^{(1)}(C^*) \geq \mu^*(C^*) - \varepsilon \geq \mu_{(N)}^* - \varepsilon$.

Let $\theta_j^{(1)}$ be the N -th order statistic of $\{\text{LCB}_j^{(1)}(C) : C \in \mathcal{S}_{\text{act}_j}^{(0)}\}$; since $T^* \subseteq \mathcal{S}_{\text{act}_j}^{(0)}$ (Lemma 9) and there are N cells with LCB at least $\mu_{(N)}^* - \varepsilon$, we have $\theta_j^{(1)} \geq \mu_{(N)}^* - \varepsilon$.

If $C \in S_\varepsilon^{(j)}$, then $\text{LCB}_j^{(1)}(C) \geq \theta_j^{(1)}$, and validity gives $\mu^*(C) \geq \text{LCB}_j^{(1)}(C) \geq \mu_{(N)}^* - \varepsilon$. $\square$

Definition 10 (ε -Uniqueness at the Top- N). *A mean function μ is ε -unique at the top- N if there exists a unique set $S^\dagger \subset \mathcal{P}$ of size N such that*

$$\min_{C \in S^\dagger} \mu^*(C) \geq \max_{C \notin S^\dagger} \mu^*(C) + 2\varepsilon.$$

Lemma 13 (Consensus under ε -Uniqueness). *Assume ε -uniqueness at the top- N and that all refined brackets have width at most ε and contain $\mu^*(C)$. Then $S_\varepsilon^{(j)} = S^\dagger$ for all $j \in [N]$.*

Proof. For any $C_{\text{good}} \in S^\dagger$ and $C_{\text{bad}} \notin S^\dagger$,

$$\text{LCB}_j^{(1)}(C_{\text{good}}) \geq \mu^*(C_{\text{good}}) - \varepsilon \geq \mu^*(C_{\text{bad}}) + \varepsilon \geq \text{LCB}_j^{(1)}(C_{\text{bad}}),$$

using ε -uniqueness and $\text{LCB}_j^{(1)}(C_{\text{bad}}) \leq \mu^*(C_{\text{bad}})$. Thus the N largest LCBs are attained exactly on $S^\dagger$ (ties do not change this inclusion). $\square$

A.2.6 B.6. Parameter choices and complexity bounds

We summarize our parameter choices ensuring width $\leq \varepsilon$ with probability at least $1 - \delta_{II}$.

Cardinalities and probabilities. By (A.11) and Lemma 9,

$$N_{\text{probe}} = \sum_{j=1}^N \sum_{C \in \mathcal{S}_{\text{act}_j}^{(0)}} |Z_C| \leq N M_{\text{act}} P_{\max} \leq N M_{\text{act}} C_d \left(\frac{h}{\eta}\right)^d.$$

Lemma 10 gives

$$q_{M_{\text{act}}, \eta} \geq \frac{1}{M_{\text{act}} P_{\max}} \left(1 - \frac{1}{N}\right)^{N-1} \geq \frac{1}{M_{\text{act}} C_d} \left(\frac{\eta}{h}\right)^d \left(1 - \frac{1}{N}\right)^{N-1}.$$

Concrete schedule for a target $\varepsilon > 0$. Choose

$$\eta = \frac{\varepsilon}{2L}, \quad \beta_1 = \log \frac{4 N_{\text{probe}}}{\delta_{II}}, \quad b \geq \max \left\{ 4 \log \frac{2 N_{\text{probe}}}{\delta_{II}}, \frac{8 \beta_1}{\varepsilon^2} \right\},$$

so that $2r_1 + L\eta \leq \varepsilon$, where $r_1 = \sqrt{\beta_1/(2b)}$. Then, using Lemma 11 and the bound on $q_{M_{\text{act}}, \eta}$,

$$T_1 \geq \frac{2b}{q_{M_{\text{act}}, \eta}} \leq \frac{2b M_{\text{act}} P_{\max}}{\left(1 - \frac{1}{N}\right)^{N-1}} \leq \frac{2b M_{\text{act}} C_d}{\left(1 - \frac{1}{N}\right)^{N-1}} \left(\frac{h}{\eta}\right)^d.$$

With $\eta = \varepsilon/(2L)$ and $b = \Theta(\beta_1/\varepsilon^2 + \log(N_{\text{probe}}/\delta_{II}))$ we obtain

$$T_1 = O\left(\frac{M_{\text{act}} (Lh)^d}{\varepsilon^{d+2}} \cdot \left[\beta_1 + \log \frac{N_{\text{probe}}}{\delta_{II}}\right]\right), \quad N_{\text{probe}} \leq O\left(N M_{\text{act}} (Lh/\varepsilon)^d\right).$$

Thus the Phase II budget T_1 is independent of the horizon T and scales (up to polylog factors) as

$$T_1 = \tilde{O}\left(M_{\text{act}} (Lh)^d \varepsilon^{-(d+2)}\right).$$

If desired, one may replace M_{act} by K using $M_{\text{act}} \leq K$ for a looser but simpler bound.

Failure budget aggregation. By Lemmas 11 and 12, the intersection of the coverage and uniform-accuracy events holds with probability at least $1 - \delta_{II}$. Proposition 9 and Theorem 23 are stated on this intersection. Lemma 13 is a deterministic consequence of the refined brackets and ε -uniqueness.

Remarks. (i) The constant $L\eta$ in (A.12) can be tightened to $L\eta/2$ using the $\eta/2$ covering radius; we keep $L\eta$ for simplicity and monotonicity with respect to η . (ii) If one prefers an anytime-in- n accuracy with an adaptive radius, use Remark 1; this modifies β_1 to include a $\log(T_1 + 1)$ factor without changing asymptotics in ε . (iii) The protocol fixes active sets during Phase II; allowing adaptive shrinking would require re-deriving $q_{M_{\text{act}}, \eta}$ or freezing a lower bound on $|\mathcal{S}_{\text{act}_j}^{(0)}|$ throughout the phase.

A.3 Appendix C: Proof Details for Phase II $\frac{1}{2}$ (Musical Chairs)

We now provide proofs for the Musical Chairs (MC) seating phase.

We know that Phase II concluded such that all players share the same N -cell target set (e.g., because the ϵ -uniqueness condition holds). The analysis below is conditioned on this event.

There are N target cells and N players. At the start of the phase, all players are unseated. In each round, every unseated player independently samples a cell uniformly at random from the N -cell target set. If exactly one unseated player chooses a currently unoccupied cell, that player becomes seated at that cell and remains seated thereafter. If a cell is chosen by two or more unseated players, or by any unseated player together with an already seated player, a collision occurs at that cell and all players who chose it obtain zero reward in that round (the seated player remains seated).

Let $U_t \in \{0, 1, \dots, N\}$ be the number of unseated players at the start of round t . Then exactly $N - U_t$ cells are occupied (by seated players) and exactly U_t cells are free. Define the drift at state u by

$$\Delta(u) := \mathbb{E}[U_t - U_{t+1} \mid U_t = u], \quad (\text{A.13})$$

i.e., the expected number of newly seated players in a round when u players are currently unseated.

A.3.1 C.1. Drift formula and basic bounds

Lemma 14 (Exact drift and a uniform lower bound). *For every $u \in \{1, \dots, N\}$,*

$$\Delta(u) = \frac{u^2}{N} \left(1 - \frac{1}{N}\right)^{u-1} \geq \frac{u^2}{eN}. \quad (\text{A.14})$$

Proof. When u players are unseated, there are u free cells.

Fix a particular free cell.

The probability that exactly one of the u unseated players chooses this cell equals $u \cdot \frac{1}{N} \cdot (1 - \frac{1}{N})^{u-1}$: choose which unseated player (there are u choices), that player picks the cell with probability $1/N$, and each of the remaining $u - 1$ unseated players avoids it with probability $(1 - 1/N)$, independently.

Each free cell with exactly one chooser yields exactly one newly seated player, and free cells are disjoint, so by linearity of expectation over the u free cells we obtain

$$\Delta(u) = u \cdot \left[u \cdot \frac{1}{N} \cdot \left(1 - \frac{1}{N}\right)^{u-1} \right] = \frac{u^2}{N} \left(1 - \frac{1}{N}\right)^{u-1}.$$

For the lower bound, $(1 - 1/N)^{u-1} \geq (1 - 1/N)^{N-1} \geq e^{-1}$ for all $N \geq 1$, yielding $\Delta(u) \geq u^2/(eN)$. $\square$

Lemma 15 (Monotonicity of the drift). *The drift $\Delta(u)$ is strictly increasing for $u \in \{1, 2, \dots, N\}$.*

Proof. For $u \in \{1, \dots, N-1\}$,

$$\frac{\Delta(u+1)}{\Delta(u)} = \frac{(u+1)^2}{u^2} \cdot \left(1 - \frac{1}{N}\right) = \left(1 + \frac{1}{u}\right)^2 \left(1 - \frac{1}{N}\right).$$

Since $u \leq N-1$, we have $1 + \frac{1}{u} \geq 1 + \frac{1}{N-1} > 1$ and hence

$$\left(1 + \frac{1}{u}\right)^2 \left(1 - \frac{1}{N}\right) \geq \left(1 + \frac{1}{N-1}\right)^2 \left(1 - \frac{1}{N}\right) = \frac{N^2}{(N-1)^2} \cdot \frac{N-1}{N} = \frac{N}{N-1} > 1.$$

Thus $\Delta(u+1) > \Delta(u)$. $\square$

A.3.2 C.2. Expected seating time

We now prove that the expected number of rounds to seat all players is linear in N .

Theorem 24 (Expected seating time). *Let $T_{MC} := \inf\{t \geq 1 : U_t = 0\}$ be the (a.s. finite) stopping time at which all players are seated. Then*

$$\mathbb{E}[T_{MC}] \leq \sum_{u=1}^N \frac{1}{\Delta(u)} \leq \frac{e\pi^2}{6} N,$$

and in particular $\mathbb{E}[T_{MC}] = O(N)$.

Proof. Define the potential

$$V(u) := \sum_{v=1}^u \frac{1}{\Delta(v)}, \quad u \in \{0, 1, \dots, N\}.$$

Note that $V(0) = 0$ and V is finite and nondecreasing.

Let $Y_t := U_t - U_{t+1} \in \{0, 1, \dots, U_t\}$ denote the number of newly seated players in round t .

Fix t and condition on $U_t = u > 0$. Using the definition of V ,

$$V(U_t) - V(U_{t+1}) = V(u) - V(u - Y_t) = \sum_{k=0}^{Y_t-1} \frac{1}{\Delta(u-k)}.$$

By Lemma 15, $\Delta(\cdot)$ is increasing, so $1/\Delta(\cdot)$ is decreasing.

Therefore

$$V(U_t) - V(U_{t+1}) \geq Y_t \cdot \frac{1}{\Delta(u)}.$$

Taking conditional expectations and using the definition of the drift (A.13),

$$\mathbb{E}[V(U_t) - V(U_{t+1}) \mid U_t = u] \geq \frac{\mathbb{E}[Y_t \mid U_t = u]}{\Delta(u)} = \frac{\Delta(u)}{\Delta(u)} = 1.$$

Taking expectations and summing over $t = 0, 1, \dots, T_{MC} - 1$, we obtain (by the tower property; no optional-stopping theorem is required)

$$\begin{aligned} \mathbb{E}[V(U_0) - V(U_{T_{MC}})] &= \sum_{t=0}^{\infty} \mathbb{E}[\mathbf{1}\{t < T_{MC}\} \cdot (V(U_t) - V(U_{t+1}))] \\ &\geq \sum_{t=0}^{\infty} \mathbb{E}[\mathbf{1}\{t < T_{MC}\}] \\ &= \mathbb{E}[T_{MC}]. \end{aligned}$$

Since $U_0 = N$ and $V(0) = 0$, we have $\mathbb{E}[T_{MC}] \leq V(N) = \sum_{u=1}^N \frac{1}{\Delta(u)}$.

Finally, using Lemma 14, $\Delta(u) \geq u^2/(eN)$, hence

$$\sum_{u=1}^N \frac{1}{\Delta(u)} \leq eN \sum_{u=1}^N \frac{1}{u^2} \leq \frac{e\pi^2}{6} N,$$

which proves the claim. □

A.3.3 C.3. Seating-phase regret

We evaluate the expected regret accumulated during the seating phase, measured against the per-round benchmark of obtaining N unit-normalized rewards (one per target cell).

Corollary 9 (Seating-phase regret: a simple bound). *Let R_{MC} denote the cumulative regret incurred during the seating phase. Then*

$$\mathbb{E}[R_{MC}] \leq N \cdot \mathbb{E}[T_{MC}] \leq \frac{e\pi^2}{6} N^2,$$

so in particular $\mathbb{E}[R_{MC}] = O(N^2)$. The bound is independent of the overall horizon T .

Proof. In each round, the per-round benchmark is at most N (one unit per cell), and actual reward is nonnegative. Therefore the instantaneous regret is at most N .

By Linearity of expectation on the random-time sum,

$$\mathbb{E}[R_{MC}] = \mathbb{E}\left[\sum_{t=1}^{T_{MC}} \text{regret}_t\right] \leq \mathbb{E}\left[\sum_{t=1}^{T_{MC}} N\right] = N \mathbb{E}[T_{MC}],$$

and the claim follows from Theorem 24. $\square$

Remark 2 (Sharper regret bound). *The $O(N^2)$ bound is conservative. One can show $\mathbb{E}[R_{MC}] = O(N \log N)$ as follows. In a round with u unseated players:*

- *The expected number of occupied cells that are hit by at least one unseated player is at most $(N - u)(1 - (1 - 1/N)^u) \leq (N - u) \cdot \frac{u}{N} \leq u$, so expected regret contributed by occupied cells is $\leq u$ (each such collision zeros that cell's reward).*
- *Among the u free cells, the expected number that are not uniquely chosen by exactly one unseated player is $u \left[1 - u \cdot \frac{1}{N} \cdot (1 - \frac{1}{N})^{u-1}\right] \leq u$, so expected regret from free cells is also $\leq u$.*

Thus $\mathbb{E}[\text{regret}_t \mid U_t = u] \leq 2u$.

Summing over the seating phase,

$$\mathbb{E}[R_{MC}] \leq 2 \mathbb{E}\left[\sum_{t=0}^{T_{MC}-1} U_t\right].$$

Define the potential $G(u) := \sum_{v=1}^u \frac{v}{\Delta(v)}$.

By the same drift argument as in Theorem 24, one shows $\mathbb{E}\left[\sum_{t < T_{MC}} U_t\right] \leq G(N) = \sum_{v=1}^N \frac{v}{\Delta(v)}$.

Using $\Delta(v) \geq v^2/(eN)$ gives $\sum_{v=1}^N \frac{v}{\Delta(v)} \leq eN \sum_{v=1}^N \frac{1}{v} \leq eN(1 + \ln N)$.

Therefore $\mathbb{E}[R_{MC}] = O(N \log N)$. We keep Corollary 1 in the main text for simplicity, as it is horizon-independent and sufficient for our overall bounds.

A.4 Appendix D: Proof of Proposition 3 (In-Cell Regret)

We prove the standard minimax rate for a single player optimizing within a fixed cell. Throughout this appendix, rewards are bounded in $[0, 1]$ (consistent with Appendix A), and μ is L -Lipschitz on the hypercube cell $C \subset \mathbb{R}^d$ of side length h in the Euclidean norm.

Reduction to the unit cube. Let $\phi : [0, 1]^d \rightarrow C$ be the affine bijection $\phi(x') = x_0 + h x'$ that maps the unit cube onto C (for some cell origin x_0). Define $\mu'(x') := \mu(\phi(x'))$ for $x' \in [0, 1]^d$. Then, for any $x', y' \in [0, 1]^d$,

$$|\mu'(x') - \mu'(y')| = |\mu(\phi(x')) - \mu(\phi(y'))| \leq L \|\phi(x') - \phi(y')\|_2 = Lh \|x' - y'\|_2.$$

Thus μ' is L' -Lipschitz on $[0, 1]^d$ with $L' := Lh$.

Known minimax rate on the unit cube. For the d -dimensional Lipschitz bandit on $[0, 1]^d$ with Lipschitz constant L' , there exist algorithms (e.g., the Zooming algorithm [36]) whose expected regret over any horizon $T' \geq 1$ satisfies

$$\mathbb{E}[R_{[0,1]^d}(T')] \leq c_d (L')^{\frac{d}{d+2}} (T')^{\frac{d+1}{d+2}} + c'_d, \quad (\text{A.15})$$

where c_d, c'_d depend only on d . This result is classical; see, e.g., Kleinberg et al. [36] or Slivkins [69].

Combining. Applying (A.15) to μ' and substituting $L' = Lh$ gives

$$\mathbb{E}[R_{\text{in}}^{(C)}(T')] \leq c_d (Lh)^{\frac{d}{d+2}} (T')^{\frac{d+1}{d+2}} + c'_d,$$

which is exactly Proposition 3. □

Remarks. (i) The proof above is intentionally short: it isolates the h -dependence via rescaling and then cites a standard unit-cube result. If desired, one may instantiate a concrete algorithm

(e.g., Zooming) and track constants; this does not affect the rate or the $(Lh)^{d/(d+2)}$ dependence.

(ii) If one prefers finite-armed baselines, an epochic discretize-and-explore scheme on grids of mesh $\asymp (Lh)^{2/(d+2)} 2^{e/(d+2)}$ per coordinate also yields the same rate by balancing exploration and discretization errors; we omit these routine details.

A.5 Appendix E: Global Regret Bounds (Theorem 3)

We assemble the end-to-end bound and reconcile expectation-level failure terms with the main-text statement. Throughout, per-round reward is in $[0, 1]$ for each player; thus the per-round system benchmark (sum of the top- N cell maxima) is at most N .

A.5.1 E.1. Clean event and failure budgeting

Let $\mathcal{E}_I$ and $\mathcal{E}_{II}$ be the Phase I and Phase II success events as defined in Appendices A and B, respectively (valid brackets and coverage/accuracy). Set

$$\delta_I = \delta_{II} = \frac{\delta_{sys}}{2N}.$$

By union bounds across players and cells (already accounted for in Appendices A and B), we have

$$\Pr(\mathcal{E}_I) \geq 1 - \frac{\delta_{sys}}{2N}, \quad \Pr(\mathcal{E}_{II}) \geq 1 - \frac{\delta_{sys}}{2N}.$$

Define the global clean event $\mathcal{E} := \mathcal{E}_I \cap \mathcal{E}_{II}$ for which

$$\Pr(\mathcal{E}) \geq 1 - \frac{\delta_{sys}}{N}. \tag{A.16}$$

We write $R_{\text{cont}}(T)$ for the total regret up to time T and decompose its expectation by indicators of $\mathcal{E}$:

$$\mathbb{E}[R_{\text{cont}}(T)] = \mathbb{E}[R_{\text{cont}}(T) \mathbf{1}\{\mathcal{E}\}] + \mathbb{E}[R_{\text{cont}}(T) \mathbf{1}\{\mathcal{E}^c\}].$$

Since per-round regret is at most N , we have

$$\mathbb{E}[R_{\text{cont}}(T) \mathbf{1}\{\mathcal{E}^c\}] \leq NT \Pr(\mathcal{E}^c) \leq \delta_{sys} T, \tag{A.17}$$

which matches the main-text term.

A.5.2 E.2. Identification and seating terms (conditioned on $\mathcal{E}$)

Condition on $\mathcal{E}$. During Phase I and Phase II, each round contributes at most N to regret:

$$\mathbb{E}[R_I(T_0) + R_{II}(T_1) \mid \mathcal{E}] \leq N(T_0 + T_1). \quad (\text{A.18})$$

We assume the ε -uniqueness condition of Definition 1 (main text) in Theorem 3, which guarantees that all players identify the same N cells (Lemma 13 in Appendix B). Musical Chairs (Appendix C) is then run on this fixed set; its randomness is independent of reward noise used to define $\mathcal{E}$. Therefore,

$$\mathbb{E}[R_{MC}(T_{MC}) \mid \mathcal{E}] = \mathbb{E}[R_{MC}(T_{MC})] \leq c_{MC} N^2, \quad (\text{A.19})$$

for an absolute constant c_{MC} (Appendix C).

A.5.3 E.3. In-cell term and end-to-end bound

On $\mathcal{E}$, the Phase III processes are collision-free and decoupled across players. For any player j , Proposition 3 gives

$$\mathbb{E}[R_{\text{in}}^{(C_j)}(T_{\text{in}}^{(j)}) \mid \mathcal{E}] \leq c_d (Lh)^{\frac{d}{d+2}} (T_{\text{in}}^{(j)})^{\frac{d+1}{d+2}} + c'_d \leq c_d (Lh)^{\frac{d}{d+2}} T^{\frac{d+1}{d+2}} + c'_d,$$

and summing over $j \in [N]$,

$$\sum_{j=1}^N \mathbb{E}[R_{\text{in}}^{(C_j)}(T_{\text{in}}^{(j)}) \mid \mathcal{E}] \leq c_d N (Lh)^{\frac{d}{d+2}} T^{\frac{d+1}{d+2}} + N c'_d. \quad (\text{A.20})$$

Combining (A.17), (A.18), (A.19), and (A.20) yields

$$\mathbb{E}[R_{\text{cont}}(T)] \leq N(T_0 + T_1) + c_{MC} N^2 + c_d N (Lh)^{\frac{d}{d+2}} T^{\frac{d+1}{d+2}} + N c'_d + \delta_{sys} T.$$

Absorbing the additive $N c'_d$ into the (horizon-independent) coordination constant completes the proof of Theorem 3. $\square$

Remarks. (i) If desired, one may pick $\delta_{sys} = \delta_{sys}(T)$ (e.g., $\leq 1/(NT)$) to keep the failure contribution ≤ 1 in expectation, at the cost of extra $\log(NT)$ factors within Phase I/II radii; these are absorbed into $\tilde{O}(\cdot)$. (ii) The proof above uses only consensus (from ε -uniqueness) to run MC on a fixed target set; no other structural gap is used in Phase III.

A.6 Appendix F: Gap-Free Analysis—Consensus, Baselines, Limits, and a Conditional Epochic Recovery

This section gives a complete treatment of the gap-free regime referenced in Section 2.10.

Our goals are:

- (1) to formalize a communication-free public dither mechanism that guarantees consensus among players in Phase II selection and to prove it correct with safe constants;
- (2) to provide fully general guarantees that hold without extra assumptions on the instance: a single-shot gap-free bound and a restart lower bound showing that restarting global identification each epoch is too costly under the Phase II sampling analyzed in Appendices A-C;
- (3) to state and prove a conditional epochic recovery theorem under an extra assumption compared to Phase II (a public coverage/scheduling property).

Throughout this appendix, we use the failure budgeting of Appendix E so that the contribution of failure events to expected regret is at most $\delta_{\text{sys}}T$.

A.6.1 F.1. Communication-free consensus via a public dither with a guaranteed gap

In the gap-free regime, the refined LCBs $\{\text{LCB}_j^{(1)}(C)\}_C$ produced in Phase II can differ slightly across players, potentially leading to different top- N sets. We enforce consensus without communication by adding a public deterministic **dither** (randomness) $\xi(C)$ to the LCBs before ranking cells. Crucially, we ensure a minimum pairwise gap in ξ so that the dither dominates cross-player LCB fluctuations for every pair of cells.

Internal vs. external precision and probe budget. Fix a target external precision $\varepsilon_{\text{main}} > 0$ for the selection of N cells at the end of Phase II. Internally, Phase II refines brackets to width

$$\varepsilon_{\text{int}} := \frac{\varepsilon_{\text{main}}}{4},$$

i.e., for all cells C and all players j ,

$$\text{UCB}_j^{(1)}(C) - \text{LCB}_j^{(1)}(C) \leq \varepsilon_{\text{int}}.$$

This is achieved in Appendix B by choosing an η -net (probe spacing η) and a per-probe success budget b so that $2r_1 + L\eta \leq \varepsilon_{\text{int}}$, where $r_1 = \sqrt{\beta_1/(2b)}$ and β_1 is the usual anytime log factor.¹

Public dither with a guaranteed minimum gap. Let $\{C_1, \dots, C_K\}$ be the cells in a fixed public order (e.g., lexicographic). Set

$$\eta_{\text{dit}} := \frac{3\varepsilon_{\text{main}}}{4}, \quad \xi(C_m) := \frac{m-1}{K-1} \eta_{\text{dit}}, \quad m = 1, \dots, K.$$

Thus the minimum pairwise dither gap is $\Delta_\xi := \eta_{\text{dit}}/(K-1)$.

We increase the per-probe success budget b by a constant (in T) factor so that

$$4r_1 \leq \Delta_\xi = \frac{\eta_{\text{dit}}}{K-1}. \quad (\text{A.21})$$

(Equivalently, we reduce r_1 by a constant factor; this changes Phase II constants but not rates in T .) Each player ranks cells by the public **score**

$$\text{Score}_j(C) := \text{LCB}_j^{(1)}(C) + \xi(C)$$

and selects the N cells with largest Scores (ties broken lexicographically).

Lemma 16 (Consensus and $\varepsilon_{\text{main}}$ -optimality with public dither). *On the Phase II accuracy event (Appendix B) with bracket width $\leq \varepsilon_{\text{int}}$ for all cells, the public dither rule above ensures, with the same high probability:*

(1) **Consensus:** All players select the same top- N set S^{dit} .

(2) **$\varepsilon_{\text{main}}$ -optimality:** Every $C \in S^{\text{dit}}$ satisfies $\mu^*(C) \geq \mu_{(N)}^* - \varepsilon_{\text{main}}$.

¹ See Appendix B (coverage lemma and anytime concentration), where we use law-of-total-probability removal of conditioning and an anytime union bound across the random counts $s \geq b$ to obtain uniform-in-time concentration at each probe.

Proof. (Consensus.) On the Phase II accuracy event, for any cell C and players j, k , $|\text{LCB}_j^{(1)}(C) - \text{LCB}_k^{(1)}(C)| \leq 2r_1$. Hence for any pair (C, C') ,

$$\sup_{j,k} \left| (\text{LCB}_j^{(1)}(C) - \text{LCB}_j^{(1)}(C')) - (\text{LCB}_k^{(1)}(C) - \text{LCB}_k^{(1)}(C')) \right| \leq 4r_1.$$

By construction $|\xi(C) - \xi(C')| \geq \Delta_\xi \geq 4r_1$ for every pair (C, C') (eq. (A.21)). Therefore, the sign of

$$(\text{LCB}_j^{(1)}(C) - \text{LCB}_j^{(1)}(C')) + (\xi(C) - \xi(C'))$$

is the same for all j , i.e., all players induce the same total order by $\text{Score}_j(\cdot)$ and select the same top- N set.

($\varepsilon_{\text{main}}$ -optimality.) Let θ^{dit} be the N -th largest Score. For any true top- N cell C^* , $\text{LCB}_j^{(1)}(C^*) \geq \mu^*(C^*) - \varepsilon_{\text{int}} \geq \mu_{(N)}^* - \varepsilon_{\text{int}}$ and $\xi(C^*) \geq 0$; hence $\theta^{\text{dit}} \geq \mu_{(N)}^* - \varepsilon_{\text{int}}$. For any selected C ,

$$\mu^*(C) \geq \text{LCB}_j^{(1)}(C) \geq \theta^{\text{dit}} - \xi(C) \geq \mu_{(N)}^* - (\varepsilon_{\text{int}} + \eta_{\text{dit}}) = \mu_{(N)}^* - \varepsilon_{\text{main}}. \quad \square$$

Data reuse across epochs. Unless stated otherwise, we assume Phase II reuses all probe data across epochs; we do not restart identification. Union over $K_{\text{ep}} = \Theta(\log T)$ epochs adds an extra $\log K_{\text{ep}}$ into the radii's β , which is absorbed by $\tilde{O}(\cdot)$.

A.6.2 F.2. A fully general, single-shot gap-free guarantee

We first prove a gap-free guarantee that requires no structural assumptions beyond Lipschitz-ness. Phase I and Phase II are run once at precision ε ; then we apply Lemma 16 to select a common N -cell set, seat via Musical Chairs (Appendix C), and run Phase III (Appendix D).

Proposition 10 (Single-shot, gap-free baseline). *For any L -Lipschitz mean on $[0, 1]^d$ and horizon T , there exists a choice of ε such that*

$$\mathbb{E}[R_{\text{cont}}(T)] = \tilde{O}\left(N (K(Lh)^d)^{\frac{1}{d+3}} T^{\frac{d+2}{d+3}}\right) = \tilde{O}\left(N L^{\frac{d}{d+3}} T^{\frac{d+2}{d+3}}\right),$$

using $K = \lceil 1/h \rceil^d$ and $Kh^d \in [1, 2^d)$, whence $K(Lh)^d = \Theta(L^d)$. The $\tilde{O}(\cdot)$ hides logarithmic factors in $N, K, 1/\delta_{\text{sys}}$.

Proof. On the clean event (Appendix E), the Phase II time at target width ε satisfies (Appendix B)

$$T_1(\varepsilon) = \tilde{O}(M_{\text{act}}(Lh)^d \varepsilon^{-(d+2)}),$$

and in the worst case $M_{\text{act}} \leq K$. Each identification round contributes at most N regret, hence $R_{\text{ID}} = \tilde{O}(NK(Lh)^d \varepsilon^{-(d+2)})$.

By Lemma 16, Phase III suboptimality is $R_{\text{Sub}} \leq N\varepsilon T$, while in-cell learning is $R_{\text{Learn}} = \tilde{O}(N(Lh)^{\frac{d}{d+2}} T^{\frac{d+1}{d+2}})$ (Appendix D).

Balancing R_{ID} and R_{Sub} gives $\varepsilon^* \asymp (K(Lh)^d/T)^{1/(d+3)}$ and

$$R_{\text{ID}} + R_{\text{Sub}} = \tilde{O}\left(N (K(Lh)^d)^{\frac{1}{d+3}} T^{\frac{d+2}{d+3}}\right).$$

Since $\frac{d+2}{d+3} > \frac{d+1}{d+2}$ for $d \geq 1$, this dominates R_{Learn} ; adding the $O(N^2)$ seating constant and the $\delta_{\text{sys}}T$ failure contribution yields the claim. $\square$

A.6.3 F.3. Why restart-style epochic identification fails

We formalize that restarting Phase II at each epoch (instead of reusing data) incurs linear identification overhead in worst-case Lipschitz instances.

Proposition 11 (Lower bound for restart-style epochic identification). *Suppose that at the start of each epoch k (of length T_k) the algorithm recomputes Phase II brackets to width ε_k by running the collision-censored sampling of Phases I/II afresh, without reusing earlier probe data. Then for worst-case L -Lipschitz instances and any epoch schedule with $\sum_k T_k = T$,*

$$\mathbb{E}\left[\sum_k R_{\text{ID}}^{(k)}\right] = \Omega(NT).$$

Proof. By Appendix B (coverage and probe-level anytime bounds), achieving width ε_k requires

$$\Omega((Lh)^d \varepsilon_k^{-(d+2)})$$

rounds (up to logs), since the per-round success probability scales as $q_{M_{\text{act}}, \eta} \asymp 1/(M_{\text{act}}P(\eta))$ with $M_{\text{act}} \asymp K$ and $P(\eta) \asymp (Lh/\varepsilon_k)^d$ in the worst case.

Each identification round contributes at most N regret, so $R_{\text{ID}}^{(k)} = \Omega(N(Lh)^d \varepsilon_k^{-(d+2)})$.

Balancing $N\varepsilon_k T_k$ and $N(Lh)^d \varepsilon_k^{-(d+2)}$ yields $\varepsilon_k \asymp T_k^{-1/(d+2)}$ and $\varepsilon_k^{-(d+2)} \asymp T_k$, hence $\sum_k R_{\text{ID}}^{(k)} = \Omega(N(Lh)^d \sum_k T_k) = \Omega(NT)$ (absorbing $(Lh)^d$ into the constant if h is fixed). $\square$

Remark (data reuse). Proposition 11 targets restart epochic identification. If Phase II aggregates probe data across epochs (our default), the incremental work per epoch is smaller; however, without further structure (next section) the cumulative identification overhead remains too large to be dominated by the Phase III learning term in the worst case.

A.6.4 F.4. Conditional epochic recovery under a public coverage/scheduling property

We now state a proof of the rate in the main text. Given the negative result in the previous subsection, we require an additional condition (public coverage/scheduling) for Phase II that is orthogonal to the reward statistics and can be viewed as a systems-level assumption. We want to emphasize that this was not assumed in Appendices A-C (which used uniform randomization over active cells and probes with collision censorship).

Assumption 1 (Public stratified coverage for Phase II). *In any epoch with probe spacing η and per-cell probe count $P(\eta)$, there exists a public, communication-free schedule with the following properties:*

- (1) *For each active probe (C, z) , there is a publicly known set $\mathcal{R}(C, z)$ of rounds with $|\mathcal{R}(C, z)| = \Theta(P(\eta))$ per $P(\eta)$ -length block, and in each $t \in \mathcal{R}(C, z)$ exactly one player samples z in C and no other player samples any point in C ; i.e., the per-round success probability for (C, z) is $\Omega(1)$ on its scheduled rounds.*
- (2) *Across the epoch, the schedule assigns $O(1)$ such (C, z) per round per player (constant load), enabling parallelism without collisions.*

Near-optimality (zooming) dimension. We use the standard instance-complexity notion: there exist $d^* \in [0, d]$ and $C > 0$ such that $\mathcal{X}_\varepsilon := \{x : \mu^* - \mu(x) \leq \varepsilon\}$ can be covered by at

most $C\varepsilon^{-d^*}$ Euclidean balls of radius $\Theta(\varepsilon/L)$ (see Kleinberg et al. [36]).

Lemma 17 (Active-cell count under d^*). *If $\mathcal{X}_\varepsilon$ admits a cover by $O(\varepsilon^{-d^*})$ balls of radius $c\varepsilon/L$ with $c\varepsilon/L \leq h/4$, then the number of partition cells that intersect $\mathcal{X}_\varepsilon$ satisfies $M_{\text{act}}(\varepsilon) = O(\varepsilon^{-d^*})$. For the finitely many coarser ε , the bound holds up to a constant factor absorbed in $\tilde{O}(\cdot)$.*

Proof. Each ball of radius $c\varepsilon/L \leq h/4$ intersects $O(1)$ cells; there are $O(\varepsilon^{-d^*})$ balls. $\square$

Proposition 12 (Per-epoch identification under Assumption 1 and d^*). *Fix an epoch with target width ε_k . Under Assumption 1 and near-optimality dimension d^* , the number of successful probe samples is $\tilde{O}(\varepsilon_k^{-(d^*+2)})$; since the per-probe success probability on scheduled rounds is $\Omega(1)$, the number of rounds is also $\tilde{O}(\varepsilon_k^{-(d^*+2)})$ (up to logarithmic factors).*

Proof. By Lemma 17, the number of probes is $O(\varepsilon_k^{-d^*})$.

Each probe needs $b = \Theta(\varepsilon_k^{-2})$ successful samples to reach noise radius $O(\varepsilon_k)$ by the anytime concentration used in Appendix B.

Assumption 1(a)–(b) guarantees $\Omega(1)$ success probability for each probe on its scheduled rounds and constant load per round, hence the round budget is $\tilde{O}(\varepsilon_k^{-d^*} \cdot \varepsilon_k^{-2})$. $\square$

Theorem 25 (Conditional epochic, gap-free recovery). *Assume $d^* \leq d - 1$ and Assumption 1. Run epochs of lengths $T_k = 2^k$ with precisions $\varepsilon_k \propto 2^{-k/(d+2)}$, reuse all data across epochs, and use public dither (Lemma 16). Then*

$$\sum_k \mathbb{E}[R_{\text{ID}}^{(k)}] = \tilde{O}\left(N T^{\frac{d^*+2}{d+2}}\right),$$

which is dominated by the Phase III learning term $\tilde{O}(N(Lh)^{\frac{d}{d+2}} T^{\frac{d+1}{d+2}})$ when $d^ \leq d - 1$. Consequently, the main-text epochic gap-free corollary holds under Assumption 1.*

Proof. By Proposition 12, per-epoch identification rounds are $\tilde{O}(\varepsilon_k^{-(d^*+2)})$, hence the identification regret is N times this.

With $\varepsilon_k \propto 2^{-k/(d+2)}$, summing a geometric series over $K_{\text{ep}} = \Theta(\log T)$ epochs yields $\tilde{O}(N T^{\frac{d^*+2}{d+2}})$.

Since $d^* \leq d - 1$, we have $\frac{d^*+2}{d+2} \leq \frac{d+1}{d+2}$, so identification is dominated by Phase III learning (Appendix D).

Add the $O(N^2)$ seating constant and the $\delta_{\text{sys}}T$ failure contribution (Appendix E).

Reusing probe data across epochs adds at most a $\log K_{\text{ep}}$ factor in the radii, absorbed by $\tilde{O}(\cdot)$.

□

A.7 Appendix G: Distance-Threshold Collisions via Packing Reduction

This appendix provides a reduction from the distance-threshold collision model to the partition-style analysis used in the main text. The regret comparator in Theorem 4 is the packing-based benchmark $\text{OPT}_{\text{pack}}(r, \sigma, N)$; without additional geometric covering assumptions, we do not claim a uniform comparison to the best ρ -separated assignment.

A.7.1 G.1. Setup and notation

Fix a collision threshold $\rho > 0$. Let $Z = \{z_1, \dots, z_M\} \subset \mathcal{X}$ be an r -packing, i.e., $\|z_i - z_{i'}\|_2 \geq r$ for all $i \neq i'$, with some $r > \rho$. We assume

$$M = |Z| \geq N, \tag{A.22}$$

which is necessary both for the comparator $\text{OPT}_{\text{pack}}(r, \sigma, N) = \sum_{m=1}^N \nu_{(m)}^*$ and for seating N players on distinct balls.

Fix a radius σ with

$$0 < \sigma < \frac{r - \rho}{2}. \tag{A.23}$$

Define the safe balls $B_i := \{x \in \mathcal{X} : \|x - z_i\|_2 \leq \sigma\}$ (balls are implicitly clipped to $\mathcal{X}$). For a Lipschitz mean μ , let $\nu_i^* := \sup_{x \in B_i} \mu(x)$ and let $\nu_{(1)}^* \geq \dots \geq \nu_{(M)}^*$ be the sorted values. The packing-based one-round benchmark is

$$\text{OPT}_{\text{pack}}(r, \sigma, N) := \sum_{m=1}^N \nu_{(m)}^*.$$

Lemma 18 (Collision safety across safe balls). *If σ satisfies (A.23), then for any $i \neq i'$ and any $x \in B_i$, $x' \in B_{i'}$, we have $\|x - x'\|_2 > \rho$. Hence inter-ball collisions are impossible.*

Proof. Triangle inequality and r -packing: $\|x - x'\|_2 \geq \|z_i - z_{i'}\|_2 - \|x - z_i\|_2 - \|x' - z_{i'}\|_2 \geq r - 2\sigma > \rho$. $\square$

A.7.2 G.2. Reduction to the partition model

We treat the index family $\{B_1, \dots, B_M\}$ as a “virtual partition.” The multi-phase protocol (Phases I–III and II $_{\frac{1}{2}}$) is run verbatim with the following substitutions:

- **Phase I (coarse identification).** Each round, a player samples a ball index $I \in [M]$ uniformly and probes its center z_I (or any fixed representative in B_I). The per-round success probability for a given (player, ball) is

$$p_M := \frac{1}{M} \left(1 - \frac{1}{M}\right)^{N-1},$$

identical to the cell-based p_K with K replaced by M . The coarse brackets mirror Appendix A with $K \mapsto M$ and with the ball geometry:

$$\text{LCB}_j^{(0)}(B_i) = \hat{\mu}_j^{(0)}(z_i) - r_j^{(0)}(B_i), \quad \text{UCB}_j^{(0)}(B_i) = \hat{\mu}_j^{(0)}(z_i) + r_j^{(0)}(B_i) + L\sigma,$$

since $\mu^*(B_i) \leq \mu(z_i) + L \max_{x \in B_i} \|x - z_i\|_2 = \mu(z_i) + L\sigma$.

- **Phase II (local peek).** For each active ball B_i , instantiate an internal η -net $Z_\eta(B_i)$; standard volumetric arguments give $|Z_\eta(B_i)| \leq C_d(\sigma/\eta)^d$. All Phase-II lemmas (coverage, anytime accuracy, refined brackets) from Appendix B carry through with the replacements $K \mapsto M$, $h \mapsto \sigma$, $D_h \mapsto D_\sigma := 2\sigma$.
- **Phase II $_{\frac{1}{2}}$ (seating).** Musical Chairs is unchanged. By Lemma 18, once one player is seated per ball, inter-ball collisions cannot occur.
- **Phase III (within-ball optimization).** With players uniquely assigned to balls, the processes decouple. Rescaling a ball of radius σ to the unit ball multiplies the Lipschitz

constant by σ ; by Appendix D, the in-ball regret over T' rounds satisfies

$$\mathbb{E}\left[R_{\text{in}}^{(B_i)}(T')\right] \leq c_d (L\sigma)^{\frac{d}{d+2}} (T')^{\frac{d+1}{d+2}} + c'_d.$$

A.7.3 G.3. Regret bound (proof of Theorem 4)

Combining Phase I/II identification costs, the seating cost (Appendix C), and the sum of in-ball regrets (Appendix D) exactly as in Appendix E yields

$$\mathbb{E}[R_{\text{cont}}(T)] \leq \underbrace{N(T_0 + T_1)}_{\text{identification}} + \underbrace{c_{MC}N^2}_{\text{seating}} + \underbrace{c_d N (L\sigma)^{\frac{d}{d+2}} T^{\frac{d+1}{d+2}}}_{\text{learning}} + \delta_{\text{sys}}T,$$

where $T_1 = \tilde{O}(M (L\sigma)^d \varepsilon^{-(d+2)})$ (Appendix B.6 with $K \mapsto M$ and $h \mapsto \sigma$). This is identical in form to Theorem 3 after the substitutions $(K, h) \mapsto (M, \sigma)$.

Remark 3 (Comparator choice). *We compare to $\text{OPT}_{\text{pack}}(r, \sigma, N)$ by design. Uniformly relating the true ρ -separated optimum to $\text{OPT}_{\text{pack}}(r, \sigma, N)$ would require a covering guarantee in addition to packing; without it, the two comparators need not be close. Even with coverage radius $O(r)$, the model-mismatch term per round can be $\Omega(NLr)$, which is not negligible for fixed r .*

A.8 Appendix H: Minimax Lower Bound

We prove Theorem 5. The ingredients are: (i) a standard $\Omega(\sqrt{K\tau})$ minimax lower bound for finite-armed stochastic bandits, and (ii) an L -Lipschitz embedding of $K' = \Theta(m^d)$ “arms” within each of N distinct cells, with a cone-shaped spike at one grid point.

A.8.1 H.1. Finite-armed minimax lower bound (cited)

For every $K \geq 2$ and $\tau \geq K$, there exists a universal constant $c > 0$ such that

$$\inf_{\text{Alg}} \sup_{\nu} \mathbb{E}_{\nu} \left[\sum_{t=1}^{\tau} (\mu^* - \mu(A_t)) \right] \geq c \sqrt{K\tau}, \quad (\text{A.24})$$

where the supremum is over all product reward distributions with means in $[0, 1]$. See, e.g., Slivkins [69].

A.8.2 H.2. Proof of Theorem 5

Partition $[0, 1]^d$ into $K = \lceil 1/h \rceil^d$ hypercubes $\mathcal{P} = \{C_1, \dots, C_K\}$ of side h . Assume

$$K = \lceil 1/h \rceil^d \geq N,$$

and select N distinct cells $\{C_{i_1}, \dots, C_{i_N}\}$. Within each chosen cell C_{i_j} , place a regular grid G of resolution m per coordinate; neighboring grid points are at Euclidean distance h/m . Let $K' = (m+1)^d$ be the number of grid points in G .

For $\theta = (\theta_1, \dots, \theta_N)$ with $\theta_j \in \{1, \dots, K'\}$, define a Lipschitz mean μ_θ as follows. In each chosen cell C_{i_j} , let $g_{j,\theta_j} \in G$ be the selected “spike” grid point and set

$$\mu_\theta(x) := \max\left\{\frac{1}{2}, \frac{1}{2} + \Delta - L \|x - g_{j,\theta_j}\|_2\right\}, \quad x \in C_{i_j},$$

and set $\mu_\theta(x) \equiv \frac{1}{2}$ for x in all other cells. Since $x \mapsto \|x - g\|_2$ is 1-Lipschitz and $\max(\cdot, \cdot)$ preserves Lipschitz constant, we have $\mu_\theta \in \mathcal{L}(L)$. At the spike g_{j,θ_j} , the value is $1/2 + \Delta$; at any other grid point $g \neq g_{j,\theta_j}$, the value is at most $1/2 + \Delta - L(h/m)$, hence the spike arm is uniquely optimal.

Let $\text{OPT}_{\text{cont}}(\mathcal{P}, N)$ be the per-round partition benchmark. In our construction, exactly the N chosen cells attain maximum $1/2 + \Delta$ and all others have maximum $1/2$, so

$$\text{OPT}_{\text{cont}}(\mathcal{P}, N) = \sum_{j=1}^N \left(\frac{1}{2} + \Delta\right) = N \left(\frac{1}{2} + \Delta\right).$$

Consider the product prior over θ in which each θ_j is independent and uniform over the K' grid points in C_{i_j} . By Yao’s minimax principle, the minimax expected regret is bounded below by the Bayes expected regret under this prior. Rewards and choices decouple across the N special cells, and the per-round benchmark is additive across cells; therefore the Bayes expected regret equals the sum of the N per-cell Bayes regrets. Each per-cell problem is a K' -armed stochastic bandit with rewards in $[0, 1]$ and horizon T , so by (A.24) the per-cell Bayes (hence minimax) regret is $\Omega(\sqrt{K'T})$. Summing over cells,

$$\mathbb{E}_{\mu_\theta}[R_{\text{cont}}(T)] \geq cN \sqrt{K'T} \asymp cN m^{d/2} \sqrt{T}.$$

Finally, choose parameters to ensure bounded rewards (take $\Delta \leq 1/6$) and to match the worst-case finite-armed scaling under the L -Lipschitz constraint. Set

$$\Delta = c_0 \frac{Lh}{m} \quad \text{with} \quad c_0 \in (0, 1/2],$$

so that the spike height is consistent with the grid spacing and the gap at the nearest other grid point is at least $\Delta/2$. Optimizing the lower bound in m (subject to $m \geq 2$ and $\Delta \leq 1/6$) yields

$$m \asymp (Lh)^{\frac{2}{d+2}} T^{\frac{1}{d+2}},$$

and hence

$$\mathbb{E}_{\mu_\theta}[R_{\text{cont}}(T)] \gtrsim N (Lh)^{\frac{d}{d+2}} T^{\frac{d+1}{d+2}}.$$

Taking the supremum over θ concludes the proof of Theorem 5. □

Remarks. (i) Disjointness of cells suffices; “non-adjacent” is inessential since collisions are intra-cell only. (ii) The same construction applies to the distance-threshold model by embedding spikes in N safe balls of a packing (Appendix G); at most one non-colliding observation per ball per round is possible, and the N -fold lower bound follows identically.

Appendix B

Supplementary Material: A Unified Framework for Locality in Scalable MARL

In this chapter, we provide supplementary material corresponding to Chapter 3.

B.1 Related work

Our work addresses the curse of dimensionality in multi-agent reinforcement learning (MARL). The challenge is that the global state and action spaces ($\mathcal{S} = \prod_i \mathcal{S}_i$, $\mathcal{A} = \prod_i \mathcal{A}_i$) grow exponentially with the number of agents, n . This problem falls into the category of “succinctly described” MDPs [5], which are known to be computationally intractable in the general case, even with structural assumptions [58].

To achieve scalability, researchers have explored various structural assumptions. One common approach in general MARL is to use function approximation [78, 46] or assume independent learners [71], though the latter can suffer from non-stationarity [50]. Other related areas include *Factored MDPs* [35, 23], which assume local states but typically a global action, and *Weakly Coupled MDPs* [51], which assume agents’ transitions are independent and coupling only occurs through the reward. Our work differs from these by focusing on systems with local, coupled transitions, which is common in networked systems.

This chapter is most related to the line of work on scalable Networked MARL [60, 62, 43]. These foundational papers were the first to show that if the system’s local transitions and rewards depend only on a local graph neighborhood, the system exhibits an Exponential Decay Property (EDP). This property, where influence decays exponentially with graph distance, justifies κ -hop

truncation and enables scalable algorithms.

A critical distinction lies in how this EDP is proven. In the challenging average-reward ($\gamma = 1$) setting, these works establish the EDP by proving a Dobrushin-style weak coupling condition*[60]. This condition is policy-independent, it relies on a worst-case (action-supremum) bound on the environment's influence, requiring $\|C\|_\infty < 1$.

This chapter builds directly on that foundation, but identifies that a policy-independent bound is inherently conservative. It can fail to certify locality in systems where the environment's action-based coupling (E^a) is strong, even if the policy itself is smooth and breaks this coupling.

Our central contribution is a more general and fundamental characterization of locality.

B.2 Proofs from Section 3.3

This appendix contains complete proofs of the results in the main text and an extension to asynchronous updates. All total variation distances are normalized: for probability measures μ and ν on a finite set, $\text{TV}(\mu, \nu) = \frac{1}{2} \sum_x |\mu(x) - \nu(x)|$. For a bounded function f on a product space, the coordinatewise oscillations are $\delta_i(f) = \sup_{x, y \in \mathcal{S}, x_{-i} = y_{-i}} |f(x) - f(y)|$ and $\delta(f)$ is the vector with entries $\delta_i(f)$.

B.2.1 Proof of Proposition 4

Fix i, j and two states s, s' that agree off coordinate i . Write

$$\mu_j = \sum_{a \in \mathcal{A}} \left(\prod_k \pi_k(a_k \mid s) \right) P_j(\cdot \mid s, a), \quad \nu_j = \sum_{a \in \mathcal{A}} \left(\prod_k \pi_k(a_k \mid s') \right) P_j(\cdot \mid s', a).$$

We claim

$$\text{TV}(\mu_j, \nu_j) \leq E_{j \leftarrow i}^s + \sum_k E_{j \leftarrow k}^a \Pi_{k \leftarrow i}(\pi).$$

The decomposition uses the triangle inequality and a coupling argument.

Define an intermediate distribution

$$\tilde{\mu}_j = \sum_{a \in \mathcal{A}} \left(\prod_k \pi_k(a_k \mid s') \right) P_j(\cdot \mid s, a).$$

By the triangle inequality,

$$\text{TV}(\mu_j, \nu_j) \leq \text{TV}(\mu_j, \tilde{\mu}_j) + \text{TV}(\tilde{\mu}_j, \nu_j).$$

We bound $\text{TV}(\tilde{\mu}_j, \nu_j)$ first. The measures $\tilde{\mu}_j$ and ν_j are mixtures with the same mixing weights $w(a) = \prod_k \pi_k(a_k \mid s')$. By convexity of total variation,

$$\text{TV}(\tilde{\mu}_j, \nu_j) \leq \sum_{a \in \mathcal{A}} w(a) \text{TV}(P_j(\cdot \mid s, a), P_j(\cdot \mid s', a)) \leq \sup_{a \in \mathcal{A}} \text{TV}(P_j(\cdot \mid s, a), P_j(\cdot \mid s', a)) = E_{j \leftarrow i}^s.$$

We now bound $\text{TV}(\mu_j, \tilde{\mu}_j)$. This is the difference between two mixtures with the same components and different mixing weights $p(a \mid s) = \prod_k \pi_k(a_k \mid s)$ and $p(a \mid s') = \prod_k \pi_k(a_k \mid s')$, but the components $P_j(\cdot \mid s, a)$ can be far apart for different a . We upper bound this difference by a coupling that changes actions along one coordinate at a time and uses the action sensitivity of the kernel.

Let (A, A') be a coupling of $p(\cdot \mid s)$ and $p(\cdot \mid s')$ constructed as follows. For each agent k , couple the marginals $\pi_k(\cdot \mid s)$ and $\pi_k(\cdot \mid s')$ by a maximal coupling so that

$$\mathbb{P}[A_k \neq A'_k] = \text{TV}(\pi_k(\cdot \mid s), \pi_k(\cdot \mid s')) = \Pi_{k \leftarrow i}(\pi).$$

Take these couplings independent across k , which is possible because the policy is of product form and the coordinates are independent under each product marginal.

Conditional on (A, A') , couple next-state coordinates for agent j by a maximal coupling of $P_j(\cdot \mid s, A)$ and $P_j(\cdot \mid s, A')$. Then

$$\mathbb{P}[S'^{(j)} \neq \tilde{S}'^{(j)} \mid A, A'] = \text{TV}(P_j(\cdot \mid s, A), P_j(\cdot \mid s, A')).$$

Taking expectation,

$$\text{TV}(\mu_j, \tilde{\mu}_j) \leq \mathbb{E} \text{TV}(P_j(\cdot \mid s, A), P_j(\cdot \mid s, A')).$$

For any a, a' differing on a set $D \subseteq [n]$ of action coordinates, the triangle inequality and the definition of E^a imply

$$\text{TV}(P_j(\cdot \mid s, a), P_j(\cdot \mid s, a')) \leq \sum_{k \in D} E_{j \leftarrow k}^a.$$

Applying this bound with the random pair (A, A') and taking expectations yields

$$\text{TV}(\mu_j, \tilde{\mu}_j) \leq \sum_{k=1}^n E_{j \leftarrow k}^a \mathbb{P}[A_k \neq A'_k] = \sum_{k=1}^n E_{j \leftarrow k}^a \Pi_{k \leftarrow i}(\pi).$$

Combining the two pieces finishes the proof:

$$\text{TV}(\mu_j, \nu_j) \leq E_{j \leftarrow i}^s + \sum_{k=1}^n E_{j \leftarrow k}^a \Pi_{k \leftarrow i}(\pi).$$

By the definition of $C_{j \leftarrow i}^\pi$ as a supremum over s, s' with $s_{-i} = s'_{-i}$, the same bound holds for $C_{j \leftarrow i}^\pi$.

B.2.2 Proof of Lemma 5

Fix i and two states s, s' with $s_{-i} = s'_{-i}$. We will couple one synchronous step from s and from s' .

First couple the actions. For each $k \in [n]$, let (A_k, A'_k) be a maximal coupling of $\pi_k(\cdot \mid s)$ and $\pi_k(\cdot \mid s')$, chosen independently across k . Then $(A, A') = ((A_k)_k, (A'_k)_k)$ is a coupling of the product measures $\pi(\cdot \mid s)$ and $\pi(\cdot \mid s')$, and

$$\mathbb{P}[A_k \neq A'_k] = \text{TV}(\pi_k(\cdot \mid s), \pi_k(\cdot \mid s')) \leq \Pi(\pi)_{k \leftarrow i}.$$

Next, conditional on $(A, A') = (a, a')$, draw the next state coordinates independently across j : let $X_j \sim P_j(\cdot \mid s, a)$ and $Y_j \sim P_j(\cdot \mid s', a')$, and couple (X_j, Y_j) by a maximal coupling of these two marginals. By construction,

$$\mathbb{P}[X_j \neq Y_j \mid A = a, A' = a'] = \text{TV}(P_j(\cdot \mid s, a), P_j(\cdot \mid s', a')).$$

Taking expectation over (A, A') gives

$$\mathbb{P}[X_j \neq Y_j] = \mathbb{E}_{A, A'} [\text{TV}(P_j(\cdot \mid s, A), P_j(\cdot \mid s', A'))].$$

For any a, a' , apply the triangle inequality:

$$\text{TV}(P_j(\cdot \mid s, a), P_j(\cdot \mid s', a')) \leq \underbrace{\text{TV}(P_j(\cdot \mid s, a), P_j(\cdot \mid s, a'))}_{\text{action change at fixed } s} + \underbrace{\text{TV}(P_j(\cdot \mid s, a'), P_j(\cdot \mid s', a'))}_{\text{state change at fixed } a'}.$$

By the definitions of E^a and E^s ,

$$\mathrm{TV}(P_j(\cdot \mid s, a), P_j(\cdot \mid s, a')) \leq \sum_{k: a_k \neq a'_k} E_{j \leftarrow k}^a, \quad \mathrm{TV}(P_j(\cdot \mid s, a'), P_j(\cdot \mid s', a')) \leq E_{j \leftarrow i}^s.$$

Therefore,

$$\mathbb{P}[X_j \neq Y_j] \leq E_{j \leftarrow i}^s + \sum_{k=1}^n E_{j \leftarrow k}^a \mathbb{P}[A_k \neq A'_k] \leq E_{j \leftarrow i}^s + \sum_{k=1}^n E_{j \leftarrow k}^a \Pi(\pi)_{k \leftarrow i}.$$

For any bounded f on the product space, $|f(X) - f(Y)| \leq \sum_{j=1}^n \delta_j(f) \mathbf{1}\{X_j \neq Y_j\}$. Taking expectations,

$$|(T^\pi f)(s) - (T^\pi f)(s')| \leq \sum_{j=1}^n \delta_j(f) \mathbb{P}[X_j \neq Y_j] \leq \sum_{j=1}^n \delta_j(f) \left(E_{j \leftarrow i}^s + \sum_{k=1}^n E_{j \leftarrow k}^a \Pi(\pi)_{k \leftarrow i} \right).$$

Taking the supremum over all s, s' that agree off i yields $\delta_i(T^\pi f) \leq \sum_j H_{j \leftarrow i}^\pi \delta_j(f)$, and the vector form follows.

The multi-step bound follow by induction.

B.2.3 Proof of Theorem 6

By Lemma 5 and Proposition 4, for each π and bounded f ,

$$\delta(T^\pi f) \leq (C^\pi)^\top \delta(f) \leq (E^s + E^a \Pi(\pi))^\top \delta(f).$$

Iterating yields

$$\delta((T^\pi)^t f) \leq \left((E^s + E^a \Pi(\pi))^\top \right)^t \delta(f).$$

Since $\rho(E^s + E^a \Pi(\pi)) \leq \lambda < 1$, for every $\varepsilon \in (0, 1 - \lambda)$ there exists a constant $C_\varepsilon < \infty$, independent of t and π , such that

$$\left\| \left((E^s + E^a \Pi(\pi))^\top \right)^t \right\|_{\infty \rightarrow \infty} \leq C_\varepsilon (\lambda + \varepsilon)^t \quad \text{for all } t \geq 0,$$

and hence $\|\delta((T^\pi)^t f)\|_\infty \leq C_\varepsilon (\lambda + \varepsilon)^t \|\delta(f)\|_\infty$.

If $\rho(E^s + E^a \Pi(\pi)) \leq \lambda < 1$, then by standard bounds for nonnegative matrices there exists $C \geq 1$ independent of t such that $\|((E^s + E^a \Pi(\pi))^\top)^t\|_{\infty \rightarrow \infty} \leq C \lambda^t$, which implies the contraction statement.

For the average-reward Poisson equation, fix π and let $g^\pi = r^\pi - \bar{r}^\pi$ where $\bar{r}^\pi$ is any constant. Consider the series $h^\pi = \sum_{t=0}^{\infty} (T^\pi)^t g^\pi$. The one-step oscillation bound and the spectral radius assumption imply absolute convergence in the oscillation seminorm:

$$\|\delta(h^\pi)\|_\infty \leq \sum_{t=0}^{\infty} \left\| \left((E^s + E^a \Pi(\pi))^\top \right)^t \delta(g^\pi) \right\|_\infty \leq \left\| (I - (E^s + E^a \Pi(\pi))^\top)^{-1} \right\|_{\infty \rightarrow \infty} \|\delta(g^\pi)\|_\infty.$$

Termwise application of $I - T^\pi$ shows $h^\pi - T^\pi h^\pi = g^\pi$.

For uniqueness up to an additive constant, suppose $h - T^\pi h = 0$. Then $\delta(h) \leq (H^\pi)^\top \delta(h)$ and iterating yields $\delta(h) \leq ((H^\pi)^\top)^t \delta(h)$ for all t . Since $((H^\pi)^\top)^t \rightarrow 0$ in operator norm, $\delta(h) = 0$, so h is constant.

This concludes the proof.

B.2.4 Spatial decay as a corollary of sparsity

Suppose that there is an undirected graph on $[n]$ with maximum degree Δ such that $E_{j \leftarrow i}^s = 0$ and $E_{j \leftarrow k}^a = 0$ whenever j and i are nonneighbors and j and k are nonneighbors. Suppose also that $\Pi(\pi)_{k \leftarrow i} = 0$ whenever k and i are nonneighbors. Then the matrix $H^\pi = (E^s + E^a \Pi(\pi))^\top$ has the same sparsity pattern as the graph. For any two sets of coordinates at graph distance at least d , the corresponding block of $(H^\pi)^t$ vanishes for $t < d$. If $\rho(H^\pi) \leq \lambda < 1$, then $\|(H^\pi)^t\|_{\infty \rightarrow \infty} \leq C(\Delta) \lambda^t$ and the influence of a perturbation decays exponentially in graph distance. This recovers the locality phenomenon used by synchronous truncated algorithms as a corollary of the geometry-free spectral condition.

B.2.5 Average-reward localized evaluation and improvement (synchronous)

We record a consequence for synchronous localized policy evaluation and policy iteration. Let $\text{Ball}_\kappa(i)$ denote the set of coordinates within distance κ of i in a given sparsity graph. Suppose the sparsity conditions in the previous subsection hold and $\rho(E^s + E^a \Pi(\pi)) \leq \lambda < 1$ uniformly over π in a policy class. Then there is a synchronous policy evaluation scheme that approximates the average-reward Poisson solution h^π using only local computations on κ -neighborhoods with

bias bounded by a constant times λ^κ . A synchronous localized policy iteration that uses these local evaluations to perform local improvements enjoys a performance loss bounded by the same exponential rate, under standard boundedness and irreducibility assumptions. The details follow from the resolvent bound for $\delta(h^\pi)$ and the sparsity argument above, and mirror standard analyses of truncated dynamic programming under Dobrushin-type decay.

B.2.6 Asynchronous updates

For completeness we state the analogue of the main results when, at each step, all agents keep their states except for a randomly selected coordinate J_t which is updated according to a site-selection distribution ν with full support.

In this model the one-step operator is $K^\pi f(s) = \mathbb{E}_\pi[f(S_{t+1}) \mid S_t = s]$ with

$$\mathbb{P}[S_{t+1} = s^{(j \rightarrow y)} \mid S_t = s] = \nu_j \sum_a \left(\prod_k \pi_k(a_k \mid s) \right) P_j(y \mid s, a) \quad \text{for } y \in \mathcal{S}_j.$$

Define E^s , E^a , and $\Pi(\pi)$ as before and set

$$M^\pi = (I - \text{diag } \nu) + \text{diag } \nu (E^s + E^a \Pi(\pi)).$$

Then for all bounded f , $\delta(K^\pi f) \leq (M^\pi)^\top \delta(f)$, and if $\rho(M^\pi) \leq \lambda < 1$ uniformly over the policy class then $\|\delta((K^\pi)^t f)\|_\infty \leq C\lambda^t \|\delta(f)\|_\infty$ and the average-reward Poisson equation has a solution with $\delta(h^\pi) \leq (I - (M^\pi)^\top)^{-1} \delta(r^\pi)$.

The proof is identical to the synchronous case, with the additional $(I - \text{diag } \nu)$ term accounting for coordinates that are not updated on a given step.

B.2.7 Extension of Theorem 6 to the weighted case

Since proofs for these are similar to the unweighted case, we skip rederiving in full detail.

Fix $w \in \mathbb{R}_{++}^n$ and $W = \text{diag}(w)$. For any bounded f and $i \in [n]$,

$$\delta_i^w(T^\pi f) = w_i \delta_i(T^\pi f) \leq w_i \sum_{j=1}^n H_{j \leftarrow i}^\pi \delta_j(f) = \sum_{j=1}^n (W^{-1} H^\pi W)_{j \leftarrow i} \delta_j^w(f).$$

This is the entrywise weighted one-step contraction. Iterating gives the multi-step bound. For the Poisson resolvent, one repeats the proof of Theorem 6 with the seminorm $\|f\|_{\delta^w} = \|\delta^w(f)\|_\infty$. The operator inequality $\delta^w(T^\pi f) \leq \tilde{H}^\pi \delta^w(f)$ implies that the Neumann series $\sum_{t=0}^\infty (\tilde{H}^\pi)^t$ converges entrywise when $\rho(\tilde{H}^\pi) < 1$ and that

$$\delta^w(h^\pi) \leq \sum_{t=0}^\infty (\tilde{H}^\pi)^t \delta^w(r^\pi) = (I - \tilde{H}^\pi)^{-1} \delta^w(r^\pi).$$

The truncated series bound and the uniform power bound over a compact policy class are identical to the unweighted case after replacing H^π by $\tilde{H}^\pi$.

B.3 Proofs from Section 3.4

We bound the policy sensitivity matrix $\Pi(\pi)$ for entropy-regularized (temperature- τ) softmax policies in terms of per-agent logit Lipschitz constants. Throughout, total variation is normalized: $\text{TV}(\mu, \nu) = \frac{1}{2} \|\mu - \nu\|_1$.

Lemma 19 (Softmax Lipschitz constant in total variation). *Fix $m \in \mathbb{N}$ and $\tau > 0$. For $u, v \in \mathbb{R}^m$, define*

$$\text{soft}_\tau(u)_a := \frac{\exp(u_a/\tau)}{\sum_{b=1}^m \exp(u_b/\tau)}, \quad a \in [m].$$

Then

$$\text{TV}(\text{soft}_\tau(u), \text{soft}_\tau(v)) \leq \frac{1}{2\tau} \|u - v\|_\infty.$$

Moreover, the constant $1/(2\tau)$ is sharp: for each τ and $m \geq 2$ there exist $u \neq v$ attaining equality.

Proof. Write $p = \text{soft}_\tau(u)$ and $q = \text{soft}_\tau(v)$. By the mean value theorem on the line segment $w(t) = v + t(u - v)$, $t \in [0, 1]$,

$$\|p - q\|_1 = \left\| \int_0^1 \frac{d}{dt} \text{soft}_\tau(w(t)) dt \right\|_1 \leq \int_0^1 \|D\text{soft}_\tau(w(t)) (u - v)\|_1 dt,$$

where $D\text{soft}_\tau(w)$ is the Jacobian. For $w \in \mathbb{R}^m$ with $r = \text{soft}_\tau(w)$,

$$D\text{soft}_\tau(w) = \frac{1}{\tau} (\text{diag}(r) - rr^\top).$$

Hence, using the induced operator norm from ℓ_∞ to ℓ_1 ,

$$\|p - q\|_1 \leq \frac{1}{\tau} \left(\sup_{t \in [0,1]} \|\text{diag}(r(t)) - r(t)r(t)^\top\|_{\infty \rightarrow 1} \right) \|u - v\|_\infty, \quad r(t) = \text{soft}_\tau(w(t)).$$

We claim that for every probability vector r ,

$$\|\text{diag}(r) - rr^\top\|_{\infty \rightarrow 1} \leq 1. \quad (\text{B.1})$$

This yields $\|p - q\|_1 \leq \frac{1}{\tau} \|u - v\|_\infty$ and therefore $\text{TV}(p, q) \leq \frac{1}{2\tau} \|u - v\|_\infty$.

It remains to prove (B.1). Let $J(r) = \text{diag}(r) - rr^\top$ and fix $\delta \in \mathbb{R}^m$ with $\|\delta\|_\infty \leq 1$. Then

$$(J(r)\delta)_i = r_i(\delta_i - \langle r, \delta \rangle), \quad i \in [m],$$

hence

$$\|J(r)\delta\|_1 = \sum_{i=1}^m r_i |\delta_i - \mu| = \mathbb{E}_{I \sim r} [|\delta_I - \mu|], \quad \mu := \langle r, \delta \rangle \in [-1, 1].$$

For fixed μ , the map $\delta \mapsto \mathbb{E}[|\delta_I - \mu|]$ is convex and is maximized over $\{\delta : \|\delta\|_\infty \leq 1\}$ at extreme points $\delta \in \{\pm 1\}^m$. Among $\{\pm 1\}$ -valued δ with mean μ , the expectation $\mathbb{E}[|\delta_I - \mu|]$ equals

$$\alpha(1 - \mu) + (1 - \alpha)(1 + \mu) = 1 - \mu^2, \quad \alpha = \frac{1+\mu}{2},$$

and the maximum over $\mu \in [-1, 1]$ is 1, attained at $\mu = 0$. Therefore $\|J(r)\delta\|_1 \leq 1$ for all δ with $\|\delta\|_\infty \leq 1$, proving (B.1). Sharpness follows by taking m even, r uniform on $[m]$, and δ with half of the entries +1 and half -1, which yields $\|J(r)\delta\|_1 = 1$ and $\mu = 0$. $\square$

We now translate Lemma 19 into a bound on the policy sensitivity matrix $\Pi(\pi)$ for product-form, temperature- τ softmax policies with local logits.

Definition 11 (Local logits and per-coordinate logit Lipschitz constants). *For each agent k , suppose there is a logit function $g_k : \mathcal{S}_{O_k} \times \mathcal{A}_k \rightarrow \mathbb{R}$ such that*

$$\pi_k(a_k \mid s_{O_k}) \propto \exp(g_k(s_{O_k}, a_k)/\tau), \quad a_k \in \mathcal{A}_k,$$

with temperature $\tau > 0$. For $i \in [n]$, define the one-coordinate logit Lipschitz constant

$$L_{k \leftarrow i} := \sup_{s_{-i} = s'_{-i}} \|g_k(s_{O_k}, \cdot) - g_k(s'_{O_k}, \cdot)\|_\infty,$$

where the sup is over $s, s' \in \mathcal{S}$ that differ only on coordinate i .

Lemma 20 (Softmax temperature controls $\Pi(\pi)$). *Under the setup above, for all $k, i \in [n]$,*

$$\Pi_{k \leftarrow i}(\pi) \leq \min \left\{ 1, \frac{L_{k \leftarrow i}}{2\tau} \right\}.$$

In particular, if $i \notin O_k$ then $L_{k \leftarrow i} = 0$ and $\Pi_{k \leftarrow i}(\pi) = 0$.

Proof. Fix k and i . If $i \notin O_k$ then $g_k(s_{O_k}, \cdot)$ is unchanged when s_i varies, hence $L_{k \leftarrow i} = 0$ and $\pi_k(\cdot \mid s_{O_k}) = \pi_k(\cdot \mid s'_{O_k})$ for all $s_{-i} = s'_{-i}$, giving $\Pi_{k \leftarrow i}(\pi) = 0$.

Assume $i \in O_k$. For s, s' with $s_{-i} = s'_{-i}$, apply Lemma 19 with $u = g_k(s_{O_k}, \cdot)$ and $v = g_k(s'_{O_k}, \cdot)$ to obtain

$$\text{TV}(\pi_k(\cdot \mid s_{O_k}), \pi_k(\cdot \mid s'_{O_k})) \leq \frac{1}{2\tau} \|g_k(s_{O_k}, \cdot) - g_k(s'_{O_k}, \cdot)\|_\infty \leq \frac{L_{k \leftarrow i}}{2\tau}.$$

Taking the supremum over such s, s' yields $\Pi_{k \leftarrow i}(\pi) \leq L_{k \leftarrow i}/(2\tau)$. The bound is trivially capped by 1 because total variation lies in $[0, 1]$. $\square$

Remark 4 (Sharpness). *The constant $1/(2\tau)$ in Lemma 20 is sharp, even for two actions with uniform logits up to a perturbation on one coordinate. Thus, controlling $\Pi(\pi)$ uniformly over a policy class can be enforced by lower-bounding the entropy temperature τ and upper-bounding the one-coordinate logit oscillations $L_{k \leftarrow i}$.*

B.4 Proofs from Section 3.5

All state and action spaces are finite. Total variation is normalized as $\text{TV}(\mu, \nu) = \frac{1}{2} \sum_x |\mu(x) - \nu(x)|$. Coordinatewise oscillations are $\delta_i(f) = \sup\{|f(x) - f(y)| : x_{-i} = y_{-i}\}$ and $\delta(f) = (\delta_i(f))_{i=1}^n$. We use Lemma 5 and Theorem 6 from the main text.

B.4.1 Proof of Theorem 7

Fix π , let $H^\pi = (E^s + E^a \Pi(\pi))^\top$ and $b^\pi = \delta(r^\pi)$. By Lemma 5, for any bounded f , $\delta((T^\pi)^t f) \leq (H^\pi)^t \delta(f)$. Taking $f = r^\pi - \bar{r}^\pi$ and summing yields

$$\delta(h^\pi) \leq \sum_{t=0}^{\infty} (H^\pi)^t b^\pi.$$

Define $U^{(\kappa)} = \sum_{t=0}^{\kappa} (H^\pi)^t b^\pi$. Then $\delta(\sum_{t=0}^{\kappa} (T^\pi)^t (r^\pi - \bar{r}^\pi)) \leq U^{(\kappa)}$, which proves the certificate for any choice of constant c_κ .

If G^π is the support graph of H^π , then $[(H^\pi)^t b^\pi]_i$ depends only on entries of b^π within graph distance $\leq t$ of i , hence $U_i^{(\kappa)}$ depends only on $\text{Ball}_\kappa(i)$, establishing locality and the message-passing implementation.

For the bias, suppose $\rho(H^\pi) \leq \lambda < 1$. For every $\varepsilon \in (0, 1 - \lambda)$ there exists $C_\varepsilon < \infty$ such that $\|(H^\pi)^t\|_{\infty \rightarrow \infty} \leq C_\varepsilon (\lambda + \varepsilon)^t$ for all $t \geq 0$ (e.g., by Gelfand's formula and compactness of the policy class, or by a uniform weighted Dobrushin condition). Then

$$\|\delta(h^\pi - \hat{h}_\kappa^\pi)\|_\infty \leq \sum_{t=\kappa+1}^{\infty} \|(H^\pi)^t\|_{\infty \rightarrow \infty} \|b^\pi\|_\infty \leq C_\varepsilon \frac{(\lambda + \varepsilon)^{\kappa+1}}{1 - (\lambda + \varepsilon)} \|b^\pi\|_\infty.$$

Finally, for any function f on a product space, $\text{osc}(f) \leq \sum_{i=1}^n \delta_i(f) \leq n \|\delta(f)\|_\infty$ and $\inf_c \|f - c\|_\infty = \frac{1}{2} \text{osc}(f)$. Applying this to $f = h^\pi - \hat{h}_\kappa^\pi$ proves the stated $\|\cdot\|_\infty$ bound (after choosing c_κ arbitrarily, since advantages are constant-invariant).

B.4.2 Preliminaries for the improvement step

We assume that for any product policy μ considered, the Markov chain under P^μ is irreducible and aperiodic, so the stationary distribution d^μ exists and is unique. Let h^π solve the Poisson equation $h^\pi - T^\pi h^\pi = r^\pi - \bar{r}^\pi$ and define

$$A^\pi(s, a) = r(s, a) - \bar{r}^\pi + \sum_y P(y \mid s, a) h^\pi(y) - h^\pi(s).$$

Lemma 21 (Advantage perturbation via value perturbation). *Let $\hat{h}^\pi$ be any function on $\mathcal{S}$ and define the surrogate advantage $\hat{A}^\pi(s, a) = r(s, a) - \bar{r}^\pi + \sum_y P(y \mid s, a) \hat{h}^\pi(y) - \hat{h}^\pi(s)$. Then*

$$|\hat{A}^\pi(s, a) - A^\pi(s, a)| \leq 2 \|\hat{h}^\pi - h^\pi\|_\infty \quad \text{for all } (s, a).$$

Lemma 22 (Per-state KL-prox duality). *Fix a finite set $\mathcal{A}$, $q \in \Delta(\mathcal{A})$, $g \in \mathbb{R}^{\mathcal{A}}$, and $\tau > 0$. Then*

$$\max_{p \in \Delta(\mathcal{A})} \{ \langle p, g \rangle - \tau \text{KL}(p \parallel q) \} = \tau \log \sum_{a \in \mathcal{A}} q(a) e^{g(a)/\tau},$$

attained uniquely at $p^*(a) \propto q(a) e^{g(a)/\tau}$, and

$$\langle p^* - q, g \rangle \geq \tau \text{KL}(p^* \| q).$$

Proof. Standard convex duality for the log-sum-exp (Donsker–Varadhan). The inequality follows from $\tau \log \sum_a q(a) e^{g(a)/\tau} \geq \langle q, g \rangle$ (Jensen). $\square$

B.4.3 Update rule and guarantee

Fix $k \in [n]$ and construct $\hat{h}_\kappa^\pi$ as in Theorem 7. Define

$$g_k^\pi(s, \cdot) := \mathbb{E}_{a_{-k} \sim \prod_{j \neq k} \pi_j(\cdot | s_{O_j})} \hat{A}_\kappa^\pi(s, (\cdot, a_{-k})).$$

The block KL-prox update (3.1) sets

$$\pi_k^{\text{new}}(\cdot | s_{O_k}) \propto \pi_k(\cdot | s_{O_k}) \exp\left(\frac{1}{\tau} g_k^\pi(s, \cdot)\right), \quad \pi_{-k}^{\text{new}} = \pi_{-k}.$$

Let μ denote the resulting product policy.

Theorem 26 (One-block improvement up to truncation). *Assume irreducibility and aperiodicity.*

Then

$$\begin{aligned} \bar{r}_\tau(\mu) - \bar{r}_\tau(\pi) &\geq \tau \mathbb{E}_{S \sim d^\mu} \left[\text{KL}(\pi_k^{\text{new}}(\cdot | S_{O_k}) \| \pi_k(\cdot | S_{O_k})) \right] \\ &\quad - 2 \|\hat{h}_\kappa^\pi - h^\pi\|_\infty. \end{aligned}$$

In particular, if $\rho(H^\pi) \leq \lambda < 1$ and $\hat{h}_\kappa^\pi$ is the truncated Neumann surrogate from Theorem 7, then

$$\bar{r}_\tau(\mu) - \bar{r}_\tau(\pi) \geq \tau \mathbb{E}_{S \sim d^\mu} \text{KL}(\pi_k^{\text{new}}(\cdot | S_{O_k}) \| \pi_k(\cdot | S_{O_k})) - \frac{nC_\varepsilon}{1 - (\lambda + \varepsilon)} (\lambda + \varepsilon)^{\kappa+1} \|\delta(r^\pi)\|_\infty,$$

for any $\varepsilon \in (0, 1 - \lambda)$ and the constant C_ε from Theorem 7.

Proof. Write $\Delta := \bar{r}_\tau(\mu) - \bar{r}_\tau(\pi)$. Insert and subtract $\hat{A}_\kappa^\pi$:

$$\Delta = \mathbb{E}_{d^\mu, \mu}[A^\pi] = \mathbb{E}_{d^\mu, \mu}[\hat{A}_\kappa^\pi] + \mathbb{E}_{d^\mu, \mu}[A^\pi - \hat{A}_\kappa^\pi] \geq \mathbb{E}_{d^\mu, \mu}[\hat{A}_\kappa^\pi] - 2\|\hat{h}_\kappa^\pi - h^\pi\|_\infty,$$

by Lemma 21. Now expand the inner expectation over $a_k \sim \pi_k^{\text{new}}(\cdot | S_{O_k})$ and $a_{-k} \sim \pi_{-k}(\cdot | S_{O_{-k}})$ and apply Lemma 22 statewise to obtain

$$\mathbb{E}_{a_{-k} \sim \pi_{-k}} \left[\langle \pi_k^{\text{new}}(\cdot | S_{O_k}) - \pi_k(\cdot | S_{O_k}), g_k^\pi(S, \cdot) \rangle \right] \geq \tau \text{KL}(\pi_k^{\text{new}}(\cdot | S_{O_k}) \| \pi_k(\cdot | S_{O_k})).$$

Averaging over $S \sim d^\mu$ yields $\mathbb{E}_{d^\mu, \mu}[\hat{A}_\kappa^\pi] \geq \tau \mathbb{E}_{S \sim d^\mu} \text{KL}(\pi_k^{\text{new}} \| \pi_k)$, and the claim follows. $\square$

Corollary 10 (Cyclic block updates). *Consider a cyclic pass $k = 1, \dots, n$ where at step k we apply the KL-prox update with baseline $\pi_k^{(k-1)}$ and keep $\pi_{-k}^{(k)} = \pi_{-k}^{(k-1)}$. Let $\pi^{(0)} = \pi$ and $\pi^{(n)}$ be the policy after the pass. Then*

$$\bar{r}_\tau(\pi^{(n)}) - \bar{r}_\tau(\pi) \geq \tau \sum_{k=1}^n \mathbb{E}_{S \sim d^{\pi^{(k)}}} \text{KL}(\pi_k^{(k)}(\cdot | S_{O_k}) \parallel \pi_k^{(k-1)}(\cdot | S_{O_k})) - 2n \|\hat{h}_\kappa^\pi - h^\pi\|_\infty.$$

Remark 5 (Recomputing vs. stale certificates). *For the cleanest guarantees, recompute $\hat{h}_\kappa^\pi$ (and H^π) at each block's current baseline $\pi^{(k-1)}$. Using a single certificate computed at the start of the outer iteration is practical but introduces a staleness penalty proportional to the drift of the policy across the pass; we omit these routine details.*

B.5 Miscellaneous Results

B.5.1 Example: A Coupled System where Policy is Ineffective

Our framework also captures the failure mode, where policy smoothing is ineffective. Suppose the environment has strong **direct state-based coupling**, where one agent's state physically determines another's.

Consider two agents, a Leader (agent 1) and a Follower (agent 2). Suppose the Follower's next state s'_2 is **deterministically** set to be a copy of the Leader's **current** state s_1 , regardless of any actions taken.

$$P_2(s'_2 = 1 \mid s, a) = \mathbf{1}\{s_1 = 1\}.$$

This models a system with direct, physical state-to-state influence (e.g., s'_2 is a sensor reading of s_1). The environment's sensitivities are:

- $E_{2 \leftarrow 1}^s = \sup_{s_{-1}=s'_{-1}, a} \text{TV}(P_2(\cdot \mid s, a), P_2(\cdot \mid s', a)) = \text{TV}(\delta_0, \delta_1) = 1$.
- $E^a = 0$ (actions have no effect on s'_2).

We can now analyze the policy-induced influence of agent 1 on agent 2, $C_{2 \leftarrow 1}^\pi$:

$$C_{2 \leftarrow 1}^\pi \leq E_{2 \leftarrow 1}^s + \sum_{k=1}^2 E_{2 \leftarrow k}^a \Pi_{k \leftarrow 1}(\pi) = 1 + (0 \cdot \Pi_{1 \leftarrow 1} + 0 \cdot \Pi_{2 \leftarrow 1}) = 1.$$

In this case, the true influence $C_{2\leftarrow 1}^\pi$ is exactly 1, and our bound correctly reflects this. The policy sensitivity $\Pi(\pi)$ is rendered completely irrelevant because the action-coupling matrix E^a is zero.

Our framework correctly shows that no amount of policy smoothing (i.e., making $\Pi(\pi)$ small by increasing τ) can create locality when the environment's direct state-to-state physics (E^s) already creates a tight, off-diagonal coupling. The system's influence matrix H^π will have an off-diagonal entry of 1, making $\rho(H^\pi) \geq 1$ and breaking the condition for spatial decay.

B.5.2 Example: policy-induced locality in a nonlocal-looking system.

Consider a hub-and-spoke system with $n \geq 3$ agents.

Let P_1 be constant and, for $j > 1$, $P_j(s'_j = 1 \mid s, a) = \mathbf{1}\{a_1 = 1\}$. Then $E^s = 0$ and $E_{j\leftarrow 1}^a = 1$ for all $j > 1$ (others zero). A policy-independent, action-supremum certificate declares the system non-local ($\|C\|_\infty = 1$) [61].

Under a temperature- τ softmax for agent 1 with logit Lipschitz constants $\{L_{1\leftarrow i}\}$, Lemma 20 gives $\Pi_{1\leftarrow i} \leq L_{1\leftarrow i}/(2\tau)$ and hence

$$\rho(H^\pi) \leq \rho\left(E^a \frac{L}{2\tau}\right) = \frac{1}{2\tau} \sum_{j>1} L_{1\leftarrow j} \leq \frac{(n-1)L_{\max}}{2\tau}.$$

Thus locality is certified whenever $\tau > (n-1)L_{\max}/2$, and if the hub ignores its inputs ($L \equiv 0$) then $\rho(H^\pi) = 0$. This illustrates how locality can be **policy-induced**, while worst-case, policy-independent tests cannot detect it.

B.5.3 Properties of the Coordinatewise Oscillation Seminorm

Let $V = B(\mathcal{S})$ be the vector space of all bounded real-valued functions on the finite state space $\mathcal{S} = \prod_{i=1}^n \mathcal{S}_i$.

Definition 12 (Coordinatewise Oscillation). *For a function $f \in V$ and a coordinate $i \in [n]$, the i -oscillation is:*

$$\delta_i(f) = \sup \left\{ |f(x) - f(y)| : x, y \in \mathcal{S}, x_{-i} = y_{-i} \right\}.$$

We define the function $p : V \rightarrow \mathbb{R}$ as the maximum oscillation:

$$p(f) = \|\delta(f)\|_\infty = \max_{i \in [n]} \delta_i(f).$$

Proposition 13. *The function $p(f) = \|\delta(f)\|_\infty$ is a seminorm on the vector space V .*

Proof. To prove that $p(f)$ is a seminorm, we must verify three properties:

- (1) **Non-negativity:** $p(f) \geq 0$ for all $f \in V$.
- (2) **Absolute Homogeneity:** $p(cf) = |c|p(f)$ for all $f \in V$ and scalar $c \in \mathbb{R}$.
- (3) **Subadditivity (Triangle Inequality):** $p(f + g) \leq p(f) + p(g)$ for all $f, g \in V$.

1. Non-negativity: The absolute value $|f(x) - f(y)|$ is always non-negative. The supremum of a set of non-negative numbers, $\delta_i(f)$, is also non-negative. The maximum of a set of non-negative numbers, $p(f)$, is therefore non-negative.

2. Absolute Homogeneity: For any $f \in V$ and $c \in \mathbb{R}$:

$$\begin{aligned} \delta_i(cf) &= \sup_{x-i=y-i} |(cf)(x) - (cf)(y)| \\ &= \sup_{x-i=y-i} |c \cdot (f(x) - f(y))| \\ &= |c| \cdot \sup_{x-i=y-i} |f(x) - f(y)| = |c| \cdot \delta_i(f). \end{aligned}$$

Taking the maximum over all i :

$$p(cf) = \max_{i \in [n]} \delta_i(cf) = \max_{i \in [n]} (|c| \cdot \delta_i(f)) = |c| \cdot \max_{i \in [n]} \delta_i(f) = |c| \cdot p(f).$$

3. Subadditivity: For any $f, g \in V$:

$$\begin{aligned} \delta_i(f + g) &= \sup_{x-i=y-i} |(f + g)(x) - (f + g)(y)| \\ &= \sup_{x-i=y-i} |(f(x) - f(y)) + (g(x) - g(y))| \\ &\leq \sup_{x-i=y-i} (|f(x) - f(y)| + |g(x) - g(y)|) \quad (\text{by the triangle inequality for } \mathbb{R}) \\ &\leq \sup_{x-i=y-i} |f(x) - f(y)| + \sup_{x-i=y-i} |g(x) - g(y)| \quad (\text{by a standard property of suprema}) \\ &= \delta_i(f) + \delta_i(g). \end{aligned}$$

Now, taking the maximum over all i :

$$p(f + g) = \max_{i \in [n]} \delta_i(f + g) \leq \max_{i \in [n]} (\delta_i(f) + \delta_i(g)).$$

For any i , we know $\delta_i(f) \leq \max_j \delta_j(f) = p(f)$ and $\delta_i(g) \leq \max_j \delta_j(g) = p(g)$. Thus:

$$p(f + g) \leq \max_{i \in [n]} (p(f) + p(g)) = p(f) + p(g).$$

This completes the proof. □

Remark 6 (Why it is a seminorm, not a norm). *A norm requires $p(f) = 0 \iff f = 0$. For our $p(f)$, if $f(x) = c$ for some non-zero constant c , then $f \neq 0$. However, for any i and any pair x, y with $x_{-i} = y_{-i}$, $f(x) = c$ and $f(y) = c$, so $|f(x) - f(y)| = 0$. This implies $\delta_i(f) = 0$ for all i , and thus $p(f) = 0$. Since $p(f) = 0$ for non-zero constant functions, $p(f)$ is a seminorm. In fact, $p(f) = 0 \iff f$ is a constant function.*

Proposition 14. *The value $p(f) = \|\delta(f)\|_\infty$ is the (best) Lipschitz constant of f with respect to the Hamming distance $d_H(\cdot, \cdot)$ on $\mathcal{S}$.*

Proof. Let $K = p(f)$. We must show that for any $x, z \in \mathcal{S}$, $|f(x) - f(z)| \leq K \cdot d_H(x, z)$.

Case 1: $d_H(x, z) = 1$. If $d_H(x, z) = 1$, then x and z differ in exactly one coordinate, say j .

This means $x_{-j} = z_{-j}$. By the definition of $\delta_j(f)$:

$$|f(x) - f(z)| \leq \sup_{x'_{-j} = y'_{-j}} |f(x') - f(y')| = \delta_j(f).$$

By the definition of $p(f)$, $\delta_j(f) \leq \max_i \delta_i(f) = p(f) = K$. Therefore, $|f(x) - f(z)| \leq K = K \cdot d_H(x, z)$. This also shows that K is the best Lipschitz constant for pairs at distance 1.

Case 2: $d_H(x, z) = k > 1$. Let the coordinates where x and z differ be $I = \{i_1, \dots, i_k\}$. We can construct a path from x to z by changing one coordinate at a time. Let $z^{(0)} = x$, and let $z^{(t)}$ be the state obtained by changing the first t coordinates in I from their values in x to their values in z . For example, $z^{(1)}$ is identical to x except $z_{i_1}^{(1)} = z_{i_1}$. In general, $z^{(t)}$ and $z^{(t+1)}$ differ only in coordinate i_{t+1} . Thus, $d_H(z^{(t)}, z^{(t+1)}) = 1$. The full path is $x = z^{(0)}, z^{(1)}, \dots, z^{(k)} = z$.

Using the triangle inequality, we “telescope” the sum:

$$\begin{aligned}
 |f(x) - f(z)| &= |f(z^{(0)}) - f(z^{(k)})| \\
 &= \left| \sum_{t=0}^{k-1} (f(z^{(t)}) - f(z^{(t+1)})) \right| \\
 &\leq \sum_{t=0}^{k-1} |f(z^{(t)}) - f(z^{(t+1)})|.
 \end{aligned}$$

For each term in the sum, $z^{(t)}$ and $z^{(t+1)}$ differ in exactly one coordinate, so $d_H(z^{(t)}, z^{(t+1)}) = 1$.

From Case 1, we know:

$$|f(z^{(t)}) - f(z^{(t+1)})| \leq p(f) = K.$$

Substituting this into the sum:

$$|f(x) - f(z)| \leq \sum_{t=0}^{k-1} K = k \cdot K.$$

Since $k = d_H(x, z)$, we have shown:

$$|f(x) - f(z)| \leq K \cdot d_H(x, z).$$

This holds for all $x, z \in \mathcal{S}$, so $p(f)$ is the Lipschitz constant of f w.r.t. d_H .

Optimality. Let L be any constant such that $|f(x) - f(y)| \leq L$, $d_H(x, y)$ for all x, y . Fix i and x, y with $x_{-i} = y_{-i}$; then $d_H(x, y) = 1$, so $|f(x) - f(y)| \leq L$. Taking the supremum over such pairs gives $\delta_i(f) \leq L$, hence $p(f) = \max_i \delta_i(f) \leq L$. Therefore $p(f)$ is the least Lipschitz constant. □

Appendix C

Supplementary Material: The Role of Generator Access in Autoregressive Post-Training

In this chapter, we provide supplementary material corresponding to Chapter 4.

C.1 Proofs for Section 4.3

The vocabulary and horizon are finite, so all path spaces here are finite as well. The definitions are nonetheless phrased with measurable kernels to preserve the generality of the main-text statements.

C.1.1 The canonical no-reset experiment

Fix a prompt $x \in \mathcal{X}$. Let $(\mathcal{Q}, \mathcal{Q})$ be a standard Borel query space and $(\mathcal{Y}, \mathcal{Y})$ a standard Borel reply space.

Definition 13. *A generator experiment at prompt x is no-reset if there exists a measurable probability kernel*

$$\Gamma_x(q, y, \mu_1, \dots, \mu_H; \cdot)$$

from $\mathcal{Q} \times \Sigma^H \times \Delta(\Sigma)^H$ to $(\mathcal{Y}, \mathcal{Y})$ such that one query is answered by first drawing $Y \sim M(\cdot \mid x)$, then setting $\mu_t := M(\cdot \mid x, Y_{<t})$ for $t = 1, \dots, H$, and finally drawing the reply from $\Gamma_x(q, Y, \mu_1, \dots, \mu_H; \cdot)$.

Definition 14. *For the fixed prompt x , let $\mathcal{W} := \Sigma^H \times \Delta(\Sigma)^H$. The experiment PathFull_x ignores its query parameter, draws*

$$Y \sim M(\cdot \mid x), \quad \mu_t := M(\cdot \mid x, Y_{<t}) \text{ for } t = 1, \dots, H,$$

and returns

$$W_x := (Y, \mu_1, \dots, \mu_H) \in \mathcal{W}.$$

Definition 15. Let $\mathcal{E}$ be an experiment with query space $(\mathcal{Q}, \mathcal{Q})$ and reply space $(\mathcal{Z}, \mathcal{Z})$. A randomized post-processing of $\mathcal{E}$ is specified by a measurable kernel $\Pi(q, z; \cdot)$ from $\mathcal{Q} \times \mathcal{Z}$ to another reply space $(\mathcal{Y}, \mathcal{Y})$. On query q , the post-processed experiment first draws the reply Z of $\mathcal{E}$ and then draws its final reply from $\Pi(q, Z; \cdot)$.

Theorem 27. Fix a prompt x . An experiment at prompt x is no-reset if and only if it is a randomized post-processing of PathFull_x .

Proof. Assume first that the experiment $\mathcal{E}$ is no-reset. By Definition 13, there is a kernel

$$\Gamma_x(q, y, \mu_1, \dots, \mu_H; \cdot)$$

such that one query to $\mathcal{E}$ is answered by drawing

$$W_x = (Y, \mu_1, \dots, \mu_H) \sim \text{PathFull}_x$$

and then drawing the final reply from that kernel. If we define

$$\Pi(q, (y, \mu_1, \dots, \mu_H); \cdot) := \Gamma_x(q, y, \mu_1, \dots, \mu_H; \cdot),$$

then $\mathcal{E}$ is a randomized post-processing of PathFull_x .

Conversely, suppose $\mathcal{E}$ is a randomized post-processing of PathFull_x . Then there exists a kernel $\Pi(q, w; \cdot)$ such that one query to $\mathcal{E}$ is answered by drawing $W_x \sim \text{PathFull}_x$ and then sampling from $\Pi(q, W_x; \cdot)$. Writing $w = (y, \mu_1, \dots, \mu_H)$, this is exactly Definition 13 with

$$\Gamma_x(q, y, \mu_1, \dots, \mu_H; \cdot) := \Pi(q, (y, \mu_1, \dots, \mu_H); \cdot).$$

So $\mathcal{E}$ is no-reset. □

C.1.2 Agreement outside a prefix set

Fix a prefix set $U \subseteq \Sigma^{<H}$. We say that two prompt-conditioned generators $M(\cdot \mid x)$ and $M'(\cdot \mid x)$ agree outside U if

$$M(\cdot \mid x, p) = M'(\cdot \mid x, p) \quad \text{for every } p \in \Sigma^{<H} \setminus U.$$

The first lemma shows that, under this assumption, both generators assign the same probability to ever entering U .

Lemma 23. *If $M(\cdot \mid x)$ and $M'(\cdot \mid x)$ agree outside U , then*

$$\text{Reach}_M(x, U) = \text{Reach}_{M'}(x, U).$$

Proof. For each prefix $p \in U$, let $|p|$ be its length and define the event

$$F_p := \left\{ Y_{1:|p|} = p \text{ and } Y_{1:s} \notin U \text{ for every } s = 0, 1, \dots, |p| - 1 \right\},$$

with the convention $Y_{1:0} = \emptyset$. The event F_p says that the rollout enters U for the first time at the prefix p .

The events $\{F_p : p \in U\}$ are pairwise disjoint, and their union is exactly the reachability event $E_U(x)$. It therefore suffices to show that $\Pr_M(F_p \mid x) = \Pr_{M'}(F_p \mid x)$ for every $p \in U$.

Fix $p \in U$ of length ℓ . On the event F_p , every proper prefix of p lies outside U . Since the two generators agree outside U ,

$$M(p_{s+1} \mid x, p_{1:s}) = M'(p_{s+1} \mid x, p_{1:s}) \quad \text{for every } s = 0, 1, \dots, \ell - 1.$$

Multiplying these transition probabilities gives

$$\Pr_M(F_p \mid x) = \prod_{s=0}^{\ell-1} M(p_{s+1} \mid x, p_{1:s}) = \prod_{s=0}^{\ell-1} M'(p_{s+1} \mid x, p_{1:s}) = \Pr_{M'}(F_p \mid x).$$

Summing over all $p \in U$ yields

$$\text{Reach}_M(x, U) = \sum_{p \in U} \Pr_M(F_p \mid x) = \sum_{p \in U} \Pr_{M'}(F_p \mid x) = \text{Reach}_{M'}(x, U).$$

□

C.1.3 One-query indistinguishability

We now show that, for the canonical experiment, the two models can differ only on rollouts that reach U .

Lemma 24. *Let $M(\cdot \mid x)$ and $M'(\cdot \mid x)$ agree outside U . Let*

$$W_x = (Y, \mu_1, \dots, \mu_H) \quad \text{and} \quad W'_x = (Y', \mu'_1, \dots, \mu'_H)$$

denote one reply of PathFull_x under M and M' , respectively. Then

$$\text{TV}(\mathcal{L}(W_x), \mathcal{L}(W'_x)) \leq \text{Reach}_M(x, U).$$

Proof. Let $E := E_U(x)$ be the event that the rollout under M reaches U , and let E' be the corresponding event under M' . By Lemma 23,

$$\Pr(E) = \Pr(E') = \text{Reach}_M(x, U).$$

We claim that the conditional laws of W_x and W'_x agree on the complements of these events:

$$\mathcal{L}(W_x \mid E^c) = \mathcal{L}(W'_x \mid (E')^c).$$

To see this, fix any trajectory $y \in \Sigma^H$ such that none of its visited prefixes lies in U . Along that trajectory, every prefix $y_{<t}$ lies outside U , so

$$M(y_t \mid x, y_{<t}) = M'(y_t \mid x, y_{<t}) \quad \text{for every } t = 1, \dots, H.$$

Therefore the trajectory y has the same probability under both models. The same observation gives

$$M(\cdot \mid x, y_{<t}) = M'(\cdot \mid x, y_{<t}) \quad \text{for every } t,$$

so the full reply $(y, \mu_1, \dots, \mu_H)$ is identical in the two worlds. Summing over all trajectories that avoid U proves the claim.

Write

$$R := \mathcal{L}(W_x \mid E^c) = \mathcal{L}(W'_x \mid (E')^c),$$

and define

$$P_1 := \mathcal{L}(W_x \mid E), \quad Q_1 := \mathcal{L}(W'_x \mid E').$$

Then

$$\mathcal{L}(W_x) = \text{Reach}_M(x, U) P_1 + (1 - \text{Reach}_M(x, U)) R$$

and

$$\mathcal{L}(W'_x) = \text{Reach}_M(x, U) Q_1 + (1 - \text{Reach}_M(x, U)) R.$$

Subtracting the two measures gives

$$\mathcal{L}(W_x) - \mathcal{L}(W'_x) = \text{Reach}_M(x, U) (P_1 - Q_1).$$

Taking total variation and using $\text{TV}(P_1, Q_1) \leq 1$ yields

$$\text{TV}(\mathcal{L}(W_x), \mathcal{L}(W'_x)) \leq \text{Reach}_M(x, U).$$

□

The same one-query bound holds for every no-reset experiment by data processing.

Corollary 11. *Let $\mathcal{E}$ be any no-reset experiment at prompt x . If the two generators agree outside U , then for every fixed query parameter q the reply laws under M and M' satisfy*

$$\text{TV}(\mathcal{L}(X_q), \mathcal{L}(X'_q)) \leq \text{Reach}_M(x, U).$$

Proof. By Theorem 27, the experiment $\mathcal{E}$ is a randomized post-processing of PathFull_x . Therefore the reply X_q is obtained by applying the same Markov kernel to W_x that produces X'_q from W'_x . Total variation cannot increase under a common Markov kernel, so Lemma 24 gives

$$\text{TV}(\mathcal{L}(X_q), \mathcal{L}(X'_q)) \leq \text{TV}(\mathcal{L}(W_x), \mathcal{L}(W'_x)) \leq \text{Reach}_M(x, U).$$

□

C.1.4 Adaptive interaction

We now pass from one query to a fully adaptive transcript.

Definition 16. Fix a query space $(\mathcal{Q}, \mathcal{Q})$ and a reply space $(\mathcal{Y}, \mathcal{Y})$. A randomized adaptive algorithm that makes at most q queries consists of a random seed together with measurable decision rules that, at each round, either stop or submit a new query based on the previous transcript. If the algorithm stops early, the remaining rounds are padded with a distinguished symbol $\perp$. The padded transcript is written as

$$T_q^A(M, x) \in ((\mathcal{Q} \times \mathcal{Y}) \cup \{\perp\})^q.$$

Theorem 28. Let $\mathcal{E}$ be any no-reset experiment at prompt x . Suppose $M(\cdot \mid x)$ and $M'(\cdot \mid x)$ agree outside U . Then every randomized adaptive algorithm A that makes at most q queries satisfies

$$\text{TV}\left(\mathcal{L}(T_q^A(M, x)), \mathcal{L}(T_q^A(M', x))\right) \leq q \text{Reach}_M(x, U).$$

Proof. We first treat the case in which A is deterministic. We construct a coupling of the two transcript laws round by round.

Let Z_r and Z'_r be the r th padded transcript entries under M and M' . At round 1, the deterministic algorithm either stops immediately or chooses the same first query in both worlds. If it stops immediately, the two transcripts are identical and there is nothing to prove. Otherwise, Corollary 11 shows that the reply laws to this common query differ in total variation by at most $\text{Reach}_M(x, U)$. By the coupling characterization of total variation, the two first replies can therefore be coupled so that

$$\Pr(Z_1 \neq Z'_1) \leq \text{Reach}_M(x, U).$$

Now suppose the coupling has been defined through round $r - 1$. On the event that the first $r - 1$ transcript entries agree, the deterministic algorithm is in the same state in the two worlds, so it either stops in both or chooses the same round- r query in both. If it stops, we couple the remaining padded symbols identically. If it continues, Corollary 11 again shows that the two reply

laws to the common query differ in total variation by at most $\text{Reach}_M(x, U)$, so we can choose the conditional coupling so that

$$\Pr(Z_r \neq Z'_r \mid Z_1 = Z'_1, \dots, Z_{r-1} = Z'_{r-1}) \leq \text{Reach}_M(x, U).$$

Let

$$D_r := \{(Z_1, \dots, Z_r) \neq (Z'_1, \dots, Z'_r)\}$$

be the event that the two transcripts have diverged by round r , and set $D_0 := \emptyset$. Then

$$D_r \setminus D_{r-1} \subseteq \{Z_1 = Z'_1, \dots, Z_{r-1} = Z'_{r-1}, Z_r \neq Z'_r\}.$$

Taking probabilities and using the conditional bound above gives

$$\Pr(D_r \setminus D_{r-1}) \leq \text{Reach}_M(x, U) \quad \text{for every } r = 1, \dots, q.$$

Summing over r yields

$$\Pr(D_q) \leq q \text{Reach}_M(x, U).$$

By the coupling characterization of total variation,

$$\text{TV}\left(\mathcal{L}(T_q^A(M, x)), \mathcal{L}(T_q^A(M', x))\right) \leq \Pr(D_q) \leq q \text{Reach}_M(x, U).$$

It remains to handle randomized algorithms. Let R denote the algorithm's random seed. Conditional on $R = r$, the algorithm becomes deterministic, so the bound above gives

$$\text{TV}\left(\mathcal{L}(T_q^A(M, x) \mid R = r), \mathcal{L}(T_q^A(M', x) \mid R = r)\right) \leq q \text{Reach}_M(x, U)$$

for every seed value r . The unconditional transcript laws are mixtures of these conditional laws with the same mixing distribution, because the seed is sampled independently of the model. Convexity of total variation under common mixing then gives the same inequality without conditioning on R . □

The testing lower bound used later is an immediate corollary.

Corollary 12. *Suppose an algorithm makes at most q adaptive queries to a no-reset experiment at prompt x and then outputs a guess in $\{M, M'\}$. If its worst-case success probability is at least $2/3$, then*

$$q \geq \frac{1}{3 \text{Reach}_M(x, U)}.$$

Proof. Let

$$P := \mathcal{L}(T_q^A(M, x)), \quad Q := \mathcal{L}(T_q^A(M', x)).$$

For any decision rule based on the transcript, the average success probability under the uniform prior on $\{M, M'\}$ is at most

$$\frac{1}{2} + \frac{1}{2} \text{TV}(P, Q).$$

By Theorem 28,

$$\text{TV}(P, Q) \leq q \text{Reach}_M(x, U).$$

If the worst-case success probability is at least $2/3$, then the average success probability under the uniform prior is also at least $2/3$. Hence

$$\frac{1}{2} + \frac{q \text{Reach}_M(x, U)}{2} \geq \frac{2}{3},$$

which rearranges to the claimed lower bound. $\square$

C.2 Proofs for Section 4.4

C.2.1 No-reset hardness on hidden paths

Recall the hidden-path family from Definition 7. For convenience, we repeat the transition rule:

$$M_z(a \mid p) = \begin{cases} p_+, & \text{if } p = z_{<t} \text{ for some } t \in \{1, \dots, H\} \text{ and } a = z_t, \\ p_-, & \text{if } p = z_{<t} \text{ for some } t \in \{1, \dots, H\} \text{ and } a \neq z_t, \\ 1/K, & \text{if } p \text{ is not a prefix of } z. \end{cases}$$

Theorem 29. Fix $K \geq 2$, $H \geq 1$, and $\lambda > 0$. Let $z, z' \in \Sigma^H$ satisfy $z_{1:H-1} = z'_{1:H-1}$ and $z_H \neq z'_H$. Let A be any randomized adaptive algorithm that makes at most q queries to any no-reset generator experiment. Then

$$\text{TV}\left(\mathcal{L}(T_q^A(M_z, x)), \mathcal{L}(T_q^A(M_{z'}, x))\right) \leq q p_+^{H-1}.$$

Consequently, any such algorithm that distinguishes M_z from $M_{z'}$ with worst-case success probability at least $2/3$ must satisfy

$$q \geq \frac{1}{3p_+^{H-1}}.$$

Proof. Let

$$c := z_{1:H-1} = z'_{1:H-1}.$$

We first show that M_z and $M_{z'}$ agree outside the singleton set

$$U := \{c\}.$$

Fix any prefix $p \in \Sigma^{<H}$ with $p \neq c$. If p is not a prefix of z , then because z and z' agree through depth $H - 1$, it is also not a prefix of z' . In that case both models are uniform at p . If instead p is a prefix of z , then since $p \neq c$ and z and z' agree on their first $H - 1$ symbols, the same prefix p is also a prefix of z' , and the favored next token is the same under both models. Thus

$$M_z(\cdot \mid p) = M_{z'}(\cdot \mid p) \quad \text{for every } p \neq c.$$

So the two generators agree outside U .

We next compute the reachability of U . A rollout reaches the prefix c if and only if its first $H - 1$ tokens are exactly c . Under either model, each of those transitions follows the hidden path and therefore has conditional probability p_+ . Hence

$$\text{Reach}_{M_z}(x, U) = \text{Reach}_{M_{z'}}(x, U) = p_+^{H-1}.$$

Theorem 28 from Appendix C.1 now gives

$$\text{TV}\left(\mathcal{L}(T_q^A(M_z, x)), \mathcal{L}(T_q^A(M_{z'}, x))\right) \leq q p_+^{H-1}.$$

The testing lower bound then follows from Corollary 12. □

C.2.2 Recovery from chosen-prefix samples

We now prove the upper bound for Algorithm 2.

Theorem 30. Fix $K \geq 2$, $H \geq 1$, $\lambda > 0$, and $\delta \in (0, 1)$. Let

$$m := \left\lceil \frac{2}{\Delta^2} \log \left(\frac{H(K-1)}{\delta} \right) \right\rceil.$$

Then Algorithm 2 recovers the hidden path z with probability at least $1 - \delta$ using at most Hm chosen-prefix samples. The queried prefixes obey the local-reset discipline.

Proof. For each stage $t \in \{1, \dots, H\}$, let

$$E_t := \{\widehat{z}_{<t} = z_{<t}\}$$

be the event that the algorithm has reconstructed the hidden path correctly through depth $t - 1$.

Fix a stage t and condition on E_t . Then the query prefix at stage t is the true hidden prefix $z_{<t}$. Therefore the replies

$$A_1^{(t)}, \dots, A_m^{(t)}$$

are independent samples from the next-token distribution at $z_{<t}$. In particular,

$$\Pr(A_j^{(t)} = z_t \mid E_t) = p_+ \quad \text{and} \quad \Pr(A_j^{(t)} = a \mid E_t) = p_- \quad \text{for every } a \neq z_t.$$

Fix any competing token $a \neq z_t$. For each sample index j , define

$$X_j^{(a,t)} := \mathbf{1}\{A_j^{(t)} = z_t\} - \mathbf{1}\{A_j^{(t)} = a\}.$$

Conditional on E_t , the variables $X_1^{(a,t)}, \dots, X_m^{(a,t)}$ are independent, each lies in the interval $[-1, 1]$, and

$$\mathbb{E}[X_j^{(a,t)} \mid E_t] = p_+ - p_- = \Delta.$$

By definition of the empirical counts,

$$N_{z_t}^{(t)} - N_a^{(t)} = \sum_{j=1}^m X_j^{(a,t)}.$$

Therefore

$$\Pr(N_a^{(t)} \geq N_{z_t}^{(t)} \mid E_t) = \Pr\left(\sum_{j=1}^m X_j^{(a,t)} \leq 0 \mid E_t\right).$$

Subtracting the mean from each summand gives

$$\Pr\left(\sum_{j=1}^m (X_j^{(a,t)} - \Delta) \leq -m\Delta \mid E_t\right).$$

Hoeffding's inequality for independent variables in $[-1, 1]$ yields

$$\Pr(N_a^{(t)} \geq N_{z_t}^{(t)} \mid E_t) \leq \exp\left(-\frac{m\Delta^2}{2}\right).$$

There are $K - 1$ competitors. A union bound therefore gives

$$\Pr(\hat{z}_t \neq z_t \mid E_t) \leq (K - 1) \exp\left(-\frac{m\Delta^2}{2}\right) \leq \frac{\delta}{H},$$

where the last inequality follows from the choice of m .

We now propagate the stagewise guarantee. Since

$$E_{t+1}^c = E_t^c \cup (E_t \cap \{\hat{z}_t \neq z_t\}),$$

we have

$$\Pr(E_{t+1}^c) \leq \Pr(E_t^c) + \Pr(\hat{z}_t \neq z_t \mid E_t) \Pr(E_t) \leq \Pr(E_t^c) + \frac{\delta}{H}.$$

Because E_1 is the sure event, $\Pr(E_1^c) = 0$. Iterating the inequality from $t = 1$ to $t = H$ gives

$$\Pr(E_{H+1}^c) \leq \delta.$$

But E_{H+1} is exactly the event $\{\hat{z} = z\}$. Hence

$$\Pr(\hat{z} = z) \geq 1 - \delta.$$

The algorithm makes exactly m chosen-prefix queries at each of the H stages, so the total number of queries is Hm .

Finally, the local-reset discipline holds because the first queried prefix is $\emptyset$, every repeated query at stage t revisits the same already queried prefix $\hat{z}_{<t}$, and the first new query at stage $t + 1$ is the one-token extension $\hat{z}_{<t}\hat{z}_t$ of a previously queried prefix. $\square$

C.2.3 Exact teacher-forced sequence scores

We also prove the sequence-scoring corollary from the main text.

Corollary 13. *In the hidden-path family, exact SeqScore recovers the hidden path using HK chosen-completion score queries.*

Proof. Fix any rule that assigns an arbitrary suffix $s^{(t)} \in \Sigma^{H-t}$ to each stage $t \in \{1, \dots, H\}$. The recovery procedure maintains a current prefix $\hat{z}_{<t}$ and, at stage t , queries the exact scores of the K completions

$$\hat{z}_{<t} a s^{(t)}, \quad a \in \Sigma.$$

It then sets $\hat{z}_t$ to the token with the largest score.

We prove by induction on t that $\hat{z}_{<t} = z_{<t}$ for every stage. The base case $t = 1$ is immediate. Assume now that $\hat{z}_{<t} = z_{<t}$. Let $c := z_{<t}$, and fix any incorrect token $a \neq z_t$. We compare the probabilities of the completions

$$y_{z_t} := c z_t s^{(t)} \quad \text{and} \quad y_a := c a s^{(t)}.$$

The completion y_a leaves the hidden path immediately at time t , so every later transition is uniform. Hence

$$\Pr_{M_z}(Y = y_a \mid x) = \Pr_{M_z}(Y_{<t} = c \mid x) \cdot p_- \cdot K^{-(H-t)}.$$

The completion y_{z_t} stays on the hidden path for at least one additional step. Suppose it remains on the hidden path for exactly $r + 1$ consecutive steps after time t , where r may be any value in $\{0, \dots, H - t\}$. Then

$$\Pr_{M_z}(Y = y_{z_t} \mid x) = \Pr_{M_z}(Y_{<t} = c \mid x) \cdot p_+^{r+1} \cdot u,$$

where u is either $p_- K^{-(H-t-r-1)}$ if the suffix eventually leaves the hidden path, or 1 if the suffix coincides with the true remaining suffix all the way to the end. In either case,

$$u \geq p_- K^{-(H-t-r)}.$$

Therefore

$$\frac{\Pr_{M_z}(Y = y_{z_t} \mid x)}{\Pr_{M_z}(Y = y_a \mid x)} \geq (Kp_+)^{r+1}.$$

Since

$$Kp_+ = \frac{Ke^\lambda}{e^\lambda + K - 1} > 1,$$

this ratio is strictly larger than 1. Thus y_{z_t} has strictly larger probability than every y_a with $a \neq z_t$.

Because SeqScore returns the exact log probability of the queried completion, the correct token z_t is the unique maximizer at stage t . Hence $\hat{z}_t = z_t$, which completes the induction.

There are K score queries at each of the H stages, so the total number of queries is HK . $\square$

C.3 Proofs for Section 4.5

C.3.1 Basic properties of the leader-trie family

Recall the leader-trie model from Definitions 8 and 9. The useful inequalities are

$$\alpha = \frac{4}{K+4}, \quad \beta = \frac{2}{K+4}, \quad \gamma = \frac{1}{K+4}, \quad \gamma_0 = \frac{1}{K+3},$$

so

$$\alpha > \beta > \gamma_0 > \gamma.$$

The relevant margins are

$$\Gamma_{\text{lead}} = \frac{\beta - \gamma_0}{2} = \frac{K+2}{2(K+4)(K+3)}$$

and

$$\gamma_{\text{lead}} = \frac{\log \beta - \log \gamma_0}{2} = \frac{1}{2} \log \left(\frac{2(K+3)}{K+4} \right) > 0.$$

C.3.2 Top-token access is useless

Theorem 31. *Assume $K \geq 3$. For every leader trie T and every queried prefix $p \in \Sigma^{<H}$, the oracle PrefixTop(p) returns the token 1. Consequently, for any adaptive algorithm interacting only with PrefixTop, the full transcript law is the same for every leader trie of depth H . In particular, no such algorithm can distinguish two different leader tries with success probability greater than $1/2$.*

Proof. Fix a leader trie T and a queried prefix $p \in \Sigma^{<H}$.

If $p \in I(T)$, then by Definition 9,

$$M_T(1 \mid p) = \alpha, \quad M_T(b_T(p) \mid p) = \beta, \quad M_T(a \mid p) = \gamma \text{ for } a \notin \{1, b_T(p)\}.$$

Since $\alpha > \beta > \gamma$, the unique maximizer is the token 1.

If instead $p \notin I(T)$, then

$$M_T(1 \mid p) = \frac{4}{K+3}, \quad M_T(a \mid p) = \gamma_0 \text{ for } a \neq 1.$$

Again the unique maximizer is the token 1.

So every query to **PrefixTop** returns the same deterministic reply, namely 1, regardless of the trie and regardless of the queried prefix.

Now let A be any adaptive algorithm. At every round, the distribution of the next query depends only on the previous transcript and the algorithm's internal randomness. Since the reply is always the constant token 1, the transcript distribution is the same for every leader trie. Therefore any decision rule based on the transcript receives identically distributed data under any two candidate tries. Under the uniform prior on a pair of different tries, no such rule can have average success probability above $1/2$, and hence no rule can have worst-case success probability above $1/2$ either. $\square$

C.3.3 Recovery from chosen-prefix logits

We now prove that the hidden branch can be recovered from chosen-prefix logit access.

Theorem 32. *Let T be a leader trie. If $\xi < \gamma_{\text{lead}}$, then Algorithm 3 recovers T exactly from $|I(T)|$ chosen-prefix logit queries. The queried prefixes obey the local-reset discipline.*

Proof. We first show that, at every queried prefix, the thresholding rule in line 4 of Algorithm 3 behaves exactly as intended.

Fix a prefix $p \in \Sigma^{<H}$.

Suppose first that $p \in I(T)$. Then the hidden child is $b_T(p) \in \{2, \dots, K\}$. For that token,

$$\log M_T(b_T(p) \mid p) = \log \beta.$$

Since the returned vector satisfies

$$\|\tilde{\ell}_p - \log M_T(\cdot \mid p)\|_\infty \leq \xi,$$

we have

$$\tilde{\ell}_p(b_T(p)) \geq \log \beta - \xi.$$

Because $\xi < \gamma_{\text{lead}}$ and $\gamma_{\text{lead}} = (\log \beta - \log \gamma_0)/2$,

$$\log \beta - \xi > \log \beta - \gamma_{\text{lead}} = \log \gamma_0 + \gamma_{\text{lead}}.$$

So the hidden child belongs to $\hat{B}(p)$.

Now fix any other nonleader token $a \in \{2, \dots, K\} \setminus \{b_T(p)\}$. Its true log probability is $\log \gamma$,

so

$$\tilde{\ell}_p(a) \leq \log \gamma + \xi.$$

Since $\gamma < \gamma_0$ and $\xi < \gamma_{\text{lead}}$,

$$\tilde{\ell}_p(a) < \log \gamma_0 + \gamma_{\text{lead}}.$$

Thus no other nonleader belongs to $\hat{B}(p)$, and we conclude that

$$\hat{B}(p) = \{b_T(p)\} \quad \text{for every } p \in I(T).$$

Suppose next that $p \notin I(T)$. Then every nonleader token has true log probability $\log \gamma_0$, so for every $a \in \{2, \dots, K\}$,

$$\tilde{\ell}_p(a) \leq \log \gamma_0 + \xi < \log \gamma_0 + \gamma_{\text{lead}}.$$

Hence

$$\hat{B}(p) = \emptyset \quad \text{for every } p \notin I(T).$$

We now prove that the queue-based procedure reconstructs the trie exactly. We argue by induction over the order in which prefixes are popped from the queue.

At initialization, the queue contains only $\emptyset$. Since every leaf of a leader trie has depth exactly H and $H \geq 1$, the root is an internal node. So the first queried prefix is correct.

Assume inductively that every prefix previously popped from the queue was a true internal node of T , and that whenever such a prefix p was processed, the algorithm recovered the correct singleton $\widehat{B}(p) = \{b_T(p)\}$ and therefore added exactly the two true children $p1$ and $p b_T(p)$ to $\widehat{T}$ and, if they were not yet at depth H , to the queue.

Let p be the next prefix popped from the queue. By construction of the queue, p was added earlier as one of the two children of a previously processed internal node. Hence $p \in T$. Since only prefixes of depth at most $H - 1$ are ever appended to the queue, we also have $|p| < H$. Because every node of T of depth strictly less than H is internal, it follows that $p \in I(T)$. Thus every queried prefix is indeed a true internal node.

Conversely, every internal node is eventually queried. This is proved by induction on depth. The root is queried first. Now fix any internal node $p \neq \emptyset$, and let p^- be its parent. By prefix-closure, $p^- \in T$, and since $|p^-| < H$, it is also an internal node. By the induction hypothesis, p^- is eventually queried. When it is processed, the algorithm adds both of its true children to the queue, including p . Therefore p is eventually queried.

We have shown that the algorithm queries exactly the internal nodes of T . At each such node it recovers the correct hidden child, and therefore it adds exactly the correct children. Thus the returned set $\widehat{T}$ is exactly T .

The number of queries is one per internal node, namely $|I(T)|$.

Finally, the queried prefixes obey the local-reset discipline. The first query is $\emptyset$. Every later queried prefix was added to the queue only after its parent had been queried, so when it is first queried it is a one-token extension of a previously queried prefix. This is exactly the local-reset condition from Section 4.2. $\square$

C.3.4 Recovery from chosen-prefix samples

For completeness we write the sample-based breadth-first procedure.

Theorem 33. *Let T be a leader trie and write $s := |I(T)|$. Suppose the learner knows an upper bound $S \geq s$. Fix $\delta \in (0, 1)$, and set*

$$m := \left\lceil \frac{1}{2\Gamma_{\text{lead}}^2} \log \left(\frac{2(K-1)S}{\delta} \right) \right\rceil.$$

Then Algorithm 4 recovers T with probability at least $1 - \delta$ using at most Sm chosen-prefix samples.

The queried prefixes obey the local-reset discipline.

Proof. Fix once and for all a breadth-first ordering of the true internal nodes of T :

$$I(T) = \{p_1, \dots, p_s\}.$$

For each $i \in \{0, \dots, s\}$, let G_i be the event that after exactly i processed prefixes, the following three conditions hold simultaneously:

the processed prefixes are exactly $p_1, \dots, p_i$,

$\hat{T}$ is correct on all discovered nodes,

and

Q contains exactly the remaining internal nodes $p_{i+1}, \dots, p_s$ in breadth-first order.

The event G_0 holds deterministically at initialization.

Fix $i \in \{1, \dots, s\}$ and condition on G_{i-1} . Then the next processed prefix is the true internal node p_i .

For each nonleader token $a \in \{2, \dots, K\}$, the empirical frequency

$$\hat{P}_{p_i}(a) = \frac{1}{m} \sum_{j=1}^m \mathbf{1}\{A_j^{(p_i)} = a\}$$

is the average of m independent Bernoulli random variables with mean $M_T(a \mid p_i)$.

There are two cases. If $a = b_T(p_i)$ is the hidden child, then

$$M_T(a \mid p_i) = \beta.$$

If $a \neq b_T(p_i)$, then

$$M_T(a \mid p_i) = \gamma.$$

In either case, the distance from the mean to the threshold $\gamma_0 + \Gamma_{\text{lead}} = (\beta + \gamma_0)/2$ is at least Γ_{lead} .

Indeed, for the hidden child,

$$\beta - (\gamma_0 + \Gamma_{\text{lead}}) = \Gamma_{\text{lead}},$$

and for every other nonleader,

$$(\gamma_0 + \Gamma_{\text{lead}}) - \gamma > (\gamma_0 + \Gamma_{\text{lead}}) - \gamma_0 = \Gamma_{\text{lead}}.$$

Hoeffding's inequality therefore gives

$$\Pr(\hat{P}_{p_i}(b_T(p_i)) \leq \gamma_0 + \Gamma_{\text{lead}} \mid G_{i-1}) \leq \exp(-2m\Gamma_{\text{lead}}^2)$$

for the hidden child, and

$$\Pr(\hat{P}_{p_i}(a) > \gamma_0 + \Gamma_{\text{lead}} \mid G_{i-1}) \leq \exp(-2m\Gamma_{\text{lead}}^2)$$

for every nonhidden nonleader a .

A union bound over the $K - 1$ nonleader tokens yields

$$\Pr(\hat{B}(p_i) \neq \{b_T(p_i)\} \mid G_{i-1}) \leq (K - 1) \exp(-2m\Gamma_{\text{lead}}^2).$$

By the definition of m ,

$$(K - 1) \exp(-2m\Gamma_{\text{lead}}^2) \leq \frac{\delta}{2S}.$$

Thus

$$\Pr(\hat{B}(p_i) \neq \{b_T(p_i)\} \mid G_{i-1}) \leq \frac{\delta}{2S}.$$

Whenever the good event $\hat{B}(p_i) = \{b_T(p_i)\}$ occurs, the algorithm processes p_i correctly and preserves the breadth-first invariant. Therefore

$$\Pr(G_i^c \mid G_{i-1}) \leq \frac{\delta}{2S}.$$

We now propagate the invariant:

$$G_i^c = G_{i-1}^c \cup (G_{i-1} \cap G_i^c),$$

so

$$\Pr(G_i^c) \leq \Pr(G_{i-1}^c) + \Pr(G_i^c \mid G_{i-1}) \Pr(G_{i-1}) \leq \Pr(G_{i-1}^c) + \frac{\delta}{2S}.$$

Starting from $\Pr(G_0^c) = 0$ and iterating gives

$$\Pr(G_s^c) \leq s \cdot \frac{\delta}{2S} \leq \frac{\delta}{2}.$$

On the event G_s , the algorithm has processed exactly the internal nodes of T , reconstructed the trie correctly, and emptied the queue after at most $s \leq S$ iterations. In that case the algorithm returns $\hat{T} = T$.

To finish the proof, we note that the bound above already controls all possible classification mistakes on true internal nodes. Since the queue only receives children of correctly processed internal nodes, the event G_s also guarantees that no off-trie prefix is ever processed. Thus

$$\Pr(\hat{T} = T) \geq 1 - \delta.$$

The query count is at most m per processed prefix and at most S processed prefixes, so the total number of chosen-prefix samples is at most Sm .

The local-reset discipline holds for the same reason as in the logit-based procedure: the first query is $\emptyset$, and every later queried prefix is a one-token extension of a previously queried prefix. $\square$

C.3.5 Summary corollary

Corollary 14. *Fix $K \geq 3$ and $\lambda > 0$. On the hidden-path family, every no-reset generator interface requires exponentially many queries on some pair of models, while chosen-prefix sampling recovers the hidden path in $O(H \log(H/\delta))$ local-reset queries. On the leader-trie family, chosen-prefix top-token access is useless, while chosen-prefix logits and chosen-prefix sampling recover the trie efficiently under the local-reset discipline. Prefix control is therefore the first boundary, and observation richness becomes a second boundary only after control is available.*

Proof. The hidden-path statement is exactly the combination of Theorems 29 and 30. For fixed K and λ , the gap Δ is a positive constant, so the upper bound

$$Hm = H \left\lceil \frac{2}{\Delta^2} \log \left(\frac{H(K-1)}{\delta} \right) \right\rceil$$

is $O(H \log(H/\delta))$.

The leader-trie statement is exactly the combination of Theorems 31, 32, and 33. The final sentence is the summary of these two witness-family statements. $\square$

C.4 Proofs for Section 4.6

C.4.1 The hard family

Fix a prompt distribution ρ on $\mathcal{X}$ and a designated hard prompt $x^* \in \mathcal{X}$ with mass $\eta := \rho(x^*) > 0$. Fix integers $D, L \geq 1$ and set $H := D + L + 1$. Let $v \in \Sigma^D$ be a known scaffold, and let $\tau_0, \tau_1 \in \Sigma$ be two distinct terminal tokens.

For each hidden suffix $s = (s_1, \dots, s_L) \in \Sigma^L$, define the hard-prompt base generator $Q_s(\cdot \mid x^*)$ as follows. Let

$$u := v s \in \Sigma^{D+L}.$$

For every prefix $p \in \Sigma^{<H}$, set

$$Q_s(a \mid x^*, p) = \begin{cases} p_+, & \text{if } p = u_{<t} \text{ for some } t \in \{1, \dots, D+L\} \text{ and } a = u_t, \\ p_-, & \text{if } p = u_{<t} \text{ for some } t \in \{1, \dots, D+L\} \text{ and } a \neq u_t, \\ 1/K, & \text{if } |p| = D+L, \\ 1/K, & \text{if } p \text{ is not a prefix of } u. \end{cases}$$

Thus $Q_s(\cdot \mid x^*)$ follows the hidden path $u = v s$ for its first $D+L$ steps and is uniform at the final step.

For every prompt $x \neq x^*$, fix once and for all an easy generator $Q_{\text{easy}}(\cdot \mid x)$, the same for all

instances. The full prompt-conditioned base generator is

$$Q_s(\cdot | x) = \begin{cases} Q_s(\cdot | x^*), & \text{if } x = x^*, \\ Q_{\text{easy}}(\cdot | x), & \text{if } x \neq x^*. \end{cases}$$

Now fix a reward scale $R > 0$. For each bit $b \in \{0, 1\}$, define the target completion

$$z_{s,b} := v s \tau_b \in \Sigma^H$$

and the outcome reward

$$r_{s,b}(x, y) := R \mathbf{1}\{x = x^*, y = z_{s,b}\}.$$

All prompts other than x^* therefore have zero reward.

The candidate target set at the hard prompt is

$$\mathcal{Z} := \{z_{s,b} : s \in \Sigma^L, b \in \{0, 1\}\},$$

whose size is

$$N := |\mathcal{Z}| = 2K^L.$$

The base-model mass of the rewarding completion is

$$q_0 := Q_s(z_{s,b} | x^*) = \frac{p_+^{D+L}}{K},$$

which does not depend on either s or b .

C.4.2 The hard-prompt objective

For a policy $\pi(\cdot | x^*)$, define the hard-prompt objective

$$J_{x^*, \beta}^{s,b}(\pi) := \mathbb{E}_{y \sim \pi(\cdot | x^*)} r_{s,b}(x^*, y) - \beta D_{\text{KL}}(\pi(\cdot | x^*) \| Q_s(\cdot | x^*)).$$

The exact optimizer at the hard prompt has the usual Gibbs form.

Lemma 25. *Fix an instance (s, b) . Let*

$$Z_{s,b} := \sum_{y \in \Sigma^H} Q_s(y | x^*) \exp(r_{s,b}(x^*, y)/\beta),$$

and define

$$\pi_{s,b}^*(y \mid x^*) := \frac{Q_s(y \mid x^*) \exp(r_{s,b}(x^*, y)/\beta)}{Z_{s,b}}.$$

Then for every policy $\pi(\cdot \mid x^*)$,

$$J_{x^*,\beta}^{s,b}(\pi) = \beta \log Z_{s,b} - \beta D_{\text{KL}}(\pi(\cdot \mid x^*) \parallel \pi_{s,b}^*(\cdot \mid x^*)).$$

In particular, $\pi_{s,b}^*$ is the unique maximizer of the hard-prompt objective, and the optimal value is $\beta \log Z_{s,b}$.

Proof. Fix any policy $\pi(\cdot \mid x^*)$. By definition of $\pi_{s,b}^*$,

$$\log \pi_{s,b}^*(y \mid x^*) = \log Q_s(y \mid x^*) + \frac{r_{s,b}(x^*, y)}{\beta} - \log Z_{s,b}.$$

Rearranging,

$$r_{s,b}(x^*, y) - \beta \log \left(\frac{\pi(y \mid x^*)}{Q_s(y \mid x^*)} \right) = \beta \log Z_{s,b} - \beta \log \left(\frac{\pi(y \mid x^*)}{\pi_{s,b}^*(y \mid x^*)} \right).$$

Taking expectation with respect to $y \sim \pi(\cdot \mid x^*)$ gives

$$J_{x^*,\beta}^{s,b}(\pi) = \beta \log Z_{s,b} - \beta D_{\text{KL}}(\pi(\cdot \mid x^*) \parallel \pi_{s,b}^*(\cdot \mid x^*)).$$

Since KL divergence is always nonnegative and vanishes only when the two distributions agree, the stated optimality properties follow. $\square$

We next show that a policy which places only small mass on the target completion must be far from optimal on the hard prompt once the reward scale has been chosen appropriately.

Lemma 26. *Assume*

$$R = \beta \log \left(\frac{4}{q_0} \right).$$

Let $\pi(\cdot \mid x^*)$ be any policy, and write

$$m := \pi(z_{s,b} \mid x^*).$$

If $m \leq 1/4$, then

$$J_{x^*,\beta}^{s,b}(\pi_{s,b}^*) - J_{x^*,\beta}^{s,b}(\pi) > \frac{\beta}{4}.$$

Proof. Let $z := z_{s,b}$ denote the target completion. By Lemma 25,

$$J_{x^\star, \beta}^{s,b}(\pi_{s,b}^\star) = \beta \log Z_{s,b}.$$

Since the reward is nonzero only at z and equals R there,

$$Z_{s,b} = 1 - q_0 + q_0 e^{R/\beta} = 1 - q_0 + 4 = 5 - q_0.$$

Hence

$$J_{x^\star, \beta}^{s,b}(\pi_{s,b}^\star) = \beta \log(5 - q_0) \geq \beta \log 4 = 2\beta \log 2.$$

We now upper bound the value of an arbitrary policy with target mass m . By data processing for KL divergence under the map that records whether $y = z$, we have

$$D_{\text{KL}}(\pi(\cdot \mid x^\star) \parallel Q_s(\cdot \mid x^\star)) \geq \text{kl}(m \parallel q_0),$$

where

$$\text{kl}(m \parallel q_0) := m \log\left(\frac{m}{q_0}\right) + (1 - m) \log\left(\frac{1 - m}{1 - q_0}\right).$$

Therefore

$$J_{x^\star, \beta}^{s,b}(\pi) \leq Rm - \beta \text{kl}(m \parallel q_0).$$

Substituting $R = \beta \log(4/q_0)$ gives

$$\begin{aligned} J_{x^\star, \beta}^{s,b}(\pi) &\leq \beta \left[m \log\left(\frac{4}{q_0}\right) - m \log\left(\frac{m}{q_0}\right) - (1 - m) \log\left(\frac{1 - m}{1 - q_0}\right) \right] \\ &= \beta [m \log 4 - m \log m - (1 - m) \log(1 - m) + (1 - m) \log(1 - q_0)] \\ &\leq \beta [m \log 4 + h_2(m)], \end{aligned}$$

where $h_2(m) := -m \log m - (1 - m) \log(1 - m)$ is the binary entropy in nats.

Define

$$g(m) := m \log 4 + h_2(m).$$

For $m \in (0, 1)$,

$$g'(m) = \log 4 + \log\left(\frac{1 - m}{m}\right) = \log\left(\frac{4(1 - m)}{m}\right).$$

Hence $g'(m) > 0$ whenever $m < 4/5$. In particular, g is increasing on $[0, 1/4]$, so

$$g(m) \leq g(1/4) = \log 2 + \frac{3}{4} \log\left(\frac{4}{3}\right).$$

Using the elementary bound $\log(1+u) \leq u$ with $u = 1/3$ gives

$$\log\left(\frac{4}{3}\right) \leq \frac{1}{3},$$

and therefore

$$g(m) \leq \log 2 + \frac{1}{4}.$$

So every policy with $m \leq 1/4$ satisfies

$$J_{x^\star, \beta}^{s,b}(\pi) \leq \beta \left(\log 2 + \frac{1}{4} \right).$$

Combining the lower bound on the optimal value with the upper bound above, we obtain

$$J_{x^\star, \beta}^{s,b}(\pi_{s,b}^\star) - J_{x^\star, \beta}^{s,b}(\pi) \geq \beta \left(2 \log 2 - \log 2 - \frac{1}{4} \right) = \beta \left(\log 2 - \frac{1}{4} \right).$$

Since $\log 2 > 1/2$, the right-hand side is strictly larger than $\beta/4$. $\square$

The hard-prompt gap lifts immediately to the prompt-distributed objective.

Corollary 15. *Assume*

$$R = \beta \log\left(\frac{4}{q_0}\right).$$

Let π be any prompt-conditioned policy. If

$$\pi(z_{s,b} \mid x^\star) \leq 1/4,$$

then

$$J_\beta(\pi_{s,b}^\star) - J_\beta(\pi) > \frac{\eta\beta}{4}.$$

Proof. At every prompt $x \neq x^\star$, the reward is zero. Therefore the contribution of such a prompt to the objective is at most 0, with equality attained by matching the base generator. The exact optimizer $\pi_{s,b}^\star$ does exactly that on all easy prompts, so

$$J_\beta(\pi_{s,b}^\star) = \eta J_{x^\star, \beta}^{s,b}(\pi_{s,b}^\star).$$

For an arbitrary policy π ,

$$J_\beta(\pi) \leq \eta J_{x^*, \beta}^{s, b}(\pi(\cdot \mid x^*)).$$

Applying Lemma 26 to the hard-prompt component and multiplying by η gives the claim. $\square$

C.4.3 Proof of the upper bound

Theorem 34. Fix $\delta \in (0, 1)$, and let

$$m := \left\lceil \frac{2}{\Delta^2} \log \left(\frac{L(K-1)}{\delta} \right) \right\rceil.$$

For the hard family above, there is an algorithm with chosen-prefix next-token sampling access to the base generator and one outcome-reward query that outputs the exact maximizer of J_β with probability at least $1 - \delta$. The algorithm uses at most $H + Lm$ generator queries and one reward query.

Proof. The algorithm works in three stages.

In the first stage, it walks down the known scaffold v in order to respect the local-reset discipline. It queries the prefixes

$$\emptyset, v_{1:1}, v_{1:2}, \dots, v_{1:D}.$$

These are $D + 1$ generator queries. Their replies are irrelevant, since the scaffold is known in advance; the only point of the stage is to build the prefix v legally.

In the second stage, it treats the hidden suffix as a length- L hidden-path problem. More precisely, consider any prefix of the form

$$v u, \quad u \in \Sigma^{t-1}, \quad t \in \{1, \dots, L\}.$$

If $u = s_{<t}$, then by construction of $Q_s(\cdot \mid x^*)$ the next-token distribution at $v u$ satisfies

$$Q_s(s_t \mid x^*, v u) = p_+, \quad Q_s(a \mid x^*, v u) = p_- \text{ for every } a \neq s_t.$$

If u is not a prefix of s , then the queried prefix lies off the hidden path and the next-token distribution is uniform. So below the known scaffold v , the unknown suffix s behaves exactly like a hidden path of length L .

The algorithm therefore applies Algorithm 2 from Section 4.4 starting at the prefix v . By Theorem 30, after Lm chosen-prefix sampling queries, it recovers the hidden suffix s with probability at least $1 - \delta$.

In the third stage, once s has been recovered, the algorithm identifies the reward bit b with one reward query. It queries the hard prompt x^* using the deterministic policy concentrated on the completion $v s \tau_0$. The observed reward equals R if and only if $b = 0$, and equals 0 otherwise. So one reward query determines b exactly.

At that point the hidden state (s, b) is known, and therefore the exact optimizer is known as well. On easy prompts it agrees with the base generator, while at the hard prompt it is given by the Gibbs formula from Lemma 25.

The algorithm fails only if the suffix-recovery stage fails, which occurs with probability at most δ . The total number of generator queries is

$$(D + 1) + Lm \leq H + Lm$$

because $H = D + L + 1$. The number of reward queries is one. $\square$

C.4.4 A posterior-symmetry lemma for the lower bound

We now prepare the no-reset lower bound. The hidden state is

$$\Theta := (S, B),$$

where S is uniform on Σ^L and B is uniform on $\{0, 1\}$, independently. Equivalently, the target completion

$$Z_\Theta := z_{S,B}$$

is uniform on the candidate set $\mathcal{Z}$ of size $N = 2K^L$.

Consider an arbitrary algorithm that may interleave generator and reward queries. Define the event A that at least one no-reset generator query made at the hard prompt x^* reaches the scaffold prefix v in its internal rollout.

Lemma 27. *For any algorithm that makes at most q_g no-reset generator queries,*

$$\Pr(A) \leq q_g p_+^D.$$

Proof. Fix one no-reset generator query made at the hard prompt x^* . Its internal rollout reaches the scaffold v if and only if its first D sampled tokens match the scaffold exactly. Under every generator in the family, each of those D transitions has conditional probability p_+ . Hence the probability that this query reaches v is p_+^D .

Generator queries made at prompts other than x^* are irrelevant for the event A . Since there are at most q_g generator queries in total, the union bound gives

$$\Pr(A) \leq q_g p_+^D.$$

□

The next lemma is the key symmetry statement. It handles arbitrary interleaving by tracking only transcripts that remain on the event A^c and have not yet received positive reward.

A safe partial transcript is an interaction history in which no generator query has reached the scaffold and no reward query has yet produced positive reward. For a safe transcript τ , define the surviving candidate set

$$C(\tau) := \mathcal{Z} \setminus \{y \in \mathcal{Z} : \text{some reward query at } x^* \text{ has sampled } y \text{ and received reward } 0\}.$$

Generator queries do not change $C(\tau)$; a negative reward query at the hard prompt removes at most one candidate.

Lemma 28. *For every safe partial transcript τ , there exists a nonnegative number $q(\tau)$ such that*

$$\Pr(\Theta = \theta, \tau) = \frac{q(\tau)}{N}$$

for every hidden state θ whose target completion lies in $C(\tau)$, and

$$\Pr(\Theta = \theta, \tau) = 0$$

for every hidden state θ whose target completion does not lie in $C(\tau)$.

Consequently, conditional on any safe transcript τ , the posterior on Θ is uniform over the surviving candidate set $C(\tau)$.

Proof. We prove the claim by induction on the length of the safe transcript.

For the empty transcript, take $q(\emptyset) = 1$. Since Θ is uniform over the N possible hidden states, the claim holds.

Now suppose the claim holds for a safe transcript τ , and consider one additional query chosen by the algorithm after observing τ .

If the next query is a generator query, there are two possibilities. Some replies correspond to an internal rollout that reaches the scaffold; those replies do not extend the transcript safely and need not be considered. For a reply g that does extend the transcript safely, the internal rollout has avoided the scaffold. Along such a rollout, every visited prefix lies outside the informative subtree below v , and all generators $Q_s(\cdot \mid x^*)$ agree there. Therefore the conditional probability of observing g , given the current safe transcript τ , is the same for every surviving hidden state θ . Call this common probability $p(g \mid \tau)$. Then

$$\Pr(\Theta = \theta, \tau, g) = \Pr(\Theta = \theta, \tau) p(g \mid \tau) = \frac{q(\tau) p(g \mid \tau)}{N}$$

for every surviving θ , while the nonsurviving states remain at probability zero. So the claim holds for the extended safe transcript (τ, g) with

$$q(\tau, g) := q(\tau) p(g \mid \tau).$$

If the next query is a reward query at a prompt $x \neq x^*$, then the observed reward is always zero, and the distribution of the sampled completion depends only on the chosen policy and not on the hidden state. Hence the same argument applies: the reply distribution is common to all surviving hidden states, so the claim is preserved and the surviving set does not change.

If the next query is a reward query at the hard prompt x^* , let $\pi_\tau(\cdot \mid x^*)$ be the policy chosen after observing τ , and let y be the sampled completion.

If $y \notin \mathcal{Z}$, then the reward is again zero for every hidden state, and the reply distribution is common to all surviving states. So the same argument applies and $C(\tau)$ is unchanged.

If $y \in \mathcal{Z}$ and the reward is positive, then the transcript is no longer safe and there is nothing to prove.

If $y \in \mathcal{Z}$ and the reward is zero, then the transcript remains safe. For a surviving hidden state θ with target completion different from y , the probability of this reply is $\pi_\tau(y \mid x^\star)$, which again does not depend on θ . For the unique hidden state whose target completion equals y , the joint probability becomes zero because that state would have produced positive reward. Therefore the new safe transcript $(\tau, y, 0)$ satisfies the same formula with

$$q(\tau, y, 0) := q(\tau)\pi_\tau(y \mid x^\star),$$

and the new surviving set is $C(\tau) \setminus \{y\}$.

This completes the induction. □

The symmetry lemma implies that reward queries can only locate the target by direct hits, and that every negative reward at the hard prompt eliminates at most one candidate.

Lemma 29. *Let $n \geq 1$ and $r \geq 0$. Consider a state in which the hidden target is uniform over a surviving candidate set C of size n , and the learner has at most r reward queries remaining. Then, regardless of how it may interleave future generator queries with those reward queries, the probability of ever receiving positive reward is at most r/n .*

Proof. We prove the claim by induction on r .

If $r = 0$, there are no reward queries left, so the probability of ever receiving positive reward is 0. This is exactly r/n .

Now suppose $r \geq 1$ and that the claim holds for $r - 1$. Condition on the current state, where the target is uniform over a set C with $|C| = n$.

Before the next reward query, the algorithm may make any number of generator queries. On the event A^c , such generator queries never reach the scaffold and therefore do not change the

posterior-symmetry structure from Lemma 28: the hidden target remains uniform over the same surviving set C . So only the next reward query matters.

Let the next reward query at the hard prompt sample a completion according to some policy. Write p_y for the probability of sampling candidate $y \in C$, and let

$$p_{\text{out}} := 1 - \sum_{y \in C} p_y$$

be the total probability of sampling a completion outside C .

If the sampled completion lies outside C , then the reward is certainly zero and no candidate is eliminated. The learner still has at most $r - 1$ reward queries remaining, and by the induction hypothesis the future chance of success is at most

$$\frac{r - 1}{n}.$$

If the sampled completion is some $y \in C$, then there are two possibilities. With probability $1/n$, the hidden target is exactly y , and the learner receives positive reward immediately. With the complementary probability $(n - 1)/n$, the hidden target lies in $C \setminus \{y\}$, the reward is zero, and the candidate y is eliminated. At that point there are at most $r - 1$ reward queries remaining and $n - 1$ surviving candidates, so the induction hypothesis bounds the future success probability by

$$\frac{r - 1}{n - 1}.$$

Combining the two cases gives

$$\begin{aligned} \Pr(\text{eventual positive reward}) &\leq p_{\text{out}} \cdot \frac{r - 1}{n} + \sum_{y \in C} p_y \left(\frac{1}{n} + \frac{n - 1}{n} \cdot \frac{r - 1}{n - 1} \right) \\ &= p_{\text{out}} \cdot \frac{r - 1}{n} + \sum_{y \in C} p_y \cdot \frac{r}{n} \\ &\leq \left(1 - \sum_{y \in C} p_y \right) \frac{r - 1}{n} + \left(\sum_{y \in C} p_y \right) \frac{r}{n} \\ &\leq \frac{r}{n}. \end{aligned}$$

This completes the induction. □

Lemma 30. *Conditional on the event A^c , the probability that some reward query ever receives positive reward is at most q_r/N .*

Proof. On the event A^c , every generator query avoids the scaffold. By Lemma 28, after every safe transcript the hidden target remains uniform over the current surviving candidate set. At the start of the reward phase, the surviving set is the full set $\mathcal{Z}$ of size N .

Now ignore the generator queries and look only at the evolution of the surviving candidate set across reward queries. Since the target is initially uniform over N candidates and there are at most q_r reward queries in total, Lemma 29 applied with $n = N$ and $r = q_r$ gives

$$\Pr(\text{some reward query receives positive reward} \mid A^c) \leq \frac{q_r}{N}.$$

□

Lemma 31. *Conditional on the event A^c and on the event that no reward query ever receives positive reward, we have*

$$\Pr\left(\widehat{\pi}(z_{S,B} \mid x^*) > \frac{1}{4} \mid A^c, \text{ no reward hit}\right) \leq \frac{4}{N - q_r}.$$

Proof. Fix any final safe transcript τ . By Lemma 28, the posterior on Θ given τ is uniform over the surviving set $C(\tau)$. Each negative reward query at the hard prompt can eliminate at most one candidate from $\mathcal{Z}$, and generator queries eliminate none. Hence

$$|C(\tau)| \geq N - q_r.$$

Let $\widehat{\pi}_\tau(\cdot \mid x^*)$ be the policy output by the algorithm after observing τ . Conditional on τ , the posterior expected mass that this policy assigns to the true target is

$$\mathbb{E}[\widehat{\pi}_\tau(z_{S,B} \mid x^*) \mid \tau] = \frac{1}{|C(\tau)|} \sum_{z \in C(\tau)} \widehat{\pi}_\tau(z \mid x^*) \leq \frac{1}{|C(\tau)|} \leq \frac{1}{N - q_r}.$$

Markov's inequality therefore implies

$$\Pr\left(\widehat{\pi}_\tau(z_{S,B} \mid x^*) > \frac{1}{4} \mid \tau\right) \leq \frac{4}{N - q_r}.$$

Since the bound is uniform over all final safe transcripts τ , averaging over τ gives the claimed inequality. □

C.4.5 Proof of the lower bound

Theorem 35. *Assume*

$$R = \beta \log\left(\frac{4}{q_0}\right).$$

Consider any algorithm that may interleave at most q_g no-reset generator queries with at most $q_r < N$ outcome-reward queries and then outputs a prompt-conditioned policy $\hat{\pi}$. Then there exists a choice of hidden suffix $s \in \Sigma^L$ and reward bit $b \in \{0, 1\}$ such that

$$\Pr\left(J_\beta(\hat{\pi}) \geq J_\beta(\pi_{s,b}^*) - \frac{\eta\beta}{4}\right) \leq q_g p_+^D + \frac{q_r}{N} + \frac{4}{N - q_r}.$$

Proof. Run the algorithm against a uniformly random hidden state

$$\Theta = (S, B),$$

and let $\pi_\Theta^* := \pi_{S,B}^*$ be the corresponding exact optimizer. Define the success event

$$\text{Succ} := \left\{ J_\beta(\hat{\pi}) \geq J_\beta(\pi_\Theta^*) - \frac{\eta\beta}{4} \right\}.$$

By Corollary 15, on every instance the event **Succ** implies

$$\hat{\pi}(z_{S,B} \mid x^*) > \frac{1}{4}.$$

Therefore

$$\begin{aligned} \text{Succ} &\subseteq A \cup \{\text{some reward query receives positive reward}\} \\ &\subseteq \left\{ A^c, \text{ no reward hit, } \hat{\pi}(z_{S,B} \mid x^*) > \frac{1}{4} \right\}. \end{aligned}$$

Taking probabilities and applying the union bound gives

$$\begin{aligned} \Pr(\text{Succ}) &\leq \Pr(A) + \Pr(\text{some reward query receives positive reward}, A^c) \\ &\quad + \Pr\left(A^c, \text{ no reward hit, } \hat{\pi}(z_{S,B} \mid x^*) > \frac{1}{4}\right). \end{aligned}$$

We now bound the three terms. By Lemma 27,

$$\Pr(A) \leq q_g p_+^D.$$

By Lemma 30,

$$\Pr(\text{some reward query receives positive reward}, A^c) \leq \frac{q_r}{N}.$$

By Lemma 31,

$$\Pr\left(A^c, \text{ no reward hit}, \hat{\pi}(z_{S,B} \mid x^*) > \frac{1}{4}\right) \leq \frac{4}{N - q_r}.$$

Combining the three bounds yields

$$\Pr(\text{Succ}) \leq q_g p_+^D + \frac{q_r}{N} + \frac{4}{N - q_r}.$$

This is the average success probability of the algorithm under the uniform prior on the hidden state. Therefore there exists at least one fixed instance (s, b) whose success probability is at most the same quantity. That instance is the one claimed in the theorem. $\square$

Corollary 16. *Let ρ be any prompt distribution with $\rho(x^*) = \eta > 0$. There is a family of KL-regularized outcome-reward post-training instances that is trivial on every prompt $x \neq x^*$ and satisfies the lower bound from Theorem 35. In particular, any guarantee of average regret below $\eta\beta/4$ must solve the hard prompt x^* .*

Proof. This is exactly the hard family defined at the start of the appendix section. Theorem 35 applies directly, and the final statement is just Corollary 15 restated in prompt-distributed language. $\square$

Corollary 17. *Fix $K \geq 2$, $\lambda > 0$, $\beta > 0$, and any constant $q_r \geq 1$. There exists a constant $c = c(K, \lambda) > 1$ such that for every sufficiently large horizon H and every prompt distribution ρ with a designated prompt x^* of mass $\eta > 0$, there is a family of KL-regularized outcome-reward post-training instances of horizon H with the following property. With chosen-prefix next-token sampling access, one reward query and*

$$O\left(H \log \frac{H}{\delta}\right)$$

generator queries suffice to output the exact optimizer with probability at least $1 - \delta$. With only no-reset generator access, any algorithm that uses at most q_r reward queries and succeeds with

probability at least $2/3$ in outputting a policy of average regret at most $\eta\beta/4$ must make

$$\Omega(c^H)$$

generator queries.

Proof. Fix the horizon H , and choose

$$D := \left\lfloor \frac{H-1}{2} \right\rfloor, \quad L := H - D - 1.$$

Then both D and L are $\Theta(H)$, and the hard family above has exactly horizon H .

The upper bound follows directly from Theorem 34. Since $L \leq H$ and $\Delta > 0$ depends only on (K, λ) ,

$$H + L \left\lceil \frac{2}{\Delta^2} \log \left(\frac{L(K-1)}{\delta} \right) \right\rceil = O \left(H \log \frac{H}{\delta} \right).$$

For the lower bound, recall that $N = 2K^L$. Since q_r is fixed and $L = \Theta(H)$, we have

$$\frac{q_r}{N} + \frac{4}{N - q_r} \rightarrow 0 \quad \text{as } H \rightarrow \infty.$$

So for all sufficiently large H ,

$$\frac{q_r}{N} + \frac{4}{N - q_r} \leq \frac{1}{6}.$$

Now suppose an algorithm succeeds with probability at least $2/3$ while using at most q_r reward queries. By Theorem 35, there exists an instance for which

$$\frac{2}{3} \leq q_g p_+^D + \frac{q_r}{N} + \frac{4}{N - q_r} \leq q_g p_+^D + \frac{1}{6}.$$

Hence

$$q_g p_+^D \geq \frac{1}{2}, \quad \text{so} \quad q_g \geq \frac{1}{2} p_+^{-D}.$$

Since $D = \Theta(H)$ and $0 < p_+ < 1$, the quantity p_+^{-D} grows exponentially in H . For example, with

$$c := p_+^{-1/2} > 1,$$

we have

$$p_+^{-D} = \Omega(c^H).$$

Therefore

$$q_g = \Omega(c^H),$$

which proves the corollary. $\square$

C.5 Extended discussion of related work

We expand here on the related work from Section 4.7.

The closest theoretical precursor is the recent line of work that makes coverage central under fixed sampling-style access. Foster, Mhammedi, and Rohatgi study exploration with language models when the learner interacts through a sampling oracle, and show that insufficient coverage of near-optimal responses lower-bounds computational efficiency [17]. Huang et al. analyze inference-time alignment and again find that coverage governs the quality-compute tradeoff of sampling-based alignment methods [30]. Chen et al. elevate this into a broader coverage principle for understanding why pre-training enables post-training and inference-time scaling [14]. Huang et al. study sharpening-style self-improvement under a fixed access model and ask when expensive verification can be amortized into a stronger policy [29]. Our results agree with these papers where the access model overlaps with ours. The difference is the question: we vary the generator interface itself and show that on-policy reachability is the governing quantity of the no-reset class rather than a universal law of post-training.

The second conceptual precursor comes from reinforcement learning and conditional-access models for sequential distributions. Reset-style interaction can change what is learnable or plannable in RL [52, 65]. Conditional queries can likewise make otherwise hard sequential models tractable to learn [48, 44]. The prefix-tree formalism in the present paper is the autoregressive version of the same distinction. Root-start interaction corresponds to ordinary episodic access. Chosen-prefix interaction corresponds to reset-like access. The point of the paper is that, for generators, this distinction should be native rather than imported only by analogy.

A substantial literature studies richer intermediate signals that are not generator-side in our

sense. Botta et al. study verifier-guided generation with prefix-level feasibility signals [6]. Amani et al. study adaptive partial rationale exposure [1]. Tuyls et al. study hidden-state-based exploration bonuses [73]. These papers support the broader point that intermediate access matters, but they study a different channel. The present paper isolates the stochastic generator itself and asks what can be learned from distributional queries without hidden states or external verifiers.

Finally, the KL bridge theorem is meant to be complementary to feedback-side theory. Mousavi-Hosseini and Erdogdu show that process rewards can avoid outcome-reward barriers under fixed generator access [54]. The theorem in Section 4.6 holds the feedback model fixed and varies only the generator interface. In that sense it fills a missing quadrant: fixed generator access with varying feedback has one kind of barrier, while fixed feedback with varying generator access has another.

The empirical literature points in the same direction. Yue et al. argue that RL with verifiable rewards often sharpens trajectories already present in the base model [77]. Karan and Du show that stronger sampling can sometimes match or exceed RL post-training while preserving diversity [34]. Tan et al. identify entropy collapse in the final-layer posterior of reasoning models and recover exploration from intermediate-layer uncertainty [72]. These papers do not prove the taxonomy in this paper, but they reinforce its premise: support, exploration, and intermediate control should be treated as part of the problem rather than as implementation details.

Algorithm 4 Recovering a leader trie with chosen-prefix samples

```

1: Initialize  $\hat{T} := \{\emptyset\}$ , queue  $Q := (\emptyset)$ , and counter  $c := 0$ 
2: while  $Q$  is nonempty and  $c < S$  do
3:   Pop the first prefix  $p$  from  $Q$  and set  $c := c + 1$ 
4:   for  $j = 1$  to  $m$  do
5:     Query  $\text{PrefixSample}(p)$  and receive  $A_j^{(p)} \in \Sigma$ 
6:   end for
7:   for each  $a \in \{2, \dots, K\}$  do
8:     Set  $\hat{P}_p(a) := \frac{1}{m} \sum_{j=1}^m \mathbf{1}\{A_j^{(p)} = a\}$ 
9:   end for
10:  Set  $\hat{B}(p) := \{a \in \{2, \dots, K\} : \hat{P}_p(a) > \gamma_0 + \Gamma_{\text{lead}}\}$ 
11:  if  $\hat{B}(p)$  is a singleton  $\{\hat{b}(p)\}$  then
12:    Add  $p1$  and  $p\hat{b}(p)$  to  $\hat{T}$ 
13:    if  $|p| + 1 < H$  then
14:      Append  $p1$  and  $p\hat{b}(p)$  to  $Q$ 
15:    end if
16:  end if
17: end while
18: if  $Q$  is empty then
19:   Return  $\hat{T}$ 
20: else
21:   Return  $\perp$ 
22: end if

```

Appendix D

Supplementary Material: Data Attribution in Adaptive Learning

In this chapter, we provide supplementary material corresponding to Chapter 5.

D.1 Additional discussion of related work

This appendix expands on the literature discussion from the main text, focusing on where the boundaries of the present contribution fall relative to nearby work that is related in spirit but answers different questions.

D.1.1 Static data attribution and deletion-by-reweighting

The classical starting point is influence-function based data attribution [37]. In that framework, one studies the effect of upweighting a training point by an infinitesimal amount and then relates that differential quantity to deletion by finite reweighting. TracIn, SGD-based tracing methods, and semivalue-based approaches such as Data Shapley all fit within the broader project of attributing model behavior to training data in fixed-dataset learning [59, 20]. The central simplification is that the learner changes the model but does not change which training examples would exist later. This is why replay and recollection collapse in the static setting.

The present paper keeps the deletion-by-reweighting convention. A perturbation with $\epsilon = -1$ suppresses the direct update contribution of the realized occurrence while leaving the realized prefix event itself intact. This is the standard convention in influence-function style attribution, and it is the natural one for occurrence-level attribution in adaptive learning. What changes in the adaptive

setting is not the meaning of deletion-by-reweighting. What changes is that after the perturbation the future data law may also change.

D.1.2 The estimand question in fixed-data attribution

A major methodological lesson from recent attribution work is that one should separate the target from the approximation used to estimate it. Bae et al. show that practical influence-function calculations can track a quantity different from literal leave-one-out retraining, namely a proximal response functional that remains meaningful even when exact leave-one-out agreement fails [3]. Schioppa et al. analyze the assumptions behind the standard leave-one-out reading of influence functions and explain why those assumptions break down in nonlinear, non-convex training [67]. Distributional training data attribution moves in a related direction by treating training randomness itself as part of the attribution object rather than as a nuisance and by asking how datasets affect the distribution of outputs over training runs [53].

The present paper adopts the same estimand-first discipline, but the source of extra counterfactual variation is different. Here the issue is not only training randomness over a fixed data source. The issue is that under an occurrence-level perturbation, the learner can change the future data stream itself. This makes the future-law term part of the target rather than a nuisance around a fixed-log computation.

D.1.3 Trajectory-specific and run-specific attribution under exogenous data

Another important line of work studies attribution along a realized optimization trajectory. Hara et al. introduced SGD-based occurrence-specific influence, where one asks about removing a point from a specific SGD step rather than from the dataset abstractly [26]. Wang et al. later formalized trajectory-specific leave-one-out influence and emphasized that the same example can have different effects when it appears at different points of training [75]. In-Run Data Shapley studies run-specific attribution for a target model produced by a single training run, rather than averaging over all runs that the learning algorithm might have produced [74].

These papers are close to the present one in two respects. First, they reject the view that data attribution must be permutation-invariant over training examples. Second, they show that a realized occurrence can be the right unit of analysis. The difference is that the data source remains exogenous. Removing or downweighting a realized occurrence changes the optimization trajectory, but it does not change the distribution of future data. In the language of the present paper, these works study refined forms of replay-side attribution. The present paper asks what happens when the future data source is itself endogenous to the learner.

D.1.4 Sequential attribution in reinforcement learning and post-training

A growing literature studies attribution in RL and other sequential learning systems, but the targets vary considerably.

Gottesman et al. analyze influential transitions for off-policy evaluation [22]. Their target is the sensitivity of an evaluation estimator to transitions in an observational dataset. That is a useful object, especially in high-stakes domains, but it is not the same as asking how a realized training occurrence changes the learned policy together with the future data the learner would collect after that perturbation.

Hiraoka et al. study influential experiences for replay-based RL agents [27]. This work is also closely related in application domain, but experience replay changes the counterfactual structure. The object of interest is typically a transition already stored in the replay buffer, and the emphasis is on how stored experiences influence the agent through replayed optimization. The present paper instead studies an on-policy occurrence-level perturbation and the way it alters future data collection.

Hu et al. are the closest nearby work in terms of motivation [28]. They begin from the observation that in online RL each experience both updates the policy and shapes future data collection, which is exactly the phenomenon emphasized here. Their framework, however, is deliberately local. It interprets recent training records relative to a nearby checkpoint and a recent buffer, especially in PPO-style training. The present paper studies a different target: a global

occurrence-level counterfactual that conditions on the realized prefix through the perturbed round and then recollects the full remaining future. The contrast is not between right and wrong targets. It is between a local target designed for nearby interpretability and a global target designed to answer the occurrence-level counterfactual question itself.

The same distinction matters in newer post-training settings for language models. Several recent works study data influence in reinforcement fine-tuning or RL-based post-training using local, off-policy, or estimator-specific surrogates. Those developments are important, but they do not by themselves settle the question of what the global attribution target should be once future rollouts depend on the perturbed learner. The present paper is aimed at that earlier step.

D.1.5 Sequential causal inference

The paper also draws a clear line to the causal inference literature on sequential interventions. Longitudinal causal inference has long studied interventions that modify later treatment mechanisms while conditioning on evolving histories, together with identification by g-computation, weighting, and doubly robust methods. The literature on longitudinal modified treatment policies is especially relevant because it treats stochastic, history-dependent interventions that change an observed treatment mechanism without forcing treatment to a fixed value [16].

We do not claim novelty at that level. The present paper does not propose a new foundation for sequential causal inference. What it contributes is an attribution-theoretic use of that perspective. In adaptive learning, the training history is both a learning trace and a data-collection trace. Once one asks for occurrence-level attribution, sequential intervention ideas become part of the right formal language. The paper then asks three questions that are specific to attribution in adaptive learning: which conditioning level identifies a realized occurrence while still leaving the future free to change, which replay-side objects fail to determine the interventional target, and which structural restrictions recover identification.

D.1.6 The contribution in context

The paper makes three technical claims. First, when future training data are endogenous, replay and recollection are genuinely different counterfactuals, and conditional intervention is the natural global occurrence-level object. Second, the interventional target is not identified by replay-side information alone, even when that information includes the baseline future law and the complete family of fixed-log replay responses for every continuation. Third, in the action-only class—where the learner affects future data only through a known action law—the perturbed future law has an exact change-of-measure form and the target is identified under overlap; unknown learner-state dependence in contexts or rewards breaks identification.

The fixed-data attribution literature has shown that attribution can depend on the realized trajectory, the specific training run, and the exact occurrence at which a point appears. The adaptive-learning literature has recognized that online RL creates a tension between attribution and endogenous future data, but has mostly worked with local or estimator-specific targets. The causal literature on sequential interventions supplies the formal language needed to state the global occurrence-level question cleanly. The present paper brings these threads together: it formulates the target, establishes the replay-side insufficiency, and identifies the structural class where logged-data attribution is possible.

D.2 Extended positioning of nearby work under the taxonomy

This appendix expands on Table 5.1. The goal is to state carefully which nearby methods are direct special cases, which ones are controlled surrogates, and which ones answer different questions.

Trajectory-specific attribution on exogenous training streams. The cleanest direct bridge is to trajectory-specific attribution in ordinary training runs with exogenous data. Hara et al. study occurrence-specific influence along SGD trajectories, and Wang et al. formalize trajectory-specific leave-one-out influence for removing a data point from a specific iteration of training [26, 75]. In our language, these works agree with the present paper on the importance of the realized

occurrence, but they lie in the exogenous limit where the perturbation does not change the law of future data. The future-law term therefore vanishes, and replay-side attribution becomes exact.

Replay-buffer experience influence. Hiraoka et al. study influential experiences stored in a replay buffer and estimate their leave-one-out effects on later RL training [27]. This is close in application domain but not literally the same process as the present paper. If the replay buffer is treated as fixed, then the counterfactual stays on the replay side. To subsume replay-buffer learning exactly, one would need to enlarge the learner state so that it includes both the policy parameters and the buffer contents, together with two event types: environment-interaction events and replay-update events.

Local and truncated online RL attribution. Snapshot studies online RL directly but adopts a deliberately local target, interpreting checkpoints with respect to records in the recent training buffer [28]. I-PPO similarly uses attribution or filtering at the level of rollout-buffer episodes in PPO-style post-training [68]. These works are best understood as moving along the future-semantics axis from full recollection toward truncated local windows, often together with a checkpoint-local target functional. Section 5.8 provides the natural formal bridge: a checkpoint-local or short-horizon version of the depth- L recollection target.

Replay-based non-local on-policy attribution. Recent workshop work on non-local attribution for on-policy RL extends the time range of attribution beyond a single recent rollout, but does so using a replay-based leave-one-out objective under fixed rollout buffers [45]. This is important to distinguish from the present target. It is non-local in time, but it remains on the replay side of the future-semantics axis. In the language of the present paper, it enlarges the replay window without recollecting the future under the perturbed learner.

Estimator sensitivity and off-policy surrogates. Gottesman et al. study influential transitions for off-policy evaluation and compute exact influence functions for fitted Q-evaluation and importance-sampling variants [22]. This changes the target functional entirely: the object is the OPE estimator rather than the final learned system after a perturbed occurrence.

CROPI is different again [79]. It studies RLVR and uses offline trajectories to estimate data

influence without fresh online rollouts. In the present taxonomy this is best classified as an **off-policy surrogate** rather than as the same counterfactual object. Conceptually it sits near the positive frontier identified by the action-only class, but theorem-level justification of such off-policy estimators belongs to a second-stage estimation paper rather than to the present foundations paper.

RLVR and RFT sample influence. RFT-Inf is especially relevant for post-training [70]. It defines influence at the level of a training sample, and measures how removing that sample changes the final training reward. This is a genuinely global influence target, but it changes the attribution unit from realized occurrence to dataset sample identity or sample presentation. The exact bridge to the present framework is to decide whether repeated presentations of the same sample are treated as distinct occurrences or collapsed into a single sample-level object. The present paper takes the former route because adaptive learning makes the exact occurrence time and learner state part of the causal object.

Conceptual ancestors and methodological support. We now highlight several papers which are conceptually important - they also study what the influence target estimator even computes in their specific setting. Bae et al. and Schioppa et al. argue, in different ways, that one must first decide which counterfactual object an influence method is approximating [3, 67]. Distributional training-data attribution asks how datasets change the distribution of outcomes across training runs, rather than only one endpoint [53]. Bottou et al. articulate the broader counterfactual-learning-systems perspective in which interventions can change later observations [7]. Longitudinal modified treatment policy work supplies the causal language for stochastic, history-dependent interventions on evolving processes [16]. These works are philosophical and methodological supports, not theorem-level reductions of the present paper.

What the taxonomy contributes. The taxonomy is useful because it separates the attribution unit, future semantics, and target functional. That makes it easier to see whether a nearby method is a direct special case of the global occurrence-level target, a controlled truncation, a replay-side surrogate, or a different question altogether. The present paper is strongest on the full-recollection corner and on the frontier between identifiable, non-identifiable, replay-side, and

truncated targets.

D.3 Proofs for Section 5.4

D.3.1 Proof of Theorem 17

Proof. Fix $t \in \{1, \dots, T\}$, fix a realized prefix $h = z_{1:t} \in \mathcal{H}_t$ with $\mathbb{P}_\nu(h) > 0$, and write $\mathcal{C}_t = \mathcal{Z}_{t+1} \times \dots \times \mathcal{Z}_T$.

For each continuation $c \in \mathcal{C}_t$, define

$$Q_\epsilon(c) := Q_{\nu,t}^\epsilon(c \mid h), \quad G_\epsilon(c) := F_\nu\left(\theta_{T+1}^{(w^{(t,\epsilon)})}(h, c)\right).$$

Then by the definition of the expected terminal target,

$$\Psi_t^\epsilon(h) = \sum_{c \in \mathcal{C}_t} Q_\epsilon(c) G_\epsilon(c).$$

Therefore, by the definition of conditional interventional influence,

$$\mathcal{I}_t^{\text{int}}(F_\nu; h) = \left. \frac{d}{d\epsilon} \sum_{c \in \mathcal{C}_t} Q_\epsilon(c) G_\epsilon(c) \right|_{\epsilon=0}.$$

The continuation space $\mathcal{C}_t$ is finite, so we may differentiate term by term:

$$\mathcal{I}_t^{\text{int}}(F_\nu; h) = \sum_{c \in \mathcal{C}_t} \left. \frac{d}{d\epsilon} [Q_\epsilon(c) G_\epsilon(c)] \right|_{\epsilon=0}.$$

Applying the ordinary product rule to each summand,

$$\mathcal{I}_t^{\text{int}}(F_\nu; h) = \sum_{c \in \mathcal{C}_t} \left. \frac{d}{d\epsilon} Q_\epsilon(c) \right|_{\epsilon=0} G_0(c) + \sum_{c \in \mathcal{C}_t} Q_0(c) \left. \frac{d}{d\epsilon} G_\epsilon(c) \right|_{\epsilon=0}.$$

We now identify the two sums.

For the first sum, by definition of $\dot{Q}_{\nu,t}(c \mid h)$ in (5.5),

$$\left. \frac{d}{d\epsilon} Q_\epsilon(c) \right|_{\epsilon=0} = \dot{Q}_{\nu,t}(c \mid h).$$

Also,

$$G_0(c) = F_\nu(\theta_{T+1}(h, c)).$$

Hence the first sum is

$$\sum_{c \in \mathcal{C}_t} \dot{Q}_{\nu,t}(c \mid h) F_{\nu}(\theta_{T+1}(h, c)).$$

For the second sum, note that

$$Q_0(c) = Q_{\nu,t}^0(c \mid h) = \mathbb{P}_{\nu}(c \mid h),$$

because $\epsilon = 0$ gives the baseline process.

Moreover, by the definition of replay influence,

$$\left. \frac{d}{d\epsilon} G_{\epsilon}(c) \right|_{\epsilon=0} = \mathcal{I}_t^{\text{rep}}(F_{\nu}; h, c).$$

Therefore the second sum is

$$\sum_{c \in \mathcal{C}_t} \mathbb{P}_{\nu}(c \mid h) \mathcal{I}_t^{\text{rep}}(F_{\nu}; h, c).$$

By the definition of conditional expectation on a finite space, this is exactly

$$\mathbb{E}_{\mathbb{P}_{\nu}}[\mathcal{I}_t^{\text{rep}}(F_{\nu}; Z_{1:T}) \mid Z_{1:t} = h].$$

Combining the two sums yields

$$\mathcal{I}_t^{\text{int}}(F_{\nu}; h) = \mathbb{E}_{\mathbb{P}_{\nu}}[\mathcal{I}_t^{\text{rep}}(F_{\nu}; Z_{1:T}) \mid Z_{1:t} = h] + \sum_{c \in \mathcal{C}_t} \dot{Q}_{\nu,t}(c \mid h) F_{\nu}(\theta_{T+1}(h, c)),$$

which is the claimed identity. $\square$

D.3.2 Proof of Proposition 6

Proof. Fix $h = z_{1:t}$ with $\mathbb{P}_{\nu}(h) > 0$. For every $\epsilon \in [-1, \rho]$, the conditional future law $Q_{\nu,t}^{\epsilon}(\cdot \mid h)$ is a probability distribution on $\mathcal{C}_t$. Hence

$$\sum_{c \in \mathcal{C}_t} Q_{\nu,t}^{\epsilon}(c \mid h) = 1.$$

Differentiating at $\epsilon = 0$ gives

$$\sum_{c \in \mathcal{C}_t} \dot{Q}_{\nu,t}(c \mid h) = 0.$$

Now subtract the conditional expected replay term from both sides of (5.6). This gives

$$\mathcal{I}_t^{\text{int}}(F_\nu; h) - \mathbb{E}_{\mathbb{P}_\nu}[\mathcal{I}_t^{\text{rep}}(F_\nu; Z_{1:T}) \mid Z_{1:t} = h] = \sum_{c \in \mathcal{C}_t} \dot{Q}_{\nu,t}(c \mid h) F_\nu(\theta_{T+1}(h, c)).$$

Because the coefficients $\dot{Q}_{\nu,t}(c \mid h)$ sum to zero, we may subtract the same constant from each summand without changing the total. Choose the constant

$$F_\nu(\theta_{t+1}(h)).$$

This depends only on the fixed prefix h , not on the continuation c . Therefore

$$\sum_{c \in \mathcal{C}_t} \dot{Q}_{\nu,t}(c \mid h) F_\nu(\theta_{T+1}(h, c)) = \sum_{c \in \mathcal{C}_t} \dot{Q}_{\nu,t}(c \mid h) [F_\nu(\theta_{T+1}(h, c)) - F_\nu(\theta_{t+1}(h))].$$

Substituting this into the previous display yields (5.7). $\square$

D.3.3 Proof of Proposition 5

Proof. Fix $t \in \{1, \dots, T\}$ and fix a realized prefix $h = z_{1:t} \in \mathcal{H}_t$ with $\mathbb{P}_\nu(h) > 0$.

For each future round $s \in \{t+1, \dots, T\}$ and each history prefix $y_{1:s-1} \in \mathcal{H}_{s-1}$, the hypothesis implies that the kernel does not depend on the learner state. Therefore there exists a single probability distribution, which we denote by

$$\bar{K}_{\nu,s}(\cdot \mid y_{1:s-1}),$$

such that

$$K_{\nu,s}^\theta(\cdot \mid y_{1:s-1}) = \bar{K}_{\nu,s}(\cdot \mid y_{1:s-1}) \quad \text{for every } \theta \in \Theta.$$

We first identify the perturbed conditional future law. Fix $\epsilon \in [-1, \rho]$, and fix a continuation

$$c = z_{t+1:T} \in \mathcal{C}_t = \mathcal{Z}_{t+1} \times \dots \times \mathcal{Z}_T.$$

By definition of the conditional future law under the one-coordinate perturbation $w^{(t,\epsilon)}$,

$$\begin{aligned} Q_{\nu,t}^\epsilon(c \mid h) &= \mathbb{P}_\nu^{(w^{(t,\epsilon)})}(z_{t+1:T} \mid z_{1:t} = h) \\ &= \prod_{s=t+1}^T K_{\nu,s}^{\theta_s^{(w^{(t,\epsilon)})}}(h, z_{t+1:s-1})(z_s \mid h, z_{t+1:s-1}). \end{aligned} \tag{D.1}$$

Because each future kernel is state-independent, every factor in (D.1) equals the common kernel $\bar{K}_{\nu,s}$. Hence

$$Q_{\nu,t}^\epsilon(c \mid h) = \prod_{s=t+1}^T \bar{K}_{\nu,s}(z_s \mid h, z_{t+1:s-1}).$$

The same calculation holds at $\epsilon = 0$, so

$$Q_{\nu,t}^0(c \mid h) = \prod_{s=t+1}^T \bar{K}_{\nu,s}(z_s \mid h, z_{t+1:s-1}).$$

Therefore

$$Q_{\nu,t}^\epsilon(c \mid h) = Q_{\nu,t}^0(c \mid h) \quad \text{for every } c \in \mathcal{C}_t.$$

Since $Q_{\nu,t}^0(c \mid h) = \mathbb{P}_\nu(c \mid h)$ by definition of the baseline process, we obtain

$$Q_{\nu,t}^\epsilon(c \mid h) = Q_{\nu,t}^0(c \mid h) = \mathbb{P}_\nu(c \mid h),$$

which is the first claim.

We now compute the finite interventional target. By definition,

$$\Psi_t^\epsilon(h) = \sum_{c \in \mathcal{C}_t} Q_{\nu,t}^\epsilon(c \mid h) F_\nu \left(\theta_{T+1}^{(w^{(t,\epsilon)})}(h, c) \right).$$

Substituting the already-proved identity for the future law gives

$$\Psi_t^\epsilon(h) = \sum_{c \in \mathcal{C}_t} \mathbb{P}_\nu(c \mid h) F_\nu \left(\theta_{T+1}^{(w^{(t,\epsilon)})}(h, c) \right).$$

Because the state space is finite and h has positive baseline probability, this is exactly the conditional expectation

$$\mathbb{E}_{\mathbb{P}_\nu} \left[F_\nu \left(\theta_{T+1}^{(w^{(t,\epsilon)})}(Z_{1:T}) \right) \mid Z_{1:t} = h \right].$$

This proves the displayed formula for $\Psi_t^\epsilon(h)$.

At $\epsilon = 0$, we similarly have

$$\Psi_t^0(h) = \mathbb{E}_{\mathbb{P}_\nu} [F_\nu(\theta_{T+1}(Z_{1:T})) \mid Z_{1:t} = h].$$

Subtracting the $\epsilon = 0$ identity from the ϵ -identity yields

$$\begin{aligned}\Delta_{t,\epsilon}^{\text{int}}(F_\nu; h) &= \Psi_t^\epsilon(h) - \Psi_t^0(h) \\ &= \mathbb{E}_{\mathbb{P}_\nu} \left[F_\nu \left(\theta_{T+1}^{(w^{(t,\epsilon)})}(Z_{1:T}) \right) - F_\nu(\theta_{T+1}(Z_{1:T})) \mid Z_{1:t} = h \right] \\ &= \mathbb{E}_{\mathbb{P}_\nu} \left[\Delta_{t,\epsilon}^{\text{rep}}(F_\nu; Z_{1:T}) \mid Z_{1:t} = h \right].\end{aligned}$$

This proves the finite-effect identity.

Finally, assume the derivative at 0 exists. Since $\mathcal{C}_t$ is finite, we may differentiate term by term:

$$\begin{aligned}\mathcal{I}_t^{\text{int}}(F_\nu; h) &= \left. \frac{d}{d\epsilon} \Psi_t^\epsilon(h) \right|_{\epsilon=0} \\ &= \left. \frac{d}{d\epsilon} \sum_{c \in \mathcal{C}_t} \mathbb{P}_\nu(c \mid h) F_\nu \left(\theta_{T+1}^{(w^{(t,\epsilon)})}(h, c) \right) \right|_{\epsilon=0} \\ &= \sum_{c \in \mathcal{C}_t} \mathbb{P}_\nu(c \mid h) \left. \frac{d}{d\epsilon} F_\nu \left(\theta_{T+1}^{(w^{(t,\epsilon)})}(h, c) \right) \right|_{\epsilon=0} \\ &= \sum_{c \in \mathcal{C}_t} \mathbb{P}_\nu(c \mid h) \mathcal{I}_t^{\text{rep}}(F_\nu; h, c) \\ &= \mathbb{E}_{\mathbb{P}_\nu} [\mathcal{I}_t^{\text{rep}}(F_\nu; Z_{1:T}) \mid Z_{1:t} = h].\end{aligned}$$

This proves the derivative identity.

Under the smoothness assumptions used later in the paper, the same state independence also implies that $Q_{\nu,t}^\epsilon(\cdot \mid h)$ is constant in ϵ , so $\dot{Q}_{\nu,t}(\cdot \mid h) \equiv 0$, and that $\nabla_\theta K_{\nu,s}^\theta(\cdot \mid z_{1:s-1}) \equiv 0$ for every $s > t$, so every stagewise future-law correction Ξ_s vanishes as well. $\square$

D.4 Proof of Theorem 18

Proof. We first construct the promised smooth class.

Take horizon $T = 2$. Let

$$\mathcal{Z}_1 = \{z_1^\star\}, \quad \mathcal{Z}_2 = \{0, 1\}, \quad \Theta = \mathbb{R}, \quad \theta_1 = 0.$$

The first interaction is deterministic:

$$K_{\nu,1}^\theta(z_1^\star \mid \emptyset) = 1 \quad \text{for every } \nu \text{ and every } \theta.$$

Define the first update map by

$$U_1(\theta, z_1^*, w) := w - 1.$$

Thus under the baseline weight $w_1 = 1$,

$$\theta_2(z_1^*) = 0,$$

while under the one-coordinate perturbation $w^{(1,\epsilon)}$,

$$\theta_2^{(w^{(1,\epsilon)})}(z_1^*) = \epsilon.$$

For the second round, define the update and target by

$$U_2(\theta, z_2, 1) := z_2, \quad F(\theta_3) := \theta_3.$$

Since $z_2 \in \{0, 1\}$, this means that the terminal value under any full history (z_1^*, z_2) is just z_2 .

For each parameter $\gamma \in \mathbb{R}$, define an environment ν_γ by the round-2 kernel

$$K_{\nu_\gamma, 2}^\theta(1 \mid z_1^*) = \sigma(\gamma\theta), \quad K_{\nu_\gamma, 2}^\theta(0 \mid z_1^*) = 1 - \sigma(\gamma\theta),$$

where $\sigma(x) = 1/(1 + e^{-x})$ is the logistic sigmoid.

All these objects are smooth in the obvious sense. The update maps are smooth in their real arguments, the round-2 kernel masses are smooth functions of θ , and F is linear.

We now compare the replay oracles at the realized prefix z_1^* .

First, we compute the baseline future law. Under the baseline process, the first update produces the state $\theta_2(z_1^*) = 0$. Therefore

$$Q_{\nu_\gamma, 1}^0(1 \mid z_1^*) = K_{\nu_\gamma, 2}^0(1 \mid z_1^*) = \sigma(0) = \frac{1}{2},$$

and similarly

$$Q_{\nu_\gamma, 1}^0(0 \mid z_1^*) = \frac{1}{2}.$$

Thus the baseline conditional future law at z_1^* is the same for every γ .

Next, we compute the replay response curves. Fix $c \in \{0, 1\}$. By construction,

$$\theta_3^{(w^{(1,\epsilon)})}(z_1^*, c) = U_2\left(\theta_2^{(w^{(1,\epsilon)})}(z_1^*), c, 1\right) = c.$$

Therefore

$$\phi_{\nu_\gamma, 1, c}(\epsilon; z_1^*) = F\left(\theta_3^{(w^{(1,\epsilon)})}(z_1^*), c\right) = c$$

for every ϵ and every γ . So the entire family of fixed-log replay response curves is also the same for every γ .

We have shown that for every $\alpha, \beta \in \mathbb{R}$,

$$\mathcal{R}_{\nu_\alpha, 1}(z_1^*) = \mathcal{R}_{\nu_\beta, 1}(z_1^*).$$

We now compute the finite interventional target. Under the perturbation $w^{(1,\epsilon)}$, the state at time 2 is $\theta_2^\epsilon = \epsilon$. Therefore, conditional on the realized prefix z_1^* ,

$$Q_{\nu_\gamma, 1}^\epsilon(1 \mid z_1^*) = K_{\nu_\gamma, 2}^\epsilon(1 \mid z_1^*) = \sigma(\gamma\epsilon).$$

Since the terminal target equals the second-round interaction,

$$\Psi_{\nu_\gamma, 1}^\epsilon(z_1^*) = \sum_{c \in \{0, 1\}} Q_{\nu_\gamma, 1}^\epsilon(c \mid z_1^*) c = Q_{\nu_\gamma, 1}^\epsilon(1 \mid z_1^*) = \sigma(\gamma\epsilon).$$

Hence for $\alpha \neq \beta$,

$$\Psi_{\nu_\alpha, 1}^\epsilon(z_1^*) \neq \Psi_{\nu_\beta, 1}^\epsilon(z_1^*)$$

whenever $\alpha\epsilon \neq \beta\epsilon$.

Differentiating at $\epsilon = 0$, we obtain

$$\mathcal{I}_{1, \nu_\gamma}^{\text{int}}(F; z_1^*) = \left. \frac{d}{d\epsilon} \sigma(\gamma\epsilon) \right|_{\epsilon=0} = \gamma\sigma'(0) = \frac{\gamma}{4}.$$

Thus for $\alpha \neq \beta$,

$$\mathcal{I}_{1, \nu_\alpha}^{\text{int}}(F; z_1^*) \neq \mathcal{I}_{1, \nu_\beta}^{\text{int}}(F; z_1^*).$$

Finally, suppose for contradiction that over this class the conditional interventional target were identified by a functional of the replay oracle. Then there would exist a measurable map Φ

such that for every γ ,

$$\mathcal{I}_{1,\nu_\gamma}^{\text{int}}(F; z_1^\star) = \Phi(\mathcal{R}_{\nu_\gamma,1}(z_1^\star)).$$

But the replay oracles agree for ν_α and ν_β , so this would imply

$$\mathcal{I}_{1,\nu_\alpha}^{\text{int}}(F; z_1^\star) = \mathcal{I}_{1,\nu_\beta}^{\text{int}}(F; z_1^\star),$$

contradicting the calculation above. Therefore the target is not identified by any functional of the replay oracle over this class. $\square$

D.5 Proofs for Section 5.6

D.5.1 Proof of Theorem 19

Proof. Fix $t \in \{1, \dots, T\}$, fix a realized prefix $h = z_{1:t} \in \mathcal{H}_t$ with $\mathbb{P}_\nu(h) > 0$, and fix a continuation

$$c = ((x_{t+1}, a_{t+1}, r_{t+1}), \dots, (x_T, a_T, r_T)) \in \mathcal{C}_t.$$

We first prove the density-ratio identity (5.10) for continuations c with $Q_{\nu,t}^0(c | h) > 0$.

By the definition of conditional future law and the factorization (5.8),

$$\begin{aligned} Q_{\nu,t}^\epsilon(c | h) &= \prod_{s=t+1}^T K_{\nu,s}^{\theta_s^{(w(t,\epsilon))}(h, c_{t+1:s-1})}(x_s, a_s, r_s | h, c_{t+1:s-1}) \\ &= \prod_{s=t+1}^T D_s(x_s | h, c_{t+1:s-1}) \pi_{\theta_s^{(w(t,\epsilon))}(h, c_{t+1:s-1})}(a_s | x_s) P_s(r_s | x_s, a_s, h, c_{t+1:s-1}). \end{aligned} \quad (\text{D.2})$$

Similarly, the baseline conditional future law is

$$Q_{\nu,t}^0(c | h) = \prod_{s=t+1}^T D_s(x_s | h, c_{t+1:s-1}) \pi_{\theta_s(h, c_{t+1:s-1})}(a_s | x_s) P_s(r_s | x_s, a_s, h, c_{t+1:s-1}). \quad (\text{D.3})$$

Because $Q_{\nu,t}^0(c | h) > 0$, every factor in (D.3) corresponding to the actually realized action a_s is positive, and therefore the ratio

$$\frac{\pi_{\theta_s^{(w(t,\epsilon))}(h, c_{t+1:s-1})}(a_s | x_s)}{\pi_{\theta_s(h, c_{t+1:s-1})}(a_s | x_s)}$$

is well defined for each $s \in \{t+1, \dots, T\}$.

Dividing (D.2) by (D.3), the context terms D_s and reward terms P_s cancel identically. We obtain

$$\frac{Q_{\nu,t}^\epsilon(c | h)}{Q_{\nu,t}^0(c | h)} = \prod_{s=t+1}^T \frac{\pi_{\theta_s^{(w^{(t,\epsilon)})}(h, c_{t+1:s-1})}(a_s | x_s)}{\pi_{\theta_s}(h, c_{t+1:s-1})(a_s | x_s)} = \Lambda_t^\epsilon(c; h).$$

Multiplying both sides by $Q_{\nu,t}^0(c | h)$ gives

$$Q_{\nu,t}^\epsilon(c | h) = \Lambda_t^\epsilon(c; h) Q_{\nu,t}^0(c | h),$$

which is (5.10).

We now prove the expectation representation (5.12) under the overlap condition (5.11). Fix $\epsilon \in (-\epsilon_0, \epsilon_0)$. By the definition of $\Psi_t^\epsilon(h)$,

$$\Psi_t^\epsilon(h) = \sum_{c \in \mathcal{C}_t} Q_{\nu,t}^\epsilon(c | h) F_\nu \left(\theta_{T+1}^{(w^{(t,\epsilon)})}(h, c) \right).$$

If $Q_{\nu,t}^\epsilon(c | h) > 0$, the overlap condition implies $Q_{\nu,t}^0(c | h) > 0$. Therefore the sum may be taken over continuations with positive baseline conditional probability only:

$$\Psi_t^\epsilon(h) = \sum_{c: Q_{\nu,t}^0(c|h) > 0} Q_{\nu,t}^\epsilon(c | h) F_\nu \left(\theta_{T+1}^{(w^{(t,\epsilon)})}(h, c) \right).$$

Using the already-proved ratio identity,

$$\Psi_t^\epsilon(h) = \sum_{c: Q_{\nu,t}^0(c|h) > 0} \Lambda_t^\epsilon(c; h) Q_{\nu,t}^0(c | h) F_\nu \left(\theta_{T+1}^{(w^{(t,\epsilon)})}(h, c) \right).$$

This is exactly the conditional expectation

$$\mathbb{E}_\nu \left[\Lambda_t^\epsilon(Z_{t+1:T}; h) F_\nu \left(\theta_{T+1}^{(w^{(t,\epsilon)})}(Z_{1:T}) \right) \mid Z_{1:t} = h \right],$$

because under the baseline law,

$$Q_{\nu,t}^0(c | h) = \mathbb{P}_\nu(c | h).$$

This proves (5.12). □

D.5.2 Proof of Corollary 7

Proof. Fix $\epsilon \in (-\epsilon_0, \epsilon_0)$. By Theorem 19,

$$\Psi_t^\epsilon(h) = \mathbb{E}_{\mathbb{P}_\nu} \left[\Lambda_t^\epsilon(Z_{t+1:T}; h) F_\nu \left(\theta_{T+1}^{(w^{(t,\epsilon)})}(Z_{1:T}) \right) \mid Z_{1:t} = h \right].$$

We now show that every quantity inside this conditional expectation is determined by $(\mathbb{P}_\nu, h)$ under the stated assumptions.

First, because the update maps $U_1, \dots, U_T$ are known, the realized prefix h and continuation c determine the replayed learner states

$$\theta_s(h, c_{t+1:s-1}) \quad \text{and} \quad \theta_s^{(w^{(t,\epsilon)})}(h, c_{t+1:s-1})$$

for every $s \in \{t+1, \dots, T+1\}$. Therefore the policy ratio

$$\Lambda_t^\epsilon(c; h)$$

is determined from h, c, ϵ , and the known action law π_θ .

Second, for every continuation c with $Q_{\nu,t}^\epsilon(c \mid h) > 0$, the terminal state

$$\theta_{T+1}^{(w^{(t,\epsilon)})}(h, c)$$

belongs to the reachable set $\mathcal{R}_{\nu,t}^{\epsilon_0}(h)$ by definition. Hence the assumption of the corollary gives

$$F_\nu \left(\theta_{T+1}^{(w^{(t,\epsilon)})}(h, c) \right) = \mathfrak{F} \left(\mathbb{P}_\nu, \theta_{T+1}^{(w^{(t,\epsilon)})}(h, c) \right).$$

Thus the terminal value appearing inside the expectation is also a measurable functional of $(\mathbb{P}_\nu, h, c, \epsilon)$.

It follows that the random variable

$$\Lambda_t^\epsilon(Z_{t+1:T}; h) F_\nu \left(\theta_{T+1}^{(w^{(t,\epsilon)})}(Z_{1:T}) \right)$$

is measurable with respect to $(\mathbb{P}_\nu, h, Z_{t+1:T})$. Since the state and continuation spaces are finite, the conditional expectation given $Z_{1:t} = h$ is itself a measurable functional of $(\mathbb{P}_\nu, h)$. Therefore $\Psi_t^\epsilon(h)$ is identified from the baseline law.

If the derivative at 0 exists, then

$$\mathcal{I}_t^{\text{int}}(F_\nu; h) = \left. \frac{d}{d\epsilon} \Psi_t^\epsilon(h) \right|_{\epsilon=0}$$

is also determined by the baseline-law function $\epsilon \mapsto \Psi_t^\epsilon(h)$, and is therefore identified from the baseline law as well. $\square$

D.5.3 Proof of the Negative Frontier

Proof. We show directly that the target is not identified from the baseline law over the enlarged class by constructing two environments with the exact same baseline law but strictly divergent interventional targets. We first demonstrate this for reward-state dependence. Take horizon $T = 2$. Let the first interaction space be a singleton $\mathcal{Z}_1 = \{z_1^\star\}$, and let the second interaction consist only of a reward, $\mathcal{Z}_2 = \{0, 1\}$. Equivalently, one may view the context and action spaces as singletons and the reward space as $\{0, 1\}$. Let the learner state space be $\Theta = \mathbb{R}$, with initial state $\theta_1 = 0$. Define the first-round kernel to be deterministic:

$$K_{\nu,1}^\theta(z_1^\star \mid \emptyset) = 1$$

Define the learner update maps and terminal target by:

$$U_1(\theta, z_1^\star, w) = w - 1, \quad U_2(\theta, r, 1) = r, \quad F(\theta_3) = \theta_3$$

Therefore, under the perturbation $w^{(1,\epsilon)}$, the learner state after round 1 is exactly the perturbation value:

$$\theta_2^{(w^{(1,\epsilon)})}(z_1^\star) = \epsilon$$

For each parameter $\gamma \in \mathbb{R}$, we define an environment ν_γ by its reward law:

$$P_\gamma^\theta(R_2 = 1 \mid z_1^\star) = \sigma(\gamma\theta), \quad P_\gamma^\theta(R_2 = 0 \mid z_1^\star) = 1 - \sigma(\gamma\theta)$$

where σ is the logistic sigmoid function. Because context and action spaces are singletons, this satisfies the factorization for reward-state dependence. We first compute the baseline law. Under

the baseline process (where $w_1 = 1$), the state entering round 2 is $\theta_2(z_1^*) = 0$. Hence:

$$P_\gamma^0(R_2 = 1 \mid z_1^*) = \sigma(0) = \frac{1}{2}, \quad P_\gamma^0(R_2 = 0 \mid z_1^*) = \frac{1}{2}$$

Since the first round is deterministic, the full baseline law on $\mathcal{H}_2$ is:

$$\mathbb{P}_{\nu_\gamma}(z_1^*, 1) = \frac{1}{2}, \quad \mathbb{P}_{\nu_\gamma}(z_1^*, 0) = \frac{1}{2}$$

This is entirely independent of γ . Thus, for any two environments $\alpha, \beta \in \mathbb{R}$, their baseline laws are identical: $\mathbb{P}_{\nu_\alpha} = \mathbb{P}_{\nu_\beta}$. We now compute the interventional target at the realized prefix z_1^* . Under the perturbation $w^{(1, \epsilon)}$, the state at round 2 is ϵ , so the perturbed future law is:

$$Q_{\nu_\gamma, 1}^\epsilon(1 \mid z_1^*) = \sigma(\gamma\epsilon)$$

Because $U_2(\theta, r, 1) = r$ and $F(\theta_3) = \theta_3$, the terminal target equals the round-2 reward. Therefore, the finite conditional interventional effect is:

$$\Psi_{\nu_\gamma, 1}^\epsilon(z_1^*) = \sigma(\gamma\epsilon)$$

Differentiating at 0 yields the influence:

$$\mathcal{I}_{1, \nu_\gamma}^{\text{int}}(F; z_1^*) = \left. \frac{d}{d\epsilon} \sigma(\gamma\epsilon) \right|_{\epsilon=0} = \frac{\gamma}{4}$$

Hence, for $\alpha \neq \beta$, their interventional targets strictly diverge:

$$\mathcal{I}_{1, \nu_\alpha}^{\text{int}}(F; z_1^*) \neq \mathcal{I}_{1, \nu_\beta}^{\text{int}}(F; z_1^*)$$

If the target were identified from the baseline law over this class, it would have to take the same value on ν_α and ν_β because their baseline laws agree exactly. Since the values are different, this is impossible.

Extension to Context-State Dependence: To prove the exact same impossibility for context-dependent environments, we use the identical mathematical construction but treat the second-round interaction as a context $X_2 \in \{0, 1\}$ rather than a reward. We set $U_2(\theta, x, 1) = x$ and define the context law as $D_\gamma^\theta(X_2 = 1 \mid z_1^*) = \sigma(\gamma\theta)$. The baseline laws again remain perfectly identical across all γ , but the interventional targets diverge exactly as shown above. Therefore, identification fails in both cases. $\square$

D.6 Proofs for section 5.7

D.6.1 Proofs for the Directional Failure of Replay (Theorem 20)

To prove the directional failure of replay in the horizon-2 bandit (Theorem 20), we must first define the update dynamics of the environment, define the intermediate policy states, and derive the exact closed-form equations for both the interventional and replay targets.

Step 1: The Bandit Update Dynamics. The learner updates its policy (the probability p_t of pulling arm 1) using two-arm entropic mirror-descent with an importance-weighted reward estimate. On the logit scale, where $\text{logit}(p) = \log(p/(1-p))$, the perturbed update rule at round t with learning weight w_t is:

$$\text{logit}(p_{t+1}^{(w)}) = \text{logit}(p_t^{(w)}) + \eta_t w_t R_t \left(\frac{\mathbf{1}\{A_t = 1\}}{p_t^{(w)}} - \frac{\mathbf{1}\{A_t = 0\}}{1 - p_t^{(w)}} \right)$$

Step 2: Horizon-2 Setup and the First-Round Derivative. We specialize to horizon $T = 2$ and condition on the realized first interaction $z_1^* := (1, 1)$ (the learner pulled arm 1 and received reward 1). The initial policy is $p_1 = q$. Under the one-coordinate perturbation at round 1 (where $w_1 = 1 + \epsilon$), the intermediate policy entering round 2 is:

$$p_2^\epsilon = \sigma \left(\text{logit}(q) + \eta_1 (1 + \epsilon) \frac{1}{q} \right)$$

where σ is the logistic sigmoid. Differentiating this with respect to ϵ at 0 yields the first-round sensitivity factor:

$$c_{q, \eta_1, p} := \left. \frac{d}{d\epsilon} p_2^\epsilon \right|_{\epsilon=0} = \frac{\eta_1}{q} p(1-p)$$

where $p := p_2^0$ is the baseline intermediate policy.

Step 3: Round-2 Policy Functions. We now express the final policy p_3 as a function of the intermediate policy p . Based on the logit update rule, if the round-2 reward is 0, the policy is unchanged. If the reward is 1, the update depends on the arm pulled. We define these two potential final policy functions:

$$f_{\eta_2}(p) := \sigma \left(\text{logit}(p) + \frac{\eta_2}{p} \right) \quad (\text{if Arm 1 is pulled})$$

$$g_{\eta_2}(p) := \sigma\left(\text{logit}(p) - \frac{\eta_2}{1-p}\right) \quad (\text{if Arm 0 is pulled})$$

Step 4: Formulating the Targets. Using these functions, we define two expectations. The first is the expected final policy after recollecting round 2 under the true environment parameters μ :

$$G_{\mu, \eta_2}(p) := p(\mu_1 f_{\eta_2}(p) + (1 - \mu_1)p) + (1 - p)(\mu_0 g_{\eta_2}(p) + (1 - \mu_0)p)$$

The second is the corresponding conditional expected replay derivative factor (which differentiates the fixed paths rather than the expectations):

$$R_{\mu, \eta_2}(p) := p(\mu_1 f'_{\eta_2}(p) + (1 - \mu_1)) + (1 - p)(\mu_0 g'_{\eta_2}(p) + (1 - \mu_0))$$

Proposition 15 (Two-step exact formulas). *In the horizon-2 bandit conditioned on $Z_1 = z_1^* = (1, 1)$, the targets resolve exactly to:*

$$\mathcal{I}_1^{\text{int}}(F; z_1^*) = c_{q, \eta_1, p} G'_{\mu, \eta_2}(p)$$

$$\mathbb{E}_{\mathbb{P}_\mu}[\mathcal{I}_1^{\text{rep}}(F; Z_{1:2}) \mid Z_1 = z_1^*] = c_{q, \eta_1, p} R_{\mu, \eta_2}(p)$$

Proof. For the interventional target, we evaluate the recollected expectation given the perturbed intermediate policy p_2^ϵ , so $\Psi_1^\epsilon(z_1^*) = G_{\mu, \eta_2}(p_2^\epsilon)$. Applying the chain rule yields $G'_{\mu, \eta_2}(p) \cdot \frac{d}{d\epsilon} p_2^\epsilon|_{\epsilon=0}$, which equals $c_{q, \eta_1, p} G'_{\mu, \eta_2}(p)$.

For the replay target, we evaluate the derivative along each fixed path and then take the baseline expectation. If $(A_2, R_2) = (1, 1)$, the final policy is $f_{\eta_2}(p_2^\epsilon)$, and its derivative at 0 is $c_{q, \eta_1, p} f'_{\eta_2}(p)$. Weighting the derivatives of all four possible round-2 outcomes by their baseline probabilities $(p\mu_1, p(1 - \mu_1), (1 - p)\mu_0, (1 - p)(1 - \mu_0))$ and factoring out $c_{q, \eta_1, p}$ yields exactly $c_{q, \eta_1, p} R_{\mu, \eta_2}(p)$. $\square$

Step 5: Auxiliary Half-Point Lemmas. To prove the sign separation, we need to evaluate these derivatives at the specific intermediate policy $p = 1/2$.

Lemma 32 (Half-point identities). *For every $\eta > 0$, $f'_\eta(1/2) = g'_\eta(1/2) = -\frac{\eta-1}{\cosh^2(\eta)}$.*

Proof. Applying the chain rule to $f_\eta(p)$ and evaluating at $p = 1/2$ (where $\text{logit}(1/2) = 0$ and $1/(p(1-p)) = 4$), we find $f'_\eta(1/2) = f_\eta(1/2)(1 - f_\eta(1/2)) \cdot 4(1 - \eta)$. Using the logistic variance identity $\sigma(x)(1 - \sigma(x)) = 1/(4 \cosh^2(x/2))$ evaluated at $x = 2\eta$, we obtain $-\frac{\eta-1}{\cosh^2(\eta)}$. The proof for g'_η is identical due to symmetry. $\square$

Lemma 33 (Closed forms at the half point). *Suppose $\mu_1 = 1$. Then for every $\eta > 0$:*

$$R_{\mu,\eta}(1/2) = \frac{1}{2} \left[(1 - \mu_0) - (1 + \mu_0) \frac{\eta - 1}{\cosh^2(\eta)} \right]$$

$$G'_{\mu,\eta}(1/2) = (1 + \mu_0)\sigma(2\eta) - \mu_0 - \frac{1 + \mu_0}{2} \frac{\eta - 1}{\cosh^2(\eta)}$$

Proof. Substitute $\mu_1 = 1$ and $p = 1/2$ into the definitions of $R_{\mu,\eta}(p)$ and $G'_{\mu,\eta}(p)$. Applying the identities from Lemma 32 isolates the stated closed-form equations. $\square$

Step 6: Proof of Theorem 20 (Strong Separation).

Proof. We force the intermediate policy to be exactly $p = 1/2$ by setting $q = 1/4$ and $\eta_1 = (\log 3)/4$. This yields a strictly positive first-round sensitivity: $c_{q,\eta_1,p} = (\log 3)/4 > 0$.

We first analyze replay. By Proposition 15 and Lemma 33, the replay factor depends on the bracketed term in $R_{\mu,\eta_2}(1/2)$. Because we fix $\eta_2 > 1$, the term $\frac{\eta_2-1}{\cosh^2(\eta_2)}$ is strictly positive. As $\mu_0 \rightarrow 1$, the bracket approaches $0 - 2\frac{\eta_2-1}{\cosh^2(\eta_2)} < 0$. By continuity, there exists a neighborhood $1 - \delta_1(\eta_2) < \mu_0 < 1$ where the conditional expected replay influence is strictly negative.

We now analyze intervention. The target is proportional to $G'_{\mu,\eta_2}(1/2)$. At the limit $\mu_0 = 1$, this evaluates to $2\sigma(2\eta_2) - 1 - \frac{\eta_2-1}{\cosh^2(\eta_2)}$, which simplifies to $\tanh(\eta_2) - \frac{\eta_2-1}{\cosh^2(\eta_2)}$. Differentiating this expression with respect to η_2 reveals its global minimum over $(0, \infty)$ occurs at $\eta_2 = 1$, where it evaluates to $\tanh(1) > 0$. Therefore, the interventional factor is strictly positive for all $\eta_2 > 1$ at $\mu_0 = 1$. By continuity, there exists a neighborhood $1 - \delta_2(\eta_2) < \mu_0 < 1$ where the conditional interventional influence is strictly positive.

Taking $\delta = \min(\delta_1, \delta_2)$ completes the proof: in this regime, replay evaluates negative while intervention is positive. $\square$

Corollary 18 (A realized sign flip). *Under the hypotheses of Theorem 20, there exists at least one second-round outcome $z_2^\dagger$ with positive baseline probability such that $\mathcal{I}_1^{\text{rep}}(F; z_1^\star, z_2^\dagger) < 0 < \mathcal{I}_1^{\text{int}}(F; z_1^\star)$.*

Proof. By Theorem 20, the conditional expected replay is strictly negative. Because an expectation is a convex combination of its realized path values, it is mathematically impossible for every positive-probability continuation to have a non-negative replay influence. Therefore, at least one realized continuation must have a strictly negative replay influence, while the interventional target remains strictly positive. $\square$

D.6.2 Proofs for the Anatomy of the Gap (Dynamic Programming and Bounds)

To rigorously prove the structural bounds discussed in Section 5.7, we must establish an exact dynamic programming recursion that calculates the interventional target for a known, smooth adaptive model. We assume the learner state space is $\Theta \subseteq \mathbb{R}^d$, and all relevant update maps U_s , kernel masses $K_{\nu,s}$, and the terminal target F_ν are continuously differentiable.

Step 1: Forward State Sensitivity. The perturbation at round t alters the learner’s state, and this alteration ripples forward through time. We define the forward sensitivity of the replayed state with respect to the round- t learning weight as Γ . At the step immediately following the perturbation, this is:

$$\Gamma_{t+1}(z_{1:t}) := \partial_w U_t(\theta_t(z_{1:t-1}), z_t, 1)$$

For all subsequent steps $s = t + 1, \dots, T$, the sensitivity propagates via the Jacobian of the update map:

$$\Gamma_{s+1}(z_{1:s}) := \partial_\theta U_s(\theta_s(z_{1:s-1}), z_s, 1) \Gamma_s(z_{1:s-1})$$

Step 2: Backward Continuation Value. We next define the baseline expected downstream reward from any given state. At the terminal step $T + 1$, this is simply the target function:

$$V_{T+1}(z_{1:T}) := F_\nu(\theta_{T+1}(z_{1:T}))$$

For prior steps $s = T, \dots, t+1$, we define this recursively by taking the expectation over the baseline next-step interaction:

$$V_s(z_{1:s-1}) := \sum_{z_s \in \mathcal{Z}_s} K_{\nu,s}^{\theta_s(z_{1:s-1})}(z_s \mid z_{1:s-1}) V_{s+1}(z_{1:s})$$

Step 3: The Exact Combined Target. We combine the forward sensitivity and backward value to compute the exact interventional target. Define the terminal gradient sequence:

$$G_{T+1}(z_{1:T}) := \nabla F_\nu(\theta_{T+1}(z_{1:T}))^\top \Gamma_{T+1}(z_{1:T})$$

And for $s = T, \dots, t+1$, define the backward recursion:

$$\begin{aligned} G_s(z_{1:s-1}) := & \sum_{z_s \in \mathcal{Z}_s} K_{\nu,s}^{\theta_s(z_{1:s-1})}(z_s \mid z_{1:s-1}) G_{s+1}(z_{1:s}) \\ & + \sum_{z_s \in \mathcal{Z}_s} \left(\nabla_\theta K_{\nu,s}^\theta(z_s \mid z_{1:s-1}) \Big|_{\theta=\theta_s(z_{1:s-1})}^\top \Gamma_s(z_{1:s-1}) \right) V_{s+1}(z_{1:s}). \end{aligned}$$

Theorem 36 (Exact model-based computation). *Under the smoothness assumptions above, the interventional target is exactly: $\mathcal{I}_t^{\text{int}}(F_\nu; h) = G_{t+1}(h)$.*

Proof. Let $\theta_s^\epsilon(g) := \theta_s^{(w^{(t,\epsilon)})}(g)$ be the perturbed replayed state. By induction on the replay dynamics, one can verify that its derivative evaluated at $\epsilon = 0$ is exactly the forward sensitivity $\Gamma_s(g)$. Next, we define the perturbed continuation value $V_s^\epsilon(g)$ by substituting θ^ϵ into the environment kernels and terminal target. By definition, $V_{t+1}^\epsilon(h) = \Psi_t^\epsilon(h)$, meaning the interventional target is exactly $\frac{d}{d\epsilon} V_{t+1}^\epsilon(h) \Big|_{\epsilon=0}$.

Let $H_s(g) := \frac{d}{d\epsilon} V_s^\epsilon(g) \Big|_{\epsilon=0}$. We show by backward induction that $H_s = G_s$. At $T+1$, the chain rule gives $H_{T+1} = \nabla F_\nu^\top \Gamma_{T+1} = G_{T+1}$. For step s , applying the product and chain rules to the recursive definition of V_s^ϵ yields exactly the two-term sum in the definition of $G_s(z_{1:s-1})$. The first term captures the fixed-law value, and the second captures the shift in the interaction law. Thus, $H_{t+1}(h) = G_{t+1}(h)$. $\square$

Interlude: Future-law score form. When the perturbed conditional future law is differentiable and its support is locally stable around $\epsilon = 0$, the future-law correction can also be written

with a score. For histories in the support of the baseline conditional future law given h , define

$$S_{\nu,t}(z_{1:T}) := \sum_{s=t+1}^T \nabla_{\theta} \log K_{\nu,s}^{\theta}(z_s \mid z_{1:s-1}) \Big|_{\theta=\theta_s(z_{1:s-1})}^{\top} \Gamma_s(z_{1:s-1}). \quad (\text{D.4})$$

Setting the score to zero on zero-probability continuations, the replay–intervention gap also admits the centered representation

$$\begin{aligned} \mathcal{I}_t^{\text{int}}(F_{\nu}; h) - \mathbb{E}_{\mathbb{P}_{\nu}}[\mathcal{I}_t^{\text{rep}}(F_{\nu}; Z_{1:T}) \mid Z_{1:t} = h] \\ = \mathbb{E}_{\mathbb{P}_{\nu}}[(F_{\nu}(\theta_{T+1}(Z_{1:T})) - F_{\nu}(\theta_{t+1}(h))) S_{\nu,t}(Z_{1:T}) \mid Z_{1:t} = h]. \end{aligned} \quad (\text{D.5})$$

Step 4: The Stagewise Decomposition. We now isolate the specific contribution of each future round to the total adaptive gap. For each future round $s \in \{t+1, \dots, T\}$, define:

$$\Xi_s(g) := \sum_{z_s \in \mathcal{Z}_s} \left(\nabla_{\theta} K_{\nu,s}^{\theta}(z_s \mid g) \Big|_{\theta=\theta_s(g)}^{\top} \Gamma_s(g) \right) (V_{s+1}(g, z_s) - V_s(g)).$$

Theorem 37 (Stagewise decomposition of the replay–intervention gap). *For every realized prefix $h = z_{1:t}$ with positive baseline probability:*

$$\mathcal{I}_t^{\text{int}}(F_{\nu}; h) - \mathbb{E}_{\mathbb{P}_{\nu}}[\mathcal{I}_t^{\text{rep}}(F_{\nu}; Z_{1:T}) \mid Z_{1:t} = h] = \sum_{s=t+1}^T \mathbb{E}_{\mathbb{P}_{\nu}}[\Xi_s(Z_{1:s-1}) \mid Z_{1:t} = h].$$

Proof. Let $M_s(g) := \mathbb{E}_{\mathbb{P}_{\nu}}[\mathcal{I}_t^{\text{rep}}(F_{\nu}; Z_{1:T}) \mid Z_{1:s-1} = g]$ be the conditional expected replay influence.

At $T+1$, $M_{T+1}(z_{1:T}) = G_{T+1}(z_{1:T})$. For prior steps, $M_s(g) = \sum_{z_s} K_{\nu,s}^{\theta_s(g)}(z_s \mid g) M_{s+1}(g, z_s)$.

Define the gap-to-go $D_s(g) := G_s(g) - M_s(g)$. Subtracting the recursion for M_s from G_s and using the identity $\sum_{z_s} \nabla_{\theta} K_{\nu,s}^{\theta}(z_s \mid g)^{\top} \Gamma_s(g) = 0$ to center the value term yields:

$$D_s(g) = \sum_{z_s \in \mathcal{Z}_s} K_{\nu,s}^{\theta_s(g)}(z_s \mid g) D_{s+1}(g, z_s) + \Xi_s(g).$$

Unrolling this recursion from $s = T+1$ (where $D_{T+1} = 0$) down to $t+1$ yields $D_{t+1}(h) = \mathbb{E}_{\mathbb{P}_{\nu}}[\sum_{s=t+1}^T \Xi_s \mid Z_{1:t} = h]$, proving the theorem. $\square$

Step 5: Quantitative Bounds. Theorem 37 mathematically formalizes the ”three gears” discussed in the main text. A future round only contributes to the gap if the forward sensitivity Γ_s is non-zero (model propagation), the kernel gradient $\nabla_{\theta} K$ is non-zero (environment sensitivity), and the downstream values V_{s+1} vary across outcomes (value oscillation). We can bound this

by defining local metrics for the environment's total variation sensitivity $L_s^{\text{TV}}(g)$ and the value oscillation $\text{osc}_s(g)$:

$$L_s^{\text{TV}}(g) := \sup_{\|u\|=1} \frac{1}{2} \sum_{z_s \in \mathcal{Z}_s} \left| \nabla_{\theta} K_{\nu,s}^{\theta}(z_s \mid g) \Big|_{\theta=\theta_s(g)}^{\top} u \right|,$$

$$\text{osc}_s(g) := \max_{z_s} V_{s+1}(g, z_s) - \min_{z_s} V_{s+1}(g, z_s).$$

Corollary 19 (General oscillation bound).

$$\begin{aligned} & \left| \mathcal{I}_t^{\text{int}}(F_{\nu}; h) - \mathbb{E}_{\mathbb{P}_{\nu}}[\mathcal{I}_t^{\text{rep}}(F_{\nu}; Z_{1:T}) \mid Z_{1:t} = h] \right| \\ & \leq \sum_{s=t+1}^T \mathbb{E}_{\mathbb{P}_{\nu}} \left[L_s^{\text{TV}}(Z_{1:s-1}) \|\Gamma_s(Z_{1:s-1})\| \text{osc}_s(Z_{1:s-1}) \mid Z_{1:t} = h \right]. \end{aligned}$$

Proof. Centering V_{s+1} around its midpoint $m_g = (\max b_g + \min b_g)/2$ ensures $|V_{s+1} - m_g| \leq \frac{1}{2} \text{osc}_s(g)$. Because the kernel gradients sum to zero, shifting by a constant does not change the sum. Factoring out $\text{osc}_s(g)$ and the norm $\|\Gamma_s(g)\|$ leaves the definition of $L_s^{\text{TV}}(g)$. Applying the triangle inequality to Theorem 37 yields the bound. $\square$

Corollary 20 (Uniform propagation bound). *Assume deterministic bounds: the initial parameter shift $\|\partial_w U_t\| \leq \bar{B}_t$, the update operator norm $\|\partial_{\theta} U_u\|_{\text{op}} \leq \bar{\rho}_u$, the environment sensitivity $L_s^{\text{TV}} \leq \bar{L}_s$, and the value oscillation $\text{osc}_s \leq \bar{\Delta}_s$. Then the gap is bounded by:*

$$\bar{B}_t \sum_{s=t+1}^T \bar{L}_s \bar{\Delta}_s \prod_{u=t+1}^{s-1} \bar{\rho}_u.$$

Proof. By the recursive definition of Γ_s in Step 1, taking norms gives $\|\Gamma_{s+1}\| \leq \bar{\rho}_s \|\Gamma_s\|$. Iterating this from $t+1$ yields $\|\Gamma_s\| \leq \bar{B}_t \prod_{u=t+1}^{s-1} \bar{\rho}_u$. Substituting this and the remaining deterministic bounds into Corollary 19 allows us to drop the conditional expectation, yielding the stated uniform bound. $\square$

D.6.3 Proofs for the Stable Small-Step Regime (Theorem 21)

To formally prove that a small learning rate mathematically neutralizes the adaptive gap (Theorem 21), we must bound the model propagation and environment sensitivity terms specifically for the two-arm Bernoulli bandit.

Step 1: Replay Sensitivity on a Fixed Log. Fix a full realized log $z_{1:T}$, where each interaction is written as $z_s = (a_s, r_s)$. Let $\mathcal{U}_s(p, a, r, w)$ be the one-step policy update map that takes the current probability p , the action a , the reward r , and the learning weight w , and outputs the next step's policy via the entropic mirror-descent update defined in Equation 5.4.

We define the replay sensitivity of the policy at any future time $s > t$ with respect to the round- t perturbation as $D_{s,t}$:

$$D_{s,t}(z_{1:T}) := \left. \frac{d}{d\epsilon} p_s^{(w^{(t,\epsilon)})}(z_{1:T}) \right|_{\epsilon=0}.$$

Because the replayed policy up to time t is independent of the perturbation, $D_{s,t} = 0$ for all $s \leq t$. At $t+1$, the sensitivity is the direct derivative of the update map: $D_{t+1,t}(z_{1:T}) = \partial_w \mathcal{U}_t(p_t(z_{1:T}), a_t, r_t, 1)$. For all subsequent steps $s > t$, the sensitivity propagates via the chain rule:

$$D_{s+1,t}(z_{1:T}) = \partial_p \mathcal{U}_s(p_s(z_{1:T}), a_s, r_s, 1) D_{s,t}(z_{1:T}).$$

Step 2: The Bandit Future-Law Score. We must translate the general environment score $S_{\nu,t}$ into the specific mechanics of the bandit. In this model, the context and reward distributions do not depend on the policy. The only term dependent on the learner's state is the action probability $\pi_p(a_s)$.

Evaluating the log-derivative $\frac{\partial}{\partial p} \log \pi_p(a_s)$ yields $1/p$ if $a_s = 1$ and $-1/(1-p)$ if $a_s = 0$. Both cases simplify elegantly to $\frac{a_s - p}{p(1-p)}$. Substituting this and our replay sensitivity $D_{s,t}$ into the general score formula from Equation D.4 yields the exact bandit score:

$$S_{\mu,t}(z_{1:T}) = \sum_{s=t+1}^T \frac{a_s - p_s(z_{1:T})}{p_s(z_{1:T})(1 - p_s(z_{1:T}))} D_{s,t}(z_{1:T})$$

Step 3: Bounding Propagation on the Logit Scale. We now impose the stable small-step conditions: the baseline policy is strictly bounded away from the edges by some constant $c \in (0, 1/2)$ such that $c \leq p_s \leq 1 - c$, and the learning rates are bounded such that $\eta_s \leq \frac{c}{1-c}$.

To track propagation cleanly, we evaluate the sensitivity on the logit scale:

$$X_{s,t} := \left. \frac{d}{d\epsilon} \text{logit}(p_s^\epsilon) \right|_{\epsilon=0}.$$

Because $p_s = \sigma(x_s)$, the chain rule dictates that $D_{s,t} = p_s(1 - p_s)X_{s,t}$.

At the perturbed round, the direct logit update derivative gives

$$X_{t+1,t} = \eta_t r_t \left(\frac{\mathbf{1}\{a_t = 1\}}{p_t} - \frac{\mathbf{1}\{a_t = 0\}}{1 - p_t} \right).$$

Because $r_t \in \{0, 1\}$ and $p_t \geq c$, we can strictly bound the initial shock:

$$|X_{t+1,t}| \leq \eta_t / c.$$

For all subsequent rounds $s > t$, if $r_s = 0$, the update is inactive and $X_{s+1,t} = X_{s,t}$. If $r_s = 1$, differentiating the logit update reveals that the new sensitivity is scaled by a contraction factor. For example, if $a_s = 1$, $X_{s+1,t} = \left(1 - \eta_s \frac{1-p_s}{p_s}\right) X_{s,t}$. Under our stable regime bounds, $0 \leq \eta_s \frac{1-p_s}{p_s} \leq \frac{c}{1-c} \cdot \frac{1-c}{c} = 1$. Because the scaling factor is bounded between 0 and 1, the magnitude of the sensitivity never grows: $|X_{s+1,t}| \leq |X_{s,t}|$.

By induction, $|X_{s,t}| \leq \eta_t / c$ for all future steps. Converting back from the logit scale using the maximum variance $p_s(1 - p_s) \leq 1/4$, we obtain a uniform bound on the model propagation:

$$|D_{s,t}| \leq \frac{\eta_t}{4c} \quad \text{for all } s > t$$

Step 4: Proof of Theorem 21 (Replay in a stable regime).

Proof. We synthesize the bounds to find the maximum gap. First, we bound the future-law score.

Since $\left| \frac{a_s - p_s}{p_s(1-p_s)} \right| \leq \frac{1}{c}$, substituting our uniform bound for $D_{s,t}$ into the score equation yields:

$$|S_{\mu,t}(z_{1:T})| \leq \sum_{s=t+1}^T \frac{1}{c} \cdot \frac{\eta_t}{4c} = \frac{T-t}{4c^2} \eta_t$$

Next, we bound the value oscillation, which depends on the total possible movement of the baseline policy. Differentiating the update map $\mathcal{U}_u$ with respect to the learning weight reveals a maximum one-step policy shift of $\eta_u / (4c)$. Summing this over all remaining steps bounds the total trajectory divergence:

$$|p_{T+1}(z_{1:T}) - p_{t+1}(h)| \leq \frac{1}{4c} \sum_{u=t+1}^T \eta_u$$

Assuming the target function F has a bounded derivative L_F over the interval $[c, 1-c]$, the maximum oscillation in the terminal target is bounded by L_F times this trajectory divergence.

Finally, we apply the centered covariance identity (Equation D.5). The gap between the interventional target and replay is the expected product of the target oscillation and the score. Taking the product of our absolute bounds yields the final, deterministic limit:

$$|\mathcal{I}_t^{\text{int}}(F; h) - \mathbb{E}_{\mathbb{P}_\mu}[\mathcal{I}_t^{\text{rep}}(F; Z_{1:T}) \mid Z_{1:t} = h]| \leq \frac{L_F(T-t)}{16c^3} \eta_t \sum_{u=t+1}^T \eta_u$$

Because this gap is bounded by the product of the initial learning rate η_t and the sum of future learning rates $\sum \eta_u$, the total error is strictly $\mathcal{O}(\eta^2)$. $\square$

D.7 Proofs for Section 5.8

D.7.1 Proof of Theorem 22

Proof. Fix $t \in \{1, \dots, T\}$, fix a realized prefix $h = z_{1:t} \in \mathcal{H}_t$ with $\mathbb{P}_\nu(h) > 0$, and fix $L \in \{0, \dots, T-t\}$. Write

$$m := t + L.$$

For notational convenience, whenever g is a prefix of length at least $u-1$, write

$$\theta_u^\epsilon(g) := \theta_u^{(w^{(t,\epsilon)})}(g).$$

Step 1: A backward recursion for the depth- L target. For each $s \in \{t+1, \dots, T+1\}$ and each prefix $g = z_{1:s-1} \in \mathcal{H}_{s-1}$, define recursively

$$W_{T+1}^{\epsilon,L}(z_{1:T}) := F_\nu(\theta_{T+1}^\epsilon(z_{1:T})),$$

and for $s = T, T-1, \dots, t+1$,

$$W_s^{\epsilon,L}(g) := \begin{cases} \sum_{z_s \in \mathcal{Z}_s} K_{\nu,s}^{\theta_s^\epsilon(g)}(z_s \mid g) W_{s+1}^{\epsilon,L}(g, z_s), & s \leq m, \\ \sum_{z_s \in \mathcal{Z}_s} K_{\nu,s}^{\theta_s^\epsilon(g)}(z_s \mid g) W_{s+1}^{\epsilon,L}(g, z_s), & s \geq m+1. \end{cases} \quad (\text{D.6})$$

Unrolling the recursion shows that

$$W_{t+1}^{\epsilon,L}(h) = \Psi_t^{\epsilon,\text{tr},L}(h).$$

Indeed, for the first L future rounds the recursion uses the perturbed kernels appearing in (5.13), and for the remaining future rounds it uses the baseline kernels appearing in (5.13); the terminal quantity is exactly the perturbed replay value in (5.14).

Step 2: Baseline reduction. We claim that for every $s \in \{t+1, \dots, T+1\}$ and every prefix $g \in \mathcal{H}_{s-1}$,

$$W_s^{0,L}(g) = V_s(g),$$

where V_s is the baseline continuation-value recursion introduced earlier.

This is immediate by backward induction on s . At $s = T+1$,

$$W_{T+1}^{0,L}(z_{1:T}) = F_\nu(\theta_{T+1}(z_{1:T})) = V_{T+1}(z_{1:T}).$$

Now suppose $W_{s+1}^{0,L} = V_{s+1}$. Since $\theta_s^0(g) = \theta_s(g)$, both branches of (D.6) reduce to

$$W_s^{0,L}(g) = \sum_{z_s \in \mathcal{Z}_s} K_{\nu,s}^{\theta_s(g)}(z_s \mid g) V_{s+1}(g, z_s) = V_s(g).$$

So the claim follows.

Step 3: Differentiate the mixed recursion. Define

$$H_s^L(g) := \left. \frac{d}{d\epsilon} W_s^{\epsilon,L}(g) \right|_{\epsilon=0}.$$

Because all interaction spaces are finite and the update maps, kernel masses, and target are continuously differentiable, all derivatives below may be passed through finite sums.

We first record the derivative of the replayed learner state. Exactly as in the forward-sensitivity induction from Theorem 36,

$$\left. \frac{d}{d\epsilon} \theta_s^\epsilon(g) \right|_{\epsilon=0} = \Gamma_s(g) \quad \text{for every } s \in \{t+1, \dots, T+1\}. \quad (\text{D.7})$$

At the terminal step,

$$H_{T+1}^L(z_{1:T}) = \nabla F_\nu(\theta_{T+1}(z_{1:T}))^\top \Gamma_{T+1}(z_{1:T}) = G_{T+1}(z_{1:T}).$$

Now fix $s \in \{t+1, \dots, T\}$.

If $s \geq m+1$, then the kernel in (D.6) is frozen at the baseline law and does not depend on ϵ .

Therefore

$$H_s^L(g) = \sum_{z_s \in \mathcal{Z}_s} K_{\nu,s}^{\theta_s(g)}(z_s | g) H_{s+1}^L(g, z_s). \quad (\text{D.8})$$

If $s \leq m$, then the kernel depends on ϵ through the perturbed learner state. By the product rule,

$$\begin{aligned} H_s^L(g) &= \sum_{z_s \in \mathcal{Z}_s} \frac{d}{d\epsilon} \left[K_{\nu,s}^{\theta_s^\epsilon(g)}(z_s | g) W_{s+1}^{\epsilon,L}(g, z_s) \right] \Big|_{\epsilon=0} \\ &= \sum_{z_s \in \mathcal{Z}_s} \frac{d}{d\epsilon} K_{\nu,s}^{\theta_s^\epsilon(g)}(z_s | g) \Big|_{\epsilon=0} W_{s+1}^{0,L}(g, z_s) + \sum_{z_s \in \mathcal{Z}_s} K_{\nu,s}^{\theta_s(g)}(z_s | g) H_{s+1}^L(g, z_s). \end{aligned} \quad (\text{D.9})$$

Using (D.7), the chain rule, and Step 2,

$$\frac{d}{d\epsilon} K_{\nu,s}^{\theta_s^\epsilon(g)}(z_s | g) \Big|_{\epsilon=0} = \nabla_\theta K_{\nu,s}^\theta(z_s | g) \Big|_{\theta=\theta_s(g)}^\top \Gamma_s(g), \quad W_{s+1}^{0,L}(g, z_s) = V_{s+1}(g, z_s).$$

So (D.9) becomes

$$\begin{aligned} H_s^L(g) &= \sum_{z_s \in \mathcal{Z}_s} K_{\nu,s}^{\theta_s(g)}(z_s | g) H_{s+1}^L(g, z_s) \\ &\quad + \sum_{z_s \in \mathcal{Z}_s} \left(\nabla_\theta K_{\nu,s}^\theta(z_s | g) \Big|_{\theta=\theta_s(g)}^\top \Gamma_s(g) \right) V_{s+1}(g, z_s). \end{aligned} \quad (\text{D.10})$$

Step 4: After the switch, the mixed recursion coincides with replay. Recall the replay-side backward recursion

$$M_{T+1}(z_{1:T}) := G_{T+1}(z_{1:T}), \quad M_s(g) := \sum_{z_s \in \mathcal{Z}_s} K_{\nu,s}^{\theta_s(g)}(z_s | g) M_{s+1}(g, z_s),$$

introduced in the proof of Theorem 37. Whenever $\mathbb{P}_\nu(g) > 0$, this quantity equals

$$M_s(g) = \mathbb{E}_{\mathbb{P}_\nu}[\mathcal{I}_t^{\text{rep}}(F_\nu; Z_{1:T}) \mid Z_{1:s-1} = g].$$

Comparing this recursion with (D.8), together with the common terminal condition $H_{T+1}^L = M_{T+1} = G_{T+1}$, shows by backward induction that

$$H_s^L(g) = M_s(g) \quad \text{for every } s \in \{m+1, \dots, T+1\}. \quad (\text{D.11})$$

If $L = 0$, then $m = t$, so (D.11) already yields

$$\mathcal{I}_t^{\text{tr},0}(F_\nu; h) = H_{t+1}^0(h) = M_{t+1}(h) = \mathbb{E}_{\mathbb{P}_\nu}[\mathcal{I}_t^{\text{rep}}(F_\nu; Z_{1:T}) \mid Z_{1:t} = h].$$

This is exactly (5.17) with an empty sum. The tail identity (5.18) then follows immediately from Theorem 37. It remains to treat the case $L \geq 1$.

Step 5: Before the switch, the remaining gap is a truncated stagewise sum. Assume now that $L \geq 1$, so $m \geq t + 1$. For $s \in \{t + 1, \dots, m\}$, define

$$E_s^L(g) := H_s^L(g) - M_s(g).$$

By (D.11),

$$E_{m+1}^L(g) = 0 \quad \text{for every } g \in \mathcal{H}_m.$$

Subtracting the replay recursion for M_s from (D.10) yields

$$\begin{aligned} E_s^L(g) &= \sum_{z_s \in \mathcal{Z}_s} K_{\nu,s}^{\theta_s(g)}(z_s \mid g) E_{s+1}^L(g, z_s) \\ &\quad + \sum_{z_s \in \mathcal{Z}_s} \left(\nabla_\theta K_{\nu,s}^\theta(z_s \mid g) \Big|_{\theta=\theta_s(g)}^\top \Gamma_s(g) \right) V_{s+1}(g, z_s). \end{aligned} \quad (\text{D.12})$$

We now center the second sum. Since $K_{\nu,s}^\theta(\cdot \mid g)$ is a probability distribution for every θ , we have

$$\sum_{z_s \in \mathcal{Z}_s} K_{\nu,s}^\theta(z_s \mid g) = 1.$$

Differentiating with respect to θ and evaluating at $\theta = \theta_s(g)$ gives

$$\sum_{z_s \in \mathcal{Z}_s} \nabla_\theta K_{\nu,s}^\theta(z_s \mid g) \Big|_{\theta=\theta_s(g)} = 0.$$

Therefore

$$\sum_{z_s \in \mathcal{Z}_s} \left(\nabla_\theta K_{\nu,s}^\theta(z_s \mid g) \Big|_{\theta=\theta_s(g)}^\top \Gamma_s(g) \right) V_s(g) = 0.$$

Subtracting this zero term from the second line of (D.12) yields

$$\sum_{z_s \in \mathcal{Z}_s} \left(\nabla_\theta K_{\nu,s}^\theta(z_s \mid g) \Big|_{\theta=\theta_s(g)}^\top \Gamma_s(g) \right) V_{s+1}(g, z_s) = \Xi_s(g),$$

where $\Xi_s(g)$ is exactly the stagewise quantity defined in (5.16). Hence, for every $s \in \{t+1, \dots, m\}$,

$$E_s^L(g) = \sum_{z_s \in \mathcal{Z}_s} K_{\nu,s}^{\theta_s(g)}(z_s | g) E_{s+1}^L(g, z_s) + \Xi_s(g), \quad E_{m+1}^L(g) = 0. \quad (\text{D.13})$$

Step 6: Unroll the truncated gap recursion. We claim that whenever $\mathbb{P}_\nu(g) > 0$,

$$E_s^L(g) = \sum_{u=s}^m \mathbb{E}_{\mathbb{P}_\nu}[\Xi_u(Z_{1:u-1}) | Z_{1:s-1} = g] \quad \text{for every } s \in \{t+1, \dots, m\}. \quad (\text{D.14})$$

We prove this by backward induction on s .

For $s = m$, the recursion (D.13) and the boundary condition $E_{m+1}^L = 0$ give

$$E_m^L(g) = \Xi_m(g),$$

which is exactly (D.14).

Now suppose (D.14) holds at step $s+1$. Using (D.13),

$$\begin{aligned} E_s^L(g) &= \Xi_s(g) + \sum_{z_s \in \mathcal{Z}_s} K_{\nu,s}^{\theta_s(g)}(z_s | g) E_{s+1}^L(g, z_s) \\ &= \Xi_s(g) + \sum_{z_s \in \mathcal{Z}_s} K_{\nu,s}^{\theta_s(g)}(z_s | g) \sum_{u=s+1}^m \mathbb{E}_{\mathbb{P}_\nu}[\Xi_u(Z_{1:u-1}) | Z_{1:s} = (g, z_s)]. \end{aligned}$$

By the tower property of conditional expectation, this equals

$$\Xi_s(g) + \sum_{u=s+1}^m \mathbb{E}_{\mathbb{P}_\nu}[\Xi_u(Z_{1:u-1}) | Z_{1:s-1} = g].$$

Since $\Xi_s(g)$ is measurable with respect to the sigma-field generated by $Z_{1:s-1}$,

$$\Xi_s(g) = \mathbb{E}_{\mathbb{P}_\nu}[\Xi_s(Z_{1:s-1}) | Z_{1:s-1} = g].$$

Thus

$$E_s^L(g) = \sum_{u=s}^m \mathbb{E}_{\mathbb{P}_\nu}[\Xi_u(Z_{1:u-1}) | Z_{1:s-1} = g],$$

which proves (D.14).

Applying (D.14) at $s = t+1$ and $g = h$ gives

$$E_{t+1}^L(h) = \sum_{u=t+1}^{t+L} \mathbb{E}_{\mathbb{P}_\nu}[\Xi_u(Z_{1:u-1}) | Z_{1:t} = h]. \quad (\text{D.15})$$

Step 7: Identify the truncated influence. By construction,

$$\mathcal{I}_t^{\text{tr},L}(F_\nu; h) = \left. \frac{d}{d\epsilon} \Psi_t^{\epsilon, \text{tr}, L}(h) \right|_{\epsilon=0} = H_{t+1}^L(h).$$

Using $H_{t+1}^L(h) = M_{t+1}(h) + E_{t+1}^L(h)$ together with (D.15), we obtain

$$\mathcal{I}_t^{\text{tr},L}(F_\nu; h) = M_{t+1}(h) + \sum_{u=t+1}^{t+L} \mathbb{E}_{\mathbb{P}_\nu}[\Xi_u(Z_{1:u-1}) \mid Z_{1:t} = h].$$

Because $\mathbb{P}_\nu(h) > 0$, the replay recursion satisfies

$$M_{t+1}(h) = \mathbb{E}_{\mathbb{P}_\nu}[\mathcal{I}_t^{\text{rep}}(F_\nu; Z_{1:T}) \mid Z_{1:t} = h].$$

This proves (5.17).

Finally, Theorem 37 gives

$$\mathcal{I}_t^{\text{int}}(F_\nu; h) = \mathbb{E}_{\mathbb{P}_\nu}[\mathcal{I}_t^{\text{rep}}(F_\nu; Z_{1:T}) \mid Z_{1:t} = h] + \sum_{u=t+1}^T \mathbb{E}_{\mathbb{P}_\nu}[\Xi_u(Z_{1:u-1}) \mid Z_{1:t} = h].$$

Subtracting (5.17) yields (5.18). This completes the proof. $\square$

D.7.2 Proof of Corollary 8

Proof. By (5.18),

$$\mathcal{I}_t^{\text{int}}(F_\nu; h) - \mathcal{I}_t^{\text{tr},L}(F_\nu; h) = \sum_{s=t+L+1}^T \mathbb{E}_{\mathbb{P}_\nu}[\Xi_s(Z_{1:s-1}) \mid Z_{1:t} = h].$$

Therefore, by the triangle inequality,

$$\left| \mathcal{I}_t^{\text{int}}(F_\nu; h) - \mathcal{I}_t^{\text{tr},L}(F_\nu; h) \right| \leq \sum_{s=t+L+1}^T \mathbb{E}_{\mathbb{P}_\nu}[|\Xi_s(Z_{1:s-1})| \mid Z_{1:t} = h]. \quad (\text{D.16})$$

We now bound $|\Xi_s(g)|$ pointwise for a fixed prefix $g = z_{1:s-1}$.

If $\Gamma_s(g) = 0$, then $\Xi_s(g) = 0$, so the desired bound is immediate. Assume therefore that $\Gamma_s(g) \neq 0$, and set

$$u_g := \frac{\Gamma_s(g)}{\|\Gamma_s(g)\|}.$$

Define the midpoint of the downstream values by

$$m_g := \frac{1}{2} \left(\max_{z_s \in \mathcal{Z}_s} V_{s+1}(g, z_s) + \min_{z_s \in \mathcal{Z}_s} V_{s+1}(g, z_s) \right).$$

Then

$$|V_{s+1}(g, z_s) - m_g| \leq \frac{1}{2} \text{osc}_s(g) \quad \text{for every } z_s \in \mathcal{Z}_s.$$

As in the proof above, the kernel gradients sum to zero:

$$\sum_{z_s \in \mathcal{Z}_s} \nabla_{\theta} K_{\nu, s}^{\theta}(z_s | g) \Big|_{\theta=\theta_s(g)} = 0.$$

Hence

$$\Xi_s(g) = \sum_{z_s \in \mathcal{Z}_s} \left(\nabla_{\theta} K_{\nu, s}^{\theta}(z_s | g) \Big|_{\theta=\theta_s(g)}^{\top} \Gamma_s(g) \right) \left(V_{s+1}(g, z_s) - m_g \right).$$

Taking absolute values and using the bound above,

$$\begin{aligned} |\Xi_s(g)| &\leq \frac{1}{2} \text{osc}_s(g) \sum_{z_s \in \mathcal{Z}_s} \left| \nabla_{\theta} K_{\nu, s}^{\theta}(z_s | g) \Big|_{\theta=\theta_s(g)}^{\top} \Gamma_s(g) \right| \\ &= \frac{1}{2} \text{osc}_s(g) \|\Gamma_s(g)\| \sum_{z_s \in \mathcal{Z}_s} \left| \nabla_{\theta} K_{\nu, s}^{\theta}(z_s | g) \Big|_{\theta=\theta_s(g)}^{\top} u_g \right| \\ &\leq L_s^{\text{TV}}(g) \|\Gamma_s(g)\| \text{osc}_s(g), \end{aligned}$$

where the last step is exactly the definition of $L_s^{\text{TV}}(g)$.

Substituting this pointwise bound into (D.16) yields (5.19).

For the uniform bound, the forward-sensitivity recursion gives

$$\|\Gamma_{t+1}(z_{1:t})\| \leq \bar{B}_t, \quad \|\Gamma_{s+1}(z_{1:s})\| \leq \bar{\rho}_s \|\Gamma_s(z_{1:s-1})\| \quad \text{for } s = t+1, \dots, T.$$

Iterating this recursion yields

$$\|\Gamma_s(z_{1:s-1})\| \leq \bar{B}_t \prod_{u=t+1}^{s-1} \bar{\rho}_u \quad \text{for every } s \in \{t+1, \dots, T\}.$$

Combining this with the deterministic bounds $L_s^{\text{TV}} \leq \bar{L}_s$ and $\text{osc}_s \leq \bar{\Delta}_s$ inside (5.19) gives

$$\left| \mathcal{I}_t^{\text{int}}(F_{\nu}; h) - \mathcal{I}_t^{\text{tr}, L}(F_{\nu}; h) \right| \leq \bar{B}_t \sum_{s=t+L+1}^T \bar{L}_s \bar{\Delta}_s \prod_{u=t+1}^{s-1} \bar{\rho}_u,$$

which is (5.20). The final tolerance claim is immediate. $\square$